MYP MATHEMATICS 1 · STUDENT COURSEBOOK

Mathematics 1

A concept-driven MYP Year 1 coursebook: investigate first, then reason, communicate and apply.

INVESTIGATE | REASON | COMMUNICATE | APPLY | REFLECT

HOW THIS BOOK WORKS

Every chapter opens with an investigation. You meet the mathematics through a task, discover a pattern, state a conclusion and only then meet the formal explanation.

SECOND EDITION

MYP YEAR 1 · CHAPTERS 1–10

MYP 1 · STUDENT COURSEBOOK

About this edition

This is the second edition of the MYP 1 mathematics coursebook. It was rebuilt from the first-edition MYP Mathematics 1 Student Coursebook, which remains the principal content reference and is preserved unchanged in the project archive.

The edition keeps the first edition's chapter scope and its visual identity, and rebuilds the teaching sequence around inquiry: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, extensive visual practice, and application, assessment and reflection.

All classroom data sets are simulated and are labelled as such. Real data would require provenance, so no claim in this book rests on unverified statistics.

No publisher, ISBN, print specification or commercial print-readiness claim is made in this file. The book is produced as an A4 review copy: fonts are embedded, pagination is continuous and every page has been checked for overflow.

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How to use this book

Each chapter follows the same sequence, so you always know what is coming next. The sequence is deliberate: you investigate before you are told the rule, because a rule you have discovered is a rule you can defend.

The chapter sequence

1. **Engagement.** A short situation that raises the question the chapter will answer.
2. **Inquiry questions.** Factual, conceptual and debatable questions that frame the chapter.
3. **Investigation.** A task you complete before the formal teaching: record evidence in a table and look for a pattern.
4. **Observation and discovery.** What you noticed, stated in your own words and compared with a partner.
5. **Mathematical conclusion.** The rule that the evidence supports, written as a general statement.
6. **Formal development.** The explanation, definitions and notation, with the reasoning made explicit.
7. **Worked examples.** Problem, visual representation, mathematical reasoning, step-by-step solution, final answer and interpretation.
8. **Practice.** Separated into Criterion A, B, C and D practice, with visual representations throughout.
9. **Application, assessment and reflection.** Real-context tasks, criterion-based assessment practice and a review of your progress.

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How to use this book

The four MYP mathematics criteria

A · Knowing and understanding

Apply knowledge to familiar and unfamiliar problems, and check results.

B · Investigating patterns

Investigate, describe patterns, generalise and justify — recognising, describing and verifying patterns at Year 1 level rather than producing formal proofs.

C · Communicating

Use correct mathematical language, notation and representations, and reason coherently.

D · Applying mathematics in real-life contexts

Apply mathematics to authentic situations, identify relevant information and comment on the accuracy and sense of the result.

Working with this book

- Practice is separated by criterion so that you can see which skill a task is training. Criterion B practice is additional genuine pattern investigation, not a repeat of the teaching.
- Visuals are part of the mathematics. Where a task supplies a diagram, grid, table, number line or model, use it; where it asks you to sketch, your drawing is part of the answer.
- Answers and student-facing reasoning are collected at the back of the book, with formula bank, glossary and references.
- Every chapter ends with project connections: a Service as Action opportunity, a Community Project connection and a Personal Project connection for MYP 5.

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MYP mathematics and this coursebook

The IB Middle Years Programme asks students to inquire into concepts through global contexts, to develop approaches to learning (ATL) skills, and to act on what they learn through service as action. In mathematics this means more than practising procedures: students investigate patterns, justify generalisations, communicate with precise language and apply mathematics to situations that matter to them.

MYP mathematics is organised into four branches — number, algebra, geometry and trigonometry, and statistics and probability — and the programme recommends at least 50 hours of teaching per subject each year. This coursebook covers the Year 1 content of those branches across ten chapters, grouped into five parts.

Assessment in MYP mathematics uses four equally weighted criteria, each with a maximum of 8: A knowing and understanding, B investigating patterns, C communicating, and D applying mathematics in real-life contexts. This book trains each criterion explicitly, and says so on the page where the training happens.

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MYP mathematics and this coursebook

Projects, service and action

Service as Action

Each chapter includes a service opportunity with a community need, a mathematical connection, a proposed action, an expected impact and a student reflection. Completing an exercise is not service: service requires action that benefits others and reflection on that action.

Community Project

MYP students completing the programme in Year 3 or Year 4 complete a community project. Each chapter offers a student-led project idea with a research question, the mathematics involved, an action or outcome, beneficiaries and evidence. Follow your school's requirements and supervision.

Personal Project (MYP 5)

Students completing MYP in Year 5 complete a personal project. Each chapter offers an adaptable MYP 5-oriented extension with a personal interest, a learning goal, a product, the mathematics involved, a process and evidence. It is deliberately different from the community project and from service as action.

What this book does not claim

- Programme requirements quoted here follow the official IB MYP curriculum, mathematics and projects pages and the MYP general regulations. Everything else — the chapter sequence, the investigations, the practice sets and the project ideas — is author-designed material built to serve those requirements, and is not an IB publication.
- No ISBN, publisher, imprint or print specification is claimed. The file is an A4 review copy with embedded fonts and continuous pagination, produced for proofing and classroom review rather than for commercial print production.

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Every chapter follows the same sequence: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, Criterion A–D practice, application, assessment and reflection, and project connections.

Number systems and place value

Numbers are read through their places, and every number belongs to a family. This chapter builds the number sense the rest of the book relies on: place value, equivalent representations, comparison, rounding and sensible accuracy.

INQUIRY QUESTIONS

- **Factual:** Which number families does a number such as −3, 0 or 0.25 belong to, and what makes each family different?
- **Conceptual:** Why does the position of a digit change its value, and why can a zero be important even when it adds nothing?
- **Debatable:** When is a rounded figure honest, and when does it hide something a reader needs to know?

01

Number systems and place value

Ten digits are enough to write any number we can name. Their power comes from position: the same symbol can mean seven, seventy or seven thousandths.

ENGAGE

A museum label says “about 49,000 objects”, while an inventory says “48,672 objects”. Which number would you put on a poster? Which would you use to check the collection? What changes when we round?

Framing the inquiry

Key concept: Form
Related concepts: Quantity, Representation
Global context: Orientation in space and time

Statement of inquiry: The position of a digit in a base-ten system encodes its value, so a small set of symbols can represent any magnitude when place and zero are used consistently.

Questions to guide us

- **Factual:** What is the value of a digit in a given position?
- **Conceptual:** Why is zero essential, and how can one number have several equivalent representations?
- **Debatable:** When is “about 2,000” more useful than the exact count?

ATL · CRITICAL THINKING. In order for a student to evaluate and improve a mathematical claim, students must test the claim on several cases, find a counterexample if one exists, and revise the statement so it is accurate.

GETTING READY

Learning intentions and prerequisite check

You are learning to

- Understand how natural numbers, whole numbers, integers and rational numbers are related.
- Know that the value of a digit depends on its place in a base-ten number.
- Write whole numbers and decimals in expanded form and return to standard form.
- Compare and order integers and decimals using place value and a number line.
- Round to a stated place and choose a reasonable level of accuracy in context.
- Estimate a result and explain whether the estimate is likely to be too high or too low.

Success criteria

- [ ] I can classify a number and explain the role of zero.
- [ ] I can state the value of any digit in a whole number or decimal.
- [ ] I can move between standard form, expanded form and a number line.
- [ ] I can order positive and negative numbers and justify the order.
- [ ] I can round to a stated place and avoid an over-precise answer.
- [ ] I can estimate, check exactly, and explain the difference.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

1.R1 Write “four thousand and sixty” using digits. READY

1.R2 Which is greater, 0.5 or 0.45? Explain using hundredths. READY

1.R3 Calculate 8×7. READY

1.R4 What is 100 more than 2,950? READY

1.R5 Write 7/10 as a decimal. READY

1.R6 Put −2, 0, 3 and −5 in order from least to greatest. READY

> HOW TO USE THIS CHAPTER
>
> Work through the examples before the practice. Use the diagrams to explain your thinking. Answers and student-facing reasoning are at the back of this proof. If more than two ready-check items are incorrect, review place value to hundreds and simple fractions before continuing.

CHAPTER 1 · INVESTIGATION · BEFORE 1.1

Investigate · Which numbers belong together?

YOUR TASK

Sort these numbers into the families below. A number can belong to more than one family, so tick every column that applies: 7, −3, 0, 3/4, 0.25.

NUMBER	NATURAL	WHOLE	INTEGER	RATIONAL
7				
−3				
0				
3/4				
0.25				

What do you notice?

- Which number belongs to every family except the natural numbers, and what does that tell you about zero?
- Which family contains all of the others, and how can you see that on a number line?

Observation. Natural numbers count from 1. Whole numbers add 0. Integers add the negatives. Rational numbers include every fraction and every terminating or repeating decimal, so 3/4 and 0.25 are both rational.

MATHEMATICAL CONCLUSION

Each family sits inside the next one, and a number line gives every number a position. Values increase from left to right and 0 is the boundary between negative and positive.

Counting numbers, zero and negatives

Natural numbers are the counting numbers $1, 2, 3, \ldots$. In this book, **whole numbers** are the natural numbers together with zero: $0, 1, 2, 3, \ldots$. **Integers** include the whole numbers and their negatives: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$.

A number line gives every number a position. Values increase from left to right. Zero is the boundary between negative and positive numbers; it is neither positive nor negative.



Values increase from left to right. The point 2.5 has a precise position even though it is not an integer.

WORKED EXAMPLE 1

Classify -4, 0 and 12.

1. -4 is an integer and a rational number, but not whole or natural.
2. 0 is whole, an integer and rational, but not natural in this book.
3. 12 is natural, whole, an integer and rational.

Each number can belong to more than one family.

CRITERION A Knowing and understanding

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

1.1.1 State every family each number belongs to: natural, whole, integer, rational. (a) 12 (b) 0 (c) −4 (d) 5/8 (e) 1.2 FLUENCY

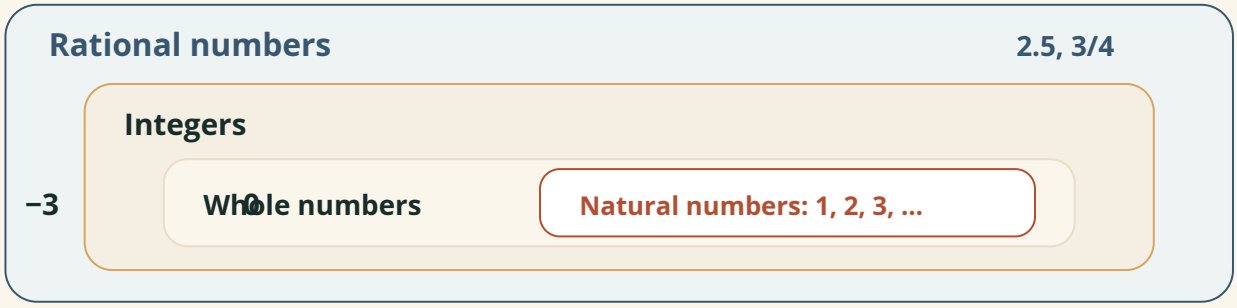
1.1.2 Write all the integers from −3 to 2 inclusive. FLUENCY

1.1.3 Replace each box with <, > or =. (a) −7 ☐ −2 (b) 0 ☐ −1 (c) −5 ☐ 5 FLUENCY

1.1.4 Order these temperatures from coldest to warmest: 3 °C, −2 °C, −7 °C, 0 °C, 5 °C. FLUENCY

Rational numbers and number families

A **rational number** can be written as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$. Terminating decimals such as 1.2 and fractions such as $\frac{5}{8}$ are rational. Every integer is rational because, for example, $-4 = \frac{-4}{1}$.



Natural numbers sit inside whole numbers; whole numbers sit inside integers; integers sit inside rational numbers, which also include fractions and terminating decimals.

MISCONCEPTION CHECK

"Zero is not a number because it means nothing." Zero is a number with a precise position. It also holds a place: 305 is not the same as 35.

WORKED EXAMPLE 2

Order $-7, -2, 0, 3, 5$ from least to greatest.

1. On a number line, more negative values lie further left.
2. Read from left to right: $-7, -2, 0, 3, 5$.

Coldest to warmest: $-7\,^\circ\text{C}, -2\,^\circ\text{C}, 0\,^\circ\text{C}, 3\,^\circ\text{C}, 5\,^\circ\text{C}$.

CRITERION A Knowing and understanding

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

1.1.5 Maya says, "Zero is a natural number because we can count zero items." Using this book's definition, explain whether her statement is correct. REASONING

1.1.6 Give a counterexample to the claim: "Every fraction is less than 1." REASONING

1.1.7 A lift starts at floor 1, travels to floor −2, then to floor 3. Describe each movement and the final position relative to the start. APPLICATION

1.1.8 Draw a number line from −5 to 5. Mark −4, 0 and 2.5. Explain why 2.5 is not an integer but still has a precise place. CHALLENGE

CHAPTER 1 · INVESTIGATION · BEFORE 1.2

Investigate · What does a zero do?

YOUR TASK

For each pair, state the value of the highlighted zero and describe the job it is doing.

PAIR	VALUE OF THE ZERO	JOB IT IS DOING
305 and 35		
0.4 and 0.40		
4.05 and 4.5		
40 and 400		
0.305 and 0.35		

What do you notice?

- In which pairs does the zero change the value, and in which does it not?
- What single rule explains the difference between the two kinds of zero?

Observation. A zero can hold a place (in 305 it holds the tens place) or mark an empty position. Adding a zero to the right of a whole number multiplies it by 10; adding a zero after the last decimal digit leaves the value unchanged, because $0.4 = 0.40$.

MATHEMATICAL CONCLUSION

Position gives a digit its value, so a zero is never simply “nothing”: it is an instruction about place.

Position gives a digit its value

In a base-ten system, each position is worth ten times the position to its right. The decimal point separates whole-number places from fractional places. A zero can be a **placeholder**: it keeps the other digits in the correct positions.

Thousands	Hundreds	Tens	Ones	Tenths	Hundredths	Thousandths
3	0	4	8	3	0	7

decimal point

The highlighted 4 is in the tens place, so it is worth 40. The zeros in the hundreds and hundredths places are placeholders.

WORKED EXAMPLE 3

State the value of the 4 in $3{,}048.307$ and calculate 36.2×10.

1. The 4 is in the tens place, so its value is 40.
2. Multiplying by 10 shifts every digit one place left: $36.2 \times 10 = 362$.

The decimal point does not "move" by itself; the digits shift relative to it.

CRITERION A Knowing and understanding

PRACTICE 1.2 · PLACE VALUE AND ZERO

1.2.1 State the value of the underlined digit. (a) 6,7<u>4</u>2 (b) 23.5<u>8</u> (c) <u>8</u>,106 (d) 0.2<u>0</u>4 FLUENCY

1.2.2 Write the number represented by 4,000 + 600 + 30 + 2. FLUENCY

1.2.3 Calculate. (a) 36.2 × 10 (b) 5.48 × 100 (c) 920 ÷ 10 (d) 7.3 ÷ 100 FLUENCY

1.2.4 In 2.06, what does the zero do? What number results if the zero is removed, and how much larger is it? REASONING

Zero, decimals and digit cards

Adding a zero to the right of a whole number multiplies it by 10, but adding a zero after a decimal may leave the value unchanged: $0.4 = 0.40$. Multiplying or dividing by powers of ten shifts digits, not just the decimal point.



Use each card once. A four-digit number cannot begin with 0.

Use each card once. A four-digit number cannot begin with 0.

WORKED EXAMPLE 4

Use 5, 0, 8 and 2 once each to make the greatest and least four-digit numbers.

1. Greatest: put the largest digits in the highest places: $8{,}520$.
2. Least: the first digit cannot be 0, so start with 2, then use the smallest remaining digits: $2{,}058$.

Greatest $8{,}520$; **least** $2{,}058$.

CRITERION A Knowing and understanding

PRACTICE 1.2 · PLACE VALUE AND ZERO

1.2.5 Which is larger, 3.5 or 3.05? Explain using place value, not the number of digits. REASONING

1.2.6 A warehouse receives 24 boxes with 100 pencils in each box. Write the total and state the value of the digit 2 in your answer. APPLICATION

1.2.7 Test the claim “adding a zero to the right of a number multiplies it by 10” on 4, 0.4 and 4.50. Then write a corrected rule. REASONING

1.2.8 Using 5, 0, 8 and 2 exactly once each, make the greatest and least four-digit numbers. Explain why the least cannot begin with 0. CHALLENGE

CHAPTER 1 · INVESTIGATION · BEFORE 1.3

Investigate · One number, many forms

YOUR TASK

Write 4,506.07 in expanded form and in words. Then rebuild a number from its parts: 3×100 + 0×10 + 7×1 + 5×0.1 + 2×0.01.

FORM	EXAMPLE
Standard form	4,506.07
Expanded form	
Word form	
From parts	3×100 + 0×10 + 7×1 + 5×0.1 + 2×0.01 =

What do you notice?

- Which form makes the value of each digit easiest to see?
- Why does the 0×10 term matter even though it adds nothing?

Observation. Expanded form writes a number as the sum of the values of its digits. The zero terms show which places are empty, so nothing is lost when the number is rebuilt: 3×100 + 0×10 + 7×1 + 5×0.1 + 2×0.01 = 307.52.

MATHEMATICAL CONCLUSION

Equivalent representations have the same value but make different features visible. Choose the form that shows the feature you need.

One number, several useful forms

Expanded form writes a number as the sum of the values of its digits. It makes the contribution of each place visible. Standard form is the usual compact way of writing the number.

WORKED EXAMPLE 5

Write 30.78 in expanded form, then return to standard form.

1. Identify each place: 3 tens, 0 ones, 7 tenths, 8 hundredths.
2. Expanded form: $30 + 0.7 + 0.08$.
3. Adding the parts gives 30.78 again.

$30.78 = 30 + 0.7 + 0.08 = 3 \times 10 + 7 \times 0.1 + 8 \times 0.01$.

WHY ZERO STILL MATTERS

$407.3 = 400 + 7 + 0.3$. The zero tens do not add value, but they keep the 4 in the hundreds place and the 7 in the ones place.

CRITERION A Knowing and understanding

PRACTICE 1.3 · EXPANDED FORM

1.3.1 Write in expanded form. (a) 2,506 (b) 30.78 (c) 0.409 FLUENCY

1.3.2 Write as a standard number. (a) $4 \times 100 + 9 \times 10 + 6 \times 1$ (b) $6 \times 1 + 5 \times 0.1 + 8 \times 0.01$ (c) $2 \times 1{,}000 + 7 \times 10 + 3 \times 0.001$ FLUENCY

1.3.3 Fill the missing value. (a) 8,000 + ☐ + 40 + 5 = 8,745 (b) 0.6 + 0.07 + ☐ = 0.679 FLUENCY

1.3.4 Explain why $2 \times 0.1 + 5 \times 0.01$ is ten times $2 \times 0.01 + 5 \times 0.001$. REASONING

Explain the structure, not just the answer

Equivalent representations have the same value but can make different features visible. A place-value chart shows positions; expanded form shows contributions; a number line shows order and distance.

WORKED EXAMPLE 6

Explain why $2 \times 0.1 + 5 \times 0.01$ is ten times $2 \times 0.01 + 5 \times 0.001$.

1. First expression: $0.2 + 0.05 = 0.25$.
2. Second expression: $0.02 + 0.005 = 0.025$.
3. Each digit has moved one place to the left, so each contribution is ten times as large.

$0.25 = 10 \times 0.025$.

CRITERION A Knowing and understanding

PRACTICE 1.3 · EXPANDED FORM

1.3.5 Student A writes 407.3 as 400 + 7 + 0.3. Student B writes 4 × 100 + 0 × 10 + 7 × 1 + 3 × 0.1. Whose form shows the tens place more clearly? Do they have the same value? REASONING

1.3.6 A label gives a length as 5 × 10 + 3 × 1 + 4 × 0.1 centimetres. Write the length in standard form and name the unit of each term. APPLICATION

1.3.7 Create a four-digit number with 6 in the hundreds place and 0 in the tens place. Swap those two digits and explain how the value changes. CHALLENGE

1.3.8 Why can 1,000 be written with one nonzero term in expanded form while its place-value chart still has four positions? REASONING

COMMUNICATE

When you explain a representation, name the place, state its value, and say what the representation makes easy to see.

CHAPTER 1 · INVESTIGATION · BEFORE 1.4

Investigate · Compare, then round

YOUR TASK

Compare each pair, then round the first number to the place stated.

PAIR	LARGER VALUE	ROUNDED VALUE
0.7 and 0.07		
2.35 and 2.4		
−3 and −5		
3.47 to the nearest tenth		
0.995 to two decimal places		

What do you notice?

- What do you compare first when the whole parts are equal?
- Why does 0.995 round to 1.00 rather than to 0.99?

Observation. Align the decimal points and compare from the highest place downwards. For negative numbers the order reverses. Rounding looks at the digit immediately to the right of the target place, and a carry can travel left through a run of 9s.

MATHEMATICAL CONCLUSION

A rounded number is a statement about a range, not an exact value, so always state the place or unit you used.

Compare place by place

To compare decimals, align the decimal points and compare digits from the highest place downwards. Add trailing zeros if helpful: $0.5 = 0.50$. For negative numbers, the number further left on the number line is smaller.



Many decimals lie between 3.5 and 3.6. The zoomed line shows 3.501 and 3.512.

WORKED EXAMPLE 7

Order $0.7, 0.07, 0.77, 0.707$ from least to greatest.

1. Write with three decimal places: $0.700, 0.070, 0.770, 0.707$.
2. Compare: $0.070 < 0.700 < 0.707 < 0.770$.

$0.07, 0.7, 0.707, 0.77$.

CRITERION A Knowing and understanding

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.1 Insert <, > or =. (a) 0.5 ☐ 0.50 (b) 2.7 ☐ 2.68 (c) −3 ☐ −5 FLUENCY

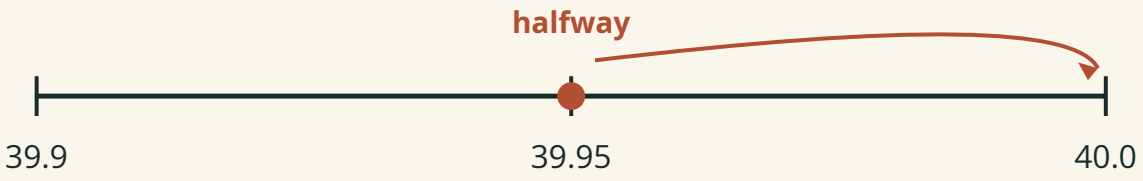
1.4.2 Order from least to greatest: 0.7, 0.07, 0.77, 0.707. FLUENCY

MISCONCEPTION CHECK

"More digits means a larger decimal." Not always: $6.09 < 6.1$ because $6.1 = 6.10$.

Round to a stated place

To round, identify the target place and look at the digit immediately to its right. If that digit is 5 or more, round up; otherwise round down. This book uses the usual convention that a 5 in the deciding place rounds up.



Rounding 39.95 to the nearest tenth gives 40.0 because the hundredths digit is 5 and the carry reaches the whole-number part.

WORKED EXAMPLE 8

Round 59.96 to the nearest tenth.

1. The target is the tenths place: 9. The deciding digit is 6.
2. Round 9 tenths up to 10 tenths, which is one whole.
3. The result is 60.0. The trailing zero records tenths.

$59.96 \approx 60.0$ **to the nearest tenth.**

CRITERION A Knowing and understanding

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.3 Round 8,647 to the nearest (a) hundred (b) ten. FLUENCY

1.4.4 Round 3.746 to the nearest hundredth and 0.897 to the nearest tenth. FLUENCY

1.4.5 Round 59.96 to the nearest tenth. Explain why the whole-number part changes. REASONING

Rounding in context

A rounded number is not the exact value; it is a statement about a range. Always state the place or unit used. In accountability contexts, give the exact figure as well as any rounded headline.

WORKED EXAMPLE 9

Give three decimals that round to 6.5 to the nearest tenth, then describe all possible values.

1. Examples: 6.45, 6.50, 6.549.
2. The lower boundary is 6.45 because the deciding digit is 5.
3. The upper boundary is just below 6.55, because 6.55 would round to 6.6.

All values from 6.45 inclusive up to but not including 6.55.

CRITERION A Knowing and understanding

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.6 One sign says a trail is 4.6 km; another says 4.56 km. Explain how both could be correct. REASONING

1.4.7 Give three decimals that round to 6.5 to the nearest tenth. Describe all possible values under the usual "5 rounds up" convention. CHALLENGE

1.4.8 A class raised 1,948 CFA francs and reports "about 2,000 CFA francs". Is this reasonable? What should an accountability report include? APPLICATION

CHAPTER 1 · INVESTIGATION · BEFORE 1.5

Investigate · Estimate first, then check

YOUR TASK

Estimate each calculation using friendly numbers, then calculate exactly and find the difference.

CALCULATION	ESTIMATE	EXACT	DIFFERENCE
48 × 31			
5.9 + 7.2			
3.87 × 4.1			
198 ÷ 9.8			

What do you notice?

- Which estimates were close enough to catch a mistake?
- What happens to the error when you round one number up and the other down?

Observation. An estimate uses friendly numbers to predict a result. Rounding one factor up and another down can cancel the errors, so the estimate can be closer than either rounding alone: 48 × 31 ≈ 50 × 30 = 1,500 and the exact value is 1,488.

MATHEMATICAL CONCLUSION

Estimate first, then check: the estimate tells you whether the exact answer is reasonable before you trust it.

Estimate first, then check

An **estimate** uses friendly numbers to predict a result or check whether an exact answer is reasonable. State the rounding you used. When one number is rounded up and another down, the estimate may be high or low; an exact check tells you which.

WORKED EXAMPLE 10

Estimate 58×31, then calculate exactly.

1. Round to the nearest ten: $60 \times 30 = 1{,}800$.
2. Exactly: $58 \times 31 = 1{,}798$.
3. The estimate is 2 high.

Estimate 1,800; exact 1,798.

CRITERION A Knowing and understanding

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

1.5.1 Estimate 692 + 3,141 by rounding each number to the nearest hundred. Then calculate exactly and compare. FLUENCY

1.5.2 Estimate 58 × 31 by rounding each factor to the nearest ten. Then calculate exactly and compare. FLUENCY

1.5.3 Estimate 9.8 × 4.2 using 10 × 4. Calculate the exact product. Is the estimate above or below? Can you tell from the rounding alone? REASONING

1.5.4 Estimate 604 ÷ 29 using 600 ÷ 30. Then compare with the exact quotient to one decimal place. FLUENCY

REASONING HABIT

Ask: “What is this estimate for?” A quick headline, a budget check and an inventory may need different accuracy.

Choose accuracy for a purpose

Extra digits are not automatically more truthful. A calculation can produce many decimal places even when the original measurement was only accurate to the nearest centimetre. Report a level of accuracy that the context can support.

WORKED EXAMPLE 11

A library has 1,247,936 books. Suggest a figure for a short public notice.

1. A headline does not need the exact count.
2. “About 1,250,000 books” to the nearest ten thousand is readable.
3. An inventory or audit should use 1,247,936 exactly.

Match the accuracy to the purpose, and state the rounding used.

CRITERION A Knowing and understanding

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

1.5.5 A shop records 397 visitors on Friday, 412 on Saturday and 289 on Sunday. Estimate the total to the nearest hundred, then calculate it exactly. APPLICATION

1.5.6 A runner completes 9.95 km and 10.08 km on two days. A poster says “about 20 km”. Is this reasonable? What accuracy is being used? REASONING

1.5.7 A library has 1,247,936 books. Suggest a rounded figure for a short public notice and state its accuracy. When should the exact figure be used? CHALLENGE

1.5.8 A calculation gives a length as 2.673913043 m. The measuring tape reads to the nearest centimetre. Suggest a sensible report and explain why the extra digits should not be presented as measured accuracy. CHALLENGE

CHECKPOINT

Before the application

Answer quickly, then explain one answer to a partner. Use the answers at the back to check both the result and the reasoning.

CHECKPOINT · BEFORE THE APPLICATION

Answer quickly, then explain one answer to a partner.

1.CK1 What is the value of the 7 in 0.0705? CHECK

1.CK2 Calculate 4.06×100. CHECK

1.CK3 Round 39.95 to the nearest tenth. CHECK

1.CK4 Estimate 49×21 using friendly numbers. CHECK

Common slips

- Forgetting a placeholder zero: 305 is not 35.
- Comparing decimals by counting digits.
- Rounding 39.95 to 39.9 instead of 40.0.
- Assuming an estimate must be high or low without checking.

Explain how you know

- Name the place before stating a digit's value.
- Use a number line to justify an order.
- State the target place before rounding.
- Compare an estimate with an exact result when possible.

ASSESSMENT PRACTICE · CRITERIA A AND C

A museum label

Simulated fact sheet. A museum collection has 48,672 objects, with 6,509 on display. The museum opened in 1894 and recorded 214,830 visitors last year.

CRITERION A · KNOWING AND UNDERSTANDING

Use accurate place value, expanded form and rounding.

CRITERION C · COMMUNICATING

Write for visitors aged 10–14 and adults who want a quick overview. Use mathematical language, forms and a clear structure.

ASSESSMENT PRACTICE · A MUSEUM LABEL

Criteria A and C. Use accurate mathematics and clear communication. Write for visitors aged 10–14 and adults who want a quick overview.

1.S1 The fact sheet gives 48,672 objects in the collection and 6,509 on display. State the value of the 8 in 48,672 and the 6 in 6,509. ASSESSMENT

1.S2 Write 6,509 in expanded form. ASSESSMENT

1.S3 Round 48,672 to the nearest thousand and 214,830 annual visitors to the nearest ten thousand. ASSESSMENT

1.S4 Write a label of no more than 70 words using the collection size, display count, opening year 1894 and annual visitors. Include one rounded value and one comparison. ASSESSMENT

1.S5 Add a two-sentence note explaining one rounding choice and one risk of rounding too heavily. ASSESSMENT

BEFORE YOU FINISH

Check that every rounded value states its accuracy, that the opening year is exact, and that your comparison is supported by the numbers.

CHAPTER REVIEW

What you should now be able to do

Key ideas

- Number families are nested: natural ⊂ whole ⊂ integer ⊂ rational.
- Place value gives each digit a value; zero can hold a place.
- Expanded form shows each digit's contribution.
- Compare decimals place by place, not by counting digits.
- Rounding records a range and a chosen accuracy.
- Estimates are useful when checked and matched to purpose.

Vocabulary

natural number · whole number · integer · rational number · place value · placeholder · decimal point · expanded form · standard form · number line · rounding · estimate · accuracy

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

1.M1 Write the value of the digit 8 in (a) 3,812 (b) 20.83 (c) 0.081. FLUENCY

1.M2 Write 5,040.207 in expanded form. FLUENCY

1.M3 Write $7 \times 1{,}000 + 3 \times 10 + 6 \times 0.01$ as a decimal. FLUENCY

1.M4 Round 64,782 to the nearest thousand, hundred and ten. FLUENCY

1.M5 Round 0.496 to the nearest hundredth and 12.95 to the nearest tenth. FLUENCY

1.M6 Calculate (a) 0.64×100 (b) $8.2 \div 10$ (c) $3.5 \times 1{,}000$. FLUENCY

Fluency and reasoning

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

1.M7 Order from least to greatest: −2.5, 0.25, −0.5, 2.05. FLUENCY

1.M8 Insert <, > or =. (a) 0.08 ☐ 0.8 (b) 1.30 ☐ 1.3 (c) −6 ☐ −2 FLUENCY

1.M9 A whole number rounds to 4,700 to the nearest hundred. Give three possible values and describe the full range. REASONING

1.M10 A student says 6.09 is larger than 6.1 because it has more digits. Correct the comparison. REASONING

1.M11 A temperature changes from −3 °C to 4 °C. Which is colder, and by how many degrees do they differ? REASONING

1.M12 A stock list gives 12,480 items. The manager says “about 12,000”. Is this an underestimate or overestimate, and by how much? Suggest a more informative rounded statement. APPLICATION

EXPLAIN HOW YOU KNOW

Choose one answer and write a short justification using place value, a number line or an exact check.

Challenge and reflect

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

1.M13 Explain the different value of the 5 in 4,507 and 457. REASONING

1.M14 Estimate 3,912 + 4,087 + 2,998 to the nearest thousand. Is the estimate above or below the exact total? Check. REASONING

1.M15 A number is $6 \times 100 + 4 \times 10 + \square \times 1$. It is even and lies between 646 and 650. Find the missing digit and explain why it is unique. CHALLENGE

1.M16 Why do 2.60 and 2.6 have the same value, but 2.60 may be a more appropriate measurement report? REASONING

1.M17 Write a number between 3.5 and 3.6 with three decimal places. Then write a second number in the same interval closer to 3.5. Justify the order. CHALLENGE

1.M18 Find four-digit numbers whose digits sum to 12, whose thousands digit is 3, and whose tens digit is twice the ones digit. Find all possibilities. CHALLENGE

Reflection

- ☐ I can explain the role of zero with an example and a counterexample.
- ☐ I can move between standard form, expanded form and a number line.
- ☐ I can round and state the range of values a rounded number could represent.
- ☐ I can choose an accuracy for a real purpose and justify it.

One idea I would explain differently next time: _____

One question I still have: _____

CHAPTER 1 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · HOW MANY NUMBERS CAN THE DIGITS WRITE?

1.B1 Using exactly n digits, with repetition allowed and no leading zero, complete the count for n = 1, 2, 3 and 4. INVESTIGATION

1.B2 Describe how the total changes when one more digit is allowed. INVESTIGATION

1.B3 State a rule for n digits and test it for n = 5. INVESTIGATION

1.B4 Explain why the first digit has 9 choices but each later digit has 10. INVESTIGATION

1.B5 Challenge: repeat the investigation in base 5 using digits 0–4. Record three cases and state a rule. CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

1.B6 How many positive whole numbers have exactly 4 digits? Exactly 5 digits? Predict a rule for exactly n digits, test your rule on n = 1 and n = 2, then justify it. INVESTIGATION

1.B7 A number is rounded to 6.5 to the nearest tenth. What is the smallest value it could have? What is the largest? Use a number line to justify both endpoints. INVESTIGATION

CHAPTER 1 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

1.C1 A classmate says “0 is not a number because it means nothing.” Write a reply that uses place value and a number line, and include one example where a zero changes a value. REASONING

1.C2 Explain to a younger student why 0.5 and 0.45 are different sizes, using hundredths and a number line. REASONING

1.C3 Write the steps you would use to check that 3,048.307 has been read correctly, and state the value of each digit 3. REASONING

1.C4 A museum label says “about 50,000 objects”. List three different exact totals that make this label correct to the nearest ten thousand, and explain the full range. REASONING

CHAPTER 1 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · A SCHOOL NOTICEBOARD

1.D1 The draft figures are: enrolment 1,284; library books 3,680; year opened 1998; annual visits 10,473; scholarship places 12. Recommend a form for each and justify your choice. APPLICATION

1.D2 Another school has enrolment 1,327. If both enrolments are rounded to the nearest hundred, what does a reader see? Is the comparison fair? Suggest a better accuracy. APPLICATION

1.D3 What must the school clarify before publishing “10,473 visits”? APPLICATION

1.D4 Identify one figure that should not be rounded and one that is most sensitive to rounding. Explain both choices. APPLICATION

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

1.D5 A school noticeboard reports “3,000 students took part” (simulated data). The exact count was 3,142. Write a one-sentence version that stays readable for a public notice and state the place you rounded to. APPLICATION

1.D6 A water tank holds 1,250 L and a display shows 1.3 kL. Estimate how many full 25 L jerrycans can be filled, then check with an exact calculation. APPLICATION

CHAPTER 1 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can name the number families a number belongs to and place it on a number line.
- [ ] I can read and write place value, including the job a zero does.
- [ ] I can compare, order and round decimals to a stated place.
- [ ] I can estimate first and judge whether an answer is accurate enough for its purpose.

Reflect

- Which representation helped you most — number line, place-value chart or expanded form — and why?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS.

CHAPTER 1 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Learners in a feeder primary school hesitate when reading large numbers and decimals aloud.
MATHEMATICAL CONNECTION	Place value, the role of zero, expanded form and rounding (sections 1.1–1.5).
PROPOSED ACTION	Plan and run a 30-minute "reading numbers" session using place-value cards and a noticeboard sorting task; keep a log of the misconceptions you meet.
EXPECTED IMPACT	Younger learners can read a number with its places named and can question a rounded figure.
STUDENT REFLECTION	Record which explanation worked, how you checked understanding, and what you would change next time.

Community Project connection

PROJECT IDEA	A "numbers you can trust" audit of public notices around the school and local market.
COMMUNITY NEED	Public notices often round without saying so, so readers cannot judge the size of the error.
RESEARCH QUESTION	How often do local notices give the exact figure alongside a rounded one, and when does the rounding mislead?
MATHEMATICS	Place value, rounding ranges and estimation.
ACTION AND OUTCOME	Collect 20 notices, classify each as exact or rounded, and produce a one-page guide for the school office.
BENEFICIARIES	School office staff, families and local shopkeepers.
EVIDENCE	Notice log, classified data set, final guide, photographs of the notices.

Personal Project connection · MYP 5

PERSONAL INTEREST	Data journalism and how reported figures are chosen.
LEARNING GOAL	To apply rounding and estimation critically to real reports.
PRODUCT OR OUTCOME	A short article or poster that checks whether three reported figures are stated to a sensible accuracy.
MATHEMATICS	Place value, rounding ranges, estimation and percentage error.
PROCESS	Collect reports, classify each figure, recompute the range it implies, and evaluate the claim.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Powers, expanded form and order of operations

Powers are a compact way to record repeated multiplication, and they explain why place value, square and cube roots, and the order of operations all behave the way they do.

INQUIRY QUESTIONS

- **Factual:** What does an exponent record, and what do 7^1 and 7^0 mean?
- **Conceptual:** Why does place value work as a system of powers of ten, and why do squares, cubes and roots reverse each other?
- **Debatable:** When does a mathematical model such as “each fold doubles the thickness” stop describing a real object?

02

Powers, expanded form and order of operations

A power is a compressed instruction: it records how many copies of a factor to multiply. A shared order of operations keeps that compression unambiguous.

ENGAGE

A storage label says “10 TB”. A computer reports the same space as “about 9.09 TiB”. Has storage disappeared? And when someone writes $2 + 3 \times 4$, what must we agree before the expression has one meaning?

Framing the inquiry

Key concept: Form
Related concepts: Simplification, Quantity
Global context: Scientific and technical innovation

Statement of inquiry: Compressing repeated multiplication into a power, and agreeing on the order in which operations apply, allows complex calculations to be recorded compactly and reproduced reliably.

Questions to guide us

- **Factual:** What does 5^3 mean? Which operation comes first in $6 + 2 \times 3$?
- **Conceptual:** Why does an exponent of 1 leave the base unchanged? How can 2^{10} and 10^3 be compared fairly?
- **Debatable:** Should a calculator decide the meaning of an expression, or should the writer make the meaning visible?

ATL · TRANSFER. In order for a student to transfer a rule to a new situation, students must check whether the structure is the same, apply the rule carefully, and test the result with an example or counterexample.

GETTING READY

Learning intentions and prerequisite check

You are learning to

- Understand a power as repeated multiplication and identify its base and exponent.
- Use powers of ten in expanded form and return to standard form.
- Recognise squares and cubes, and find square roots and cube roots.
- Apply the order of operations, including brackets, indices and roots.
- Substitute values into expressions and formulas containing powers.
- Compare quantities written as powers and justify the comparison.

Success criteria

- [ ] I can expand and evaluate a power without reading the exponent as a multiplier.
- [ ] I can explain why $a^1 = a$ and, for $a \neq 0$, $a^0 = 1$.
- [ ] I can move between standard form and expanded form with powers of ten.
- [ ] I can reverse squaring and cubing with roots.
- [ ] I can show one operation per line and respect brackets.
- [ ] I can explain how brackets change the base of a power.

READY CHECK

Try these without a calculator. They check multiplication, place value and brackets.

2.R1 Calculate 6 × 7 and 9 × 9. READY

2.R2 Write 4,000 + 500 + 30 + 6 as a single number. READY

2.R3 Calculate 12 + 3 × 4. READY

2.R4 Write 3 × 3 × 3 as a product and evaluate it. READY

2.R5 What is the area of a square with side 8 cm? READY

2.R6 Insert brackets so that 2 + 4 × 3 has the value 18. READY

> HOW TO USE THIS CHAPTER
>
> Try the guided practice immediately after the matching example. Use the diagrams to explain what the notation means. Answers and student-facing reasoning are at the back of this proof. If more than two ready-check items are incorrect, review multiplication facts and brackets before continuing.

CHAPTER 2 · INVESTIGATION · BEFORE 2.1

Investigate · How many factors?

YOUR TASK

Write each power as repeated multiplication, then evaluate it. Continue the table to 2^7.

POWER	REPEATED MULTIPLICATION	VALUE
2^1		
2^2		
2^3		
2^4		
2^5		
7^1		
5^0		

What do you notice?

- What pattern connects each row to the row above it?
- Why is 5^0 defined to be 1 rather than 0?

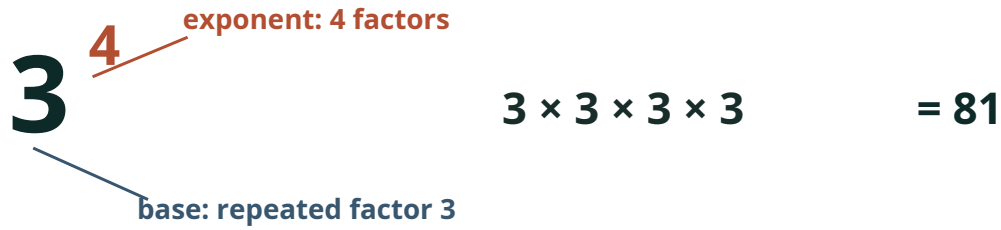
Observation. The exponent counts how many times the base is used as a factor. Each step down divides by the base, so continuing that pattern below 2^1 gives 2^0 = 1; the same argument gives 5^0 = 1 for any nonzero base.

MATHEMATICAL CONCLUSION

Index notation records two things: the base (the repeated factor) and the exponent (the number of copies). An exponent of 1 means one factor, and a nonzero base to the power 0 is defined to be 1.

The exponent counts factors

Writing $3 \times 3 \times 3 \times 3$ is correct but long. Index notation records two things: the repeated factor, called the **base**, and the number of copies, called the **exponent** or index.



The exponent counts factors; it is not a multiplier.

In 3^4, the base 3 is repeated four times. The value 81 is the result, not part of the notation.

WORKED EXAMPLE 1

Write each power as repeated multiplication and evaluate it: 2^5, 5^2, 10^4, 4^3.

1. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.
2. $5^2 = 5 \times 5 = 25$. This is a square.
3. $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$.
4. $4^3 = 4 \times 4 \times 4 = 64$. This is a cube.

Check: the number of factors must equal the exponent.

CRITERION A Knowing and understanding

GUIDED PRACTICE · COPY AND COMPLETE

2.G1 In 8^3, identify the base and exponent, write the expanded product, and evaluate it. What would the value be if the exponent were 1? GUIDED

CRITERION A Knowing and understanding

PRACTICE 2.1 · POWERS AND INDEX NOTATION

2.1.1 Write each product as a power. (a) $4 \times 4 \times 4$ (b) 7×7 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ FLUENCY

2.1.2 Evaluate. (a) 2^3 (b) 3^2 (c) 5^3 (d) 10^3 (e) 2^6 (f) 4^3 FLUENCY

2.1.3 Find the missing exponent. (a) $3^{\square} = 81$ (b) $2^{\square} = 64$ (c) $10^{\square} = 1{,}000$ (d) $7^{\square} = 1$ FLUENCY

Special cases and common errors

An exponent of 1 means one factor: $7^1 = 7$. For a nonzero base, an exponent of 0 is defined to be 1. The pattern of dividing by the base explains the convention:

$$5^3 = 125, \quad 5^2 = 25, \quad 5^1 = 5, \quad 5^0 = 1.$$

MISCONCEPTION CHECK

$3^4 \neq 3 \times 4$. The exponent tells how many times to use the base as a factor. Likewise, $5^2 = 25$, not 10.

WORKED EXAMPLE 2

Write each product as a power and evaluate it: $6 \times 6 \times 6 \times 6$, 9×9, $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

1. Four factors of 6: $6^4 = 1{,}296$.
2. Two factors of 9: $9^2 = 81$.
3. Seven factors of 2: $2^7 = 128$.

Count the factors before writing the exponent.

CRITERION A Knowing and understanding

PRACTICE 2.1 · POWERS AND INDEX NOTATION

2.1.4 Explain the difference between 4^2 and 4×2. Include the value of each. REASONING

2.1.5 A square tile has side length 9 cm. Write its area using a power and evaluate it. State the unit. APPLICATION

2.1.6 Compare 2^4 and 4^2. Are they equal? Compare 2^3 and 3^2. What does this suggest about the claim ‘swapping base and exponent never changes the value’? REASONING

2.1.7 A rumour starts with one person. In round 1, that person tells 3 new people. In round 2, each of those 3 tells 3 new people, and the pattern continues. Write the number of new people told in round 4 as a power, evaluate it, and find the total number of new people told across rounds 1 to 4. APPLICATION

2.1.8 Decide which is larger, 2^{10} or 10^3, first by reasoning with factors and then by checking the values. CHALLENGE

CHAPTER 2 · INVESTIGATION · BEFORE 2.2

Investigate · Place value is built from powers of ten

YOUR TASK

Complete the table, then write 5,207 and 60.09 using powers of ten.

PLACE	POWERS OF TEN	DIGIT IN 5,207	CONTRIBUTION
Ten thousands	10^4		
Thousands	10^3		
Hundreds	10^2		
Tens	10^1		
Ones	10^0		

What do you notice?

- What happens to the value of a place each time you move one place to the left?
- Which places are empty in 5,207, and how does expanded form show that?

Observation. Each place is a power of ten: the ones place is 10^0, the tens place is 10^1 and so on. Moving one place left multiplies by 10 and moving one place right divides by 10, so 5,207 = 5×10^3 + 2×10^2 + 0×10^1 + 7×10^0.

MATHEMATICAL CONCLUSION

Expanded form with powers of ten exposes every place, including the empty ones, so a number can be rebuilt without losing a zero.

Place value is built from powers of ten

The ones place is 10^0, the tens place is 10^1, the hundreds place is 10^2, and so on. Each step left multiplies by 10; each step right divides by 10.

Millions	Hundred thousand	Ten thousands	Thousands	Hundreds	Tens	Ones
10^6	10^5	10^4	10^3	10^2	10^1	10^0
1,000,000	100,000	10,000	1,000	100	10	1

Each step left multiplies by 10; each step right divides by 10.

Place values from millions to ones written as powers of ten.

WORKED EXAMPLE 3

Write 5,207 in expanded form using powers of ten.

1. Read the places: 5 thousands, 2 hundreds, 0 tens, 7 ones.
2. Replace each place by a power of ten: $5 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$.
3. Check: $5{,}000 + 200 + 0 + 7 = 5{,}207$.

The zero term contributes nothing but preserves the place structure.

CRITERION A Knowing and understanding

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

2.2.1 Evaluate. (a) 10^2 (b) 10^4 (c) 10^6 (d) 10^0 FLUENCY

2.2.2 Write 8,304 in expanded form using powers of ten. FLUENCY

2.2.3 Write $4 \times 10^3 + 7 \times 10^2 + 0 \times 10^1 + 6 \times 10^0$ in standard form. FLUENCY

Return to standard form without losing zeros

Expanded form is longer than standard form because it exposes the value of every digit. That makes it useful for checking place value and for seeing the effect of multiplying or dividing by 10.

WORKED EXAMPLE 4

Write $6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$ in standard form. State the value of the digit 3.

1. Evaluate nonzero terms: $60{,}000$, 300, 90.
2. Keep the empty thousands and ones places: 60,390.
3. The 3 is in the hundreds place, so it is worth 300.

60,390; the digit 3 contributes 300.

MISCONCEPTION CHECK

10^4 is not 40. It is $10 \times 10 \times 10 \times 10 = 10{,}000$.

CRITERION A Knowing and understanding

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

2.2.4 Write 90,050 in expanded form using powers of ten, then state the value of the digit 5. FLUENCY

2.2.5 A number is written as $2 \times 10^4 + 5 \times 10^1$. Explain why the standard form contains zeros, and write the number. REASONING

2.2.6 A simulated school roll is 12,480 students. Write this number in expanded form using powers of ten and explain which digit contributes the most to the total. APPLICATION

2.2.7 Compare 3×10^3 and 3×10^2. How many times larger is the first expression? Explain using place value. REASONING

2.2.8 Write a five-digit number whose expanded form contains exactly one zero term and whose digits add to 18. Explain how you constructed it and whether more than one answer is possible. CHALLENGE

CHAPTER 2 · INVESTIGATION · BEFORE 2.3

Investigate · Squares, cubes and their reverses

YOUR TASK

Complete the table for n = 1 to 6, then use it to find the square root and cube root of 64.

N	N^2	N^3
1		
2		
3		
4		
5		
6		

What do you notice?

- Which numbers in your table are both square numbers and cube numbers?
- Test the claim “the square root of a sum is the sum of the square roots” on 9 + 16.

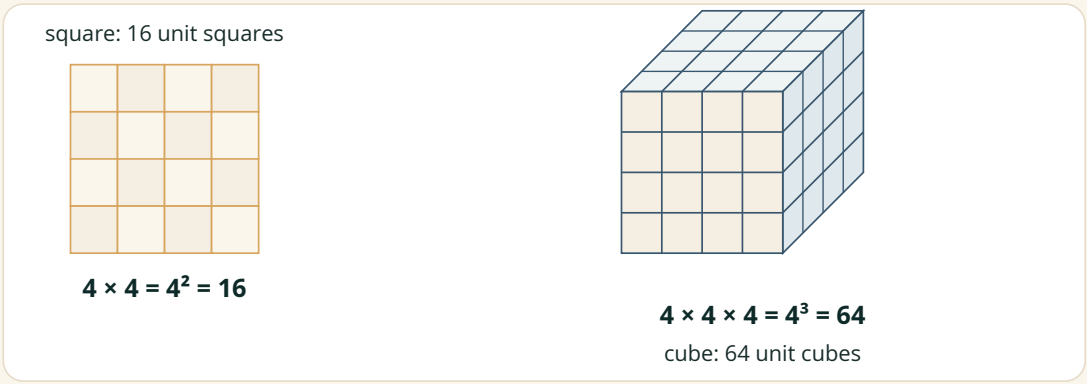
Observation. Square numbers grow by consecutive odd numbers (1, 3, 5, 7, …) and cube numbers grow faster. The root symbol reverses the power, but it does not split sums: $\sqrt{25} = 5$ while $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$.

MATHEMATICAL CONCLUSION

A square number is a whole number multiplied by itself and a cube number is a whole number multiplied by itself twice. Roots reverse powers; they never distribute over addition.

Models for squares and cubes

A **square number** is a whole number multiplied by itself. The positive square numbers begin 1, 4, 9, 16, … A **cube number** is a whole number multiplied by itself twice: 1, 8, 27, 64, …



A square shows two equal dimensions; a cube shows three. The cube can be seen as 4 layers of 16 unit cubes.

WORKED EXAMPLE 5

Evaluate 12^2, 7^3, $\sqrt{81}$, $\sqrt{144}$ and $\sqrt[3]{64}$.

1. $12^2 = 144$; $7^3 = 343$.
2. $9^2 = 81$, so $\sqrt{81} = 9$.
3. $12^2 = 144$, so $\sqrt{144} = 12$.
4. $4^3 = 64$, so $\sqrt[3]{64} = 4$.

Reverse a root by squaring or cubing the result.

CRITERION A Knowing and understanding

GUIDED PRACTICE · COPY AND COMPLETE

2.G2 Since $13^2 = 169$, what is $\sqrt{169}$? Since $6^3 = 216$, what is $\sqrt[3]{216}$? A square has area 196 cm²; find its side length. GUIDED

CRITERION A Knowing and understanding

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

2.3.1 List the first twelve square numbers. FLUENCY

2.3.2 Evaluate. (a) 13^2 (b) 15^2 (c) 6^3 (d) 10^3 FLUENCY

2.3.3 Evaluate. (a) $\sqrt{49}$ (b) $\sqrt{100}$ (c) $\sqrt{169}$ (d) $\sqrt[3]{8}$ (e) $\sqrt[3]{27}$ (f) $\sqrt[3]{125}$ FLUENCY

Roots reverse powers; they do not split sums

The square root symbol asks for the nonnegative value whose square is the number under the root. The cube root asks for the value whose cube is the number. In geometry, roots recover a side or edge from an area or volume.

WORKED EXAMPLE 6

A square has area 121 cm^2. A cube has volume 125 cm^3. Find the side or edge. If the cube's edge doubles, what happens to its volume?

1. $\sqrt{121} = 11$, so the square's side is 11 cm.
2. $\sqrt[3]{125} = 5$, so the cube's edge is 5 cm.
3. Doubling all three dimensions multiplies volume by $2^3 = 8$.

The new cube volume is 8 times the original.

MISCONCEPTION CHECK

$\sqrt{36 + 64} = \sqrt{100} = 10$, but $\sqrt{36} + \sqrt{64} = 6 + 8 = 14$. A root applies to the whole value underneath it.

CRITERION A Knowing and understanding

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

2.3.4 A square mosaic uses 144 identical tiles. How many tiles are along each side? APPLICATION

2.3.5 Find the positive value of x. (a) $x^2 = 64$ (b) $x^3 = 216$ (c) $x^2 = 225$ FLUENCY

2.3.6 Compare √81 + √16 with √(81 + 16). Are they equal? Explain the difference. REASONING

2.3.7 Between which two consecutive whole numbers does √50 lie? Explain using square numbers, then state which whole number it is closer to. CHALLENGE

2.3.8 A cube has volume 343 cm^3. Find its edge length. If the edge length is doubled, calculate the new volume and explain how your answer connects to 2^3. CHALLENGE

CHAPTER 2 · INVESTIGATION · BEFORE 2.4

Investigate · Which order?

YOUR TASK

Evaluate each expression twice: once left to right, and once using the agreed order of operations.

EXPRESSION	LEFT TO RIGHT	AGREED ORDER
$3 + 4 \times 2$		
$(3 + 4) \times 2$		
$20 - 6 \div 3$		
2×3^2		
$12 \div (2 + 4)$		

What do you notice?

- Which pairs of answers differ, and why?
- What do brackets tell you to do, and what do they change about the meaning?

Observation. Without an agreed order the same expression can give different answers. Brackets are evaluated first, then indices, then multiplication and division from left to right, then addition and subtraction: $3 + 4 \times 2 = 11$ but $(3 + 4) \times 2 = 14$.

MATHEMATICAL CONCLUSION

The order of operations is a shared convention that makes one written expression mean exactly one value. Brackets are the tool for changing it deliberately.

A shared convention makes expressions readable

Without an agreed order, $2 + 3 \times 4$ could mean 20 or 14. Mathematics uses a convention so that the same expression has the same value for every reader.

PRIORITY	OPERATIONS	HOW TO APPLY IT
1	Brackets and grouping symbols	Work from the innermost bracket outwards. A fraction bar groups the numerator and denominator.
2	Indices and roots	Evaluate powers and roots before multiplication, division, addition or subtraction.
3	Multiplication and division	Same priority: work left to right.
4	Addition and subtraction	Same priority: work left to right.

WORKED EXAMPLE 7

Evaluate $7 + 2 \times 3^2$, $(7 + 2) \times 3^2$, $24 \div 4 \times 2$ and $24 \div (4 \times 2)$.

1. $7 + 2 \times 9 = 7 + 18 = 25$.
2. $9 \times 9 = 81$.
3. Left to right: $24 \div 4 = 6$, then $6 \times 2 = 12$.
4. Brackets first: $24 \div 8 = 3$.

25, 81, 12, 3.

CRITERION A Knowing and understanding

PRACTICE 2.4 · ORDER OF OPERATIONS

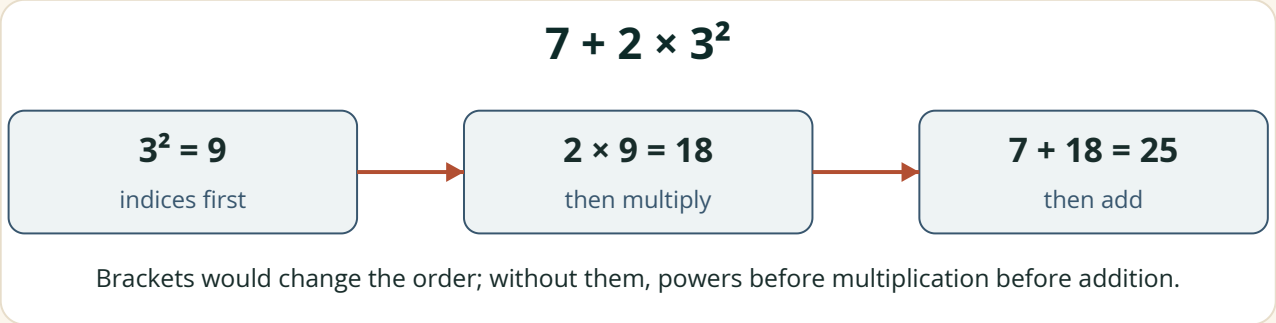
2.4.1 Evaluate $8 + 2 \times 5$. FLUENCY

2.4.2 Evaluate $20 - 3 \times 4$. FLUENCY

2.4.3 Evaluate. (a) $6 + 4^2$ (b) $(6 + 4)^2$ (c) $2^3 + 3^2$ FLUENCY

Brackets, fraction bars and the unary minus

Brackets are the writer's tool for changing the order. A fraction bar acts like brackets around the whole numerator and the whole denominator.



Evaluate the power first, then multiply, then add. Brackets could change this order.

WORKED EXAMPLE 8

Evaluate $(18 - 4 \times 2)^2 \div 5$ and $\frac{36-6}{2+4}$.

1. Inside the bracket: $4 \times 2 = 8$, so $18 - 8 = 10$. Then $10^2 \div 5 = 100 \div 5 = 20$.
2. Numerator: $36 - 6 = 30$. Denominator: $2 + 4 = 6$. Then $30 \div 6 = 5$.

20 and 5.

BOUNDARY NOTE

A minus sign in front of a power is not part of the base unless brackets include it: $-3^2 = -(3^2) = -9$, while $(-3)^2 = 9$.

CRITERION A Knowing and understanding

GUIDED PRACTICE · COPY AND COMPLETE

2.G4 Boundary note: evaluate -3^2 and $(-3)^2$. Explain what the brackets include in the base. GUIDED

CRITERION A Knowing and understanding

PRACTICE 2.4 · ORDER OF OPERATIONS

2.4.4 Evaluate. (a) $24 \div 4 \times 2$ (b) $24 \div (4 \times 2)$ (c) $36 \div 6 + 3$ FLUENCY

2.4.5 Evaluate $5 \times 2^3 - 3^2$, showing each stage. FLUENCY

Diagnose the error, then repair the calculation

When a calculation goes wrong, identify the first line where the convention was broken. Do not only change the final answer.

Incorrect

$12 + 6 \div 3 = 18 \div 3 = 6$

The student added before dividing.

Correct

$12 + 6 \div 3 = 12 + 2 = 14$

Division has higher priority than addition.

CRITERION A Knowing and understanding

PRACTICE 2.4 · ORDER OF OPERATIONS

2.4.6 Insert brackets to make each statement true. (a) 3 + 5 × 2 = 16 (b) 12 − 4 × 2 = 16 (c) 18 ÷ 3 + 6 = 2 REASONING

2.4.7 A student calculates 12 + 6 ÷ 3 as 18 ÷ 3 = 6. Identify the mistake, give the correct value, and explain which operation should be performed first. REASONING

2.4.8 Use the digits 2, 3 and 4 once each, with any operations and brackets, to make at least four different values. One expression must contain an index. CHALLENGE

COMMUNICATE

Write one operation per line. If a reader cannot see which rule you used, add a bracket or a short label rather than more words.

CHAPTER 2 · INVESTIGATION · BEFORE 2.5

Investigate · Substitute, then simplify

YOUR TASK

Take a = 3 and b = 5. Evaluate each expression by substituting first, then check with a second method.

EXPRESSION	SUBSTITUTION	VALUE
$2a + 3b$		
$a^2 + b^2$		
$(a + b)^2$		
2^a		
$ab - a$		

What do you notice?

- Compare $a^2 + b^2$ with $(a + b)^2$. Why are the answers different?
- What does 2^a mean when a = 3, and how is it different from 2a?

Observation. Substitution replaces a letter with its value, then the order of operations applies: $2a + 3b = 2\times3 + 3\times5 = 21$. But $(a + b)^2 = 64$ while $a^2 + b^2 = 34$, because the brackets are part of the meaning.

MATHEMATICAL CONCLUSION

A letter stands for a number. Substitute carefully, write the multiplication signs, and respect brackets — they carry meaning that the shorthand hides.

Substitute first, then ask what is squared

A letter stands for a number. Substitution means replacing the letter with its value and evaluating using the order of operations. Keep brackets around the replacement so the structure stays visible.

$3x^2$ when $x = 4$

$3 \times 4^2 = 3 \times 16 = 48$

Only x is squared.

$(3x)^2$ when $x = 4$

$(3 \times 4)^2 = 12^2 = 144$

The whole product 3x is squared.

The brackets determine whether only x is squared or the whole product 3x is squared.

WORKED EXAMPLE 9

Evaluate $3x^2$ and $(3x)^2$ when $x = 4$.

1. $3x^2 = 3 \times 4^2 = 3 \times 16 = 48$.
2. $(3x)^2 = (3 \times 4)^2 = 12^2 = 144$.
3. The second expression squares the whole product, so it is three times the first.

48 and 144.

CRITERION A Knowing and understanding

GUIDED PRACTICE · COPY AND COMPLETE

2.G3 For $x = 5$, compare $2x^2$ and $(2x)^2$. For $x = 2$, compare $x^2 + 3^2$ and $(x + 3)^2$. Explain what the brackets change. GUIDED

CRITERION A Knowing and understanding

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.1 Evaluate $2n^2$ for (a) $n = 3$ (b) $n = 5$ (c) $n = 10$. FLUENCY

2.5.2 Evaluate $(2n)^2$ for (a) $n = 3$ (b) $n = 5$ (c) $n = 10$. Compare each result with the matching result in question 2.5.1. FLUENCY

2.5.3 For $a = 3$ and $b = 4$, evaluate $a^2 + b^2$ and $(a + b)^2$. Are they equal? FLUENCY

Formulas connect powers to measurement

Area and volume formulas use powers because they multiply equal dimensions. A model may use a power to describe growth or motion; always check what the variables and units mean.

WORKED EXAMPLE 10

Find the area of a square with $s = 7$ cm, the volume of a cube with $s = 4$ cm, and the model distance $d = 5t^2$ when $t = 3$.

1. $A = s^2 = 7^2 = 49$ cm^2.
2. $V = s^3 = 4^3 = 64$ cm^3.
3. $d = 5 \times 3^2 = 5 \times 9 = 45$ m. The index is evaluated before multiplying by 5.

49 cm², 64 cm³, 45 m. The distance formula is a simplified model.

CRITERION A Knowing and understanding

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.4 A square has side length s = 12 cm. Use A = s^2 to find its area, with units. APPLICATION

2.5.5 Use V = s^3 to find the volume of a cube with edge (a) 3 cm (b) 5 cm. FLUENCY

2.5.6 In a simplified falling-object model, d = 5t^2, where d is in metres and t is in seconds. Find d when (a) t = 2 (b) t = 4. Explain why doubling t does not double d. REASONING

REASONING HABIT

Before calculating, say what the exponent applies to: a single variable, a product, or a sum.

Test a transferred rule

A rule that works in one structure may fail in another. Test the claim $(a+b)^2 = a^2 + b^2$ before using it.

WORKED EXAMPLE 11

Test the claim with $a = 3, b = 4$, then with $a = 2, b = 5$.

1. $(3+4)^2 = 49$, but $3^2 + 4^2 = 25$.
2. $(2+5)^2 = 49$, but $2^2 + 5^2 = 29$.
3. One counterexample is enough to reject a universal claim.

The claim is false. Brackets change the base of the power.

CRITERION A Knowing and understanding

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.7 Insert one pair of brackets to make each statement true. (a) $2x^2 = 36$ when $x = 3$ (b) $a + b^2 = 49$ when $a = 2$ and $b = 5$. Explain how the brackets change the base. REASONING

2.5.8 A square has area 81 cm^2. Find its side length. If the side length is tripled, find the new area and explain how 3^2 appears in your answer. CHALLENGE

ATL · TRANSFER. In order for a student to decide whether a familiar rule applies, students must compare the structure of the new expression with the old one and verify the result with a substitution.

CHECKPOINT

Before the application

Answer quickly, then explain one answer to a partner. Use the answers at the back to check both the result and the reasoning.

CHECKPOINT · BEFORE THE APPLICATION

Answer quickly, then explain one answer to a partner.

2.CK1 In 6^4, name the base and exponent, then evaluate the power. CHECK

2.CK2 Write 5,020 in expanded form using powers of ten. CHECK

2.CK3 Evaluate $\sqrt{144}$ and $\sqrt[3]{64}$. CHECK

2.CK4 Evaluate $18 - 2 \times 3^2$. CHECK

Common slips

- Reading an exponent as a multiplier: $5^2 \neq 10$.
- Forgetting that $a^0 = 1$ only for nonzero a.
- Doing multiplication before division just because it appears first in a mnemonic.
- Squaring only part of a bracketed product or sum.

Explain how you know

- Count the factors in an expanded power.
- Name the place before writing a power-of-ten term.
- Reverse a root by squaring or cubing.
- Label the first operation you perform.

ASSESSMENT PRACTICE · CRITERIA A AND C

A power primer for the science lab

The science department wants a one-page reference card beside the microscopes. Write for students who have not yet studied this chapter. The card should be readable from about one metre away.

CRITERION A · KNOWING AND UNDERSTANDING
Use accurate definitions, notation and calculations.

CRITERION C · COMMUNICATING
Use mathematical language, diagrams, notation and a clear structure for the intended audience.

ASSESSMENT PRACTICE · A POWER PRIMER FOR THE SCIENCE LAB

Criteria A and C. Design a one-page reference card for students who have not yet studied this chapter. Use no more than 180 words of explanation, plus diagrams and one fully worked example.

2.S1 Define power, base and exponent using $5^3 = 125$ as your example. ASSESSMENT

2.S2 Explain 5^1 and 5^0 with a pattern that makes clear why $5^0 = 1$ rather than 0. ASSESSMENT

2.S3 Show $4^2 = 16$ as a square of 16 unit squares. Add a sketch or description explaining why 4^3 is called a cube. ASSESSMENT

2.S4 Work through $4 + 3 \times (8 - 2)^2 \div 6$, with one line per operation and a label naming the rule used at each step. ASSESSMENT

2.S5 Explain the difference between $3x^2$ and $(3x)^2$ when $x = 2$. Then ask a classmate to use your card to evaluate $2 + 5 \times 3^2$. Record one strength, one limitation and one revision. ASSESSMENT

PEER TEST

Ask a classmate to use your card to evaluate $2 + 5 \times 3^2$. If they cannot follow a step, revise the card rather than blaming the reader.

CHAPTER REVIEW

What you should now be able to do

Key ideas

- In a^n, the base is repeated and the exponent counts factors.
- $a^1 = a$; for $a \neq 0$, $a^0 = 1$.
- Powers of ten encode place value.
- Square roots and cube roots reverse squaring and cubing.
- Brackets, indices, then multiplication/division left to right, then addition/subtraction left to right.
- Substitution preserves structure: brackets change the base.

Vocabulary

power · base · exponent · index · square number · cube number · square root · cube root · expanded form · standard form · order of operations · substitution · formula

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

2.M1 Evaluate. (a) 3^4 (b) 2^7 (c) 10^6 (d) 6^3 (e) 11^2 (f) 4^4 FLUENCY

2.M2 Write as a single power. (a) $9 \times 9 \times 9$ (b) 12×12 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $5 \times 5 \times 5 \times 5 \times 5 \times 5$ FLUENCY

2.M3 Write 60,304 in expanded form using powers of ten. FLUENCY

2.M4 Write $8 \times 10^4 + 2 \times 10^2 + 5 \times 10^0$ in standard form. FLUENCY

2.M5 Evaluate. (a) $\sqrt{81}$ (b) $\sqrt{196}$ (c) $\sqrt[3]{27}$ (d) $\sqrt[3]{1{,}000}$ FLUENCY

2.M6 Calculate. (a) $6 + 2 \times 5$ (b) $(6 + 2) \times 5$ (c) $3^2 \times 4 - 10$ (d) $48 \div (4 + 4)$ (e) $5 + 12 \div 3$ (f) $(7 - 2)^2$ FLUENCY

Fluency and reasoning

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

2.M7 Substitute $x = 3$ and $y = 2$. (a) $x^2 + y^2$ (b) $(x + y)^2$ (c) $2x^3$ (d) $(2x)^2$ FLUENCY

2.M8 Insert one pair of brackets to make each statement true. (a) $2 + 3 \times 4 = 20$ (b) $24 - 8 \div 4 = 4$ (c) $3 \times 2^2 = 36$ (d) $10 - 4^2 = 36$ REASONING

2.M9 Explain why $2^5 = 32$ but $5^2 = 25$. Which is larger, and why does the position of the exponent matter? REASONING

2.M10 A student writes $7^2 = 14$. Identify the error and give the correct value with a one-sentence explanation. REASONING

2.M11 Find the missing whole number. (a) $2^n = 64$ (b) $10^n = 100{,}000$ (c) $3^n = 81$ (d) $n^3 = 125$ REASONING

2.M12 Explain why $10^0 = 1$ using the pattern $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$. REASONING

EXPLAIN HOW YOU KNOW

Choose one answer and write a short justification using factors, place value, inverse operations or the order of operations.

Challenge and reflect

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

2.M13 Evaluate $10^3 + 10^2 + 10^1 + 10^0$. Describe the pattern in the digits of the answer and explain why it occurs. REASONING

2.M14 A square rug has area 169 cm^2. Find its side length and perimeter. If the side length is doubled, find the new area and perimeter. Explain why the area changes by a factor of 2^2 while the perimeter changes by a factor of 2. APPLICATION

2.M15 A cube has edge 2 cm. Find its volume. Repeat for a cube with edge 4 cm. How many times larger is the second volume? Explain using 2^3. APPLICATION

2.M16 Using the numbers 4, 3 and 2 in that order, insert operations and brackets to make (a) 14 and (b) 20. Check each result with the order of operations. CHALLENGE

2.M17 A rumour spreads in rounds. One student tells 3 friends in round 1. In round 2, each of those 3 tells 3 new people, and the pattern continues. (a) How many new people hear it in round 4? (b) How many people in total have heard it after 4 rounds, including the original student? (c) Explain why the answer to (b) is not 3^5. CHALLENGE

2.M18 Explain why $\sqrt{(36 + 64)}$ is not $\sqrt{36} + \sqrt{64}$. Evaluate both expressions and state the rule that the example disproves. REASONING

Reflection

- ☐ I can explain why an exponent is not a multiplier.
- ☐ I can use powers of ten to justify expanded form.
- ☐ I can reverse squares and cubes with roots.
- ☐ I can show the order of operations one step at a time.
- ☐ I can explain how brackets change the base of a power.

One rule I transferred correctly, and how I checked it: _____

One model limit I can explain: _____

CHAPTER 2 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · THE DOUBLING SHEET

2.B1 Fold a sheet of paper as many times as you safely can and record the number of folds you achieve. Then, under the ideal assumptions that every fold doubles the layers and the sheet is 0.1 mm thick, complete the layers and thickness for n = 0 to 5. INVESTIGATION

2.B2 Describe the pattern in words and write a rule for the thickness T millimetres after n folds. INVESTIGATION

2.B3 Verify your rule for n = 10. How many times thicker is the stack than before any folds? INVESTIGATION

2.B4 Find the least number of folds for which the ideal stack is thicker than 1 m. Show the values on either side of your answer. INVESTIGATION

2.B5 People often say that a sheet of paper cannot be folded more than about eight times. Does the table prove that claim false? State two assumptions of the model and one way to improve the thickness estimate. REASONING

2.B6 Extension: a different ideal model triples in thickness with each fold, so $T = 0.1 \times 3^n$. Find the thickness after 5 folds and compare it with the doubling model. Which model grows faster, and why? CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

2.B7 A sheet of paper is about 0.1 mm thick. In an ideal model each fold doubles the number of layers. Complete a table for 1 to 6 folds, predict the thickness after 10 folds, then explain why the model fails for a real sheet. INVESTIGATION

2.B8 Test the rule “the square of an even number is even” on n = 2, 4, 6, 8 and 10. Then explain why the rule must hold for every even number. INVESTIGATION

CHAPTER 2 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

2.C1 Explain why 3^4 is not 3 × 4, and why 5^2 is not 10. Use one sentence for each error. REASONING

2.C2 Write a short explanation, with one example, of why √(9 + 16) is not the same as √9 + √16. REASONING

2.C3 A science lab display reads "2 × 10^3 mm". Explain in one sentence what the display means and convert it to metres. REASONING

2.C4 Two students evaluate 8 + 2 × 5 and get 50 and 18. Explain which answer is correct and what the other student probably did. REASONING

CHAPTER 2 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE SCHOOL DATA STORE

2.D1 The club plans to use three drives labelled 4 TB, 4 TB and 2 TB. Find the total capacity in TB and in GB, and write the GB total as a power of ten. APPLICATION

2.D2 The system reserves 15% of the total capacity. Each finished project needs 5 GB for a master file and 1 GB for a backup. How many complete projects can the club store after the reserve is removed, and how much space is left over? APPLICATION

2.D3 Explain why the answer in 2.D2 must be rounded down rather than to the nearest whole number. REASONING

2.D4 Extension: some software reports storage in binary units. In this task, 1 TB = 10^{12} bytes and 1 TiB = 2^{40} bytes. Use a calculator to explain why 10 TB is about 9.09 TiB. Is any storage missing? Write one honest label for a poster. CHALLENGE

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

2.D5 A data store saves 4 files of 2^30 bytes each (simulated figures). Estimate the total in gigabytes, then state the exact total in bytes. APPLICATION

2.D6 A phone battery loses 20% of its charge each hour. Write a rule for the charge after n hours starting from 100%, and use it to predict the charge after 5 hours. APPLICATION

2.D7 A rectangular room is 4.2 m by 3.5 m. Estimate its area to one significant figure, then calculate the area to two decimal places and comment on the accuracy that is sensible for ordering flooring. APPLICATION

CHAPTER 2 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can write a power as repeated multiplication and evaluate it, including powers of 0 and 1.
- [ ] I can use powers of ten to read, write and convert large and small numbers.
- [ ] I can evaluate squares, cubes, square roots and cube roots.
- [ ] I can apply the order of operations and substitute values into an expression.

Reflect

- Which convention in this chapter would be hardest to explain to someone new — and how would you explain it?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS.

CHAPTER 2 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Younger learners commonly read 3^2 as 3 × 2 and lose marks they do not need to lose.
MATHEMATICAL CONNECTION	Index notation, powers of ten, roots and the order of operations (sections 2.1–2.5).
PROPOSED ACTION	Create a set of six "spot the error" cards with worked corrections and run a lunchtime clinic; record which errors recur.
EXPECTED IMPACT	Learners can read index notation correctly and check a calculation using the order of operations.
STUDENT REFLECTION	Log the errors you saw, how you explained each one, and what you would change in the cards.

Community Project connection

PROJECT IDEA	A school "big numbers" display for the library: storage, population and distance figures written in index notation.
COMMUNITY NEED	Large figures are often quoted without units or powers, so they cannot be compared.
RESEARCH QUESTION	Which school and local figures are easiest to compare when written with powers of ten?
MATHEMATICS	Powers of ten, expanded form, estimation and unit conversion.
ACTION AND OUTCOME	Collect 12 figures, rewrite each with powers of ten, and build a labelled display with sources.
BENEFICIARIES	Students, library staff and visitors.
EVIDENCE	Source list, rewritten figures with checks, photographs of the display.

Personal Project connection · MYP 5

PERSONAL INTEREST	Computing and how devices report storage.
LEARNING GOAL	To explain device storage figures accurately using powers of two.
PRODUCT OR OUTCOME	A one-page explainer or short video comparing "GB" as advertised and as reported by a device.
MATHEMATICS	Powers of two, index notation, estimation and unit conversion.
PROCESS	Collect device figures, recompute them in bytes, and explain the difference.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Factors, multiples and primes

Factors, multiples and prime products reveal how numbers are built, so exact grouping and shared schedules can be planned and justified.

INQUIRY QUESTIONS

- **Factual:** How do you know when a search for factor pairs is complete, and which divisibility tests are shortcuts?
- **Conceptual:** Why does every composite number have a prime factor no larger than its square root, and why do different factor trees give the same prime product?
- **Debatable:** When is “equal teams” a fair way to organise a group, and when does it exclude someone?

CHAPTER 3 · OPENER

Factors, multiples and primes

Factors, multiples and prime products reveal how numbers are built, so exact grouping and shared schedules can be planned and justified.

ENGAGEMENT

Can 192 students be placed into identical mixed teams without leaving anyone out? What must be true about the number of teams?

Learning goals

- List factors and multiples systematically.
- Apply divisibility tests and classify primes.
- Write prime products and find HCF and LCM.
- Choose a grouping or schedule method and justify it.

MYP lens

Key concept: Relationships

Related concepts: System, Representation

Global context: Fairness and development

ATL FOCUS. In order for a student to justify a generalisation, students must organise evidence, test a proposed rule and explain why it works beyond the examples.

Language bank: factor · multiple · factor pair · divisible · prime · composite · prime factorisation · HCF · LCM

CHAPTER 3 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

3.R1 Calculate 7×8 and $96 \div 8$. READY

3.R2 List the first five positive multiples of 4. READY

3.R3 Write 100 as a product of two equal factors. READY

3.R4 Which is larger, 3^2 or 2^3? READY

3.R5 Does 7 divide 56 exactly? Explain. READY

Chapter map

1. **3.1 Factors, multiples and factor pairs** — Search for factor pairs in order, then stop when the partners would repeat.
2. **3.2 Divisibility tests** — Use digit tests as shortcuts, then confirm with an exact multiplication when needed.
3. **3.3 Prime and composite numbers** — Classify by the number of factors, and stop a prime search at the square-root boundary.
4. **3.4 Prime factorisation** — Break a composite into factors until every branch ends at a prime.
5. **3.5 Highest common factor and lowest common multiple** — Choose HCF for largest equal groups and LCM for the next shared repeat.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 3 · INVESTIGATION · BEFORE 3.1

Investigate · Factor pairs meet

YOUR TASK

List the factor pairs of each number, then count the factors. Record the pair where the partners meet.

N	FACTOR PAIRS	NUMBER OF FACTORS	PAIRS MEET AT
12			
16			
36			
49			
60			

What do you notice?

- Which numbers have an odd number of factors, and what is special about them?
- How do you know a pair search is finished without testing every number up to n?

Observation. Pairs are listed in order and the search stops when the partners would repeat: 36 = 1×36, 2×18, 3×12, 4×9, 6×6. A repeated partner such as 6×6 is counted once, so square numbers have an odd number of factors.

MATHEMATICAL CONCLUSION

A factor divides a number exactly and a multiple is that number times a whole number. Pair searches are systematic, and squares are the numbers whose factor count is odd.

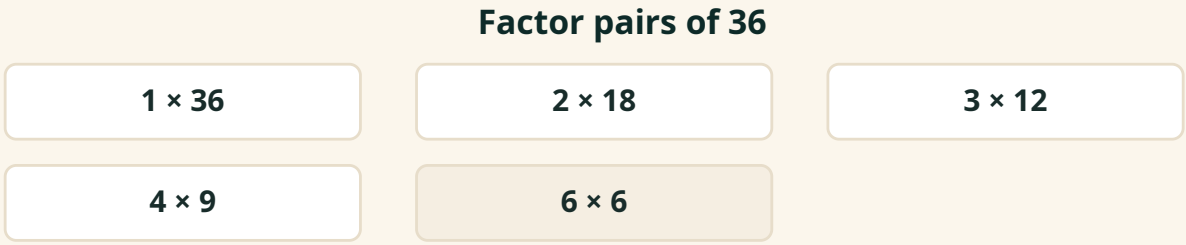
3.1 Factors, multiples and factor pairs

BIG IDEA

Search for factor pairs in order, then stop when the partners would repeat.

A factor divides a positive whole number exactly. A multiple is that number times a whole number. Since $7 \times 12 = 84$, 7 and 12 are factors of 84, and 84 is a multiple of both.

For 36, the pairs are 1×36, 2×18, 3×12, 4×9 and 6×6. The repeated 6 is counted once. A pair search is complete when the next possible first factor has already appeared as a partner.



Factor pairs meet at 6×6. Search in order so no pair is missed.

WORKED EXAMPLE · FIND EVERY FACTOR OF 48

Use factor pairs rather than guessing.

1. 1×48, 2×24, 3×16, 4×12 and 6×8.
2. 5 and 7 do not divide 48. The next partner would repeat.
3. Collect the factors in order.

1, 2, 3, 4, 6, 8, 12, 16, 24, 48: ten factors.

Criterion A practice · 3.1 Factors, multiples and factor pairs

WATCH FOR THIS MISCONCEPTION

A factor is usually no larger than the number; a positive multiple is at least as large. Do not confuse the two directions.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

3.1.1 List all factor pairs of 36. FLUENCY

3.1.2 List the first five positive multiples of 7. FLUENCY

3.1.3 Is 56 a multiple of 8? Is 8 a factor of 56? Explain. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.1.4 A student lists 1, 2, 3, 6, 9, 18 as all factors of 54. Find the missing factors. REASONING

3.1.5 Which rectangular arrays can be made with exactly 24 counters? Count rotations as the same array. APPLICATION

3.1.6 A number between 30 and 50 has factors 6 and 8. Find it and explain uniqueness. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · INVESTIGATION · BEFORE 3.2

Investigate · Digit tests as shortcuts

YOUR TASK

Test 4,728 against each divisor using only its digits. Record pass or fail, then confirm one result by dividing.

DIVISOR	TEST USED	PASS OR FAIL	CHECK
2			
3			
4			
5			
8			
9			

What do you notice?

- Which tests use the last digit, and which use a digit sum?
- 4,728 passes the tests for 2 and 3. What else can you conclude, and why?

Observation. Tests for 2, 5 and 10 use the last digit; tests for 4 and 8 use the last two and three digits because 100 is divisible by 4 and 1,000 by 8; tests for 3 and 9 use the digit sum. Passing 2 and 3 together means 6 also divides the number.

MATHEMATICAL CONCLUSION

Digit tests are shortcuts justified by place value. They give evidence, and an exact division confirms it when the stakes are high.

3.2 Divisibility tests

BIG IDEA

Use digit tests as shortcuts, then confirm with an exact multiplication when needed.

Tests for 2, 5 and 10 use the last digit. Tests for 4 and 8 use the last two and three digits because 100 is divisible by 4 and 1,000 by 8.

For 3 and 9, add the digits. A number is divisible by 6 exactly when it is divisible by both 2 and 3. These tests concern positive whole numbers here; a passing test is evidence, not a guess.

WORKED EXAMPLE · TEST 4,728

Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide it?

1. Last digit 8: passes 2, fails 5 and 10. Digit sum 21: passes 3, fails 9.
2. Last two digits 28: passes 4. Last three digits 728 = 8 × 91: passes 8.
3. Passing 2 and 3 means it also passes 6.

2, 3, 4, 6 and 8.

Criterion A practice · 3.2 Divisibility tests

WATCH FOR THIS MISCONCEPTION

A digit sum divisible by 3 does not prove divisibility by 6: the number must also be even.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

3.2.1 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 4,728? FLUENCY

3.2.2 Test 315 for divisibility by 3, 5, 9 and 10. FLUENCY

3.2.3 Explain why 513 is not divisible by 6 even though its digit sum is 9. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.2.4 Why does testing the last two digits decide divisibility by 4? REASONING

3.2.5 A teacher has 126 pencils. Can they be shared equally into 6 packs? Into 9 packs? APPLICATION

3.2.6 Use digits 4, 5 and 6 once each to make the largest three-digit number divisible by 4. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · INVESTIGATION · BEFORE 3.3

Investigate · Prime or composite?

YOUR TASK

Test each number by dividing only by primes up to its square root. Record the first factor you find, or write “prime”.

N	√N (APPROX.)	PRIMES TO TEST	FIRST FACTOR OR PRIME
37			
51			
57			
91			
97			

What do you notice?

- Which numbers looked prime but were not, and which test caught them?
- Why is it enough to test primes no larger than √n?

Observation. If n = a × b and both factors were larger than √n, the product would be larger than n, so a composite must have a factor at or below that boundary. Testing only primes is enough: 51 = 3×17, 57 = 3×19 and 91 = 7×13, while 37 and 97 are prime.

MATHEMATICAL CONCLUSION

A prime has exactly two positive factors, 1 and itself; 1 has only one factor, so it is neither prime nor composite. Trial division stops at √n.

3.3 Prime and composite numbers

> **BIG IDEA**
>
> Classify by the number of factors, and stop a prime search at the square-root boundary.

A prime is a whole number greater than 1 with exactly two positive factors: 1 and itself. A composite has more than two. The number 1 has one factor, so it is neither.

To test n, try prime divisors no larger than √n. If n = a × b and both factors exceeded √n, their product would exceed n. Thus a composite must have a factor at or below that boundary.

WORKED EXAMPLE · IS 97 PRIME?

Test only the primes up to √97.

1. √97 is between 9 and 10, so test 2, 3, 5 and 7.
2. 97 is not divisible by any of them.

97 is prime.

Criterion A practice · 3.3 Prime and composite numbers

WATCH FOR THIS MISCONCEPTION

2 is prime even though it is even. It is the only even prime; every larger even number has 2 as a factor.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

3.3.1 List the primes below 30. FLUENCY

3.3.2 Classify 1, 2, 21 and 29 as prime, composite or neither. FLUENCY

3.3.3 Which prime divisors must be tested to decide whether 97 is prime? Is it prime? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.3.4 Give a counterexample to “all odd numbers are prime”. REASONING

3.3.5 Find a twin-prime pair between 50 and 70 and give its sum. APPLICATION

3.3.6 Does checking that $28 = 11 + 17$ prove that every even number above 2 is a sum of two primes? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · INVESTIGATION · BEFORE 3.4

Investigate · Different trees, same primes

YOUR TASK

Factorise 60 with two different starting splits, then factorise 84. Write each as a product of primes.

STARTING SPLIT	TREE BRANCHES	PRIME PRODUCT
$60 = 2 \times 30$		
$60 = 6 \times 10$		
$84 = 4 \times 21$		
$84 = 7 \times 12$		

What do you notice?

- Do the two trees for 60 give the same prime product? What about the two trees for 84?
- How do you use index form to write the product more compactly?

Observation. Different starting pairs lead to the same prime factors apart from order: $60 = 2^2 \times 3 \times 5$ and $84 = 2^2 \times 3 \times 7$. Index form groups equal primes, and multiplying the prime product checks the original number.

MATHEMATICAL CONCLUSION

A factor tree records repeated splitting until every endpoint is prime. The prime factorisation of a number is unique apart from the order of the factors.

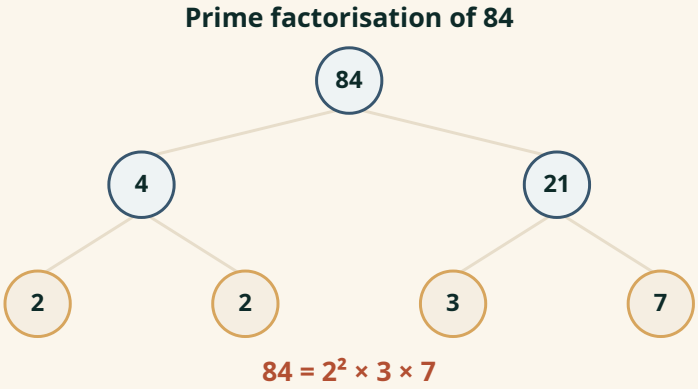
3.4 Prime factorisation

BIG IDEA

Break a composite into factors until every branch ends at a prime.

A factor tree records repeated splitting. Different starting pairs can lead to the same prime factors, apart from order. Index form groups equal primes.

For 84, split 84 into 4×21, then 4 into 2×2 and 21 into 3×7. Thus $84 = 2^2 \times 3 \times 7$. Multiply the prime product to check the original number.



Split composite branches until every endpoint is prime: $84 = 2^2 \times 3 \times 7$.

WORKED EXAMPLE · FACTORISE 72

Use a factor tree or repeated division.

1. $72 = 8 \times 9$.
2. $8 = 2^3$ and $9 = 3^2$.
3. Check $8 \times 9 = 72$.

$72 = 2^3 \times 3^2$.

Criterion A practice · 3.4 Prime factorisation

WATCH FOR THIS MISCONCEPTION

Stop only when all terminal factors are prime. A product containing 4 or 9 is not yet a prime factorisation.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

3.4.1 Write 36, 54 and 72 as products of primes in index form. FLUENCY

3.4.2 Find the number represented by $2^3 \times 3 \times 5$. FLUENCY

3.4.3 A tree for 84 ends in 4, 3 and 7. Is it complete? Repair it. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.4.4 Why can two different factor trees for 84 give the same final prime product? REASONING

3.4.5 A display has 126 tiles. Write 126 as a prime product and use it to find a 6-by-21 array. APPLICATION

3.4.6 Find the smallest positive integer that makes 180n a perfect square. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · INVESTIGATION · BEFORE 3.5

Investigate · Shared groups and shared repeats

YOUR TASK

Two buses leave together every 12 minutes and every 18 minutes. List the common multiples up to 100 and the common factors of 12 and 18.

QUANTITY	VALUES UP TO 100	SMALLEST OR LARGEST
Common multiples of 12 and 18		
Common factors of 12 and 18		
Prime products	$12 = 2^2 \times 3$; $18 = 2 \times 3^2$	

What do you notice?

- Which quantity answers “when do they leave together again?” and which answers “how large can equal teams be?”
- Check HCF × LCM = 12 × 18 for this pair.

Observation. The common multiples begin 36, 72, so the buses meet again after 36 minutes; the common factors are 1, 2, 3, 6, so the largest equal teams have 6 members. HCF(12, 18) = 6, LCM(12, 18) = 36 and 6 × 36 = 216 = 12 × 18.

MATHEMATICAL CONCLUSION

Choose the HCF for the largest equal groups and the LCM for the next shared repeat. For two numbers the HCF and LCM multiply to the product of the numbers.

3.5 Highest common factor and lowest common multiple

BIG IDEA

Choose HCF for largest equal groups and LCM for the next shared repeat.

The highest common factor (HCF) is the largest number dividing each given number. The lowest common multiple (LCM) is the smallest positive number that is a multiple of each.

Using prime products, take the smallest exponent of each shared prime for the HCF and the largest exponent of each prime appearing for the LCM. For $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, HCF = 6 and LCM = 36.

WORKED EXAMPLE · RIBBONS AND SCHEDULES

48 cm and 60 cm ribbons are cut into the longest equal pieces. Two bells repeat every 15 and 20 minutes.

1. HCF(48,60) = 12, so pieces are 12 cm long.
2. There are 4 + 5 = 9 pieces.
3. LCM(15,20) = 60, so the bells next coincide after 60 minutes.

12 cm, 9 pieces; 60 minutes.

Criterion A practice · 3.5 Highest common factor and lowest common multiple

WATCH FOR THIS MISCONCEPTION

HCF cannot exceed the smaller number; a positive LCM cannot be below the larger number. Use these bounds to check.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

3.5.1 Find the HCF and LCM of 12 and 18. FLUENCY

3.5.2 Find the HCF and LCM of 8, 12 and 18. FLUENCY

3.5.3 Cut 48 cm and 60 cm ribbons into the longest equal pieces with no waste. Find the length and total pieces. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.5.4 Buses leave every 15 and 20 minutes and coincide at 08:00. When do they next coincide? APPLICATION

3.5.5 A student uses HCF to find when two repeating events coincide. Explain the correct choice. REASONING

3.5.6 Find two different unordered pairs of positive integers with HCF 6 and LCM 36. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

3.CK1 List all factors of 24. CHECK

3.CK2 Is 91 prime? CHECK

3.CK3 Write 90 as a prime product. CHECK

3.CK4 Find HCF(18,24) and LCM(18,24). CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The number museum

MYP ASSESSMENT PRACTICE

Criteria A and C. Create a visitor card for 84 and a second number, 60. Use precise language, a factor tree and a short explanation.

Your display should help a younger student see the structure of a number, not merely list calculations.

ASSESSMENT PRACTICE · THE NUMBER MUSEUM

Criteria A and C. Create a visitor card for 84 and a second number, 60. Use precise language, a factor tree and a short explanation.

3.S1	Show all factor pairs of 84 and write its prime factorisation.	ASSESSMENT
3.S2	Find HCF(84,60) and LCM(84,60), and explain which answers “largest equal groups?”.	ASSESSMENT
3.S3	Write a short museum label and one visitor question with its answer.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 3

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 3

Connect ideas. Give units and reasons where requested.

3.M1 List the factors of 42. FLUENCY

3.M2 Write the first six positive multiples of 9. FLUENCY

3.M3 Which of 2,3,4,5,6,8,9,10 divide 360? FLUENCY

3.M4 Explain why 1 is neither prime nor composite. REASONING

3.M5 Prime-factorise 96 and 150. FLUENCY

3.M6 Two lights flash every 18 and 24 seconds. After how many seconds do they next flash together? APPLICATION

3.M7 Find the longest equal pieces that can be cut from 42 cm and 70 cm strings without waste. APPLICATION

3.M8 Does a larger number always have more factors? Compare 28 and 29. REASONING

3.M9 Find the smallest number with prime factorisation containing 2^2, 3 and 5, then list its factor count. CHALLENGE

3.M10 Two numbers have HCF 4 and LCM 60. One number is 12. Find the other and check. CHALLENGE

> **FINAL REFLECTION**
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 3 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · WHY DO SQUARES HAVE AN ODD FACTOR COUNT?

3.B1 Complete the factor counts for 4, 9, 16, 25 and 36. INVESTIGATION

3.B2 Which numbers from 1 to 30 have an odd number of factors? INVESTIGATION

3.B3 Explain why a non-square has an even number of factors. INVESTIGATION

3.B4 Verify the square pattern on 49 and compare factor counts of 24 and 25. INVESTIGATION

3.B5 Find every number below 100 with exactly three factors. Explain the rule. CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

3.B6 List the number of factors of 1 to 20 in a table. Describe the pattern for square numbers, predict the number of factors of 144 and check your prediction. INVESTIGATION

3.B7 Investigate the statement: “if a number is divisible by 4 and by 6, then it is divisible by 24.” Test several numbers and justify your conclusion. INVESTIGATION

CHAPTER 3 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

3.C1 Explain to a partner why the factor search for 97 can stop at 9, using one sentence and one calculation. REASONING

3.C2 Write the steps for finding the HCF of 84 and 60 using prime factorisation, and explain why you take the lower power of each shared prime. REASONING

3.C3 A student writes $84 = 2 \times 42$ and stops, calling it a prime factorisation. Explain the error and correct it. REASONING

3.C4 Describe how you would justify that 1 is neither prime nor composite to someone who thinks every whole number is one or the other. REASONING

CHAPTER 3 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE FAIRNESS RELAY

3.D1 What is the largest possible number of identical mixed teams? How many students are in each? APPLICATION

3.D2 If instead each year forms teams of 12, how many teams are there altogether? APPLICATION

3.D3 Both plans include everyone. Explain why “fair” needs a definition. REASONING

3.D4 Could 16 identical mixed teams be made? Explain. CHALLENGE

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

3.D5 A school must split 192 students into identical mixed teams with nobody left out (simulated data). List the possible team sizes that give more than 8 but fewer than 30 teams, and recommend one with a reason. APPLICATION

3.D6 Two suppliers deliver supplies every 15 days and every 25 days. Both delivered today. On which day number will they next deliver together, and how many deliveries has each made by then? APPLICATION

3.D7 A charity packs 1,008 water bottles into identical boxes for three shelters, with each shelter receiving a whole number of boxes. Suggest two packing plans and state the number of boxes each shelter receives. APPLICATION

CHAPTER 3 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can list factors and multiples systematically and know when a pair search is complete.
- [ ] I can apply divisibility tests and classify numbers as prime or composite.
- [ ] I can write a prime factorisation and use it to find the HCF and LCM.
- [ ] I can choose a grouping or scheduling method and justify it.

Reflect

- Which method did you trust most in this chapter — digit tests, factor trees or prime products — and why?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to justify a generalisation, students must organise evidence, test a proposed rule and explain why it works beyond the examples.

CHAPTER 3 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Younger learners struggle to organise a fair rota because they cannot spot common factors.
MATHEMATICAL CONNECTION	Factors, HCF and LCM (sections 3.1 and 3.5).
PROPOSED ACTION	Design a rota-planning aid for a school club, then teach two feeder classes how to use it with a worked example.
EXPECTED IMPACT	Club organisers can build rotas that treat every member equally and learners can explain the arithmetic.
STUDENT REFLECTION	Record the questions learners asked, how you checked the rota was fair, and what you would simplify.

Community Project connection

PROJECT IDEA	A "fair share" audit of how school resources are divided between year groups.
COMMUNITY NEED	Resources are often shared by custom rather than by a stated rule.
RESEARCH QUESTION	When resources are divided equally between year groups, which group sizes actually allow an equal share?
MATHEMATICS	Factors, HCF, LCM and prime factorisation.
ACTION AND OUTCOME	Model three real sharing situations, compute the possible equal divisions and present a recommendation.
BENEFICIARIES	School leadership, club organisers and students.
EVIDENCE	Sharing models, calculations and the final recommendation with its justification.

Personal Project connection · MYP 5

PERSONAL INTEREST	Event organisation and scheduling.
LEARNING GOAL	To design a scheduling method that guarantees fair rotation.
PRODUCT OR OUTCOME	A scheduling tool or guide for a club, sports team or community group.
MATHEMATICS	Factors, HCF, LCM and prime factorisation.
PROCESS	Collect the constraints, compute the possible cycles and test the plan against them.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Fractions and decimals

The same quantity can be written as a fraction, a decimal or a measurement; choosing a representation makes comparison and calculation more reliable.

INQUIRY QUESTIONS

- **Factual:** What makes two fractions equivalent, and which representations of the same quantity are exact?
- **Conceptual:** Why must parts be the same size before fractions are added, and why can dividing by a proper fraction make an answer larger?
- **Debatable:** When is a fraction a better description than a decimal, and when is the reverse true?

CHAPTER 4 · OPENER

Fractions and decimals

The same quantity can be written as a fraction, a decimal or a measurement; choosing a representation makes comparison and calculation more reliable.

ENGAGEMENT

A recipe must serve nine people instead of six. How can every ingredient change proportionally without changing the taste?

Learning goals

- Simplify and compare fractions.
- Use all four operations with fractions and decimals.
- Round with stated precision and convert representations.
- Scale a recipe and justify exact versus rounded measurements.

MYP lens

Key concept: Relationships

Related concepts: Equivalence, Representation

Global context: Scientific and technical innovation

ATL FOCUS. In order for a student to transfer mathematical knowledge, students must connect representations and apply a known method in a new measurement or recipe context.

Language bank: numerator · denominator · equivalent · simplest form · common denominator · reciprocal · terminating · repeating · precision

CHAPTER 4 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

4.R1 Simplify 6/8. READY

4.R2 Calculate 24÷6 and 7×8. READY

4.R3 Which is larger, 0.3 or 0.25? READY

4.R4 Find half of 18. READY

4.R5 Write 1 as a fraction with denominator 7. READY

Chapter map

1. **4.1 Equivalent fractions and simplest form** — Multiply or divide numerator and denominator by the same nonzero number.
2. **4.2 Comparing and ordering fractions** — Use a common whole, a common denominator, a benchmark or a common numerator.
3. **4.3 Adding and subtracting fractions** — Make the parts the same size before combining or comparing amounts.
4. **4.4 Multiplying and dividing fractions** — Use “a fraction of” for multiplication and “how many groups?” for division.
5. **4.5 Decimal place value, ordering and rounding** — Read place value, compare from left to right and round only at the requested stage.
6. **4.6 Fractions, decimals and measurement** — A fraction bar means division; choose the representation that makes the task clearer.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 4 · INVESTIGATION · BEFORE 4.1

Investigate · Same value, different names

YOUR TASK

Shade each fraction of the same bar, then write the missing equivalent fractions.

FRACTION	EQUIVALENT WITH DENOMINATOR 12	SIMPLEST FORM	DECIMAL
1/2			
2/3			
3/4			
5/6			
36/48			

What do you notice?

- What single operation turns 1/2 into 6/12, and why does it not change the value?
- How do you know 36/48 is not in simplest form?

Observation. Multiplying or dividing the numerator and denominator by the same nonzero number gives an equivalent fraction, because the whole is cut into more, smaller parts. Dividing by the highest common factor gives the simplest form: 36/48 = 3/4.

MATHEMATICAL CONCLUSION

Equivalent fractions describe the same quantity with different-sized parts. Simplest form is reached when the numerator and denominator have no common factor greater than 1.

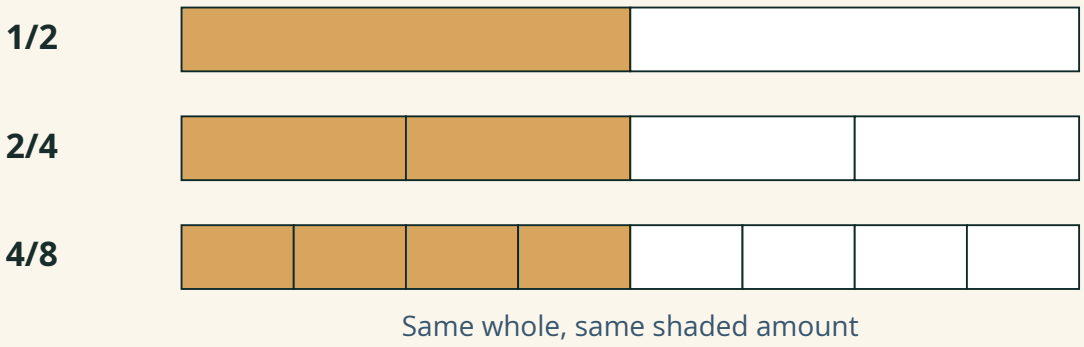
4.1 Equivalent fractions and simplest form

BIG IDEA

Multiply or divide numerator and denominator by the same nonzero number.

Fractions describe equal parts of one whole. 1/2, 2/4 and 4/8 name the same amount when the wholes are the same size. The number of pieces changes, but the covered amount does not.

To simplify, divide top and bottom by a common factor. A fraction is in simplest form when its numerator and denominator have HCF 1.



Three equal wholes. The same amount is shaded in each bar.

WORKED EXAMPLE · SIMPLIFY 36/48

Find a common factor, or divide by the HCF.

1. HCF(36,48) = 12.
2. 36÷12 = 3 and 48÷12 = 4.

3/4.

Criterion A practice · 4.1 Equivalent fractions and simplest form

WATCH FOR THIS MISCONCEPTION

Equivalent fractions use the same whole. Half of a small pizza is not the same amount as half of a large pizza.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.1.1 Complete $2/3 = \square/12$. FLUENCY

4.1.2 Simplify 18/24, 35/49 and 36/48. FLUENCY

4.1.3 A student changes 3/5 to 5/7 by adding 2 to both parts. Is it equivalent? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.1.4 Write three fractions equivalent to 4/6, including its simplest form. REASONING

4.1.5 A recipe uses 12 cups of flour out of 16 cups of dry ingredients. What fraction is flour, in simplest form? APPLICATION

4.1.6 A fraction simplifies to 3/5 and its denominator is 40. Find the numerator. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 4 · INVESTIGATION · BEFORE 4.2

Investigate · Which fraction is bigger?

YOUR TASK

Compare each pair using a benchmark, a common denominator or a diagram. Record the method you used.

PAIR	LARGER	METHOD USED
2/3 and 3/5		
5/8 and 3/5		
7/10 and 3/4		
4/9 and 5/12		
11/20 and 1/2		

What do you notice?

- Which pairs could you decide without any calculation, and why?
- Did a larger denominator ever mean a larger fraction? Explain what happened.

Observation. A benchmark such as 1/2 settles some pairs immediately (5/8 > 1/2 > 3/5 needs care, but 7/10 > 1/2). A common denominator always works: 2/3 = 10/15 and 3/5 = 9/15. A larger denominator does not mean a larger fraction, because the size of the parts also depends on the numerator.

MATHEMATICAL CONCLUSION

Compare fractions with a common whole, then use a benchmark, a common denominator or a common numerator — whichever is quickest for the pair in front of you.

4.2 Comparing and ordering fractions

BIG IDEA

Use a common whole, a common denominator, a benchmark or a common numerator.

With equal denominators, more parts means a larger fraction. With equal numerators, more parts means each part is smaller. For 3/7 and 3/5, fifths are larger than sevenths.

For unlike parts, use a common denominator: 3/4 = 9/12 and 5/6 = 10/12, so 5/6 is larger. A benchmark such as 1/2 can make comparison faster.

WORKED EXAMPLE · ORDER 2/3, 3/5 AND 7/10

Use denominator 30.

1. 2/3 = 20/30; 3/5 = 18/30; 7/10 = 21/30.

3/5 < 2/3 < 7/10.

Criterion A practice · 4.2 Comparing and ordering fractions

WATCH FOR THIS MISCONCEPTION

A larger denominator does not automatically mean a larger fraction. Always check the numerator and the size of the whole.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.2.1 Compare 3/7 and 3/5 using <, > or =. FLUENCY

4.2.2 Compare 3/4 and 5/6. FLUENCY

4.2.3 Order 2/3, 3/5 and 7/10 from smallest to largest. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.2.4 Which is closer to 1: 7/8 or 8/9? Explain. REASONING

4.2.5 Amina reads 5/8 of a book and Ben reads 2/3 of an identical book. Who has read more? APPLICATION

4.2.6 Find a fraction strictly between 1/3 and 1/2 with denominator 12. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 4 · INVESTIGATION · BEFORE 4.3

Investigate · Make the parts the same size

YOUR TASK

Use a diagram to add 1/2 and 1/3, then check the claim that 1/2 + 1/3 = 2/5.

STEP	WORKING	RESULT
Draw both fractions on the same bar		
Rewrite with a common denominator		
Add the parts		
Compare with 2/5		

What do you notice?

- Why can the denominators not simply be added?
- How can you tell from the answer alone that 1/2 + 1/3 cannot be 2/5?

Observation. Halves and thirds are different-sized parts, so they must be rewritten with a common denominator: 1/2 = 3/6 and 1/3 = 2/6, giving 5/6. The result 2/5 is impossible because it is smaller than 1/2, and adding a positive amount to 1/2 must give more than 1/2.

MATHEMATICAL CONCLUSION

Addition and subtraction of fractions need parts of the same size. Always rewrite with a common denominator, then combine the numerators.

4.3 Adding and subtracting fractions

BIG IDEA

Make the parts the same size before combining or comparing amounts.

For like denominators, add or subtract numerators and keep the denominator. For unlike denominators, first write equivalent fractions with a common denominator.

Mixed numbers can be handled as whole and fractional parts or as improper fractions. Borrowing one whole means borrowing denominator/denominator, not 1/10.

WORKED EXAMPLE · ADD 3/4 AND 5/6

Use denominator 12.

1. 3/4 = 9/12 and 5/6 = 10/12.
2. 9/12 + 10/12 = 19/12.

19/12 = 1 7/12.

Criterion A practice · 4.3 Adding and subtracting fractions

WATCH FOR THIS MISCONCEPTION

Do not add denominators: 1/2 + 1/3 is not 2/5. It must be more than 1/2.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.3.1 Calculate 3/8 + 1/8 and 7/9 − 2/9. FLUENCY

4.3.2 Calculate 3/4 + 5/6. FLUENCY

4.3.3 Calculate 5 1/3 − 2 3/4. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.3.4 Explain why 1/2 + 1/3 cannot equal 2/5. REASONING

4.3.5 A jug contains 3/4 litre. Another 1/3 litre is added. How much is in the jug? APPLICATION

4.3.6 Find the missing addend: 2/5 + ☐ = 1 1/10. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 4 · INVESTIGATION · BEFORE 4.4

Investigate · A fraction of a fraction

YOUR TASK

Shade 1/2 of a bar, then 1/3 of the shaded part. Afterwards, work out how many halves fit into 2/3.

TASK	MODEL	CALCULATION	RESULT
1/2 of 1/3			
2/3 of 3/4			
2/3 ÷ 1/2			
3/4 ÷ 2/3			

What do you notice?

- What do you notice about the numerators and denominators when you multiply?
- Why does dividing by 1/2 give an answer larger than 2/3?

Observation. Multiplication takes a fraction of a quantity: numerators multiply and denominators multiply, so $1/2 \times 1/3 = 1/6$. Division asks how many groups fit: $2/3 \div 1/2 = 2/3 \times 2/1 = 4/3$, which is larger than 2/3 because a half is smaller than the amount being shared.

MATHEMATICAL CONCLUSION

Use “a fraction of” for multiplication and “how many groups?” for division. Multiplying by a proper fraction shrinks a positive amount; dividing by one can make it larger.

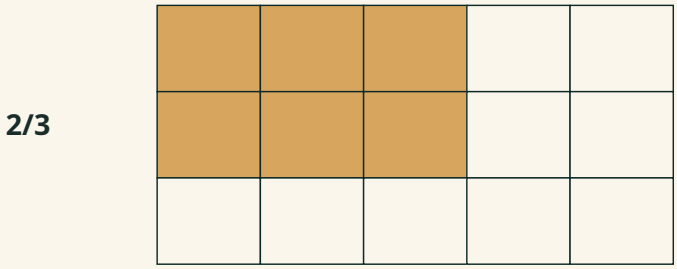
4.4 Multiplying and dividing fractions

BIG IDEA

Use “a fraction of” for multiplication and “how many groups?” for division.

To multiply fractions, multiply numerators and denominators: $a/b \times c/d = ac/bd$. An area model shows why: a rectangle split into b by d equal parts has ac shaded parts.

Dividing by a nonzero fraction asks how many of those fractional groups fit. Dividing by c/d is multiplying by d/c. For example, $3/4 \div 1/8 = 6$.



3/5

overlap = 6/15 = 2/5

The overlap is 6 of 15 cells: 2/3 × 3/5 = 2/5.

WORKED EXAMPLE · FIND 2/3 OF 3/5

Use a rectangle split into thirds one way and fifths the other.

1. The whole has 15 equal cells.
2. The overlap has $2 \times 3 = 6$ cells.
3. 6/15 simplifies to 2/5.

2/3 × 3/5 = 2/5.

Criterion A practice · 4.4 Multiplying and dividing fractions

WATCH FOR THIS MISCONCEPTION

Multiplying by a proper fraction makes a positive amount smaller. Dividing by a proper fraction can make it larger.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.4.1 Calculate $2/3 \times 3/5$ and $4/7 \times 7/8$. FLUENCY

4.4.2 Calculate $3/4 \div 1/8$ and $5/6 \div 5/3$. FLUENCY

4.4.3 A recipe needs 3/4 cup of milk. How much is needed for half a recipe? APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.4.4 How many 1/4-litre cups fill a 3/2-litre jug? APPLICATION

4.4.5 Explain why $6 \div 1/2$ is 12, not 3. REASONING

4.4.6 Calculate $(2/3 + 1/4) \div 5/6$. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 4 · INVESTIGATION · BEFORE 4.5

Investigate · Decimals, place by place

YOUR TASK

Order 0.4, 0.35, 0.405 and 0.45, then round 0.995 to two decimal places.

TASK	WORKING	RESULT
Order the four decimals		
Write them with three decimal places		
Round 0.995 to two decimal places		
Round 3.4789 to two decimal places		

What do you notice?

- Which digit decides the order when the tenths are equal?
- Why does 0.995 round to 1.00 rather than to 0.99?

Observation. Comparing from the highest place downwards puts the order at 0.35, 0.4, 0.405, 0.45. Rounding 0.995 to two decimal places looks at the third decimal digit 5, so the hundredths digit rounds up from 9 to 10 and the carry travels left, giving 1.00.

MATHEMATICAL CONCLUSION

Decimals are compared place by place and rounded at the requested stage only. A carry can travel through a run of 9s, so keep every digit until the final step.

4.5 Decimal place value, ordering and rounding

BIG IDEA

Read place value, compare from left to right and round only at the requested stage.

Digits right of the decimal point represent tenths, hundredths, thousandths and so on. Trailing zeros can record precision: 0.40 and 0.4 have equal value but different stated precision.

Compare equal place values from left to right. To round, inspect the next digit: 0–4 rounds down, 5–9 rounds up. Keep trailing zeros when they show the requested decimal places.

WORKED EXAMPLE · ROUND 0.995 TO TWO DECIMAL PLACES

The third decimal digit is 5.

1. The hundredths digit 9 rounds up and carries through the tenths digit.
2. Write two digits after the point.

1.00.

Criterion A practice · 4.5 Decimal place value, ordering and rounding

WATCH FOR THIS MISCONCEPTION

0.35 is larger than 0.4? Compare tenths first: 3 tenths is less than 4 tenths. Length in digits is not size.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.5.1 Write the value of the 7 in 3.072. FLUENCY

4.5.2 Order 0.35, 0.4, 0.305 and 0.45 from smallest to largest. FLUENCY

4.5.3 Calculate 3.7 + 0.48 and 5 − 2.36. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.5.4 Calculate 0.6 × 0.4 and 2.4 ÷ 0.3. FLUENCY

4.5.5 Round 0.995 to two decimal places and explain the zeros. APPLICATION

4.5.6 A length is 4.7 cm to one decimal place. Give the interval of possible exact lengths under the usual half-up convention. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 4 · INVESTIGATION · BEFORE 4.6

Investigate · A fraction bar means division

YOUR TASK

Convert each fraction to a decimal and describe what happens to the digits.

FRACTION	DIVISION	DECIMAL	STOPS OR REPEATS?
3/8			
1/3			
7/40			
5/6			
2/7			

What do you notice?

- Which fractions give a decimal that stops, and which repeat?
- How could you predict the behaviour before dividing?

Observation. 3/8 = 0.375 and 7/40 = 0.175 stop, while 1/3 = 0.333..., 5/6 = 0.8333... and 2/7 = 0.285714... repeat. In simplest form, a fraction stops exactly when its denominator has no prime factors other than 2 and 5.

MATHEMATICAL CONCLUSION

A fraction bar is a division instruction. Some quotients terminate and some repeat, and the prime factors of the denominator predict which.

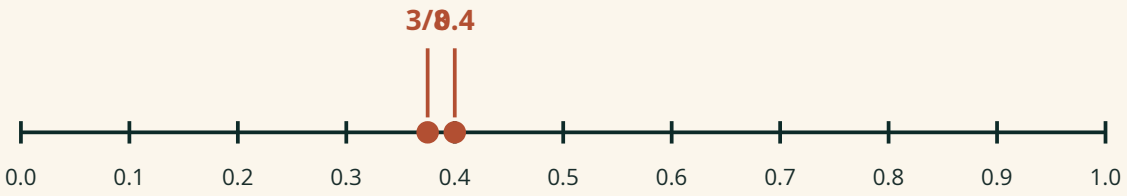
4.6 Fractions, decimals and measurement

BIG IDEA

A fraction bar means division; choose the representation that makes the task clearer.

To convert a fraction to a decimal, divide numerator by denominator, or rewrite with denominator 10, 100 or 1,000 when possible. 3/8 = 0.375.

A reduced fraction terminates exactly when its denominator has no prime factors except 2 and 5. Otherwise its decimal repeats: 1/3 = 0.333.... Use an exact fraction when a rounded decimal would hide the value.



A common scale makes 3/8 = 0.375 and 0.4 directly comparable.

WORKED EXAMPLE · COMPARE 3/8 INCH AND 0.4 INCH

Convert to the same representation.

1. 3/8 = 0.375.
2. 0.4 = 0.400.

0.4 inch is larger by 0.025 inch.

Criterion A practice · 4.6 Fractions, decimals and measurement

WATCH FOR THIS MISCONCEPTION

A calculator display is often rounded. 0.333 is an approximation to 1/3, not exactly equal to it.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

4.6.1 Write 3/8, 7/20 and 2/5 as decimals. FLUENCY

4.6.2 Write 0.24 and 1.125 as fractions in simplest form. FLUENCY

4.6.3 Which of 3/8, 7/20 and 5/6 has a repeating decimal? Explain. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

4.6.4 Compare 3/8 inch and 0.4 inch. Which is larger and by how much? APPLICATION

4.6.5 One cup is defined here as 240 mL. How many mL are in 3/4 cup and 1 1/2 cups? APPLICATION

4.6.6 Calculate 0.75 + 2/5 − 0.3 exactly. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

4.CK1 Simplify 24/36. CHECK

4.CK2 Calculate 1/2 + 1/3. CHECK

4.CK3 Calculate 2/3 ÷ 4/9. CHECK

4.CK4 Write 5/8 as a decimal. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The measurement explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Use a diagram, exact fractions and a clear comparison for a workshop card.

A fitting is labelled 3/8 inch; another is labelled 0.4 inch. A learner must choose the larger without rounding too early.

ASSESSMENT PRACTICE · THE MEASUREMENT EXPLAINER

Criteria A and C. Use a diagram, exact fractions and a clear comparison for a workshop card.

4.S1	Convert 3/8 inch to a decimal and compare with 0.4 inch.	ASSESSMENT
4.S2	Calculate 2/3 + 1/4 and explain a diagram that checks the result.	ASSESSMENT
4.S3	Write one misconception warning and one check question with its answer.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 4

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 4

Connect ideas. Give units and reasons where requested.

4.M1	Simplify 45/60.	FLUENCY
4.M2	Compare 5/9 and 4/7.	FLUENCY
4.M3	Calculate 2 1/2 + 3/4.	FLUENCY
4.M4	Calculate 7/10 − 1/4.	FLUENCY
4.M5	Calculate 3/5 × 10/9 and 4/5 ÷ 2/3.	FLUENCY
4.M6	Calculate 1.25 × 0.8 and 3.6 ÷ 0.6.	FLUENCY
4.M7	Round 2.675 to two decimal places using half-up rounding.	APPLICATION
4.M8	Explain why 0.333 is not exactly 1/3.	REASONING
4.M9	Find 3/4 of a number if the result is 18.	CHALLENGE
4.M10	Calculate (1/2 + 1/3) × 6/5.	CHALLENGE

> FINAL REFLECTION
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 4 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · WHICH FRACTIONS STOP?

4.B1 Convert 3/8, 7/20, 5/6 and 7/15 to decimals. Identify which terminate. INVESTIGATION

4.B2 Factor the denominators 8, 20, 6 and 15. INVESTIGATION

4.B3 State the rule for a fraction already in simplest form. INVESTIGATION

4.B4 Predict whether 13/40 and 13/30 terminate, then check. INVESTIGATION

4.B5 Why must the fraction be reduced before applying the denominator rule? Give an example. CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

4.B6 Investigate which fractions with denominators 2 to 12 give terminating decimals. State a rule, test it on 7/8 and 9/12, and justify the rule using prime factors. INVESTIGATION

4.B7 Test the claim “multiplying always makes numbers bigger” using fractions. Find two counterexamples and state a correct version of the claim. INVESTIGATION

CHAPTER 4 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

4.C1 Explain why 1/2 + 1/3 cannot be 2/5, without calculating the correct answer. REASONING

4.C2 A recipe for 6 people uses 2/3 cup of oil. Write the steps to scale it for 9 people and explain why the amount is not simply doubled. REASONING

4.C3 Compare 3/8 inch and 0.4 inch for a workshop measurement. Show two methods and say which you would put on a work card. REASONING

4.C4 Explain, with one example each, the difference between “0.5 of a pizza” and “0.5 of a different pizza”. REASONING

CHAPTER 4 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE SCALING KITCHEN

4.D1 Scale the recipe to 9 people. Give each ingredient in cups. APPLICATION

4.D2 Convert the 9-person milk and oil amounts to mL. APPLICATION

4.D3 A cook adds the scale factors for two ingredients instead of multiplying each amount. Explain the error. REASONING

4.D4 For 4 people, the cook has 100 mL milk. Is it enough? Find the surplus or shortage. CHALLENGE

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

4.D5 A market seller mixes 3/4 kg of rice at 480 XOF per kg with 1/2 kg at 600 XOF per kg (simulated prices). Find the total cost and the cost per kg of the mixture. APPLICATION

4.D6 A rainwater tank is 2/5 full and holds 1,600 L when full. After rain it is 3/4 full. How many litres were added, and what fraction of the full tank is that? APPLICATION

4.D7 Two joggers run 5/8 km and 0.75 km (simulated times). Which ran further, by how many metres, and what would you write in a training log? APPLICATION

CHAPTER 4 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can find equivalent fractions and write a fraction in simplest form.
- [ ] I can compare and order fractions using a benchmark or a common denominator.
- [ ] I can add, subtract, multiply and divide fractions and justify each method.
- [ ] I can move between fractions and decimals and choose the better representation for a task.

Reflect

- Which model — bars, areas or number lines — explains fractions best to you, and why?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to transfer mathematical knowledge, students must connect representations and apply a known method in a new measurement or recipe context.

CHAPTER 4 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Younger learners often add denominators when adding fractions, and the error persists without a visual model.
MATHEMATICAL CONNECTION	Equivalent fractions, comparison and addition (sections 4.1–4.3).
PROPOSED ACTION	Build a set of bar and area models, then run a workshop that starts from the misconception and corrects it with a diagram.
EXPECTED IMPACT	Learners can explain why parts must be the same size before adding, and can check an answer with a benchmark.
STUDENT REFLECTION	Record the misconceptions you met, the model that worked best, and how you checked understanding.

Community Project connection

PROJECT IDEA	A community kitchen scaling guide for local dishes.
COMMUNITY NEED	Recipes are written for fixed numbers of people, so households rescale them unsafely.
RESEARCH QUESTION	How can a scaling guide keep proportions exact for the dish sizes actually cooked locally?
MATHEMATICS	Equivalent fractions, multiplication of fractions and measurement.
ACTION AND OUTCOME	Collect 6 local recipes, produce a scaling table for 4, 6, 9 and 12 servings, and test two recipes.
BENEFICIARIES	Families, community cooks and school catering staff.
EVIDENCE	Recipe sources, scaling tables with checks, and photographs of the tested dishes.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, nutrition or food technology.
LEARNING GOAL	To build a reliable scaling method for recipes.
PRODUCT OR OUTCOME	A scaling guide or calculator for a set of recipes.
MATHEMATICS	Fractions, equivalence, multiplication and division of fractions.
PROCESS	Collect recipes, compute scaled quantities, test the results and evaluate the accuracy.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Percentages and applications

Percentages express change and comparison relative to a whole; identifying that whole makes financial and fairness decisions more reliable.

INQUIRY QUESTIONS

- **Factual:** What does “per cent” mean, and how do you convert between fractions, decimals and percentages?
- **Conceptual:** Why do successive percentage changes not simply add, and why does every percentage need a stated whole?
- **Debatable:** When does a discount advertisement mislead, even when every number on it is correct?

CHAPTER 5 · OPENER

Percentages and applications

Percentages express change and comparison relative to a whole; identifying that whole makes financial and fairness decisions more reliable.

ENGAGEMENT

Two shops advertise different discount sequences. Which offer is genuinely cheaper, and what must a price card make clear?

Learning goals

- Convert between fractions, decimals and percentages.
- Find percentages and percentage changes.
- Use multipliers for successive changes.
- Interpret discounts, taxes, costs and profit in a budget.

MYP lens

Key concept: Change

Related concepts: Quantity, Proportion

Global context: Fairness and development

ATL FOCUS. In order for a student to think critically, students must identify the reference quantity, test a claim with calculations and explain the limits of a financial decision.

Language bank: per cent · reference amount · multiplier · percentage change · discount · tax · revenue · cost · gross profit · net profit

CHAPTER 5 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

5.R1 Find 10% of 90. READY

5.R2 Write 1/2 as a decimal and percentage. READY

5.R3 Calculate 0.2×50. READY

5.R4 Find 60−12. READY

5.R5 Write 75/100 in simplest form. READY

Chapter map

1. **5.1 Per cent and equivalent representations** — Read “per cent” as “out of one hundred”, then connect fractions, decimals and percentages.
2. **5.2 Finding a percentage of a quantity** — Translate “of” into multiplication and identify the reference amount first.
3. **5.3 Percentage increase and decrease** — Find the change from the original amount, or use a single multiplier.
4. **5.4 Discounts, taxes, profit and budgets** — Separate revenue, costs and profit; apply percentages to the correct line of a budget.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 5 · INVESTIGATION · BEFORE 5.1

Investigate · Out of one hundred

YOUR TASK

Shade each percentage on a 100-square grid, then write it as a fraction in simplest form and as a decimal.

PERCENTAGE	SQUARES SHADED	FRACTION	DECIMAL
25%			
30%			
60%			
12.5%			
7%			

What do you notice?

- What does the grid make obvious that the symbol % hides?
- Which conversion did you use most often, and why?

Observation. A 100-square grid shows that a percentage is a number of parts out of 100, so 25% = 25/100 = 1/4 = 0.25 and 12.5% = 12.5/100 = 1/8 = 0.125. Dividing by 100 converts a percentage to a decimal.

MATHEMATICAL CONCLUSION

"Per cent" means "out of one hundred". Percentages, fractions and decimals are three representations of the same quantity, and the grid makes the equivalence visible.

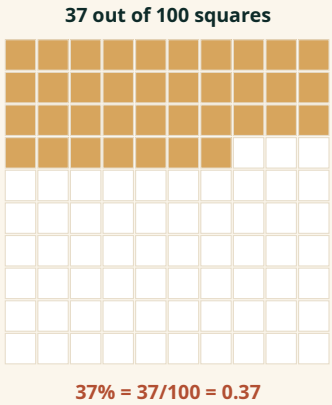
5.1 Per cent and equivalent representations

BIG IDEA

Read “per cent” as “out of one hundred”, then connect fractions, decimals and percentages.

A percentage is a ratio with denominator 100. 37% means 37/100 = 0.37. A percentage can exceed 100 when the amount is more than the reference whole.

To write a fraction as a percentage, find an equivalent fraction out of 100 or multiply by 100%. To write a percentage as a fraction, put it over 100 and simplify.



37 of 100 equal squares: 37/100 = 0.37 = 37%.

WORKED EXAMPLE · CONNECT 3/5, 0.6 AND 60%

Scale the fraction to hundredths.

1. 3/5 = 60/100.
2. 60/100 = 0.60 = 60%.

All three name the same proportion.

Criterion A practice · 5.1 Per cent and equivalent representations

WATCH FOR THIS MISCONCEPTION

Percentages compare to a stated whole. 50% of different totals need not be equal amounts.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

5.1.1 Write 37%, 5% and 125% as decimals. FLUENCY

5.1.2 Write 3/5, 7/20 and 1/8 as percentages. FLUENCY

5.1.3 Write 24% and 150% as fractions in simplest form. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.1.4 A student says 0.5% = 0.5. Correct the statement. REASONING

5.1.5 In a group of 40, 18 choose football. What percentage is that? APPLICATION

5.1.6 Can a quantity be 120% of a reference amount? Give an example. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · INVESTIGATION · BEFORE 5.2

Investigate · A percentage needs a whole

YOUR TASK

Find each percentage of each amount. Then compare your two answers for 20%.

PERCENTAGE	OF 50	OF 200	OF 1,250
10%			
20%			
5%			
12.5%			
1%			

What do you notice?

- Why are the two answers for 20% different even though the percentage is the same?
- Which building-block percentages did you use to find 5% and 12.5%?

Observation. A percentage is not an amount until its whole is known: 20% of 50 is 10 while 20% of 200 is 40. Building blocks make the work easier: 10% then halve for 5%, and 10% plus a quarter of it for 12.5%, so 12.5% of 320 = 40.

MATHEMATICAL CONCLUSION

Translate “of” into multiplication by the percentage as a fraction or decimal, and always identify the reference amount first.

5.2 Finding a percentage of a quantity

BIG IDEA

Translate “of” into multiplication and identify the reference amount first.

p% of A means p/100 × A. Useful benchmarks are 10% (divide by 10), 1% (divide by 100), 50% (halve) and 25% (quarter). Build other percentages from these.

For 15% of 240, find 10% = 24 and 5% = 12, then add: 36. Or use the multiplier 0.15 × 240 = 36.

WORKED EXAMPLE · FIND 12.5% OF 320

Use a fraction benchmark.

1. 12.5% = 1/8.
2. 320 ÷ 8 = 40.

40.

Criterion A practice · 5.2 Finding a percentage of a quantity

WATCH FOR THIS MISCONCEPTION

A percentage is not an amount until its whole is known. 20% of 50 and 20% of 200 are different.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

5.2.1 Find 10%, 25% and 15% of 240. FLUENCY

5.2.2 Find 12.5% of 320 and 1% of 750. FLUENCY

5.2.3 Explain two ways to find 35% of 80. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.2.4 A £60 jacket is reduced by 20%. Find the reduction and sale price. APPLICATION

5.2.5 Is 30% of 50 equal to 50% of 30? Explain. REASONING

5.2.6 15% of a number is 27. Find the number. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · INVESTIGATION · BEFORE 5.3

Investigate · Do two discounts add?

YOUR TASK

Apply 20% off and then 10% off to a 250 XOF price (simulated prices). Compare the result with a single 30% discount.

STEP	CALCULATION	PRICE	MULTIPLIER
Original			1
After 20% off			
After a further 10% off			
Single 30% off			

What do you notice?

- What single multiplier equals the two discounts together?
- Which offer is cheaper, and by how much?

Observation. The multipliers compose: $0.8 \times 0.9 = 0.72$, so two successive discounts of 20% and 10% leave 72% of the price, which is 180 XOF. A single 30% discount leaves 70%, which is 175 XOF, so the single discount is cheaper by 5 XOF.

MATHEMATICAL CONCLUSION

Successive percentage changes multiply, because the second change uses a new base. The combined change is not the sum of the two percentages.

5.3 Percentage increase and decrease

BIG IDEA

Find the change from the original amount, or use a single multiplier.

An increase of p% uses multiplier 1 + p/100; a decrease uses 1 − p/100. The original amount is the reference unless the question says otherwise.

Percentage change = change/original × 100%. A 20% increase followed by a 20% decrease uses 1.2 × 0.8 = 0.96, ending at 96% of the start.



Each percentage uses the current amount as its base.

Each multiplier acts on the current amount: 250 → 200 → 180.

WORKED EXAMPLE · £250 WITH 20% OFF, THEN 10% OFF

Apply each multiplier to the current price.

1. 250 × 0.8 = 200.
2. 200 × 0.9 = 180.
3. A single 30% discount would give 175.

£180; total discount 28%, not 30%.

Criterion A practice · 5.3 Percentage increase and decrease

WATCH FOR THIS MISCONCEPTION

The second percentage uses a new base. Do not add successive percentage changes unless they apply to the same unchanged whole.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

5.3.1 Increase 80 by 15% and decrease 80 by 15%. FLUENCY

5.3.2 A price rises from 50 to 60. Find the percentage increase. FLUENCY

5.3.3 Apply 20% off then 10% off to £250. Compare with 30% off. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.3.4 Increase £100 by 20%, then decrease the result by 20%. Explain the final amount. REASONING

5.3.5 Does reversing 10% off and 20% off change the final price? REASONING

5.3.6 A value falls by 20%. What percentage increase restores the original? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · INVESTIGATION · BEFORE 5.4

Investigate · Revenue, cost and profit

YOUR TASK

Complete the budget for a market stall (simulated figures), then express each line as a percentage of revenue.

LINE	AMOUNT (XOF)	PERCENTAGE OF REVENUE
Revenue	480	100%
Goods cost	264	
Gross profit		
Fixed costs	90	
Net profit		

What do you notice?

- Which percentage base did you use for each line, and why does it matter?
- What happens to the net profit percentage if fixed costs rise to 140 XOF?

Observation. Gross profit is revenue minus goods cost: 480 − 264 = 216, which is 45% of revenue. Net profit also subtracts fixed costs: 216 − 90 = 126, which is 26.25% of revenue. If fixed costs rise to 140, net profit falls to 76, or about 15.8%.

MATHEMATICAL CONCLUSION

Separate revenue, costs and profit, and apply each percentage to the correct base. A percentage without its denominator cannot be interpreted.

5.4 Discounts, taxes, profit and budgets

BIG IDEA

Separate revenue, costs and profit; apply percentages to the correct line of a budget.

A discount reduces a marked price. A tax or tip usually increases a stated base. Revenue is money received; cost is money spent; profit is revenue minus all relevant costs.

For a budget, calculate each line, then add. Whole-item purchases may require rounding up. A lower unit price is not automatically a better decision if extra stock cannot be sold.

WORKED EXAMPLE · A MARKET-STALL BUDGET

Sell 20 honey jars at £8.10, 36 soaps at £3 and 25 jams at £5.

1. Revenue = 162 + 108 + 125 = £395.
2. Goods cost = 20×6 + 36×2.50 + 25×3.20 = £290.
3. Fixed costs = 45 + 18 = £63.

Net profit = 395 − 290 − 63 = £42.

Criterion A practice · 5.4 Discounts, taxes, profit and budgets

WATCH FOR THIS MISCONCEPTION

Gross profit subtracts goods cost only; net profit also subtracts fixed costs. State which profit you mean.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

5.4.1 A £9 item has 10% off. A £4 item has 25% off. Find both sale prices. FLUENCY

5.4.2 A meal costs £40 before 10% tax. Find the total. APPLICATION

5.4.3 Revenue is £395, goods cost £290 and fixed costs £63. Find gross and net profit. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.4.4 A shop buys an item for £6 and sells it for £9. Is the profit 50% of cost or of selling price? Calculate both. REASONING

5.4.5 A group needs 23 jars at £6 each. A supplier offers 30 jars at 10% off the £6 cost. Compare total outlay. APPLICATION

5.4.6 Fixed costs of £63 rise by 8%. With revenue £395 and goods cost £290, find the new net profit. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

5.CK1 Find 15% of 200. CHECK

5.CK2 Increase 50 by 12%. CHECK

5.CK3 A price falls from 80 to 60. Find the percentage decrease. CHECK

5.CK4 Find the single multiplier for 10% off then 20% off. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The fair-price card

MYP ASSESSMENT PRACTICE

Criteria A and C. Design a short price card that shows the original price, discount, final price and one check.

Choose one item priced at £80. Compare “25% off” with “20% off, then 5% off”.

ASSESSMENT PRACTICE · THE FAIR-PRICE CARD

Criteria A and C. Design a short price card that shows the original price, discount, final price and one check.

5.S1 Calculate both final prices and the better offer. ASSESSMENT

5.S2 Write the equivalent single discount for 20% then 5%. ASSESSMENT

5.S3 Give a one-sentence warning that prevents a common percentage error. ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 5

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 5

Connect ideas. Give units and reasons where requested.

5.M1	Write 0.08 as a percentage.	FLUENCY
5.M2	Find 35% of 60.	FLUENCY
5.M3	Increase 120 by 5%.	FLUENCY
5.M4	A £50 item has 30% off. Find the sale price.	APPLICATION
5.M5	A value rises from 40 to 50. Find the percentage increase.	REASONING
5.M6	Explain why 50% off then 50% off is not free.	REASONING
5.M7	Revenue £200, costs £145. Find profit and profit as a percentage of revenue.	APPLICATION
5.M8	After a 10% increase a price is £88. Find the original price.	CHALLENGE
5.M9	Find the single multiplier for a 10% increase followed by a 10% decrease.	CHALLENGE
5.M10	A group needs 18 bottles. Packs of 6 cost £4.50. Find the minimum cost.	APPLICATION

> **FINAL REFLECTION**
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 5 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · DO TWO DISCOUNTS ADD UP?

5.B1 Complete the table for £250, £40 and £80. INVESTIGATION

5.B2 What single discount is equivalent to 20% then 10%? Does order matter? INVESTIGATION

5.B3 Find the combined discount for 25% then 20%. INVESTIGATION

5.B4 Is there a distinct unordered pair of whole-number percentage discounts, other than 20% and 10%, giving exactly 28% off? Justify your answer. CHALLENGE

5.B5 Write a rule for successive discounts a% and b%, and explain the base. INVESTIGATION

CRITERION B Investigating patterns

INVESTIGATE FURTHER

5.B6 Investigate whether a 20% increase followed by a 20% decrease returns a price to its original value. Test three starting prices, then explain the result using multipliers. INVESTIGATION

5.B7 A shop claims “up to 50% off” while only one item is reduced by 50% (simulated claim). Collect five reductions of your own, compute the average reduction and compare it with the claim. INVESTIGATION

CHAPTER 5 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

5.C1 Explain why a 25% increase followed by a 25% decrease does not return a price to its start. Use a multiplier for each step. REASONING

5.C2 A friend says “20% of 60 and 60% of 20 are the same because 20 and 60 appear in both.” Check the claim and explain the general rule. REASONING

5.C3 Write a short explanation of the difference between gross profit and net profit for a market stall, with one calculation of each. REASONING

5.C4 Design a price card that shows the original price, the discount and the final price for one item, and explain in two sentences how a reader can check it. REASONING

CHAPTER 5 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

> **CRITERION D** Applying mathematics in real-life contexts

> **CRITERION D** Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE COMMUNITY MARKET STALL

5.D1 Find each sale price and profit per item. APPLICATION

5.D2 Find total revenue, goods cost, gross profit and net profit. APPLICATION

5.D3 Next time, buy 15% more of each item, rounding up to whole items. Find the new goods cost. CHALLENGE

5.D4 Should the group accept 30 honey jars at 10% off cost instead of the 23 jars needed? Give a financial reason. REASONING

> **CRITERION D** Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

5.D5 A stall sells 120 items at 350 XOF each (simulated data). Goods cost 60% of revenue and fixed costs are 9,000 XOF. Find the gross profit, the net profit and the net profit as a percentage of revenue. APPLICATION

5.D6 A phone costs 80,000 XOF. One shop adds 15% tax to the shelf price; another shows tax included. Compare the final prices and state which display is clearer for a customer. APPLICATION

5.D7 A school raises 250,000 XOF and spends 30% on books, 45% on repairs and the rest on transport. Find each amount and check that the total is correct. APPLICATION

CHAPTER 5 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can convert between percentages, fractions and decimals and explain what “per cent” means.
- [ ] I can find a percentage of a quantity using building blocks or a multiplier.
- [ ] I can calculate percentage increase and decrease, including successive changes.
- [ ] I can interpret revenue, cost and profit and state the base of every percentage.

Reflect

- Which percentage building blocks (10%, 5%, 1%) will you reuse, and how did they help?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to think critically, students must identify the reference quantity, test a claim with calculations and explain the limits of a financial decision.

CHAPTER 5 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Families in the school community compare discount offers without a shared method.
MATHEMATICAL CONNECTION	Percentages, multipliers and profit (sections 5.1–5.4).
PROPOSED ACTION	Produce a one-page "which offer is cheaper?" guide in plain language and present it at a parents' evening.
EXPECTED IMPACT	Families can compare successive discounts and spot a misleading base.
STUDENT REFLECTION	Record the questions asked, the examples you used and how you checked that the guide was understood.

Community Project connection

PROJECT IDEA	A transparent pricing display for a school fair or community market stall.
COMMUNITY NEED	Stall prices are often advertised with discounts but no stated base.
RESEARCH QUESTION	How can a stall display prices so that every customer can check the discount?
MATHEMATICS	Percentages, multipliers, revenue, cost and profit.
ACTION AND OUTCOME	Design and run a stall with price cards showing base, discount and final price, then report revenue and profit.
BENEFICIARIES	Customers, stall volunteers and the school fund.
EVIDENCE	Price cards, sales records and a short report with percentages of revenue.

Personal Project connection · MYP 5

PERSONAL INTEREST	Small business, retail or personal finance.
LEARNING GOAL	To build a pricing model that stays profitable under discounts.
PRODUCT OR OUTCOME	A pricing and discount planner for a small enterprise or fundraiser.
MATHEMATICS	Percentages, multipliers, revenue, cost and profit.
PROCESS	Set a cost base, model discount options, test the plan and evaluate the outcome.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Patterns, expressions and equations

A pattern can be described by a rule; expressions, equations and formulas make the relationship precise and allow prediction and design.

INQUIRY QUESTIONS

- **Factual:** How do you describe how a pattern grows, and what does a letter in an expression stand for?
- **Conceptual:** Why does expanding brackets use the distributive law, and why must an equation stay balanced?
- **Debatable:** Is a rule that fits the first few terms of a pattern proved, or only suggested?

CHAPTER 6 · OPENER

Patterns, expressions and equations

A pattern can be described by a rule; expressions, equations and formulas make the relationship precise and allow prediction and design.

ENGAGEMENT

How can a gardener order exactly the right number of border slabs for a square pond of any whole-number side length?

Learning goals

- Describe and generalise a growing pattern.
- Substitute into expressions and simplify or expand.
- Solve linear equations and check solutions.
- Use formulas and budget constraints in a design.

MYP lens

Key concept: Relationships

Related concepts: Pattern, Equivalence

Global context: Scientific and technical innovation

ATL FOCUS. In order for a student to transfer mathematical knowledge, students must move between a diagram, a table, a rule and an equation, then check the result in context.

Language bank: sequence · term · position · variable · expression · coefficient · like terms · expand · factorise · equation · formula · solution

CHAPTER 6 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

6.R1	Calculate 3×4+5.	READY
6.R2	Continue 2, 5, 8, 11 by one term.	READY
6.R3	Find 4×7 and 28÷4.	READY
6.R4	Simplify 2x+3x.	READY
6.R5	What number added to 7 gives 15?	READY

Chapter map

1. **6.1 Patterns, rules and sequences** — Describe how a pattern grows, then connect the picture to a rule for any position.
2. **6.2 Expressions and substitution** — A letter stands for a number. Replace it carefully and follow the order of operations.
3. **6.3 Simplifying and expanding expressions** — Combine only like terms; use the distributive law to remove brackets.
4. **6.4 Equations, formulas and checking solutions** — An equation states equality. Keep both sides balanced, then substitute the solution to check.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 6 · INVESTIGATION · BEFORE 6.1

Investigate · Growing patterns

YOUR TASK

Draw or build stages 1 to 4 of the matchstick chain, then complete the table.

STAGE N	1	2	3	4	5	10
Matches						
Change from previous stage	—					

What do you notice?

- What stays the same at every stage, and what changes?
- Write a rule for the number of matches at stage n and test it on stage 5.

Observation. Each stage adds the same number of matches, so the growth is linear. If the first stage uses 4 matches and each later stage adds 3, the rule is matches = 3n + 1, which gives 4, 7, 10, 13, 16 and 31 for stages 1, 2, 3, 4, 5 and 10.

MATHEMATICAL CONCLUSION

Describe how a pattern grows, then connect the picture to a rule for any position. A rule that explains the structure can be trusted beyond the drawn stages.

6.1 Patterns, rules and sequences

BIG IDEA

Describe how a pattern grows, then connect the picture to a rule for any position.

A sequence has terms in positions 1, 2, 3, ... A term-to-term rule says how to get the next term. A position-to-term rule uses the position n to find a term directly.

For 4, 7, 10, 13, ..., add 3 each time. The nth term is 3n + 1 because position 1 gives 4. A rule must match every given term, not only the first.



1 square: 4 matches **2 squares: 7 matches** **3 squares: 10 matches**

Each extra square shares one side and adds 3 matches.

Each new square shares a side: 4, 7, 10 matches. The rule is 3n + 1.

WORKED EXAMPLE · A GROWING MATCHSTICK CHAIN

One square uses 4 matches; each extra square shares one side and adds 3.

1. n squares use 4 + 3(n−1) matches.
2. Simplify to 3n + 1.
3. For 10 squares: 31 matches.

3n + 1; 31 matches for 10 squares.

Criterion A practice · 6.1 Patterns, rules and sequences

WATCH FOR THIS MISCONCEPTION

A pattern in the first few terms does not prove a unique rule. Explain the structure that makes the rule continue.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

6.1.1 Continue 4, 7, 10, 13 by two terms and state the term-to-term rule. FLUENCY

6.1.2 Find the nth term of 4, 7, 10, 13, ... and the 20th term. FLUENCY

6.1.3 Explain why n squares in a shared-side chain need 3n + 1 matches. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.1.4 A square pond of side n metres has a one-slab border. Find the border rule and the count for n=7. APPLICATION

6.1.5 Does 50 matches make a whole number of squares in the 3n+1 chain? REASONING

6.1.6 What is the largest whole number of squares that can be made with 100 matches in this chain? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · INVESTIGATION · BEFORE 6.2

Investigate · A letter stands for a number

YOUR TASK

Take a = 4 and b = 3. Evaluate each expression by substituting first, then check with a second method.

EXPRESSION	AFTER SUBSTITUTION	VALUE
$2a + 3b$		
ab		
a^2		
$(a + b)^2$		
2^a		

What do you notice?

- What is the difference between ab and $a + b$? Between a^2 and $2a$?
- Why do the answers for $a^2 + b^2$ and $(a + b)^2$ differ?

Observation. Substitution replaces each letter with its value, then the order of operations applies: $2a + 3b = 2\times4 + 3\times3 = 17$ and $ab = 12$. But $a^2 + b^2 = 25$ while $(a + b)^2 = 49$, because the brackets are part of the meaning.

MATHEMATICAL CONCLUSION

A letter stands for a number. Write multiplication signs when substituting, and treat brackets as instructions that change the value.

6.2 Expressions and substitution

BIG IDEA

A letter stands for a number. Replace it carefully and follow the order of operations.

An expression such as 3n + 5 combines numbers, letters and operations. It has no equals sign. A term is a part added or subtracted; a coefficient multiplies a letter.

To evaluate, substitute the value and use brackets where needed. If n=4, 3n+5 = 3×4+5 = 17. If x=−2, x^2 = 4 but $-x^2$ = −4.

WORKED EXAMPLE · EVALUATE 2A + 3B

Let a=5 and b=2.

1. 2×5 + 3×2.
2. 10 + 6.

16.

Criterion A practice · 6.2 Expressions and substitution

WATCH FOR THIS MISCONCEPTION

3n means 3 × n, not 3 + n. Write the multiplication sign when substituting to avoid losing it.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

6.2.1 Evaluate $3n+5$ for $n=4$ and $n=10$. FLUENCY

6.2.2 Evaluate $2a+3b$ for $a=5$, $b=2$. FLUENCY

6.2.3 Evaluate x^2 and $-x^2$ for $x=-2$. Explain the difference. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.2.4 A taxi charges £3 plus £2 per km. Write the cost for d km and evaluate for 8 km. APPLICATION

6.2.5 A student says $4n=4+n$ when $n=3$. Correct the statement. REASONING

6.2.6 Find n if $2n+7=25$, then check by substitution. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · INVESTIGATION · BEFORE 6.3

Investigate · Combine and expand

YOUR TASK

Simplify $4x + 3 + 2x - 5$, then expand $(x + 2)(x + 3)$ using an area model.

TASK	WORKING	RESULT
Collect like terms in $4x + 3 + 2x - 5$		
Area model for $(x + 2)(x + 3)$		
Expand $3(x + 4)$		
Compare $x + x$ and $x \times x$		

What do you notice?

- Which terms may be combined, and which may not?
- How does the area model show that $(x + 2)(x + 3) = x^2 + 5x + 6$?

Observation. Only like terms combine: $4x + 2x = 6x$ and $3 - 5 = -2$, giving $6x - 2$. An area model splits a rectangle into four parts, giving $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$, and $3(x + 4) = 3x + 12$ by the distributive law.

MATHEMATICAL CONCLUSION

Combine only like terms and use the distributive law to remove brackets. Addition and multiplication are different operations: $x + x = 2x$ while $x \times x = x^2$.

6.3 Simplifying and expanding expressions

BIG IDEA

Combine only like terms; use the distributive law to remove brackets.

Like terms have the same letter part. 3x + 5x = 8x, but 3x + 5 cannot be combined into one term. Rearranging terms does not change their value.

Expanding uses a(b+c)=ab+ac. For 3(2x+4), multiply every term inside: 6x+12. Factorising reverses this: 6x+12 = 3(2x+4).

WORKED EXAMPLE · SIMPLIFY 4X + 3 + 2X − 5

Group like terms.

1. 4x+2x = 6x.
2. 3−5 = −2.

6x − 2.

Criterion A practice · 6.3 Simplifying and expanding expressions

WATCH FOR THIS MISCONCEPTION

$x + x = 2x$, not x^2. $x \times x = x^2$. Addition and multiplication are different operations.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

6.3.1 Simplify $4x+3+2x-5$ and $7a-2a+4$. FLUENCY

6.3.2 Expand $3(2x+4)$ and $5(y-2)$. FLUENCY

6.3.3 Explain why $x+x$ is not x^2. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.3.4 A rectangle has length $2x+3$ and width 4. Write and expand its perimeter. APPLICATION

6.3.5 Is $3(x+2)=3x+2$ true for every x? Test and correct it. REASONING

6.3.6 Factorise $12x+18$ completely and check by expanding. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 6 · INVESTIGATION · BEFORE 6.4

Investigate · Balanced equations

YOUR TASK

Solve 4n + 4 = 44 using the balance model, then check by substitution. Repeat with 3n + 5 = 20.

STEP	BALANCE ACTION	RESULT
Start		4n + 4 = 44
Remove 4 from each side		
Divide each side by 4		
Check by substitution		

What do you notice?

- What must happen to both sides of the balance at every step?
- Why is subtracting 5 before dividing by 3 necessary in 3n + 5 = 20?

Observation. An equation states that two expressions are equal, so the same operation must be applied to both sides: 4n + 4 = 44 gives 4n = 40 and n = 10, and 4(10) + 4 = 44 confirms it. In 3n + 5 = 20, dividing by 3 first would leave n + 5/3, so the 5 is removed first.

MATHEMATICAL CONCLUSION

Keep both sides balanced, apply inverse operations in a sensible order, and always substitute the solution back to check.

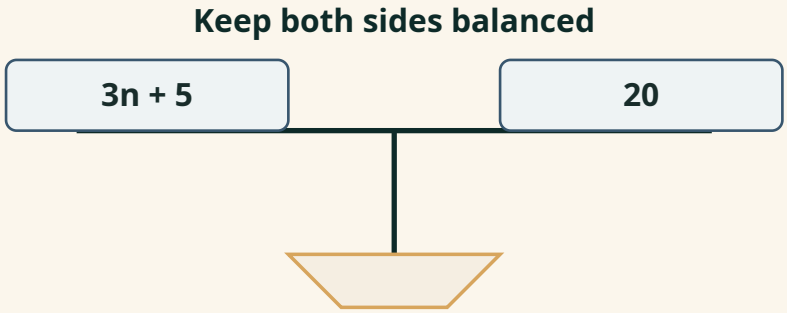
6.4 Equations, formulas and checking solutions

BIG IDEA

An equation states equality. Keep both sides balanced, then substitute the solution to check.

To solve, undo operations in reverse order. For 3n+5=20, subtract 5 from both sides, then divide by 3: n=5. Whatever you do to one side, do to the other.

A formula connects quantities, such as C=3+2d. To find d when C=19, solve 19=3+2d. A solution should satisfy the original equation and make sense in context.



Subtract 5 from both sides, then divide both sides by 3.

Subtract 5 from both sides, then divide both sides by 3: n = 5.

WORKED EXAMPLE · SOLVE 4N + 4 = 44

Undo addition, then multiplication.

1. 4n = 40.
2. n = 10.
3. Check: 4×10+4 = 44.

n=10.

Criterion A practice · 6.4 Equations, formulas and checking solutions

WATCH FOR THIS MISCONCEPTION

Do not divide only one term. In 3n+5=20, subtract 5 before dividing the whole side by 3.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

6.4.1 Solve n+7=15 and 4n=28. FLUENCY

6.4.2 Solve 3n+5=20 and check. FLUENCY

6.4.3 A border uses 4n+4 slabs. If 44 slabs are used, find the pond side n. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

6.4.4 C=3+2d. Find d when C=19. APPLICATION

6.4.5 A student solves 2(x+3)=18 as x=6. Is this correct? Show a check. REASONING

6.4.6 Solve 5n−7=2n+11 and check. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

6.CK1 Find the 10th term of 3n+1. CHECK

6.CK2 Evaluate 2a+3 for a=6. CHECK

6.CK3 Expand 4(x+2). CHECK

6.CK4 Solve 2n+3=13. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The pattern explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Produce a short explanation with a diagram, a rule, a substitution and a check.

Explain the border rule 4n+4 to a learner who keeps counting the four corners twice.

ASSESSMENT PRACTICE · THE PATTERN EXPLAINER

Criteria A and C. Produce a short explanation with a diagram, a rule, a substitution and a check.

6.S1	Give two equivalent expressions for the border and explain one.	ASSESSMENT
6.S2	Use the rule for n=9 and check by drawing or area subtraction.	ASSESSMENT
6.S3	Write one misconception warning and a check question with answer.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 6

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 6

Connect ideas. Give units and reasons where requested.

6.M1 Find the next two terms of 5, 9, 13, 17. FLUENCY

6.M2 Evaluate $4n-3$ for $n=6$. FLUENCY

6.M3 Simplify $5x+2-3x+7$. FLUENCY

6.M4 Expand $2(3x-5)$. FLUENCY

6.M5 Solve $5n=45$ and $n-8=12$. FLUENCY

6.M6 Solve $3n+2=23$. APPLICATION

6.M7 Explain why $2x+3$ cannot be simplified to $5x$. REASONING

6.M8 A formula is $C=5+2n$. Find n when $C=31$. APPLICATION

6.M9 Solve $4(n-2)=28$. CHALLENGE

6.M10 Find the largest n with $3n+1 \leq 100$. CHALLENGE

> FINAL REFLECTION
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 6 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · THE MATCHSTICK CHAIN

6.B1 Complete the match counts for n=1 to 5. INVESTIGATION

6.B2 Write a term-to-term rule and a position-to-term rule. INVESTIGATION

6.B3 How many matches for 20 squares? Explain why counting one by one is inefficient. INVESTIGATION

6.B4 With 100 matches, what is the largest chain? Does 50 matches fit a complete chain? CHALLENGE

6.B5 Justify the rule from the structure, not only the table. INVESTIGATION

CRITERION B Investigating patterns

INVESTIGATE FURTHER

6.B6 A pattern of dots gives 5, 9, 13, 17 for stages 1 to 4. Predict stage 10, then investigate whether a different rule could also fit the first four stages. INVESTIGATION

6.B7 Investigate the claim: “the sum of two consecutive whole numbers is always odd.” Test several pairs, then justify the claim using algebra. INVESTIGATION

CHAPTER 6 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

6.C1 Explain why $3n$ means $3 \times n$ and not $3 + n$, and describe one habit that prevents the error when substituting. REASONING

6.C2 A classmate solves $3n + 5 = 20$ by dividing every term by 3 first and then subtracting 5. Explain what goes wrong and give the correct order. REASONING

6.C3 Write a short explanation of the difference between an expression and an equation, with one example of each. REASONING

6.C4 Explain how you would check that $n = 10$ solves $4n + 4 = 44$, and why the check is not optional. REASONING

CHAPTER 6 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE COURTYARD POND

6.D1 Find the border rule and the slabs for n=7 and n=12. APPLICATION

6.D2 A design uses 44 slabs. Find n and write the cost expression. APPLICATION

6.D3 With a £200 budget, find the largest whole-number pond side and check the next size. CHALLENGE

6.D4 Laying each slab takes 12 minutes, plus 2 hours preparation. Find the working time for the largest pond. APPLICATION

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

6.D5 A gardener borders a square pond of side n metres with slabs, one slab every 0.5 m plus 4 corner slabs. Write a rule for the number of slabs, then find the number needed for a pond of side 6 m (simulated design). APPLICATION

6.D6 A taxi charges 500 XOF plus 180 XOF per kilometre (simulated fares). Write a formula for the fare, find the fare for 12 km, and state the fixed and variable parts. APPLICATION

6.D7 A phone plan costs 3,000 XOF per month plus 25 XOF per minute of calls (simulated plan). A second plan costs 4,500 XOF with unlimited calls. Find the number of minutes at which the plans cost the same. APPLICATION

CHAPTER 6 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can describe how a pattern grows and write a rule for any position.
- [ ] I can substitute values into an expression and respect the order of operations.
- [ ] I can simplify and expand expressions using like terms and the distributive law.
- [ ] I can solve an equation and check the solution by substitution.

Reflect

- Which step in solving an equation do you most often get wrong, and what check catches it?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to transfer mathematical knowledge, students must move between a diagram, a table, a rule and an equation, then check the result in context.

CHAPTER 6 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Younger learners can compute but cannot explain why a pattern rule works.
MATHEMATICAL CONNECTION	Patterns, expressions and equations (sections 6.1–6.4).
PROPOSED ACTION	Create a set of pattern cards with a "show the structure" challenge and run a workshop that requires learners to justify, not just extend, a pattern.
EXPECTED IMPACT	Learners can describe growth, write a rule and defend it with the structure of the pattern.
STUDENT REFLECTION	Record which justifications learners found convincing and how you prompted them to go further.

Community Project connection

PROJECT IDEA	A costing model for a school fundraiser stall.
COMMUNITY NEED	Fundraisers often set prices by guesswork, so the profit is unpredictable.
RESEARCH QUESTION	What price and volume does a stall need in order to cover its costs and reach a target profit?
MATHEMATICS	Expressions, substitution, formulas and solving equations.
ACTION AND OUTCOME	Build a cost model, solve for the break-even point and present a recommended price.
BENEFICIARIES	Stall volunteers, the school fund and customers.
EVIDENCE	Cost model, break-even calculation and the final recommendation.

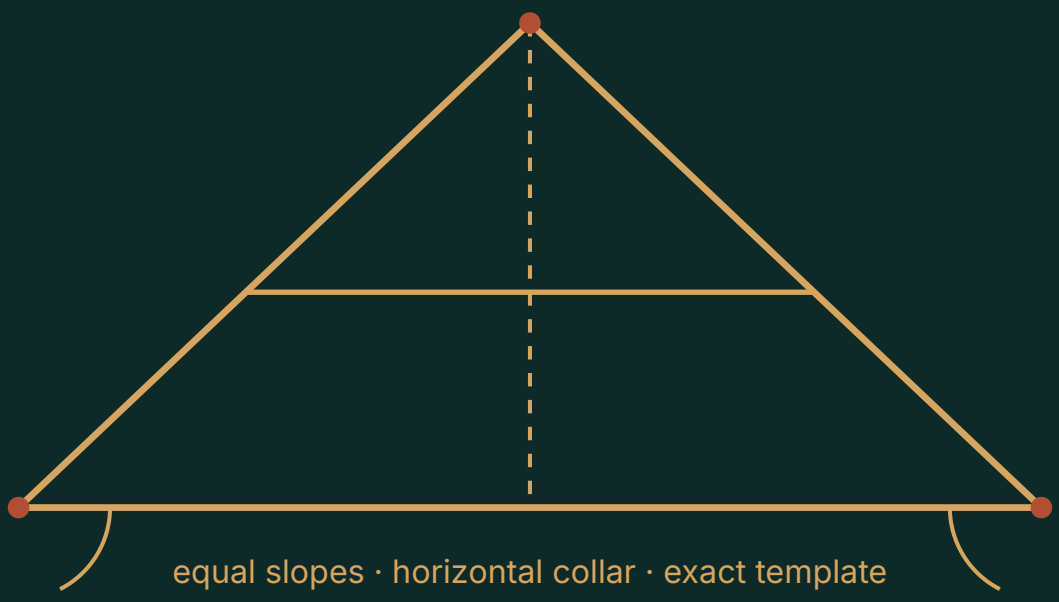
Personal Project connection · MYP 5

PERSONAL INTEREST	Design, engineering or gaming mechanics.
LEARNING GOAL	To model a real or designed system with a formula and solve for an unknown.
PRODUCT OR OUTCOME	A model or simulator for a system of personal interest.
MATHEMATICS	Patterns, expressions, formulas and equations.
PROCESS	Identify the variables, build the formula, test it against data and solve for a target.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, an entire system of shape reasoning follows. Constructions turn that reasoning into a repeatable physical procedure.



KEY CONCEPT · RELATIONSHIPS

RELATED · SPACE

RELATED · REPRESENTATION

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

A workshop needs a triangular roof truss with equal slopes and a horizontal collar beam. Which angles must the carpenter set, how can the shape be checked without measuring every angle, and how can a full-size template be reproduced exactly?

CHAPTER 7 · OPENER

Reasoning with straight lines

Geometry begins with a few reliable facts and becomes powerful when those facts are connected. In this chapter you will name angle relationships, explain triangle and polygon sums, and use equal-radius constructions to force a property rather than guess it.

ENGAGEMENT

Before cutting timber, a carpenter can test a design with a scale drawing. Which measurements are enough to guarantee the shape? Which checks protect against a drawing that only looks correct?

Inquiry questions

- **Factual:** What do angles on a straight line and around a point add to?
- **Conceptual:** Why must the two shorter sides of a triangle together exceed the longest side?
- **Debatable:** When is a construction reliable enough to build from?

Learning goals

- Use angle facts on lines, at points and with parallel lines.
- Reason with triangle and polygon angle sums.
- Apply the triangle inequality and regular-polygon facts.
- Construct perpendicular bisectors, angle bisectors and SSS triangles.

MYP lens

Key concept: Relationships

Related concepts: Space, Representation

Global context: Orientation in space and time

Criteria: A, B, C and D in separate tasks

ATL FOCUS. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable.

Language bank: acute · right · obtuse · reflex · vertically opposite · corresponding · alternate · co-interior · isosceles · exterior angle · polygon · perpendicular bisector · angle bisector · construction

CHAPTER 7 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the map from concrete angle facts to constructions and design decisions.

Ready check

Before starting, check that you can: name acute, right and obtuse angles; draw and label an angle ABC; use a ruler to draw a 6 cm segment; open compasses to a given radius. If any skill is uncertain, ask for a quick demonstration before beginning 7.1.

Chapter map

1. **7.1 Lines, angles and angle facts** — name the relationship first: straight line, full turn, vertically opposite or right angle.
2. **7.2 Parallel lines and triangles** — use corresponding, alternate and co-interior angles only when the lines are parallel.
3. **7.3 Polygons and the triangle inequality** — split polygons into triangles; test whether three lengths can close a triangle.
4. **7.4 Ruler-and-compass constructions** — use equal radii to force a property, keep arcs visible and check the defining relationship.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 7 · HOW TO USE THE PRACTICE

Six kinds of mathematical thinking

The practice is organised by concept, then by the kind of thinking it develops. The labels describe the task, not a separate mark scheme.

Understand

Identify, classify, state, complete and interpret definitions, notation and properties.

Fluency

Calculate, solve, complete diagrams and use procedures accurately.

Reasoning

Justify, compare methods, find errors and explain why a statement is true or false.

Investigation

Observe, organise evidence, make conjectures, test them and draw conclusions.

Application

Use mathematics in design, measurement and everyday contexts, then interpret the result.

Challenge

Combine ideas, consider alternatives and explain deeper relationships.

WORKING SPACE

Use the space provided for diagrams, equations and checks. Construction tasks ask you to draw in your notebook at the stated size; printed figures are models, not always full-size templates.

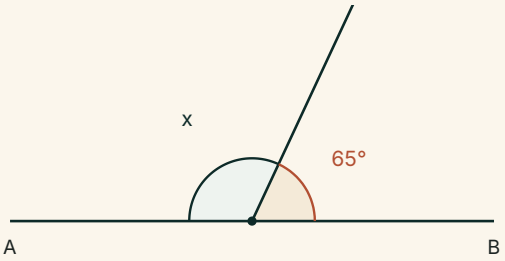
ATL EVIDENCE. As you work, record one place where you chose a method, one place where you tested a result, and one place where you explained why the method is reliable.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

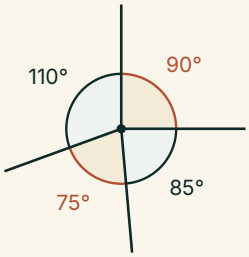
7.1 Lines, angles and angle facts

Name the relationship first: straight line, full turn, vertically opposite or right angle.

An angle measures a turn. A half-turn along a straight line is 180°, so adjacent angles on a straight line are supplementary. A full turn about a point is 360°. When two straight lines cross, vertically opposite angles are equal.



Straight line: the two angles complete 180°.



Full turn: the four angles complete 360°.

Key idea

Straight line: 180°. Full turn: 360°. Vertically opposite: equal. Right angle: 90°. Complementary angles sum to 90°; supplementary angles sum to 180°. The middle letter of angle ABC names the vertex B.

COMMON ERROR

Do not swap the totals: a straight line is 180°, a full turn is 360°. Do not treat vertically opposite angles as supplementary; they are equal. The adjacent pair is supplementary.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.1

Missing angle on a straight line

PROBLEM

Two adjacent angles on a straight line are 47° and x. Find x.

VISUAL REPRESENTATION

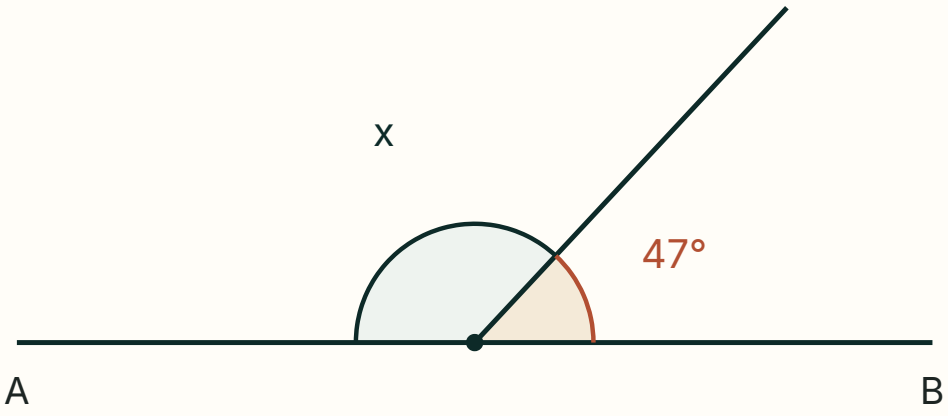


Figure 7.1 · The two angles share a vertex and lie on a straight line.

MATHEMATICAL REASONING

A straight line is a half-turn, so adjacent angles on it are supplementary: their sum is 180°.

STEP-BY-STEP SOLUTION

1. Identify the relationship: the two angles lie on one straight line.
2. Write the equation: $47° + x = 180°$.
3. Subtract 47° from both sides: $x = 180° - 47° = 133°$.
4. Check: $47° + 133° = 180°$.

Final answer. x = 133°.

Interpretation. The unknown angle is obtuse. The diagram and the equation agree: the two angles complete a half-turn.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.2

Angles around a point

PROBLEM

Four angles meet at a point: 95°, 120°, 85° and x. Find x.

VISUAL REPRESENTATION

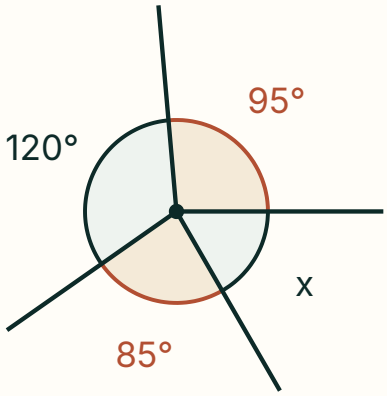


Figure 7.2 · Four sectors fill one full turn.

MATHEMATICAL REASONING

A full turn around a point is 360°. The four angles must fill the turn exactly.

STEP-BY-STEP SOLUTION

1. Add the known angles: 95° + 120° + 85° = 300°.
2. Subtract from the full turn: x = 360° − 300° = 60°.
3. Check: 95° + 120° + 85° + 60° = 360°.

Final answer. x = 60°.

Interpretation. The missing sector is acute. The visual model shows why the total must be 360°, not 180°.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.3

Vertically opposite angles with algebra

PROBLEM

Two lines cross. Vertically opposite angles are labelled $3x + 10°$ and $x + 50°$. Find x and one adjacent angle.

VISUAL REPRESENTATION

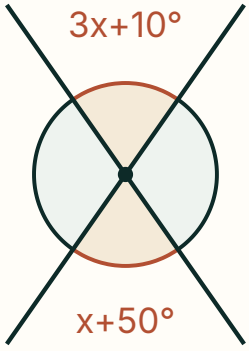


Figure 7.3 · Opposite angles are equal; adjacent angles sum to 180°.

MATHEMATICAL REASONING

Vertically opposite angles are equal. Adjacent angles on a straight line are supplementary.

STEP-BY-STEP SOLUTION

1. Set the opposite angles equal: $3x + 10 = x + 50$.
2. Collect terms: $2x = 40$, so $x = 20°$.
3. Each marked angle is $3(20) + 10 = 70°$.
4. An adjacent angle is $180° − 70° = 110°$. Check: $70° + 110° = 180°$.

Final answer. $x = 20°$; marked angles $70°$; adjacent angle $110°$.

Interpretation. The algebra confirms the visual symmetry: opposite sectors match, and neighbouring sectors complete a line.

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

Practice 7.1 · Angle language and turn facts

7.1.01–7.1.03

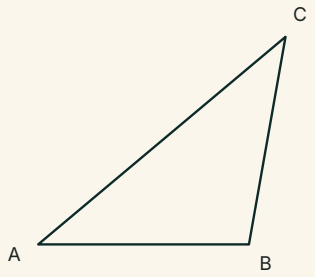
Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.01 UNDERSTAND

Classify each angle as acute, right, obtuse, straight or reflex: 35°, 90°, 125°, 180°, 210°. State the boundary values you used.

7.1.02 UNDERSTAND

In angle ABC, which point is the vertex? Which two rays form the angle? Draw a labelled sketch of any obtuse angle ABC.



7.1.03 UNDERSTAND

Complete the fact table and add one labelled sketch for each fact.

CONFIGURATION	FACT
Angles on a straight line	sum = ___°
Angles around a point	sum = ___°
Vertically opposite angles	they are ___

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

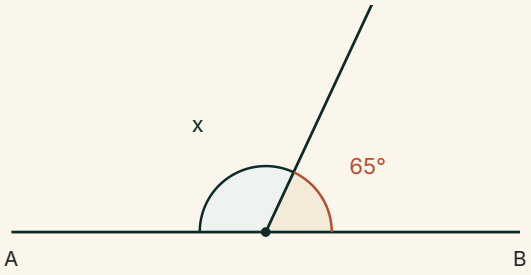
Practice 7.1 · Angle language and turn facts

7.1.04–7.1.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

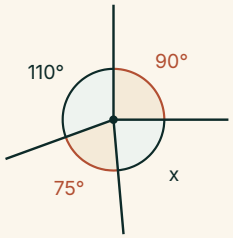
7.1.04 FLUENCY

Find x. State the angle fact used and check by addition.



7.1.05 FLUENCY

Three angles around a point are 90°, 110° and 75°. Find the fourth angle and check the full turn.



CHAPTER 7.1 · PRACTICE

Angle language and turn facts

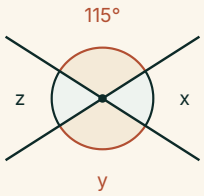
Practice 7.1 · Angle language and turn facts

7.1.06–7.1.07

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

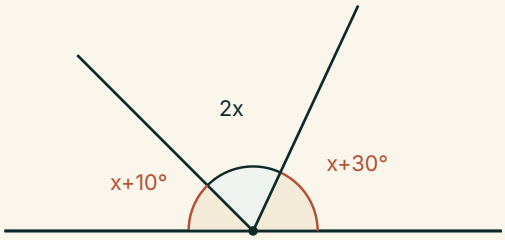
7.1.06 FLUENCY

Two lines cross. One angle is 115°. Find x, y and z, naming the relationship used for each.



7.1.07 FLUENCY

Three adjacent angles on a straight line are x + 30°, 2x and x + 10°. Find x and all three angles.



CHAPTER 7.1 · PRACTICE

Angle language and turn facts

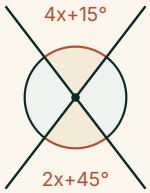
Practice 7.1 · Angle language and turn facts

7.1.08–7.1.10

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

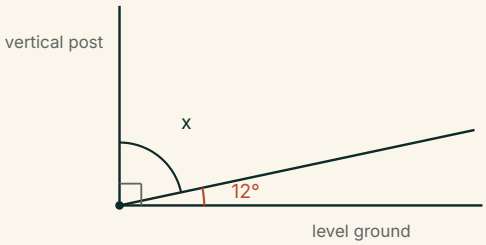
7.1.08 FLUENCY

The vertically opposite angles $4x + 15°$ and $2x + 45°$ are marked. Find x, the marked angles and one adjacent angle.



7.1.09 APPLICATION

A ramp makes $12°$ with level ground. A vertical post meets the ground at the same point, and the ramp lies between the ground and the post. Find the angle between the ramp and the post, and explain why the ground–post angle is $90°$.



7.1.10 APPLICATION

A turning sensor records three consecutive turns of $118°$, $97°$ and $85°$ before completing one full turn. Find the fourth turn. Would the answer change if the turns were recorded in a different order?

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

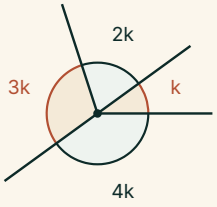
Practice 7.1 · Angle language and turn facts

7.1.11–7.1.13

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.11 APPLICATION

A logo divides a full turn into four angles in the ratio 1 : 2 : 3 : 4. Find the four angles. Is any pair complementary?



7.1.12 REASONING

A student writes two claims. Identify the incorrect claim, correct it, and draw a counterexample to the original claim.

CLAIM	STATEMENT
1	Adjacent angles on a straight line sum to 180°.
2	Angles around a point sum to 180°.

7.1.13 REASONING

True or false: if two adjacent angles on a straight line are equal, each must be 90°. Justify with an equation and a sketch.

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

Practice 7.1 · Angle language and turn facts

7.1.14–7.1.16

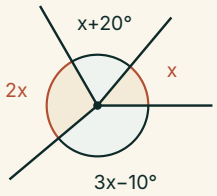
Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.14 REASONING

Complete the missing steps: for angles 2x, x + 40° and 80° around a point, __ + __ + __ = 360°; so __ x = __; x = __. Check the three angles.

7.1.15 CHALLENGE

Four angles around a point are x, x + 20°, 2x and 3x − 10°. Find x and the largest angle. Explain why the largest angle is not reflex.



7.1.16 CHALLENGE

Draw two different pairs of crossing lines with one angle 70°. Label all four angles in each diagram. Compare your diagrams with a partner: what stays the same and what changes?

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.2 Parallel lines and transversals

A transversal crosses two or more lines. When the crossed lines are parallel, the angles fall into equal or supplementary families.

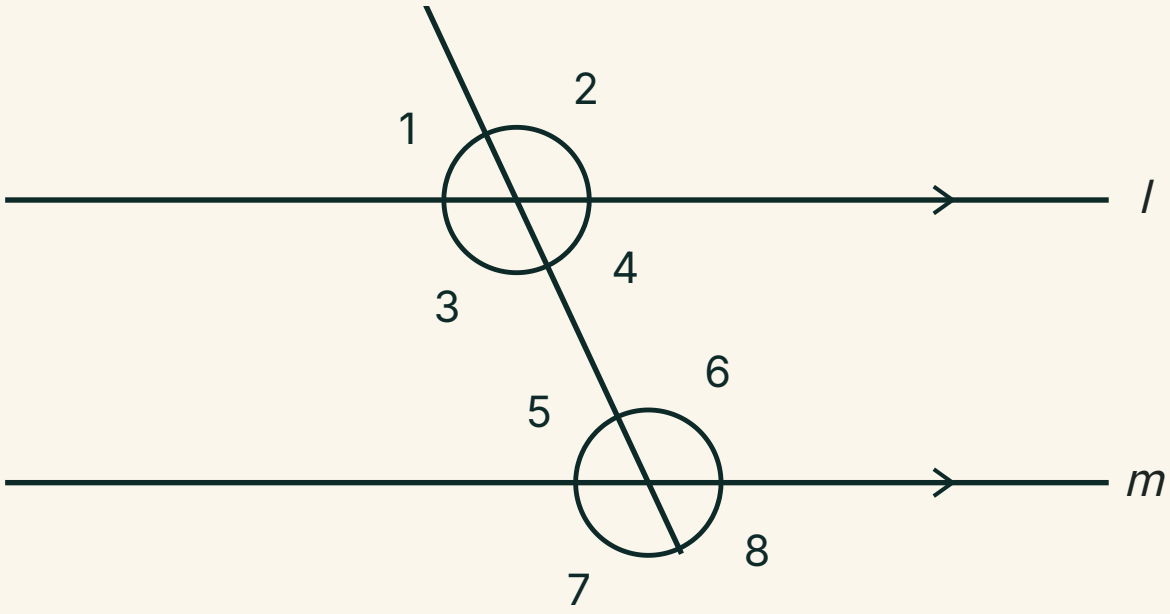


Figure 7.4 · With l parallel to m: corresponding and alternate angles are equal; co-interior angles are supplementary.

Key idea

Corresponding angles sit in the same position at each intersection. Alternate angles lie on opposite sides of the transversal between the parallel lines. Co-interior angles lie on the same side between the parallel lines.

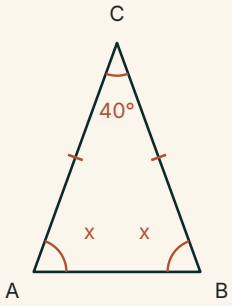
WATCH FOR THIS MISCONCEPTION

Do not assume lines are parallel or angles equal unless marked or stated. Conversely, equal alternate angles or supplementary co-interior angles can show that two lines are parallel.

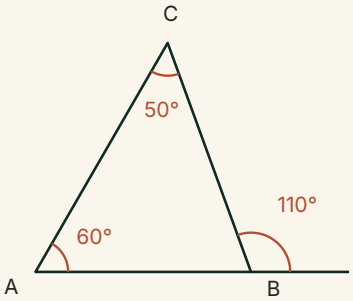
CHAPTER 7 · MATHEMATICAL DEVELOPMENT

Triangle and exterior-angle facts

Every triangle has angle sum 180°. An exterior angle equals the sum of the two opposite interior angles.



Equal sides give equal base angles.



Exterior angle = sum of opposite interior angles.

Key idea

Triangles may be classified by sides (equilateral, isosceles, scalene) and by angles (acute, right, obtuse). In an isosceles triangle the angles opposite the equal sides are equal; conversely, equal angles face equal sides. In a right triangle the two acute angles are complementary.

COMMON ERROR

The exterior angle is equal to the sum of the two opposite interior angles. It is not added to them to make 180°. The adjacent interior angle is the one supplementary to the exterior angle.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.1

Parallel-line angle chase

PROBLEM

Lines l and m are parallel. The marked angle is 68°. Find a, b and c.

VISUAL REPRESENTATION

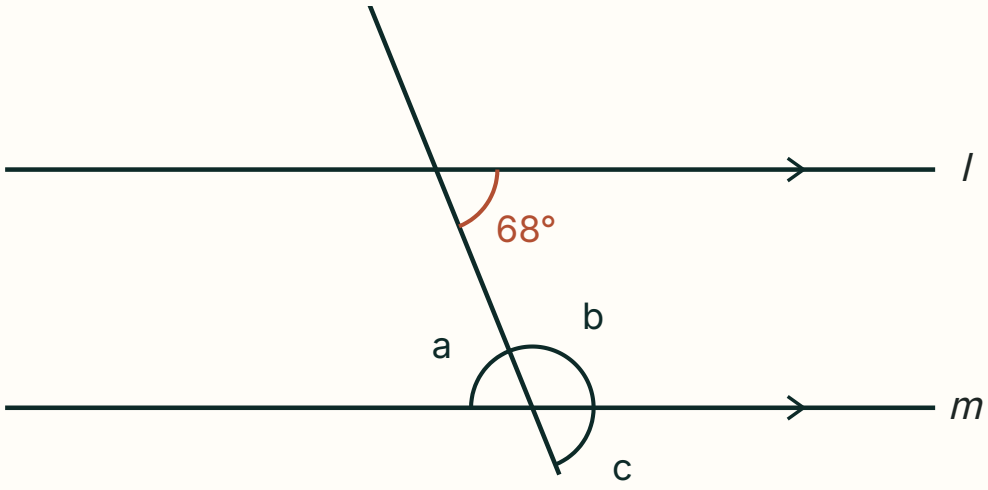


Figure 7.5 · Parallel marks justify the angle relationships.

MATHEMATICAL REASONING

With parallel lines, alternate angles are equal, co-interior angles are supplementary, and corresponding angles are equal.

STEP-BY-STEP SOLUTION

1. a is alternate to 68°, so a = 68°.
2. b is co-interior with 68°, so b = 180° − 68° = 112°.
3. c corresponds to the 68° angle (both are in the lower-right position), so c = 68°.
4. Check: a + b = 68° + 112° = 180°.

Final answer. a = 68°, b = 112°, c = 68°.

Interpretation. The transversal creates two angle families: 68° and 112°. Naming the relationship prevents guessing from appearance.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.2

Exterior angle of a triangle

PROBLEM

An exterior angle of a triangle is 115°. One opposite interior angle is 48°. Find the other opposite interior angle and the adjacent interior angle.

VISUAL REPRESENTATION

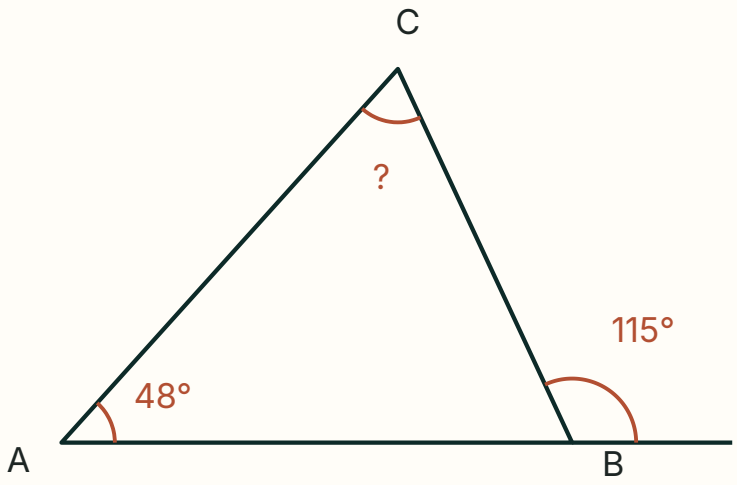


Figure 7.6 · The exterior angle is outside the triangle at the extended base.

MATHEMATICAL REASONING

An exterior angle equals the sum of the two opposite interior angles. It is supplementary to the interior angle at the same vertex.

STEP-BY-STEP SOLUTION

1. Other opposite interior angle = 115° − 48° = 67°.
2. Adjacent interior angle = 180° − 115° = 65°.
3. Check the triangle sum: 48° + 67° + 65° = 180°.

Final answer. Other opposite angle 67°; adjacent interior angle 65°.

Interpretation. The exterior angle is larger than either opposite interior angle. The diagram shows which angles belong to the exterior-angle relationship.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.3

Isosceles base angles

PROBLEM

An isosceles triangle has apex angle 40°. Find each base angle.

VISUAL REPRESENTATION

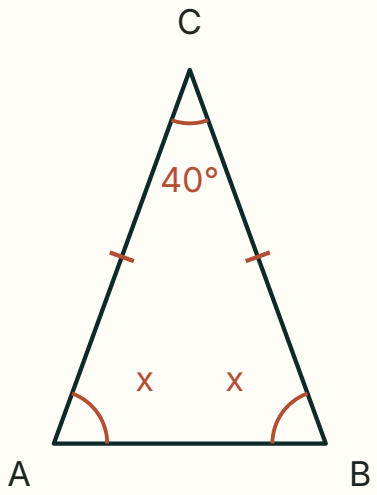


Figure 7.7 · Equal sides force equal base angles.

MATHEMATICAL REASONING

Equal sides have equal opposite angles. The triangle angle sum is 180°.

STEP-BY-STEP SOLUTION

1. The tick marks show two equal sides, so the base angles are equal. Call each x.
2. Write $2x + 40° = 180°$.
3. Then $2x = 140°$, so $x = 70°$.
4. Check: $70° + 70° + 40° = 180°$.

Final answer. Each base angle is 70°.

Interpretation. The equal-side marks are the evidence for equal base angles; the calculation then uses the angle sum.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.4

Why every triangle has angle sum 180°

PROBLEM

Use a line through the apex parallel to the base to explain why the angles of a triangle sum to 180°.

VISUAL REPRESENTATION

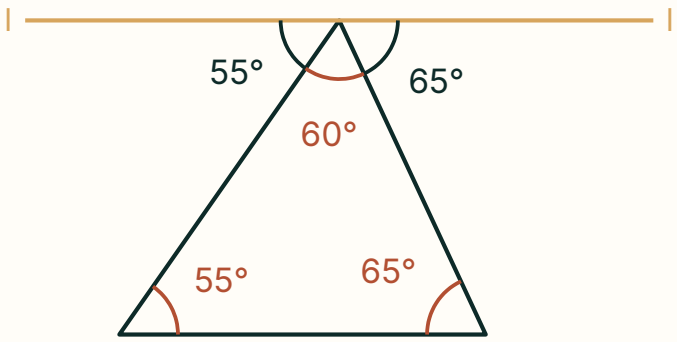


Figure 7.8 · The parallel line transfers the base angles to the apex.

MATHEMATICAL REASONING

Alternate angles between parallel lines are equal. Angles on a straight line sum to 180°.

STEP-BY-STEP SOLUTION

1. Draw a line through the apex parallel to the base.
2. The left base angle equals the left angle at the apex (alternate angles). The right base angle equals the right angle at the apex.
3. The three angles on the straight line through the apex are therefore the three triangle angles.
4. They sum to 180°, so the triangle angles sum to 180°.

Final answer. A + B + C = 180° for every triangle.

Interpretation. This is a visual argument, not a measurement of one triangle. It explains why the rule is general.

CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

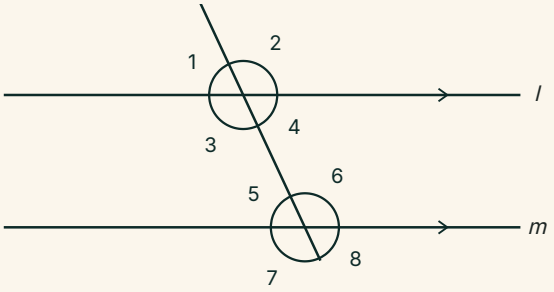
Practice 7.2 · Parallel lines and transversals

7.2.01–7.2.02

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

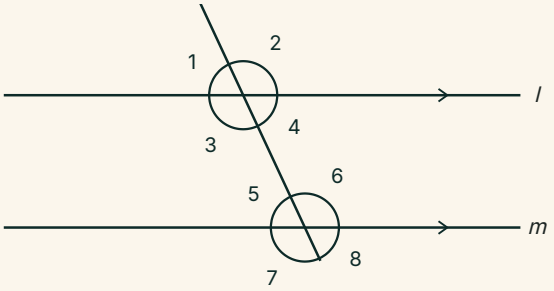
7.2.01 UNDERSTAND

Use the numbered diagram. Name one pair of corresponding angles, one pair of alternate angles and one pair of co-interior angles. State which pairs are equal and which are supplementary.



7.2.02 UNDERSTAND

True or false: any two angles between parallel lines are equal. Use the diagram to justify your answer.



CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

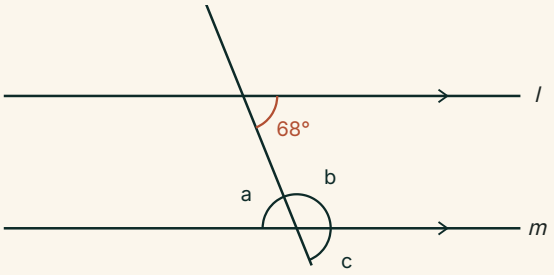
Practice 7.2 · Parallel lines and transversals

7.2.03–7.2.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

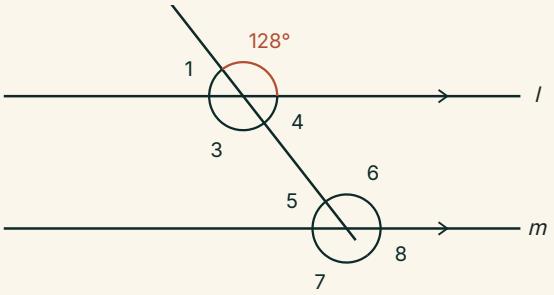
7.2.03 FLUENCY

Find a, b and c. Give a reason for each value.



7.2.04 FLUENCY

In the diagram, angle 2 = 128°. Find angle 5, angle 6 and angle 7, naming each relationship.



7.2.05 REASONING

Two lines are cut by a transversal. A pair of co-interior angles measures 112° and 68°. Are the lines parallel? Justify. Then draw a non-parallel pair and explain why the same conclusion cannot be drawn from appearance alone.

CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

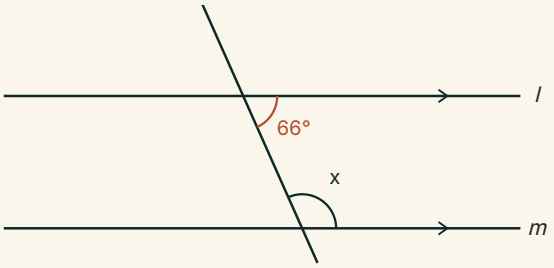
Practice 7.2 · Parallel lines and transversals

7.2.11–7.2.19

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

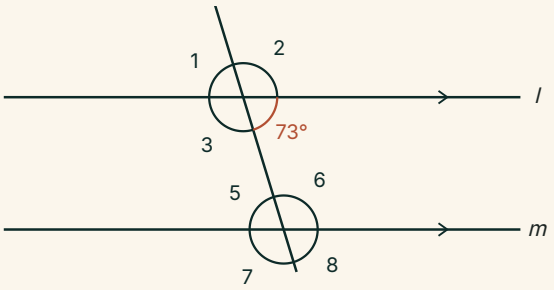
7.2.11 APPLICATION

A collar is parallel to the base. At a rafter joint the co-interior angle inside the frame is 66°. Find the angle x on the same side of the transversal at the other joint.



7.2.19 APPLICATION

Two parallel shelves are crossed by a support. One acute angle is 73°. Complete a table showing all eight angles, then state how many are acute and how many are obtuse.



CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

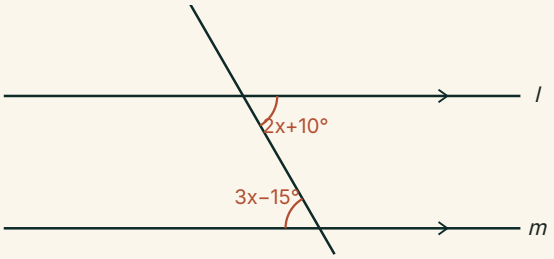
Practice 7.2 · Parallel lines and transversals

7.2.20–7.2.20

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.20
REASONING

The alternate angles marked $2x + 10°$ and $3x − 15°$ are formed by parallel lines. Find x, the marked angles, and the co-interior partner of one marked angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

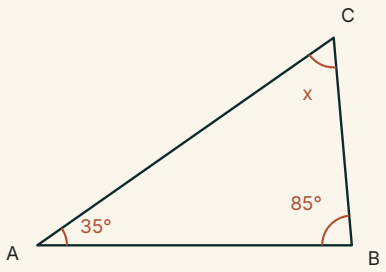
Practice 7.2 · Triangle and exterior-angle facts

7.2.06–7.2.07

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

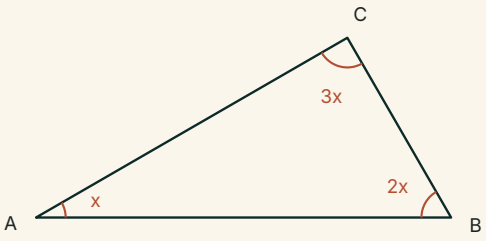
7.2.06 UNDERSTAND

A triangle has angles 35° and 85°. Find the third angle and classify the triangle by angles and by sides.



7.2.07 FLUENCY

The angles of a triangle are x, 2x and 3x. Find x and classify the triangle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

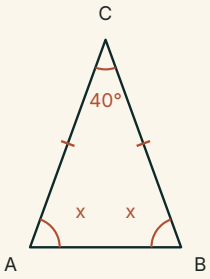
Practice 7.2 · Triangle and exterior-angle facts

7.2.08–7.2.09

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

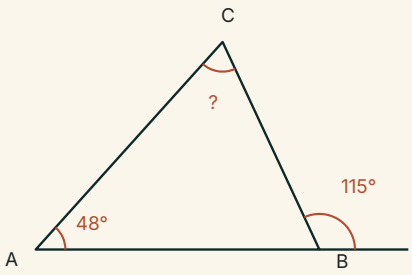
7.2.08
FLUENCY

Find each base angle x. Which marks tell you the base angles are equal?



7.2.09
FLUENCY

An exterior angle is 115° and one opposite interior angle is 48°. Find the other opposite interior angle and the interior angle adjacent to the exterior angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

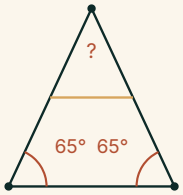
Practice 7.2 · Triangle and exterior-angle facts

7.2.10–7.2.12

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.10 APPLICATION

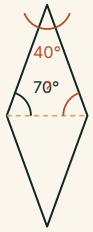
The truss has equal base angles of 65°. Find the apex angle and the exterior angle at a base. Classify the triangle and state what this implies about the rafters.



8 m base

7.2.12 APPLICATION

The kite is made from two identical isosceles triangles joined along their bases. The upper triangle has apex 40°. Find the marked base angle, then find the whole angle at the left tip of the kite.



upper and lower triangles are identical

CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

Practice 7.2 · Triangle and exterior-angle facts

7.2.13–7.2.15

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.13 APPLICATION

A right-triangular ramp frame has a 14° angle with the ground. A handrail continues the sloping side beyond the top. Find the top interior angle and the exterior angle between the handrail extension and the vertical post.

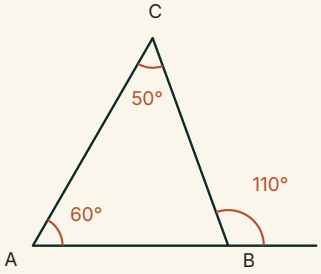


7.2.14 REASONING

Can a triangle have two obtuse angles? Explain using the angle sum, and sketch what goes wrong if you try.

7.2.15 REASONING

A student says the exterior angle 110° should be added to the opposite interior angles 60° and 50° to make 180°. Identify the error, correct it, and find the adjacent interior angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

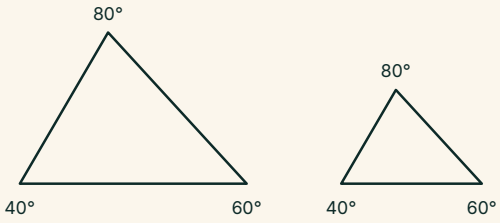
Practice 7.2 · Triangle and exterior-angle facts

7.2.16–7.2.17

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

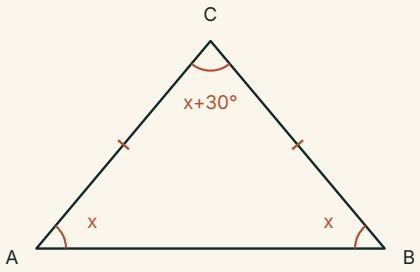
7.2.16 REASONING

Compare the two triangles. What is the same and what is different? Does having the same three angles force the side lengths to be the same? Justify.



7.2.17 CHALLENGE

An isosceles triangle has base angles x and apex x + 30°. Find all three angles. Show two different checks.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

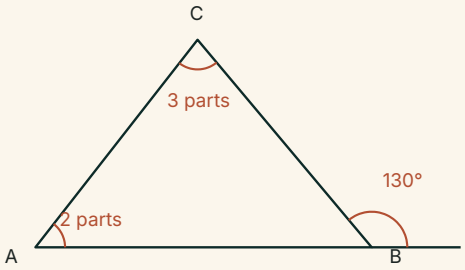
Practice 7.2 · Triangle and exterior-angle facts

7.2.18–7.2.18

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.18 CHALLENGE

An exterior angle of a triangle is 130°. The two opposite interior angles are in the ratio 2 : 3. Find all three interior angles and classify the triangle.



CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.3 Polygons and their angles

A polygon is a closed plane shape made from straight sides. Its interior angle sum depends on the number of sides.

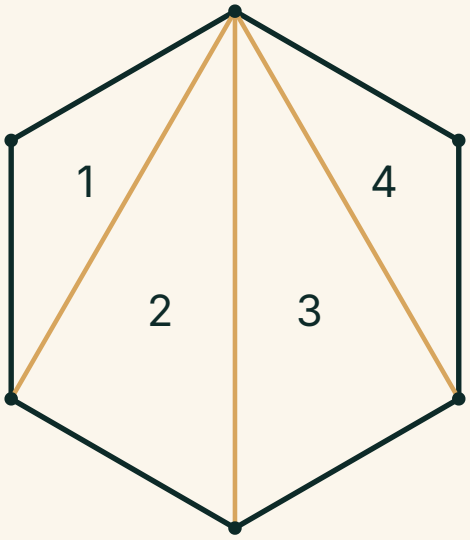


Figure 7.9 · A hexagon splits into four triangles from one vertex.

Choose one vertex and draw diagonals to every non-adjacent vertex. An n-sided polygon divides into n−2 triangles, so its interior angle sum is (n−2)×180°. The exterior angles of any convex polygon sum to 360°.

Key idea

Interior sum: (n−2)×180°. Exterior sum: 360°. Regular polygon: each exterior = 360°÷n; each interior = 180°−360°÷n.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.1

Missing angle in a quadrilateral

PROBLEM

A quadrilateral has angles 85°, 95°, 110° and x. Find x.

VISUAL REPRESENTATION

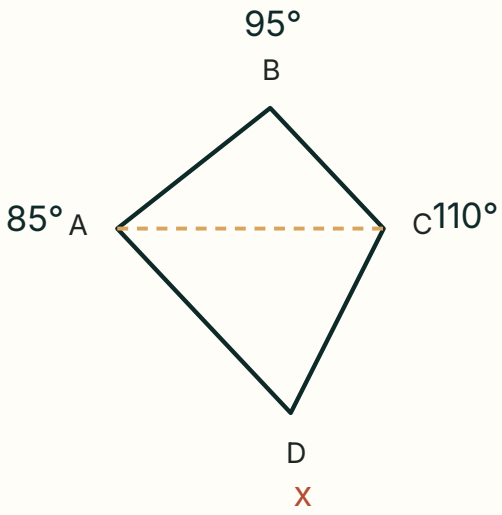


Figure 7.10 · One diagonal divides the quadrilateral into two triangles.

MATHEMATICAL REASONING

A quadrilateral can be split into two triangles, so its interior angle sum is $2 \times 180° = 360°$.

STEP-BY-STEP SOLUTION

1. Add the known angles: $85° + 95° + 110° = 290°$.
2. Subtract from 360°: $x = 360° - 290° = 70°$.
3. Check: $85° + 95° + 110° + 70° = 360°$.

Final answer. x = 70°.

Interpretation. The diagonal in the diagram shows where the 360° comes from: two triangle sums.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.2

Regular polygon from an exterior angle

PROBLEM

A regular polygon has exterior angle 40°. Find the number of sides and each interior angle.

VISUAL REPRESENTATION

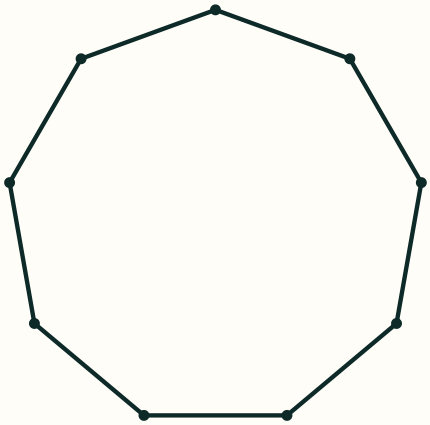


Figure 7.11 · A regular nonagon has nine equal exterior turns.

MATHEMATICAL REASONING

The exterior angles of any convex polygon sum to 360°. In a regular polygon they are equal. Interior and exterior angles at a vertex are supplementary.

STEP-BY-STEP SOLUTION

1. Number of sides $n = 360° \div 40° = 9$.
2. Each interior angle $= 180° - 40° = 140°$.
3. Check with the interior-sum formula: $(9 - 2) \times 180° = 1260°$, and $1260° \div 9 = 140°$.

Final answer. 9 sides; each interior angle 140°.

Interpretation. A small exterior angle means many sides. The polygon is a regular nonagon.

CHAPTER 7.3 · PRACTICE

Polygon angle sums

Practice 7.3 · Polygons

7.3.01–7.3.02

7.3.01 UNDERSTAND

Name the polygons and complete the number of sides. Then state what makes a polygon regular.

NAME	SIDES
Pentagon	___
Hexagon	___
Octagon	___

7.3.02 UNDERSTAND

Compare a regular hexagon with an irregular hexagon. Sketch each and label one property that is the same and one that may differ.

CHAPTER 7.3 · PRACTICE

Polygon angle sums

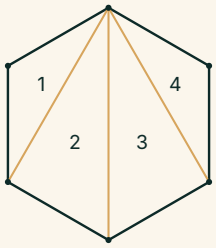
Practice 7.3 · Polygons

7.3.03–7.3.04

7.3.03
FLUENCY

Complete the triangulation table. Use the hexagon diagram to check the last row.

POLYGON	TRIANGLES	INTERIOR SUM
Quadrilateral	___	___°
Pentagon	___	___°
Hexagon	___	___°



7.3.04
FLUENCY

Find the interior angle sum of a heptagon, a nonagon and a decagon. Show the number of triangles first.

POLYGON	SIDES	TRIANGLES	SUM
Heptagon	7	___	___°
Nonagon	9	___	___°
Decagon	10	___	___°

CHAPTER 7.3 · PRACTICE

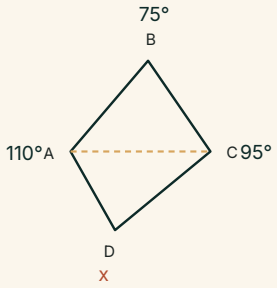
Polygon angle sums

Practice 7.3 · Polygons

7.3.05–7.3.06

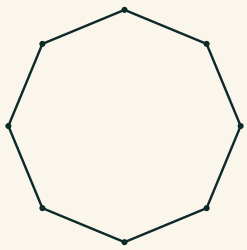
7.3.05 FLUENCY

Find the missing angle x in the quadrilateral. Check by adding all four angles.



7.3.06 FLUENCY

A regular polygon has exterior angle 45°. Find the number of sides and each interior angle. Sketch the polygon and mark one interior–exterior pair.



CHAPTER 7.3 · PRACTICE

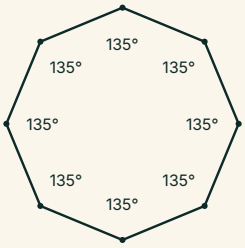
Polygon angle sums

Practice 7.3 · Polygons

7.3.07–7.3.08

7.3.07
APPLICATION

A floor-tile design uses a regular polygon whose interior angle is 135°. Identify the polygon. Can copies of this tile alone meet around a point with no gap? Justify.



7.3.08
REASONING

A student walks once around a convex polygon, turning through the exterior angle at each vertex. Complete the table and explain why the total turn is 360° for every convex polygon.

POLYGON	NUMBER OF EXTERIOR TURNS	TOTAL TURN
Triangle	3	___°
Quadrilateral	4	___°
Decagon	10	___°

CHAPTER 7 · INQUIRY LAUNCH

Which lengths close a triangle?

Three strips joined end to end may or may not close a triangle. Predict first, then test. Do not state a rule until the evidence forces it.

SIDE LENGTHS (CM)	PREDICTION	ARCS MEET ABOVE THE BASE?	TRIANGLE?
3, 4, 5			
3, 4, 8			
4, 5, 9			
6, 7, 10			
2, 2, 3			

> OBSERVE
>
> When the arcs meet on the base, the “triangle” has zero area. When they do not meet, there is a gap. What must be true about the two shorter lengths for a real apex to exist?

Keep this prediction. You will return to it after testing more cases.

CHAPTER 7 · INVESTIGATION · CRITERION B PRACTICE

Test lengths with compass arcs

Choose a longest side c and two shorter sides a and b. Set the base to c, then draw an arc of radius a from one endpoint and an arc of radius b from the other. Record what happens.

A, B, C (CM)	A+B	COMPARE A+B WITH C	ARCS MEET ABOVE BASE?	TRIANGLE?
4, 5, 6				
2, 3, 6				
5, 8, 13				
6, 8, 9				
your own case				

1. Complete the table. Include at least one borderline case of your own.
2. Compare a+b with c for each row. Write a conjecture that predicts whether a triangle can be made.
3. Test your conjecture on 5, 8, 12 and 7, 7, 14. Does it still work?
4. Explain why the arcs behave as they do. Use a labelled sketch.

ATL EVIDENCE. Choose your own extra case, organise the evidence, test the conjecture and explain the result.

CHAPTER 7 · INVESTIGATION · DISCOVERY

From evidence to a rule

Write your conjecture before comparing it with the formal statement below.

DISCOVERY

The two shorter sides must together be **strictly greater** than the longest side. If a+b=c, the arcs meet on the base and the triangle has zero area. If a+b<c, the arcs do not meet.

Conclusion

For sides a, b, c with c longest, a triangle exists if and only if a+b>c. The other two inequalities then hold automatically. This is the triangle inequality.



Figure 7.12 · Equality gives a degenerate straight line.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.3

Possible third sides

PROBLEM

Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values.

VISUAL REPRESENTATION

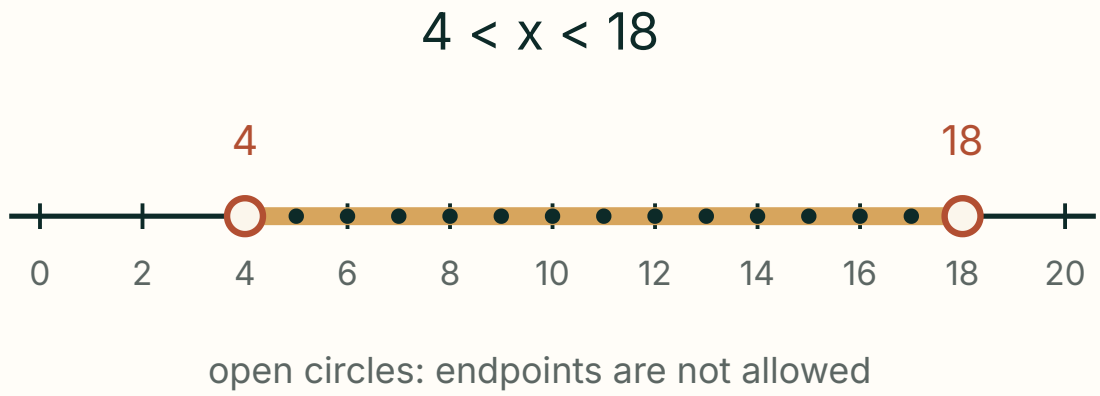


Figure 7.13 · The number line represents 4 < x < 18.

MATHEMATICAL REASONING

The third side must be greater than the difference of the other two sides and less than their sum. Endpoints give degenerate cases.

STEP-BY-STEP SOLUTION

1. Lower boundary: 11 − 7 = 4, so x > 4.
2. Upper boundary: 11 + 7 = 18, so x < 18.
3. Whole-number values are 5, 6, 7, ..., 17.
4. Check endpoints: x = 4 gives 4 + 7 = 11; x = 18 gives 7 + 11 = 18. Both are straight-line cases, not triangles.

Final answer. x = 5, 6, 7, ..., 17 cm.

Interpretation. The open circles on the number line show that 4 and 18 are excluded. The integer dots are the possible whole-centimetre lengths.

CHAPTER 7.3 · PRACTICE

Triangle inequality

Practice 7.3 · Possible triangles

7.3.09–7.3.11

7.3.09 INVESTIGATION

Test whether each triple can form a triangle. Record the sum of the two shorter sides and compare it with the longest side.

SIDES (CM)	SHORTER SUM	LONGEST	TRIANGLE?
3, 4, 8	___	___	___
5, 5, 9	___	___	___
6, 7, 10	___	___	___

7.3.10 INVESTIGATION

Complete the table. Include one row of your own that is a borderline case. What pattern separates possible triangles from impossible ones?

A, B, C (CM)	A+B	COMPARE WITH C	TRIANGLE?
4, 5, 6	___	___	___
2, 3, 6	___	___	___
5, 8, 13	___	___	___
6, 8, 9	___	___	___
your row	___	___	___

7.3.11 FLUENCY

A frame has sides 7 cm, 9 cm and 15 cm. Can it be a triangle? If not, change only the longest side to the smallest whole-number length that makes a triangle.

CHAPTER 7.3 · PRACTICE

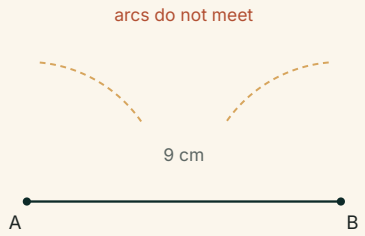
Triangle inequality

Practice 7.3 · Possible triangles

7.3.12–7.3.14

7.3.12 APPLICATION

A student says a triangular brace with sides 4 cm, 4 cm and 9 cm is possible because 4+9>4. Correct the reasoning and sketch what happens when the two 4 cm sides are joined to the 9 cm base.



7.3.13 REASONING

Use the arcs to explain why sides 5 cm, 8 cm and 13 cm do not form a triangle. What changes if the 13 cm side becomes 12 cm?



7.3.14 CHALLENGE

Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values and represent them on a number line.

CHAPTER 7.3 · PRACTICE

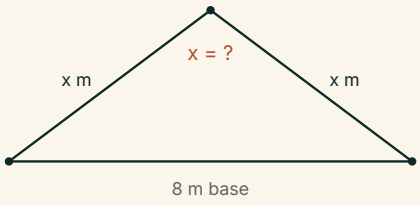
Triangle inequality

Practice 7.3 · Possible triangles

7.3.15–7.3.17

7.3.15 CHALLENGE

A truss model has an 8 m base and equal rafters of length x m. Find the smallest whole-number value of x that makes a triangle. Explain why x = 4 is not enough.

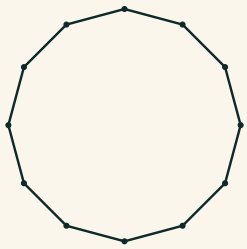


7.3.16 CHALLENGE

Use compass arcs to explain why the two shorter sides of a triangle must together exceed the longest side. Include the cases a+b<c, a+b=c and a+b>c in your explanation.

7.3.17 CHALLENGE

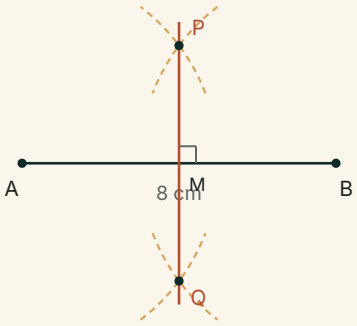
A regular polygon has each interior angle five times its exterior angle. Find the number of sides. Show the relationship between one interior and exterior pair.



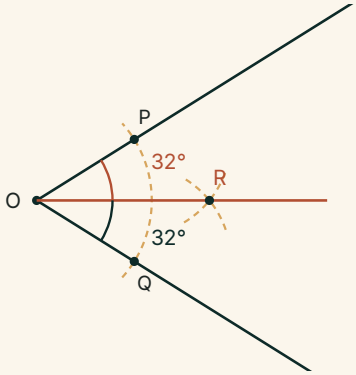
CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.4 Constructions with ruler and compass

A construction is an argument made with instruments. Each arc records a fixed distance from a named point.



Perpendicular bisector: equidistant from two endpoints.



Angle bisector: equidistant from the two arms.

CONSTRUCTION	COMPASS SETTING	HOW TO TEST
Perpendicular bisector of AB	One radius greater than half AB, unchanged from A and B	Crosses AB at its midpoint and makes 90°
Angle bisector	Arc from vertex cuts both arms; same radius from those points	Two half-angles are equal
Triangle from three sides	Base equal to one side; radii equal to the other two sides	Arcs meet; two shorter sides exceed the longest

COMMON ERROR

Never change the compass between a pair of arcs. Check the defining property, not the appearance of the drawing.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

Constructing triangles from sides

To construct a triangle from three sides, fix the base and locate the apex with two arcs.

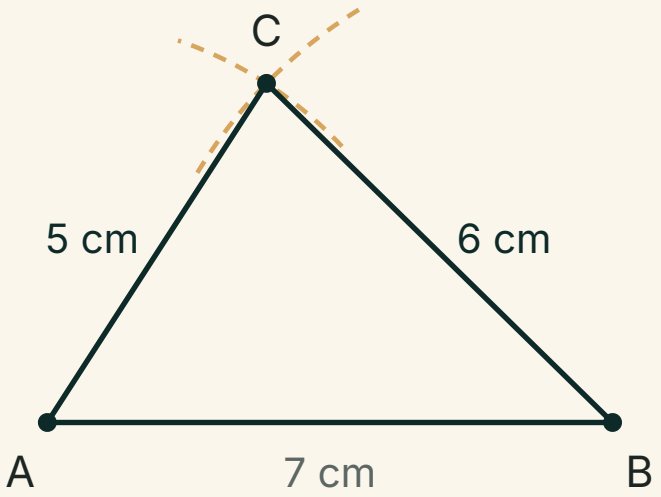


Figure 7.14 · The arc intersection fixes the third vertex.

The triangle inequality predicts whether the arcs meet above the base. If the two shorter sides together equal the longest side, the apex lies on the base; if they are shorter, the arcs do not meet.

Key idea

SSS construction: base AB, arc from A with radius AC, arc from B with radius BC. The intersection is C. Keep the arcs visible as evidence.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.1

Perpendicular bisector

PROBLEM

Construct the perpendicular bisector of AB = 8 cm. State the midpoint length and the check.

VISUAL REPRESENTATION

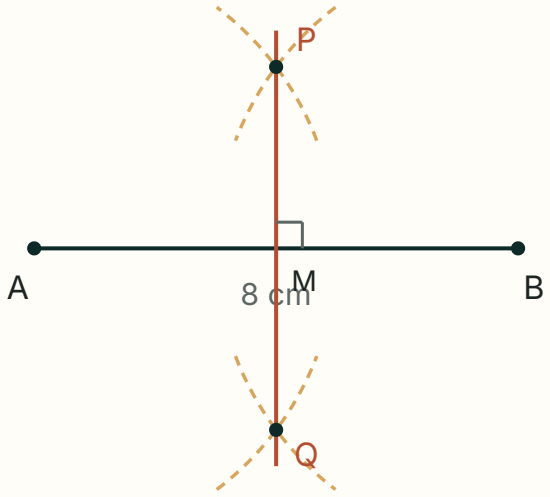


Figure 7.15 · The same compass radius is used from both endpoints.

MATHEMATICAL REASONING

Points on the perpendicular bisector are equidistant from A and B. Equal-radius arcs locate two such points.

STEP-BY-STEP SOLUTION

1. Draw AB = 8 cm.
2. Choose one radius greater than half of AB, for example 5 cm. Keep it fixed.
3. Draw arcs from A and from B above and below AB. Label the intersections P and Q.
4. Draw PQ. It crosses AB at its midpoint M, so AM = MB = 4 cm, and PQ is perpendicular to AB.

Final answer. AM = MB = 4 cm; PQ is perpendicular to AB.

Interpretation. The arcs are evidence: P and Q are each the same distance from A and B, so the line through them is the perpendicular bisector.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.2

Angle bisector

PROBLEM

Bisect a 64° angle with ruler and compasses. Find each half.

VISUAL REPRESENTATION

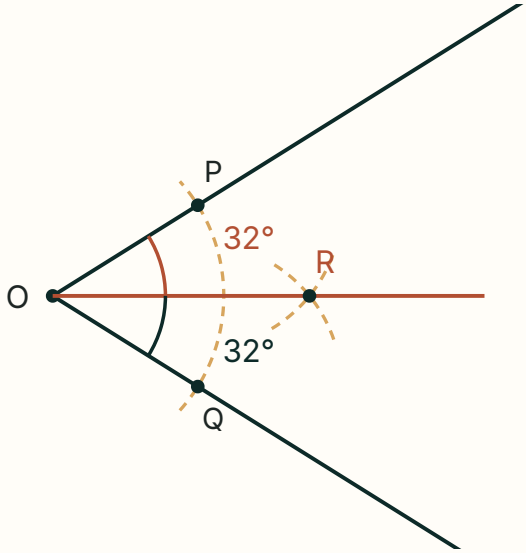


Figure 7.16 · The bisector is the ray OR.

MATHEMATICAL REASONING

An angle bisector divides an angle into two equal angles. Equal-radius arcs from the arms locate a point on the bisector.

STEP-BY-STEP SOLUTION

1. Draw an arc from the vertex O, cutting the arms at P and Q.
2. With the same radius from P and Q, draw arcs meeting at R.
3. Draw OR. This is the angle bisector.
4. Each half is $64° \div 2 = 32°$. Check: $32° + 32° = 64°$.

Final answer. Each half is 32°.

Interpretation. The construction forces equality; measuring is only a check. Do not confuse this with the perpendicular bisector of a segment.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.3

Triangle from three sides (SSS)

PROBLEM

Construct a triangle with sides 5 cm, 6 cm and 7 cm.

VISUAL REPRESENTATION

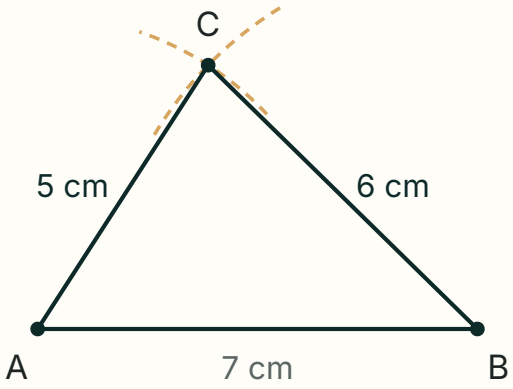


Figure 7.17 · The arc intersection fixes the third vertex.

MATHEMATICAL REASONING

If two sides and the base are fixed, the third vertex is the intersection of two circles/arcs. The triangle inequality predicts whether the arcs meet.

STEP-BY-STEP SOLUTION

1. Check: 5 + 6 > 7, so a triangle is possible.
2. Draw base AB = 7 cm.
3. From A draw an arc of radius 5 cm; from B draw an arc of radius 6 cm.
4. Label the intersection C and join AC and BC. Keep the arcs visible.

Final answer. Triangle ABC has AB = 7 cm, AC = 5 cm and BC = 6 cm.

Interpretation. The arcs locate the only possible apex on each side of the base. The two choices are mirror images.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.4

Isosceles triangle and a locus

PROBLEM

Construct an isosceles triangle with equal sides 6 cm and base 8 cm. Explain why the perpendicular bisector of the base passes through the apex.

VISUAL REPRESENTATION

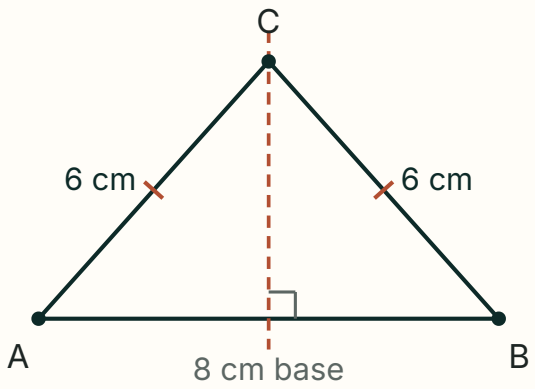


Figure 7.18 · Equal sides place the apex on the perpendicular bisector.

MATHEMATICAL REASONING

The apex is equidistant from the base endpoints. The perpendicular bisector is the locus of all points equidistant from the endpoints.

STEP-BY-STEP SOLUTION

1. Draw base AB = 8 cm.
2. With radius 6 cm from A and from B, draw arcs meeting at C.
3. Join AC and BC. Then AC = BC = 6 cm.
4. Because C is equidistant from A and B, it lies on the perpendicular bisector of AB.

Final answer. The perpendicular bisector of AB passes through C.

Interpretation. The equal sides are not just a picture feature; they place the apex on a precise locus. This connects construction with reasoning.

CHAPTER 7.4 · PRACTICE

Bisectors and loci

Practice 7.4 · Bisectors and loci

7.4.01–7.4.03

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.4.01 UNDERSTAND

Select the construction that locates all points equally distant from two points A and B: angle bisector, perpendicular bisector, parallel line, or circle centre A. Contrast this with the locus of points inside the angle between two intersecting roads that are equally distant from the roads.

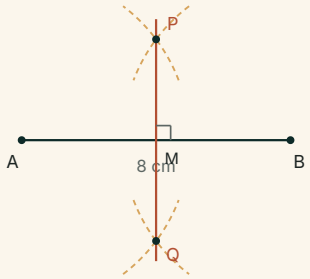
GIVEN	LOCUS NEEDED
Two points A and B	___
Inside the angle between two roads	___

7.4.02 UNDERSTAND

In a ruler-and-compass construction, what do equal-radius arcs record? Why must the compass setting stay fixed between a pair of arcs?

7.4.03 FLUENCY

Order the steps for a perpendicular bisector of AB = 8 cm: (i) draw arcs from A and B with the same radius; (ii) draw AB; (iii) join the arc intersections; (iv) open the compass to more than half AB. State the radius condition.



CHAPTER 7.4 · PRACTICE

Bisectors and loci

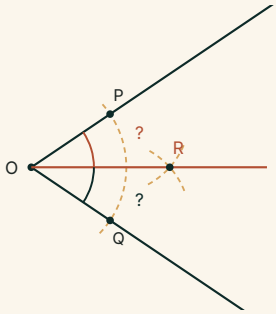
Practice 7.4 · Bisectors and loci

7.4.04–7.4.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

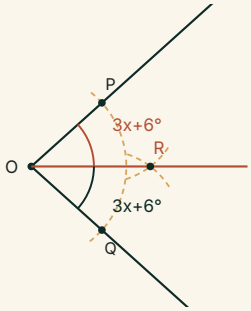
7.4.04 FLUENCY

An angle of 68° is bisected. Find each half and describe how the arcs show the two halves are equal.



7.4.05 FLUENCY

An 84° angle is bisected. Each half is labelled 3x + 6°. Find x and check.



CHAPTER 7.4 · PRACTICE

Bisectors and loci

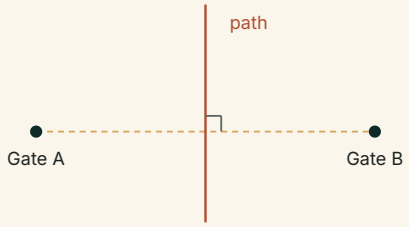
Practice 7.4 · Bisectors and loci

7.4.06–7.4.08

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

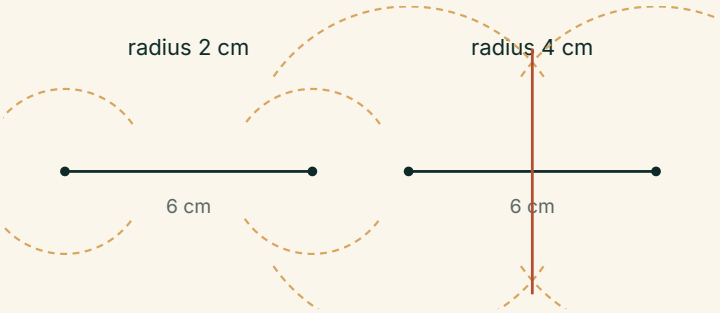
7.4.06 APPLICATION

A path must stay equally distant from gates A and B. Which construction locates the path? Explain how a partner could check one point on the path.



7.4.07 REASONING

A student uses a 2 cm radius to bisect a 6 cm segment. Compare the two diagrams. Explain why the first fails and the second works, and state the general radius condition.



7.4.08 REASONING

A construction looks perpendicular but the arcs were erased. Can you rely on appearance? Explain what evidence you would ask for and how you would check the result.

CHAPTER 7.4 · PRACTICE

Constructing triangles

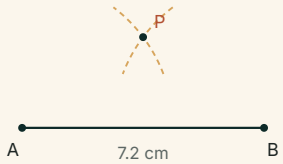
Practice 7.4 · Constructing triangles

7.4.09–7.4.10

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

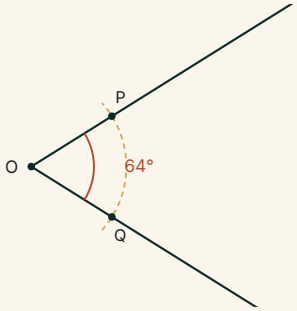
7.4.09 FLUENCY

Construct AB = 7.2 cm and its perpendicular bisector. Measure each half and state the check.



7.4.10 FLUENCY

Construct a 64° angle and bisect it with ruler and compasses. Measure both halves and state what they should be.



CHAPTER 7.4 · PRACTICE

Constructing triangles

Practice 7.4 · Constructing triangles

7.4.11–7.4.12

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

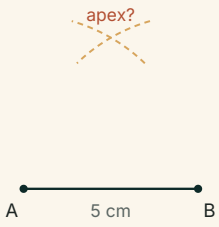
7.4.11 FLUENCY

Construct a triangle with sides 4 cm, 5 cm and 6 cm. Keep the arcs visible and label the vertices.



7.4.12 FLUENCY

Construct an equilateral triangle of side 5 cm. Explain why every angle must be 60°.



CHAPTER 7.4 · PRACTICE

Constructing triangles

Practice 7.4 · Constructing triangles

7.4.13–7.4.14

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

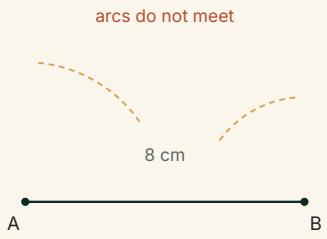
7.4.13 CHALLENGE

Construct an isosceles triangle with equal sides 6 cm and base 8 cm. Then construct the perpendicular bisector of the base. Explain why it passes through the apex.



7.4.14 REASONING

Attempt to construct a triangle with sides 3 cm, 4 cm and 8 cm. Describe what the arcs do and connect this to the triangle inequality.



CHAPTER 7.4 · PRACTICE

Constructing triangles

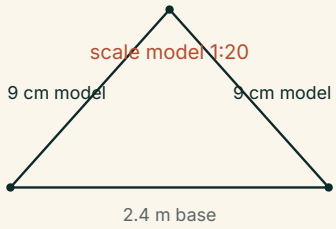
Practice 7.4 · Constructing triangles

7.4.15–7.4.16

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

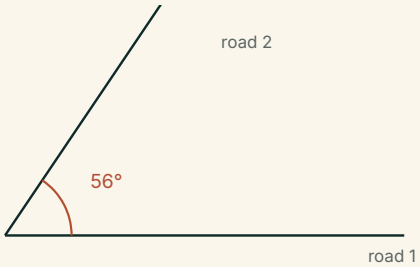
7.4.15 APPLICATION

A scale drawing uses 1 : 20. The real base of a triangular frame is 2.4 m. Find the model base in centimetres, then construct an isosceles model with equal sides 9 cm. What real rafter length does this represent?



7.4.16 CHALLENGE

A sprinkler is inside the angle between two straight roads that meet at 56° and must be equally distant from the roads. Sketch the locus and find the angle between the locus and each road. How would the answer change if the requirement were equal distance from two points instead?



CHAPTER 7 · MIXED PRACTICE

Connect ideas across the chapter

This route revisits items from the bank in a mixed order. Do not use a calculator. Draw a sketch, mark known facts and give a reason for each step.

ROUTE	ITEMS	FOCUS
Fluency	7.1.04, 7.1.05, 7.2.08, 7.3.06	Angle facts and regular polygons
Visual reasoning	7.2.03, 7.2.15, 7.3.05, 7.2.10	Diagrams, triangles and trusses
Construction	7.4.03, 7.4.06, 7.4.07, 7.4.08	Equal radii and evidence
Inquiry and challenge	7.3.13, 7.3.14, 7.3.15, 7.3.16	Triangle inequality and design

CHECK YOUR PROGRESS

Match the fact to the position: a line gives 180°, a point gives 360°, and co-interior angles are supplementary, not equal. Use the correct pair: alternate and corresponding angles are equal only when the crossed lines are parallel.

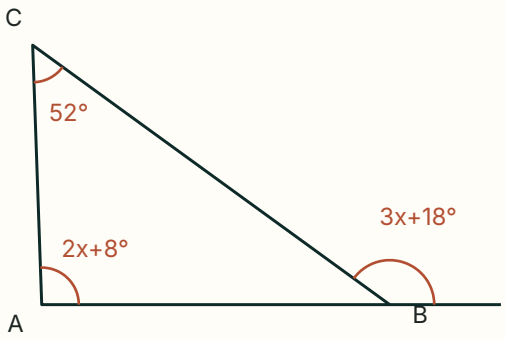
CHAPTER 7 · ASSESSMENT · CRITERION A

Knowing and understanding

Select appropriate mathematics, apply it successfully and solve problems in familiar and unfamiliar situations. Show your method and a check.

A1 · The badge design

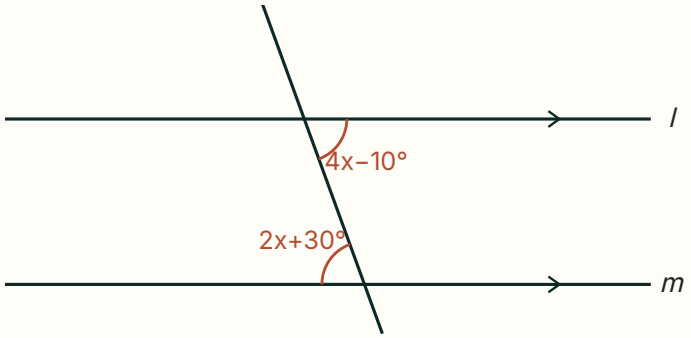
A badge is made from a triangle with one side extended. The exterior angle is $3x + 18°$, and the two opposite interior angles are $2x + 8°$ and $52°$. Find x, all three interior angles, and classify the triangle.



Assessment figure A1

A2 · Parallel supports

Two parallel supports are crossed by a brace. Alternate angles are labelled $4x - 10°$ and $2x + 30°$. Find x and the co-interior partner of one marked angle.



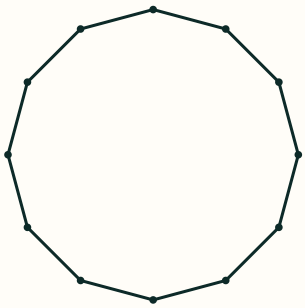
Assessment figure A2

CHAPTER 7 · ASSESSMENT · CRITERION A

Knowing and understanding · continued

A3 · A tile and a frame

A regular tile has interior angle 150°. Identify the polygon. A triangular frame has two sides 5 cm and 9 cm. Find all whole-centimetre lengths for the third side. Explain why the endpoints are excluded.



Assessment figure A3

Criterion A evidence: your choice of method, successful application and correct solutions. Do not just state answers; show the mathematics that supports them.

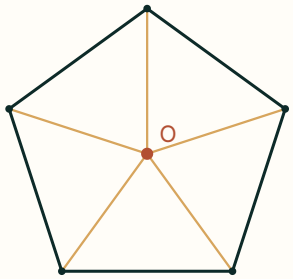
CHAPTER 7 · ASSESSMENT · CRITERION B

Investigating patterns

Work independently. Choose your own method and representation. Your task is to discover a relationship, describe it as a general rule, and verify it on new examples.

B1 · Triangles from an interior point

Choose a convex polygon. Mark a point O inside it and join O to every vertex. Investigate how the number of sides affects the interior angle sum.



One possible starting diagram

1. Investigate at least a triangle, quadrilateral, pentagon and hexagon. Organise your findings in a way of your choice.
2. Describe a general rule for an n-sided polygon using this interior-point method.
3. Verify your rule for an octagon using a different method.
4. Explain why the angles around O must be subtracted. What changes if O lies on a side instead of inside?

ATL EVIDENCE. Choose a technique, look for a pattern, generalise, verify and explain.

CHAPTER 7 · ASSESSMENT · CRITERION B

Workspace for investigation

Criterion B evidence: organised cases, a rule consistent with your findings, verification on a new example, and an explanation. A table is optional; choose a representation that makes the pattern visible.

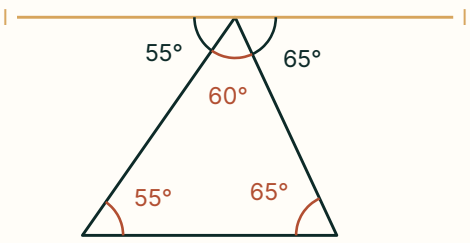
CHAPTER 7 · ASSESSMENT · CRITERION C

Communicating

Write for a partner who knows angle facts but has not seen this argument. Use mathematical language, a labelled diagram and a logical sequence.

C1 · Explain the triangle angle sum

Use the diagram to explain why the angles of any triangle sum to 180°. Your explanation must include the parallel line, alternate angles and the straight line through the apex.



Assessment figure C1

C2 · Improve a flawed explanation

A student writes: “The angles are 55°, 65° and 60°, so they add to 180°. Therefore every triangle adds to 180°.” Explain why this is not a general argument, then rewrite it using the parallel-line model.

A strong response uses correct notation, names the relationships, and moves logically from the diagram to the conclusion.

CHAPTER 7 · ASSESSMENT · CRITERION C

Workspace for communication

Self-check before submitting

- Have I used terms such as alternate, parallel and supplementary correctly?
- Does my diagram support each step?
- Could a partner follow the reasoning without seeing my working?

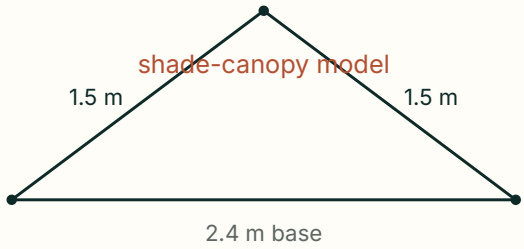
CHAPTER 7 · ASSESSMENT · CRITERION D

Applying mathematics in real-life contexts

Design brief: a simplified shade-canopy frame for a school courtyard. The dimensions below are illustrative, not construction specifications.

D1 · Check and draw the canopy frame

The frame is an isosceles triangle with base 2.4 m and equal rafters 1.5 m. A scale drawing uses 1 : 20.



Assessment figure D1 · simplified geometry only

1. Identify the relevant mathematical elements of the brief.
2. Select and apply strategies to check whether the triangle is possible and to find the scale-drawing lengths.
3. Construct the scale triangle with ruler and compasses. Measure the apex angle to the nearest degree and check the angle sum.
4. Explain the degree of accuracy of your drawing and measurements.
5. Describe whether the solution makes sense in context, and state one limitation of the model.

CHAPTER 7 · ASSESSMENT · CRITERION D

Workspace for application and interpretation

Context check

This task assesses geometry, scale and interpretation. It does not certify a real structure. Real canopies also need material thickness, joints, loads, wind and safety checks.

Criterion D evidence: relevant elements, appropriate strategies, a successful solution, accuracy, and interpretation in context.

CHAPTER 7 · REVIEW

Summary and connections

Angle facts

- Straight line: 180°.
- Full turn: 360°.
- Vertically opposite: equal.
- Parallel lines: corresponding and alternate equal; co-interior supplementary.

Triangles and polygons

- Triangle sum: 180°.
- Exterior angle: sum of opposite interior angles.
- n-gon interior sum: $(n-2)\times180°$.
- Exterior sum: 360°.

Constructions

- Perpendicular bisector: equal radii greater than half the segment.
- Angle bisector: equal arcs from the arms.
- SSS triangle: arcs meet when the triangle inequality holds.

Inquiry

- Triangle inequality: two shorter sides > longest.
- Equality gives a degenerate straight line.
- Keep arcs visible as evidence.

CONNECTIONS · DRAWING IS NOT PROVING

A construction is an argument made with instruments. Each arc records a fixed distance from a named point, so the finished drawing already contains evidence: equal radii, equal distances, and a property that can be checked. A picture that looks right is not enough; the defining relationship must hold.

Where this leads: Chapter 8 turns classification into a precise language. Sets describe which objects share a property, and Venn diagrams show relationships of overlap and exclusion. Chapter 9 then measures the shapes whose boundaries you have learned to reason about.

CHAPTER 7 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	A local workshop or school design club produces layouts that cannot be checked before cutting materials.
MATHEMATICAL CONNECTION	Angle facts, the triangle inequality and ruler-and-compass constructions (sections 7.1–7.4).
PROPOSED ACTION	Produce a one-page checking card — angle sum, parallel-line test and the triangle inequality — and teach it to the club so every design is checked before cutting.
EXPECTED IMPACT	Designs are checked with measurements rather than by eye, so less material is wasted.
STUDENT REFLECTION	Record which check caught the most errors, how you taught it and what you would add to the card.

Community Project connection

PROJECT IDEA	A measured plan for a community shade, ramp or garden frame.
COMMUNITY NEED	Small community builds are often laid out by eye, so joints do not fit.
RESEARCH QUESTION	Which measurements guarantee that a triangular frame is rigid and fits its site?
MATHEMATICS	Angle sums, parallel lines, the triangle inequality and constructions.
ACTION AND OUTCOME	Produce a scale drawing and a construction card, then check a prototype with a partner.
BENEFICIARIES	Community group members and the people who use the structure.
EVIDENCE	Scale drawing, construction card, checking log and photographs of the prototype.

Personal Project connection · MYP 5

PERSONAL INTEREST	Design, carpentry, engineering or architecture.
LEARNING GOAL	To produce a design whose geometry can be verified without a protractor.
PRODUCT OR OUTCOME	A scaled design with a construction and checking card.
MATHEMATICS	Angle facts, polygon sums, the triangle inequality and constructions.
PROCESS	Choose a structure, compute its measurements, construct a prototype and evaluate the fit.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

CHAPTER 7 · CHECK YOUR PROGRESS

Reflection and next step

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: ITEM OR TASK	NEXT STEP
I can name and classify angles.			
I can use angle facts with reasons.			
I can calculate triangle and polygon angles.			
I can carry out and explain constructions.			
I can decide whether three lengths form a triangle.			
I can produce and evaluate a scale drawing.			

REFLECTION

Which inquiry question changed your thinking most: the factual question about angle totals, the conceptual question about why the inequality must hold, or the debatable question about when a construction is reliable enough to build from? Use one item from the bank as evidence.

ATL EVIDENCE. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable. Point to the place in your work where you did this.

Review the answer key only after attempting each question. A correct answer without a reason is only half an explanation.

Sets and classification

Sets organise membership and overlap; clear definitions and representations prevent double counting and make survey claims testable.

INQUIRY QUESTIONS

- **Factual:** What is the difference between an element of a set and a subset of a set, and what do union and intersection mean?
- **Conceptual:** Why must a Venn diagram be filled from the overlap outwards, and how do you classify a number by the sets it belongs to?
- **Debatable:** When can a survey category be defined so precisely that its counts can be trusted?

CHAPTER 8 · OPENER

Sets and classification

Sets organise membership and overlap; clear definitions and representations prevent double counting and make survey claims testable.

ENGAGEMENT

When two groups overlap, what can we know about the missing intersection, and how does that change a decision?

Learning goals

- Use set notation, elements and subsets.
- Represent union, intersection and complement with Venn diagrams.
- Classify number families and place values on the real line.
- Interpret survey counts and justify decisions with stated denominators.

MYP lens

Key concept: Form

Related concepts: Representation, Systems

Global context: Fairness and development

ATL FOCUS. In order for a student to organise information, students must define categories, represent overlap, check totals and explain how a definition changes a conclusion.

Language bank: set · element · subset · empty set · universal set · union · intersection · complement · Venn diagram · natural · whole · integer · rational · irrational · real

CHAPTER 8 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

8.R1 List the even numbers from 1 to 8. READY

8.R2 Calculate 17−6 and 30−25. READY

8.R3 Find 2^3. READY

8.R4 Write −3, 0, 2 in order. READY

8.R5 What is 6/80 as a percentage? READY

Chapter map

1. **8.1 Sets, elements and subsets** — Define a set clearly, list each element once and distinguish “element of” from “subset of”.
2. **8.2 Venn diagrams and set operations** — Place the overlap first, then the “only” regions, then the outside region.
3. **8.3 Number sets and the real line** — Classify a number by the sets it belongs to, not by the way it is written.
4. **8.4 Sets in surveys and decisions** — Define the universal set and categories before counting; state whether “or” includes overlap.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 8 · INVESTIGATION · BEFORE 8.1

Investigate · Elements and subsets

YOUR TASK

List every subset of {a, b} and of {a, b, c}. Count them and look for a pattern.

SET	NUMBER OF ELEMENTS	NUMBER OF SUBSETS	SUBSETS
{a}			
{a, b}			
{a, b, c}			
{a, b, c, d} (predict)			

What do you notice?

- How does the number of subsets change when one element is added?
- Is the empty set a subset of every set? How can you tell from your list?

Observation. A set with 1 element has 2 subsets, with 2 elements it has 4, and with 3 elements it has 8, so a set with n elements has 2^n subsets. Each element is either in or out of a subset, which explains the doubling. The empty set appears in every list.

MATHEMATICAL CONCLUSION

An element belongs to a set; a subset is a set whose every element belongs to another set. Every set has 2^n subsets, including the empty set and the set itself.

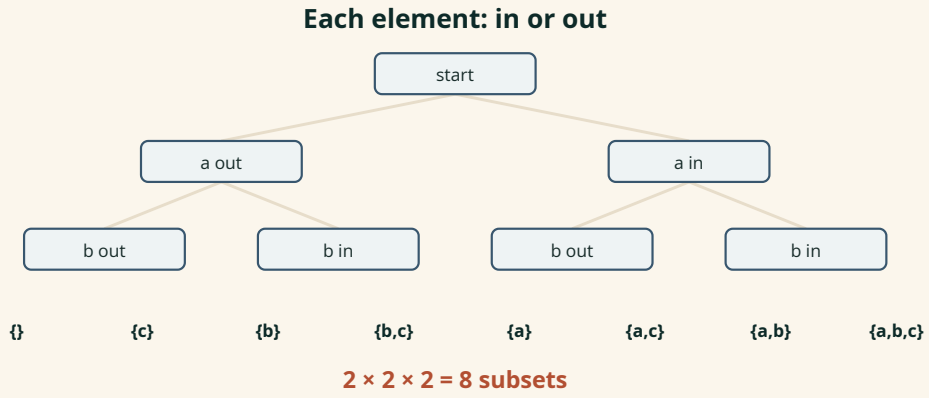
8.1 Sets, elements and subsets

BIG IDEA

Define a set clearly, list each element once and distinguish “element of” from “subset of”.

A set is a well-defined collection. $x \in A$ means x is an element of A; $B \subseteq A$ means every element of B is also in A. Order and repeated listings do not change a set.

The empty set $\emptyset$ has no elements and is a subset of every set. A set with n distinct elements has 2^n subsets, because each element is either included or excluded.



Each element has two choices: in or out. Three elements give $2^3 = 8$ subsets.

WORKED EXAMPLE · SUBSETS OF {A,B,C}

Choose included or excluded for each element.

1. 2×2×2 = 8 subsets.
2. $\emptyset$, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c}.

8 subsets.

Criterion A practice · 8.1 Sets, elements and subsets

WATCH FOR THIS MISCONCEPTION

$\in$ relates an element to a set; $\subseteq$ relates one set to another. $\{a\} \subseteq \{a,b\}$, but $a \in \{a,b\}$.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

8.1.1 List the elements of A = {even numbers from 1 to 10}. FLUENCY

8.1.2 How many subsets has a 3-element set? A 4-element set? FLUENCY

8.1.3 Are {1,2,2,3} and {3,2,1} the same set? Explain. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.1.4 For A={a,b}, decide: $a \in A$, $\{a\} \subseteq A$, $\emptyset \subseteq A$, $\{a\} \in A$. REASONING

8.1.5 A club chooses any subset of 5 activities, including none. How many choices? APPLICATION

8.1.6 How many subsets of {1,2,3,4,5} contain 1? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · INVESTIGATION · BEFORE 8.2

Investigate · Fill the overlap first

YOUR TASK

In a group of 30 students, 18 play football, 14 play music and 7 do both. Complete the Venn diagram counts.

REGION	COUNT	CALCULATION
Both football and music		
Football only		
Music only		
Neither		
Total		

What do you notice?

- Why is the overlap placed first, before the “only” regions?
- How many students are in at least one club, and how did you avoid double counting?

Observation. The overlap of 7 is counted in both totals, so it must be placed first: football only is $18 - 7 = 11$ and music only is $14 - 7 = 7$. At least one club is $11 + 7 + 7 = 25$, so 5 students are in neither.

MATHEMATICAL CONCLUSION

Place the overlap first, then the “only” regions, then the outside region. The union counts every element once, which is why the overlap is subtracted.

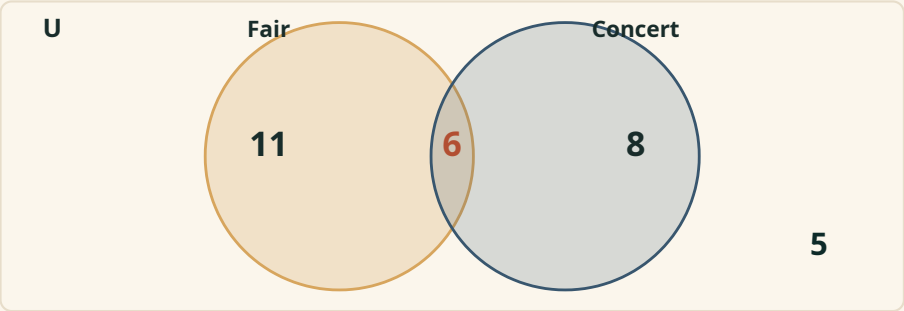
8.2 Venn diagrams and set operations

BIG IDEA

Place the overlap first, then the “only” regions, then the outside region.

Union A ∪ B contains elements in A or B or both. Intersection A ∩ B contains elements in both. Complement A′ contains elements in the universal set ξ that are not in A.

For finite sets, n(A∪B)=n(A)+n(B)−n(A∩B). Subtract the overlap once because it was counted in both totals.



Total = 11 + 6 + 8 + 5 = 30

Overlap first: 11 + 6 + 8 + 5 = 30.

WORKED EXAMPLE · COMPLETE A VENN DIAGRAM

ξ has 30 students; 17 help at a fair, 14 at a concert, 6 at both.

1. Both: 6. Fair only: 17−6=11. Concert only: 14−6=8.
2. At least one: 11+6+8=25. Neither: 30−25=5.

Regions 11, 6, 8, 5.

Criterion A practice · 8.2 Venn diagrams and set operations

WATCH FOR THIS MISCONCEPTION

"A or B" in mathematics includes both unless the question says "exactly one".

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

8.2.1 $A=\{1,2,3,4\}$, $B=\{3,4,5\}$. Find $A\cup B$ and $A\cap B$. FLUENCY

8.2.2 In a 30-student group, 17 like F, 14 like C and 6 like both. Find all four Venn regions. FLUENCY

8.2.3 Why is $n(A\cup B)$ not always $n(A)+n(B)$? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.2.4 Of 40 people, 22 take tea, 18 take coffee and 7 take both. How many take neither? APPLICATION

8.2.5 If $A\subseteq B$, what are $A\cap B$ and $A\cup B$? REASONING

8.2.6 $n(A)=12$, $n(B)=9$ and $n(A\cup B)=17$. Find $n(A\cap B)$. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · INVESTIGATION · BEFORE 8.3

Investigate · Classify by properties

YOUR TASK

Place each number in the correct region of the number-set diagram: −3, 0, 0.75, 1/3, −2.5, √2.

NUMBER	WHOLE?	INTEGER?	RATIONAL?	IRRATIONAL?
−3				
0				
0.75				
1/3				
√2				

What do you notice?

- Which numbers belong to more than one set, and what does that show?
- Why is √2 not rational, even though it is written with a symbol?

Observation. Whole numbers sit inside integers, integers sit inside rational numbers, and irrational numbers such as √2 sit outside them. A number can belong to several sets at once: −3 is an integer and a rational number, and 0.75 and 1/3 are both rational.

MATHEMATICAL CONCLUSION

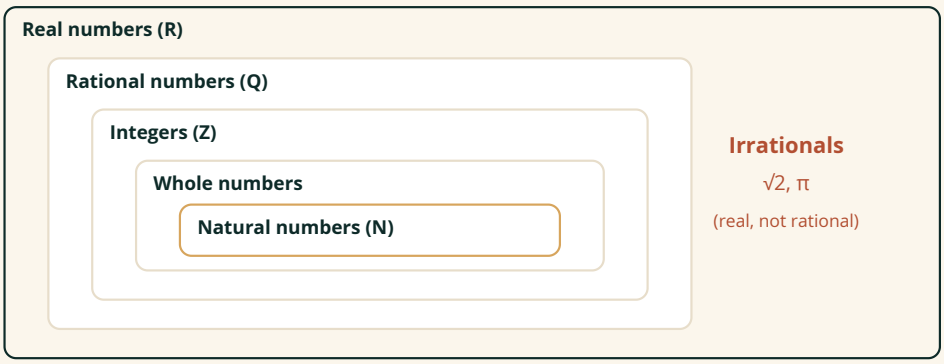
Classify a number by the properties it has, not by the way it is written. Rational numbers are exactly those that can be written as a fraction of two integers.

8.3 Number sets and the real line

> **BIG IDEA**
> Classify a number by the sets it belongs to, not by the way it is written.

Natural numbers $\mathbb{N}$ are counting numbers (here 1,2,3,...); whole numbers add 0; integers $\mathbb{Z}$ add negatives; rationals $\mathbb{Q}$ are ratios of integers with nonzero denominator. Real numbers $\mathbb{R}$ include rationals and irrationals.

Thus natural numbers are a subset of whole numbers, which are a subset of integers, which are a subset of rationals, which are a subset of reals. A terminating or repeating decimal is rational. Non-repeating, non-terminating decimals such as $\sqrt{2}$ and π are irrational.



Natural numbers sit inside whole numbers, integers, rationals and reals. Irrationals are real but not rational.

WORKED EXAMPLE · CLASSIFY −3, 0, 0.75 AND √2

Use the most specific familiar set, then note wider membership.

1. −3 is an integer and rational.
2. 0 is whole, integer and rational.
3. 0.75 = 3/4 is rational.
4. √2 is irrational and real.

All four are real; only √2 is irrational.

Criterion A practice · 8.3 Number sets and the real line

WATCH FOR THIS MISCONCEPTION

A number can belong to several sets. “Rational” does not mean “fraction-looking”; 0.5 and 1/2 are both rational.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

8.3.1 Classify -4, 0, 5 and 1/2 as natural, whole, integer, rational or real (give the most specific listed). FLUENCY

8.3.2 Which are irrational: $\sqrt{2}$, π, 0.25, 1/3? FLUENCY

8.3.3 Is every integer rational? Is every rational an integer? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.3.4 Place -2, 0, 1/2 and 2 on a number line in order. APPLICATION

8.3.5 Explain why 0.333… is rational. REASONING

8.3.6 Find a rational and an irrational number between 1 and 2. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · INVESTIGATION · BEFORE 8.4

Investigate · Define before you count

YOUR TASK

A sports centre has 80 members (simulated data). 46 use the pool, 37 use the gym and 19 use both. Answer the questions below.

QUESTION	REGION NEEDED	ANSWER
How many use the pool or the gym?		
How many use exactly one facility?		
How many use neither?		
What percentage of members use both?		

What do you notice?

- Why must the universal set be stated before any count?
- What percentage of the 80 members use at least one facility?

Observation. With 80 members in the universal set: pool only is $46 - 19 = 27$ and gym only is $37 - 19 = 18$, so 64 use at least one facility and 16 use neither. Exactly one facility is $27 + 18 = 45$, and 19 of 80 is 23.75%.

MATHEMATICAL CONCLUSION

Define the universal set and the categories before counting, and state whether “or” includes the overlap. Every percentage needs its denominator.

8.4 Sets in surveys and decisions

BIG IDEA

Define the universal set and categories before counting; state whether “or” includes overlap.

A survey count is only meaningful with its universal set. “Pool users” may include people who also use the gym; “pool only” excludes them.

Use a Venn diagram or table to organise counts, check that regions sum to the total, and translate the result into a decision with a stated assumption.

WORKED EXAMPLE · A CENTRE AUDIT

80 members: 44 pool, 37 gym, 18 both.

1. Pool only 26; gym only 19; both 18; neither 17.
2. At least one: 63, which is 78.75% of 80.

Regions 26, 18, 19, 17; 78.75% use at least one.

Criterion A practice · 8.4 Sets in surveys and decisions

WATCH FOR THIS MISCONCEPTION

A percentage needs its denominator. 37/80 and 19/80 answer different questions about gym use.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

8.4.1 80 members: 44 pool, 37 gym, 18 both. Find the four regions. APPLICATION

8.4.2 What percentage use at least one facility? APPLICATION

8.4.3 Is “more than a third use the gym” true? Discuss both meanings. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.4.4 Pool-only pass $4, gym-only $3, combined $6. Find income if each member buys the relevant pass. APPLICATION

8.4.5 Why should a survey report state the universal set? REASONING

8.4.6 50 people: 34 like vanilla, 29 chocolate, 7 neither. Find both and vanilla only. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

8.CK1 How many subsets has {a,b,c,d}? CHECK

8.CK2 A={1,2,3}, B={2,3,4}. Find A∩B. CHECK

8.CK3 n(A)=10, n(B)=8, overlap 3. Find n(A∪B). CHECK

8.CK4 Classify 0.6 as rational or irrational. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The classification poster

MYP ASSESSMENT PRACTICE

Criteria A and C. Create a poster that distinguishes membership, subsets and number families, with one checked example.

Your poster should prevent the common confusion between $\in$ and $\subseteq$ and between rational and irrational.

ASSESSMENT PRACTICE · THE CLASSIFICATION POSTER

Criteria A and C. Create a poster that distinguishes membership, subsets and number families, with one checked example.

8.S1 Give one correct $\in$ statement and one correct $\subseteq$ statement using $\{1,2,3\}$. ASSESSMENT

8.S2 Classify -2, 0, $3/4$ and $\sqrt{2}$, and state one nesting fact. ASSESSMENT

8.S3 Write a check question that exposes double counting, with its answer. ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 8

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 8

Connect ideas. Give units and reasons where requested.

8.M1 How many subsets has a 5-element set? FLUENCY

8.M2 A={2,4,6}, B={4,6,8}. Find union and intersection. FLUENCY

8.M3 In 50 people, 28 like A, 22 like B, 9 both. Find neither. APPLICATION

8.M4 Is 0.121212... rational? Explain. REASONING

8.M5 Classify √9 and √10. FLUENCY

8.M6 If A⊆B, n(A)=7 and n(B)=12, find n(A∩B) and n(A∪B). REASONING

8.M7 60 people: 35 use X, 28 use Y, 10 neither. Find both. APPLICATION

8.M8 n(A)=20, n(B)=15, universal 30. Find the possible overlap range. CHALLENGE

8.M9 Explain why {1,2} and {1,1,2,2} are equal. REASONING

8.M10 A set has 6 elements. How many subsets contain a particular element? APPLICATION

> FINAL REFLECTION
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 8 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · HOW MUCH OVERLAP IS POSSIBLE?

8.B1 With overlap 6, find all four regions and the number helping at least one event. INVESTIGATION

8.B2 With unknown overlap x, write n(F∪C) and the neither count. INVESTIGATION

8.B3 Find the smallest possible x and explain why x=0 fails. INVESTIGATION

8.B4 Find the largest possible x and describe that diagram. INVESTIGATION

8.B5 The committee needs 20 helpers for a combined event. With overlap 6, are there enough? Could the answer change across the possible range? CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

8.B6 Investigate the largest and smallest possible overlap between two groups: 46 of 80 people swim and 37 of 80 cycle. Find the range of the number who do both and justify both endpoints. INVESTIGATION

8.B7 Test the claim: "if A ⊂ B and B ⊂ C, then A ⊂ C." Try several sets of your own, then justify the claim. INVESTIGATION

CHAPTER 8 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

8.C1 Explain the difference between $a \in \{a, b\}$ and $\{a\} \subset \{a, b\}$, using one sentence for each symbol. REASONING

8.C2 Write the steps you would use to complete a Venn diagram for two overlapping groups, and explain why the overlap is placed first. REASONING

8.C3 A report says “46 members swim” without stating the total. Explain what extra information is needed before any percentage can be calculated. REASONING

8.C4 Explain why 0.5 and 1/2 are both rational but are written differently, and what this shows about classifying numbers. REASONING

CHAPTER 8 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE CENTRE AUDIT

8.D1 Complete the four regions and check the total. APPLICATION

8.D2 Find pass income and the percentage using at least one facility. APPLICATION

8.D3 Test “more than a third use the gym” under two meanings. REASONING

8.D4 A combined pass costs $6 instead of $4+$3. Give one member argument, one centre argument and a recommendation. CHALLENGE

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

8.D5 A class of 32 students chooses a trip (simulated data): 18 choose the museum, 15 choose the farm and 6 choose either. How many students have no preference recorded, and what percentage is that? APPLICATION

8.D6 A library survey of 120 borrowers (simulated data) finds 70 borrow fiction, 55 borrow non-fiction and 30 borrow both. Produce a Venn diagram summary and state the percentage who borrow exactly one type. APPLICATION

8.D7 A sports club has 90 members (simulated data). 52 use the pool, 41 use the gym and the club wants at least 20 members using neither. Find the range of possible overlaps and recommend a target. APPLICATION

CHAPTER 8 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can describe a set, list its elements and identify subsets.
- [ ] I can complete and interpret a Venn diagram, placing the overlap first.
- [ ] I can classify numbers into the number sets and explain the classifications.
- [ ] I can state the universal set and the denominator behind any survey count.

Reflect

- Which was harder — placing elements or counting regions — and what strategy helped?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to organise information, students must define categories, represent overlap, check totals and explain how a definition changes a conclusion.

CHAPTER 8 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	Younger learners confuse "element of" with "subset of" and misread survey tables.
MATHEMATICAL CONNECTION	Sets, subsets, Venn diagrams and number sets (sections 8.1–8.3).
PROPOSED ACTION	Produce a sorting activity with physical objects and a survey table, then teach two feeder classes and record the errors that persist.
EXPECTED IMPACT	Learners can place elements correctly, read a Venn diagram and state the region a count refers to.
STUDENT REFLECTION	Log the misconceptions, the activity that worked and how you verified understanding.

Community Project connection

PROJECT IDEA	A community needs survey with clearly defined categories.
COMMUNITY NEED	Local surveys often use overlapping categories without stating the population.
RESEARCH QUESTION	How can a short survey be designed so that every count has a stated denominator and unambiguous categories?
MATHEMATICS	Sets, Venn diagrams, counting and percentages.
ACTION AND OUTCOME	Design a survey of 30 to 50 respondents, present the results with a Venn diagram and a stated denominator.
BENEFICIARIES	Community organisers, school leadership and respondents.
EVIDENCE	Survey questions, raw data table, Venn summary and a short report.

Personal Project connection · MYP 5

PERSONAL INTEREST	Information design or data handling.
LEARNING GOAL	To design a classification system that supports clear reporting.
PRODUCT OR OUTCOME	A classification scheme and report for a data set of personal interest.
MATHEMATICS	Sets, subsets, Venn diagrams and percentages.
PROCESS	Define categories, collect data, build the diagram and evaluate whether the categories are unambiguous.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Perimeter, area, surface area and volume

Choosing the right measure—length, area, surface area or volume—connects geometric structure to material use, capacity and cost.

INQUIRY QUESTIONS

- **Factual:** Which formulas give the perimeter, area, surface area and volume of common shapes, and what units does each use?
- **Conceptual:** Why does a triangle have half the area of a parallelogram with the same base and height, and why is a prism's volume a cross-section times a length?
- **Debatable:** When is an estimate more useful than an exact measurement, and how much rounding is sensible when buying materials?

CHAPTER 9 · OPENER

Perimeter, area, surface area and volume

Choosing the right measure—length, area, surface area or volume—connects geometric structure to material use, capacity and cost.

ENGAGEMENT

How can a community project estimate turf, paving, fencing, soil and paint without confusing boundary, surface and space?

Learning goals

- Distinguish perimeter, area, surface area and volume.
- Calculate with triangles, quadrilaterals, circles and composites.
- Use nets and prism formulas with correct units.
- Convert capacity, compare packaging and evaluate a project budget.

MYP lens

Key concept: Form

Related concepts: Measurement, Space

Global context: Globalization and sustainability

ATL FOCUS. In order for a student to apply mathematics in real-life contexts, students must select the correct measure, convert units, calculate costs and evaluate a budget with stated assumptions.

Language bank: perimeter · area · surface area · volume · capacity · radius · diameter · circumference · semicircle · composite · net · prism · cross-section · litre

CHAPTER 9 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

9.R1 Find 7×4 and 2(7+4). READY

9.R2 Find half of 8×5. READY

9.R3 Calculate 6^2. READY

9.R4 Convert 2 L to mL. READY

9.R5 Find 1.2×0.6. READY

Chapter map

1. **9.1 Perimeter and area of triangles and quadrilaterals** — Choose the quantity the task needs: boundary length, covered surface or space inside.
2. **9.2 Circles and composite shapes** — Identify radius and diameter, then split a composite shape into familiar pieces.
3. **9.3 Nets and surface area** — A net shows every outside face; surface area adds face areas, not the space inside.
4. **9.4 Prisms, volume and capacity** — Volume measures space; capacity measures how much a container can hold.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 9 · INVESTIGATION · BEFORE 9.1

Investigate · Boundary or surface?

YOUR TASK

For a 6 cm by 4 cm rectangle, find the perimeter and the area. Then cut it along a diagonal and find the perimeter and area of each triangle.

SHAPE	PERIMETER (CM)	AREA (CM^2)	UNITS USED
Rectangle 6 × 4			
Each triangle from the diagonal			
Parallelogram, base 6, height 4			
Triangle, base 6, height 4			

What do you notice?

- What happens to the perimeter when the rectangle is cut, and what happens to the total area?
- Why does the triangle have half the area of the parallelogram with the same base and height?

Observation. The rectangle has perimeter 20 cm and area 24 cm^2. Cutting along a diagonal keeps the total area at 24 cm^2 but increases the total perimeter, because the cut edge is new boundary. A triangle on base 6 and height 4 has area 12 cm^2, half of the parallelogram.

MATHEMATICAL CONCLUSION

Perimeter adds lengths in ordinary units; area multiplies two dimensions in square units. Choose the quantity the task needs before calculating.

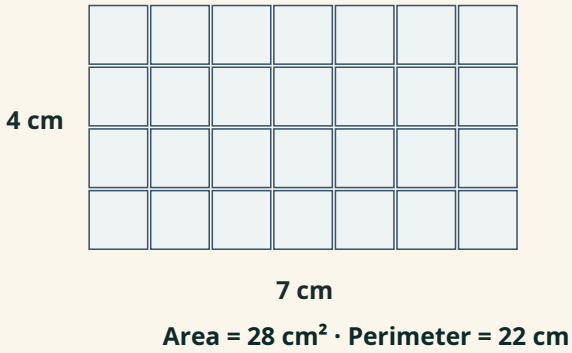
9.1 Perimeter and area of triangles and quadrilaterals

BIG IDEA

Choose the quantity the task needs: boundary length, covered surface or space inside.

Perimeter is the total boundary length. Area measures surface in square units. Rectangle area = length×width; triangle area = ½×base×perpendicular height; parallelogram area = base×height.

A trapezoid with parallel sides a and b and perpendicular height h has area ½(a+b)h. The height must be perpendicular to the parallel sides, not a slant edge.



Area counts square units (28 cm²); perimeter counts boundary units (22 cm).

WORKED EXAMPLE · A TRIANGLE AND A TRAPEZOID

Triangle: base 8 cm, height 5 cm. Trapezoid: parallel sides 6 and 10 cm, height 4 cm.

1. Triangle: ½×8×5 = 20 cm².
2. Trapezoid: ½×(6+10)×4 = 32 cm².

20 cm² and 32 cm².

Criterion A practice · 9.1 Perimeter and area of triangles and quadrilaterals

WATCH FOR THIS MISCONCEPTION

Do not use a slant side as the height. Do not report area in cm or perimeter in cm^2.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

9.1.1 Find the area and perimeter of a 7 cm by 4 cm rectangle. FLUENCY

9.1.2 Find the area of a triangle with base 8 cm and perpendicular height 5 cm. FLUENCY

9.1.3 Find the area of a trapezoid with parallel sides 6 cm and 10 cm and height 4 cm. FLUENCY

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.1.4 A triangle has sides 5, 5, 6 cm. A student uses ½×6×5. Explain the error and find the correct height and area. REASONING

9.1.5 A rectangular lawn is 18 m by 12 m. Find its area and the cost of turf at \$7.50/m^2. APPLICATION

9.1.6 A rectangle has area 48 cm^2 and whole-number sides. Which dimensions give the smallest perimeter? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · INVESTIGATION · BEFORE 9.2

Investigate · Around and inside a circle

YOUR TASK

Measure the diameter and circumference of three circular objects, then divide the circumference by the diameter.

OBJECT	DIAMETER D (CM)	CIRCUMFERENCE C (CM)	C ÷ D
Class average			

What do you notice?

- What value does C ÷ d approach, and what is it called?
- If the radius is 3 cm, find the circumference and the area using $\pi \approx 3.14$.

Observation. Every object gives C ÷ d close to 3.14, so $C = \pi d = 2\pi r$. For r = 3 cm, $C \approx 18.84$ cm and $A = \pi r^2 \approx 28.26$ cm^2.

MATHEMATICAL CONCLUSION

The ratio of circumference to diameter is constant for every circle. Use $C = \pi d$ and $A = \pi r^2$, and label results as approximate when π is rounded.

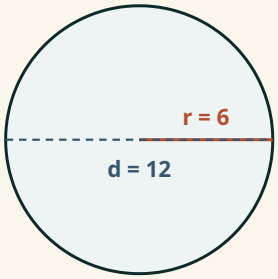
9.2 Circles and composite shapes

BIG IDEA

Identify radius and diameter, then split a composite shape into familiar pieces.

For a circle, diameter d=2r. Circumference C=πd=2πr; area A=πr^2. In this chapter use π≈3.14 unless told otherwise, and label results as approximate.

A semicircle has half the circle area and a curved boundary of πr. For a composite plan, add areas of non-overlapping pieces and trace only the outer boundary for fencing.



C = 2πr = πd · A = πr² · π ≈ 3.14

d = 2r. Area uses r^2; circumference uses 2r. Use π ≈ 3.14 here.

WORKED EXAMPLE · A SEMICIRCULAR STAGE

Diameter 12 m, so radius 6 m.

1. Area = ½×3.14×6^2 = 56.52 m^2.
2. Curved boundary = 3.14×6 = 18.84 m.

Area ≈56.52 m^2; arc ≈18.84 m.

Criterion A practice · 9.2 Circles and composite shapes

WATCH FOR THIS MISCONCEPTION

The diameter is not part of the outer boundary when the semicircle is attached to a rectangle along that edge.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

9.2.1 Find the circumference and area of a circle with radius 5 cm, using π=3.14. FLUENCY

9.2.2 A circle has diameter 12 m. Find its radius, circumference and area (π=3.14). FLUENCY

9.2.3 Find the area and curved boundary of a semicircle of diameter 12 m (π=3.14). APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.2.4 An 18 m by 12 m rectangle has a semicircle of diameter 12 m attached to one short end. Find total area and outer boundary (π=3.14). APPLICATION

9.2.5 Why is the 12 m diameter not included in the fencing length of the composite plan? REASONING

9.2.6 If the radius of a circle doubles, what happens to its circumference and area? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · INVESTIGATION · BEFORE 9.3

Investigate · Nets and surface area

YOUR TASK

Sketch a net for a cuboid measuring 3 cm by 2 cm by 4 cm, then find its surface area. Repeat for the same cuboid with the top face removed.

SOLID	FACES	FACE AREAS (CM²)	SURFACE AREA (CM²)
Closed cuboid			
Open-topped cuboid			
Cube of side 3 cm			

What do you notice?

- How many faces does each solid have, and which face is missing in the open-topped version?
- Why is surface area not the same as volume?

Observation. A closed cuboid has six faces in three matching pairs: $2(3\times2) + 2(3\times4) + 2(2\times4) = 12 + 24 + 16 = 52$ cm². Removing the top face (12 cm²) leaves 40 cm². A cube of side 3 cm has surface area $6 \times 9 = 54$ cm².

MATHEMATICAL CONCLUSION

A net shows every outside face; surface area adds the face areas, not the space inside. Identify which faces are actually exposed before adding.

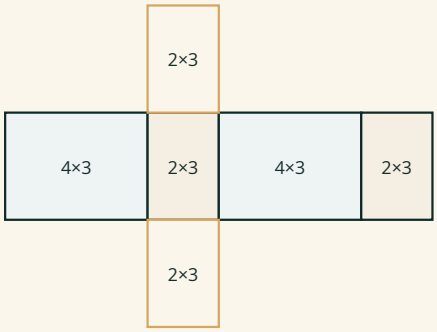
9.3 Nets and surface area

BIG IDEA

A net shows every outside face; surface area adds face areas, not the space inside.

A net is a flat pattern that folds into a solid. The surface area of a closed cuboid is 2(lw+lh+wh). For an open-topped box, omit the top face.

Use square units for surface area. Painting an outside surface depends on area; filling the box depends on volume. Keep the two quantities separate.



Surface area = 2(4×3 + 4×2 + 3×2) = 52 square units

A net exposes all six faces. Surface area = 2(lw + lh + wh).

WORKED EXAMPLE · AN OPEN-TOPPED RAISED BED

Length 1.2 m, width 0.6 m, height 0.5 m.

1. Base: $1.2 \times 0.6 = 0.72$ m^2.
2. Two long sides: $2 \times 1.2 \times 0.5 = 1.20$ m^2. Two short sides: $2 \times 0.6 \times 0.5 = 0.60$ m^2.

Outside area = 2.52 m^2.

Criterion A practice · 9.3 Nets and surface area

WATCH FOR THIS MISCONCEPTION

A closed cuboid has six faces; an open-topped bed has five. Do not paint an imaginary top.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

9.3.1 Find the surface area of a closed cuboid 4 cm by 3 cm by 2 cm. FLUENCY

9.3.2 Find the surface area of a cube with side 6 cm. FLUENCY

9.3.3 Find the outside area of an open-topped bed 1.2 m by 0.6 m by 0.5 m. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.3.4 A student uses the closed-cuboid formula for an open box. What must be subtracted? REASONING

9.3.5 Three beds each have outside area 2.52 m^2. Paint covers 9 m^2 per litre. How many litres are needed, and how many 1 L tins must be bought? APPLICATION

9.3.6 A cuboid has volume 24 cm^3 and whole-number edges. Compare surface areas of 1×1×24 and 2×3×4. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · INVESTIGATION · BEFORE 9.4

Investigate · Volume and capacity

YOUR TASK

Fill a 4 cm by 3 cm by 2 cm box with 1 cm cubes. Count the cubes, then work out how many 1 L bottles fill a 1 m^3 tank.

QUANTITY	CALCULATION	RESULT	UNITS
Cubes in the box			
Volume of the box			
Litres in 1 m^3			
Litres in 1 cm^3			

What do you notice?

- How does the number of cubes relate to the formula $V = l \times w \times h$?
- Why is 1 m^3 equal to 1,000 L and not 100 L?

Observation. The box holds $4 \times 3 \times 2 = 24$ cubes, so its volume is 24 cm^3. A 1 m cube is 100 cm on each side, so it holds $100 \times 100 \times 100 = 1{,}000{,}000$ cm^3, and since 1 cm^3 = 1 mL, that is 1,000 L.

MATHEMATICAL CONCLUSION

Volume measures space in cubic units and capacity measures how much a container holds. Lengths scale by 100, areas by 10,000 and volumes by 1,000,000 between metres and centimetres.

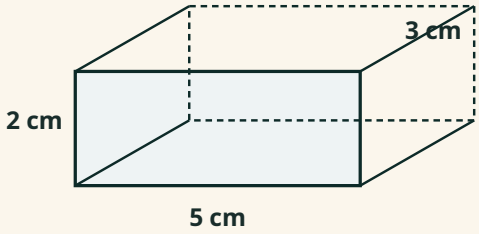
9.4 Prisms, volume and capacity

BIG IDEA

Volume measures space; capacity measures how much a container can hold.

A prism has a constant cross-section. Volume = cross-section area × length. For a cuboid, V=lwh; for a triangular prism, V=(½×base×height)×length.

Capacity conversions: 1 cm^3 = 1 mL; 1,000 cm^3 = 1 L; 1 m^3 = 1,000 L. Convert units before multiplying or dividing.



Volume = 5 × 3 × 2 = 30 cm^3 = 30 mL

Volume counts cubic units: 5 × 3 × 2 = 30 cm^3 = 30 mL.

WORKED EXAMPLE · A TANK AND A PRISM

Tank 50 cm × 30 cm × 40 cm; triangular prism with cross-section base 8 cm, height 5 cm, length 14 cm.

1. Tank volume 60,000 cm^3 = 60 L.
2. Prism volume ½×8×5×14 = 280 cm^3.

60 L; 280 cm^3.

Criterion A practice · 9.4 Prisms, volume and capacity

WATCH FOR THIS MISCONCEPTION

1 m^3 is 1,000 L, not 100 L. Length units scale by 100, area by 10,000 and volume by 1,000,000 between metres and centimetres.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

9.4.1 Find the volume of a cuboid 20 cm by 10 cm by 5 cm. FLUENCY

9.4.2 Find the volume of a triangular prism: cross-section base 8 cm, height 5 cm, length 14 cm. FLUENCY

9.4.3 A tank is 50 cm by 30 cm by 40 cm. Find its capacity in litres. APPLICATION

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.4.4 Convert 2.5 L to cm^3, 450 mL to cm^3 and 0.8 m^3 to litres. FLUENCY

9.4.5 A student says 1 m^3=100 L because 1 m=100 cm. Correct the reasoning. REASONING

9.4.6 A prism has volume 360 cm^3 and cross-section area 30 cm^2. Find its length. If every dimension doubles, what happens to volume? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

9.CK1 Find the area of a triangle with base 10 cm and height 6 cm. CHECK

9.CK2 Find the area of a circle with radius 3 cm (π=3.14). CHECK

9.CK3 Find the surface area of a cube with side 4 cm. CHECK

9.CK4 Find the capacity in litres of a 10 cm cube. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The measure explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Explain which measure a purchase needs and check units with a small example.

A learner confuses perimeter, area, surface area and volume. Use a 4 cm by 3 cm by 2 cm box to make the distinctions clear.

ASSESSMENT PRACTICE · THE MEASURE EXPLAINER

Criteria A and C. Explain which measure a purchase needs and check units with a small example.

9.S1	Find the box volume and closed surface area, with units.	ASSESSMENT
9.S2	Which measure determines cardboard, paint and filling?	ASSESSMENT
9.S3	Write a unit-check warning and one conversion fact.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 9

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 9

Connect ideas. Give units and reasons where requested.

9.M1	Find the area of a parallelogram with base 9 cm and height 4 cm.	FLUENCY
9.M2	Find the circumference of a circle with diameter 10 cm (π=3.14).	FLUENCY
9.M3	Find the surface area of a closed cuboid 5 cm by 2 cm by 3 cm.	FLUENCY
9.M4	Find the capacity in litres of a tank 40 cm by 25 cm by 20 cm.	APPLICATION
9.M5	A student uses a slant side as triangle height. Explain the correction.	REASONING
9.M6	A semicircle has radius 4 m. Find its area and curved boundary (π=3.14).	APPLICATION
9.M7	A cuboid has volume 60 cm^3 and base area 12 cm^2. Find its height.	CHALLENGE
9.M8	If all lengths of a solid double, what happens to surface area and volume?	REASONING
9.M9	Paint covers 9 m^2/L. How many 1 L tins for 20 m^2?	APPLICATION
9.M10	Compare closed boxes 1×1×36 and 3×3×4 (both volume 36). Which uses less cardboard and by how much?	CHALLENGE

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 9 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · THE PACKAGING PROBLEM

9.B1 Complete the surface areas for the six designs. INVESTIGATION

9.B2 Which design uses least and most cardboard? INVESTIGATION

9.B3 Test volume 27 cm^3 with 3×3×3, 1×3×9 and 1×1×27. Does the cube use least? INVESTIGATION

9.B4 Design two closed boxes of volume 36 cm^3 and recommend one by material use. CHALLENGE

9.B5 Give two practical reasons a manufacturer might not choose the minimum-area box, and one limitation of the pattern. INVESTIGATION

CRITERION B Investigating patterns

INVESTIGATE FURTHER

9.B6 Investigate the rectangles with whole-number sides and perimeter 24 cm. Which has the greatest area, and what happens to the area as the shape becomes less square? INVESTIGATION

9.B7 Two cuboids both have volume 24 cm^3 with whole-number edges. Find at least three and compare their surface areas. Which has the smallest surface area, and why? INVESTIGATION

CHAPTER 9 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

9.C1 Explain, with one calculation each, the difference between the perimeter and the area of a 4 cm by 3 cm rectangle. REASONING

9.C2 A learner reports the area of a rectangle as “12 cm”. Explain the error and write the corrected statement. REASONING

9.C3 Describe how a net of a cuboid helps you find its surface area, and state what changes when the box has no top. REASONING

9.C4 Explain why 1 m^3 of water is 1,000 L, using one calculation. REASONING

CHAPTER 9 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE COURTYARD PROJECT

9.D1 Find lawn area, stage area and total area. APPLICATION

9.D2 Find fencing length and the costs of turf, paving and fencing. APPLICATION

9.D3 Find soil needed for three beds and the number and cost of 40 L bags. APPLICATION

9.D4 Find paint needed for the three open beds and the tin cost. APPLICATION

9.D5 Find the total cost and budget position. Then add 10% contingency and state the new position. CHALLENGE

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

9.D6 A rectangular courtyard is 12 m by 8 m (simulated project). Tiles cover 0.25 m^2 each and cost 1,800 XOF each. Estimate the number of tiles needed, then find the cost including 10% extra for cutting. APPLICATION

9.D7 An open-topped raised bed measures 2 m by 1 m by 0.4 m (simulated design). Find the volume of soil needed and the interior surface area to be painted, ignoring the base. APPLICATION

9.D8 A cylindrical tank has radius 0.5 m and height 1.2 m (simulated design, $\pi \approx 3.14$). Estimate its capacity in litres and state the accuracy of your estimate. APPLICATION

CHAPTER 9 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can find the perimeter and area of triangles and quadrilaterals and choose the right formula.
- [ ] I can use π to find the circumference and area of a circle and of composite shapes.
- [ ] I can use a net to find a surface area and identify the exposed faces.
- [ ] I can find the volume of a prism and convert between volume and capacity.

Reflect

- Which quantity — perimeter, area, surface area or volume — is easiest to confuse with another, and how do you keep them apart?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to apply mathematics in real-life contexts, students must select the correct measure, convert units, calculate costs and evaluate a budget with stated assumptions.

CHAPTER 9 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	A school garden has no reliable way to order soil, mulch or paint, so materials are over-ordered.
MATHEMATICAL CONNECTION	Perimeter, area, surface area and volume (sections 9.1–9.4).
PROPOSED ACTION	Measure the beds, compute the soil volume and paint area, and produce an ordering sheet the gardener can reuse each term.
EXPECTED IMPACT	The garden orders the right quantities, reducing waste and cost.
STUDENT REFLECTION	Record how you checked the measurements, how much allowance you made for error, and what you would measure differently.

Community Project connection

PROJECT IDEA	A water-harvesting estimate for a community building.
COMMUNITY NEED	Roof catchment volume is often guessed, so tanks are sized badly.
RESEARCH QUESTION	How much rain can a given roof area collect in a season, and what tank size does that imply?
MATHEMATICS	Area, volume, capacity and unit conversion.
ACTION AND OUTCOME	Measure a roof, model the collected volume for a stated rainfall and recommend a tank size.
BENEFICIARIES	Community building users and maintenance staff.
EVIDENCE	Measurements, volume model, tank recommendation and the assumptions stated.

Personal Project connection · MYP 5

PERSONAL INTEREST	Architecture, product design or gardening.
LEARNING GOAL	To design an object or space whose measurements are calculated rather than guessed.
PRODUCT OR OUTCOME	A scaled design with an ordering list for materials.
MATHEMATICS	Perimeter, area, surface area, volume and unit conversion.
PROCESS	Measure or specify dimensions, compute each quantity, cost the materials and evaluate the design.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

Data, tables and graphs

Data displays and summaries make patterns visible, but choices of grouping, scale and denominator shape the claims that seem reasonable.

INQUIRY QUESTIONS

- **Factual:** What is a variable, and which display suits categorical, discrete and continuous data?
- **Conceptual:** Why can two graphs of the same data support different claims, and how do averages summarise a distribution?
- **Debatable:** When is a statistic fair to publish, and when does the choice of display do the arguing?

CHAPTER 10 · OPENER

Data, tables and graphs

Data displays and summaries make patterns visible, but choices of grouping, scale and denominator shape the claims that seem reasonable.

ENGAGEMENT

How can the same survey support a fair decision without letting a misleading graph or an unstated denominator do the arguing?

Learning goals

- Classify variables and organise frequency or two-way tables.
- Construct and interpret bar and pie charts.
- Summarise distributions with dot plots, histograms and averages.
- Describe trends and detect misleading scales or claims.

MYP lens

Key concept: Relationships

Related concepts: Representation, Pattern

Global context: Fairness and development

ATL FOCUS. In order for a student to process and communicate information, students must organise data, choose an appropriate representation, calculate summaries and justify a claim with its limitations.

Language bank: variable · categorical · nominal · ordinal · discrete · continuous · frequency · two-way table · bar chart · pie chart · sector angle · dot plot · histogram · mean · median · mode · range · outlier · trend

CHAPTER 10 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

10.R1 Find 96+74+30. READY

10.R2 Find 96/200 as a percentage. READY

10.R3 Find the mean of 2, 4, 6. READY

10.R4 Find 360×0.25. READY

10.R5 Order 5, 2, 8, 3. READY

Chapter map

1. **10.1 Data, variables and tables** — Identify what was measured, the unit, the population and the type of variable before summarising.
2. **10.2 Bar charts and pie charts** — Use bars to compare categories and sectors to show parts of one whole.
3. **10.3 Dot plots, histograms and averages** — Describe shape, centre and spread; choose an average that fits the data and the question.
4. **10.4 Trends, comparisons and misleading graphs** — Read the scale, denominator and time order before accepting a visual claim.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 10 · INVESTIGATION · BEFORE 10.1

Investigate · What kind of variable?

YOUR TASK

Classify each item as categorical (nominal or ordinal) or numerical (discrete or continuous), and state the unit where there is one.

ITEM	CATEGORICAL OR NUMERICAL	SUB-TYPE	UNIT
Lunch choice			
Number of siblings			
Height			
Shirt size code			
Rating from 1 to 5			

What do you notice?

- Which items are numbers used only as labels?
- Why does the type of variable decide which average makes sense?

Observation. Lunch choice is nominal, a 1 to 5 rating is ordinal, the number of siblings is discrete numerical, height is continuous numerical, and a shirt size code is categorical even though it uses digits. Averages only make sense for numerical variables.

MATHEMATICAL CONCLUSION

Identify what was measured, the unit, the population and the type of variable before summarising anything.

10.1 Data, variables and tables

BIG IDEA

Identify what was measured, the unit, the population and the type of variable before summarising.

Data are recorded values. A categorical variable names groups (nominal) or ordered categories (ordinal). A numerical variable is discrete if it counts separate values, or continuous if it measures on a scale.

A frequency table lists categories or intervals and their counts. Frequencies must sum to the number of observations. A two-way table shows how two categories overlap.

Frequency table

Lunch choice	Frequency
School	96
Packed	74
Off-site	30
Total	**200**

A frequency table organises categories; the counts sum to the 200 students surveyed.

WORKED EXAMPLE · CLASSIFY VARIABLES

Lunch choice, year group, number of siblings and height.

1. Lunch choice: nominal categorical.
2. Year group: ordinal categorical (or discrete numerical code).
3. Siblings: discrete numerical. Height: continuous numerical.

The variable type guides the display and summary.

Criterion A practice · 10.1 Data, variables and tables

WATCH FOR THIS MISCONCEPTION

A number used as a label (such as a shirt size code) is not automatically a quantity to average.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

10.1.1 Classify: favourite fruit, school year, number of pets, height. FLUENCY

10.1.2 A table has frequencies 38, 31, 27 for school lunch in Years 7–9. Find the total. FLUENCY

10.1.3 Why is “year group” not usually averaged as if it were height? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.1.4 In a two-way table, Year 7 has 38 school, 24 packed, 8 off-site. Find the row total and the school-lunch share of Year 7. APPLICATION

10.1.5 A survey records “0, 1, 2, 3” as codes for four lunch choices. Can these codes be added meaningfully? REASONING

10.1.6 A frequency table has total 200 and known frequencies 96 and 74. Find the missing frequency and its percentage. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · INVESTIGATION · BEFORE 10.2

Investigate · Choosing a display

YOUR TASK

Display the lunch survey as a bar chart and as a pie chart: school 96, packed 74, off-site 30, out of 200 students (simulated data).

DISPLAY	WHAT IT SHOWS BEST	SECTOR ANGLE OR BAR HEIGHT	LIMITATION
Bar chart			
Pie chart			
Check: sector angle for school			

What do you notice?

- How do you compute the sector angle for “school”?
- Which display would you use to compare year groups, and why?

Observation. School is 96/200 = 48% of the total, so its sector angle is $0.48 \times 360° = 172.8°$. Bars compare categories directly; a pie shows parts of one whole but is poor for many similar categories or several groups side by side.

MATHEMATICAL CONCLUSION

Use bars to compare categories and sectors to show parts of one whole. Label axes, start the frequency axis at zero and state the denominator.

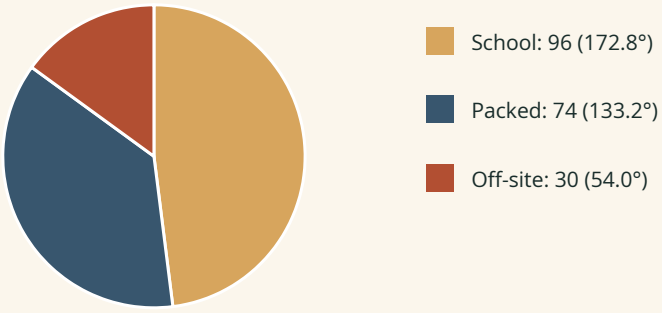
10.2 Bar charts and pie charts

BIG IDEA

Use bars to compare categories and sectors to show parts of one whole.

A bar chart uses equal-width bars with heights proportional to frequency. Label axes, start the frequency axis at zero and keep spacing consistent.

A pie chart represents a whole by 360°. Sector angle = category frequency/total × 360°. Percentages should sum to 100% apart from stated rounding.



Sector angles are proportional to frequency: 172.8°, 133.2° and 54°.

WORKED EXAMPLE · LUNCH CHOICES FOR 200 STUDENTS

School 96, packed 74, off-site 30.

1. School: 96/200×360 = 172.8°.
2. Packed: 74/200×360 = 133.2°.
3. Off-site: 30/200×360 = 54°.

Angles sum to 360°.

Criterion A practice · 10.2 Bar charts and pie charts

WATCH FOR THIS MISCONCEPTION

A pie chart is poor for comparing many similar categories or several groups side by side; a clustered bar chart is clearer.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

10.2.1 Find pie-chart angles for frequencies 96, 74 and 30 out of 200. FLUENCY

10.2.2 What percentage of 200 is 96? FLUENCY

10.2.3 Why should a bar-chart frequency axis start at zero? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.2.4 Year 7 has 70 students, Year 8 and Year 9 have 65 each. Which display compares lunch choices across years? APPLICATION

10.2.5 Three sector percentages are 48%, 37% and 15%. Do they fit one whole? REASONING

10.2.6 A category has sector angle 54° in a survey of 200. Find its frequency. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · INVESTIGATION · BEFORE 10.3

Investigate · Centre and spread

YOUR TASK

For the values 4, 5, 5, 6, 7, 8, 9, 20, find the mean, median, mode and range. Then remove the 20 and recompute.

SUMMARY	WITH 20	WITHOUT 20	CHANGE
Mean			
Median			
Mode			
Range			

What do you notice?

- Which summary changed most when the 20 was removed, and why?
- Which average best describes the remaining seven values, and for what question?

Observation. With the 20 the mean is $64 \div 8 = 8$ and the median is 6.5; without it the mean is $44 \div 7 \approx 6.29$ and the median is 6. The mean moved much more than the median because the 20 pulls it upwards.

MATHEMATICAL CONCLUSION

Describe shape, centre and spread together. An outlier moves the mean a lot and the median very little, so choose the average that fits the data and the question.

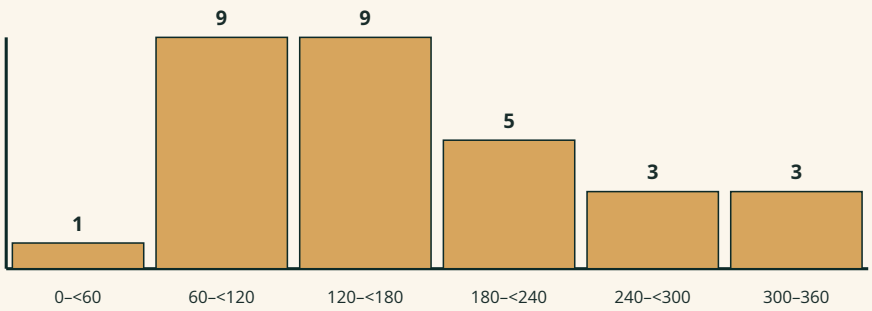
10.3 Dot plots, histograms and averages

BIG IDEA

Describe shape, centre and spread; choose an average that fits the data and the question.

A dot plot shows individual values. A histogram groups continuous or finely measured numerical data into intervals; with equal widths, bar height is frequency. State the boundary rule.

Mean = sum÷count; median is the middle ordered value (or mean of the two middle values); mode is most frequent; range = maximum−minimum. Outliers pull the mean more than the median.



Equal interval widths; heights show frequency.

Equal 60-minute intervals. Boundary rule: left endpoint included, right excluded; the final interval includes 360.

WORKED EXAMPLE · SUMMARISE 30 SCREEN-TIME VALUES

The ordered data have sum 4,740; the 15th and 16th values are 135 and 140; 120 occurs four times.

1. Mean = 4740/30 = 158 min.
2. Median = (135+140)/2 = 137.5 min.
3. Mode = 120 min; range = 360−45 = 315 min.

Mean 158; median 137.5; mode 120; range 315 minutes.

Criterion A practice · 10.3 Dot plots, histograms and averages

WATCH FOR THIS MISCONCEPTION

For grouped data, a midpoint estimate of the mean is an estimate, not the exact mean of the raw values.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

10.3.1 Find the mean, median, mode and range of 2, 3, 3, 8. FLUENCY

10.3.2 For the 30 screen-time values, sum 4,740, middle values 135 and 140, mode 120, min 45, max 360. Find all four summaries. APPLICATION

10.3.3 Why is the median lower than the mean for the screen-time data? REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.3.4 Group the screen data into 0–<60, 60–<120, 120–<180, 180–<240, 240–<300, 300–360 (last includes 360). Find frequencies. APPLICATION

10.3.5 Why must a histogram state its interval boundaries? REASONING

10.3.6 Estimate the mean from grouped intervals with midpoints 30, 90, 150, 210, 270, 330 and frequencies 1, 9, 9, 5, 3, 3. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · INVESTIGATION · BEFORE 10.4

Investigate · Read the scale

YOUR TASK

Two graphs show the same five values: 88, 72, 80, 90, 96 (simulated data). One starts its axis at 0 and one at 70. Describe what each graph suggests.

GRAPH	AXIS START	VISUAL IMPRESSION	IS IT MISLEADING?
Graph A	0		
Graph B	70		
Same data, table	—		

What do you notice?

- What claim does each graph support, and which claim is better supported by the data?
- What three things should you check before accepting a visual claim?

Observation. Graph B makes a small rise from 88 to 96 look dramatic because the axis starts at 70; Graph A shows the change in proportion. The data themselves show a rise of 8 over five years, which is a modest change on a scale of 0 to 100.

MATHEMATICAL CONCLUSION

Read the scale, the denominator and the time order before accepting a visual claim. A truncated axis can make a small change look large without any number being wrong.

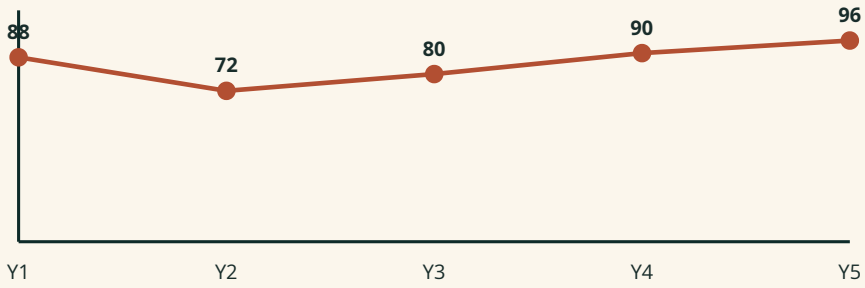
10.4 Trends, comparisons and misleading graphs

BIG IDEA

Read the scale, denominator and time order before accepting a visual claim.

A line graph or time-ordered bar chart can show change over time. Describe direction, size of change and exceptions; do not call every fluctuation a trend.

Misleading displays include truncated axes, unequal intervals, 3-D distortion, area used for a one-dimensional value, missing denominators and selective time windows. Compare like with like.



Zero-based axis; time order left to right.

A zero-based axis shows the fall from 88 to 72 and the later recovery to 96 without exaggeration.

WORKED EXAMPLE · TEST "SALES DOUBLED"

Sales rose from 50 to 90.

1. Increase = 40.
2. Percentage increase = 40/50×100 = 80%.

The claim is false: sales rose by 80%, not 100%.

Criterion A practice · 10.4 Trends, comparisons and misleading graphs

WATCH FOR THIS MISCONCEPTION

A percentage can be correct while the headline is misleading if the base is tiny or the denominator is hidden.

CRITERION A Knowing and understanding

GUIDED PRACTICE · FIRST STEPS

10.4.1 Uptake over five years is 88, 72, 80, 90, 96. Describe the overall change and the largest fall. FLUENCY

10.4.2 Find the percentage fall from 88 to 72. APPLICATION

10.4.3 A headline says “sales doubled” after 50 to 90. Test it. REASONING

CRITERION A Knowing and understanding

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.4.4 A pictogram doubles both height and width for a value twice as large. Explain the distortion. REASONING

10.4.5 Year 9 off-site is 13/65; Year 7 is 8/70. Compare percentages fairly. APPLICATION

10.4.6 Incidents rise from 2 to 4. Is “doubled” correct? What else should a reader ask? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

10.CK1 Classify “number of siblings”. CHECK

10.CK2 Find the sector angle for 30 out of 200. CHECK

10.CK3 Find the median of 1, 3, 3, 6, 7. CHECK

10.CK4 A value rises from 50 to 90. Find the percentage increase. CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

The data explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Choose a display, compute one summary and write a claim that the data actually support.

Use the lunch survey to explain why a truncated bar chart or a single pie chart could mislead.

ASSESSMENT PRACTICE · THE DATA EXPLAINER

Criteria A and C. Choose a display, compute one summary and write a claim that the data actually support.

10.S1 Which display best compares lunch choices across the three year groups? Explain. ASSESSMENT

10.S2 Calculate the school-lunch share and one check on the pie chart. ASSESSMENT

10.S3 Write one supported claim and one warning about a misleading display. ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

Mixed practice 10

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 10

Connect ideas. Give units and reasons where requested.

10.M1 Classify “height in centimetres”. FLUENCY

10.M2 Find the sector angle for 74 out of 200. FLUENCY

10.M3 Find mean, median, mode and range of 4, 4, 6, 10. FLUENCY

10.M4 A frequency table has counts 12, 18, 20. Find the percentage for 18. APPLICATION

10.M5 Why can a truncated vertical axis mislead? REASONING

10.M6 Uptake falls from 90 to 72. Find the percentage fall. APPLICATION

10.M7 A data set has mean 50 and median 30. What might explain the gap? REASONING

10.M8 Grouped intervals 0–<10 and 10–20 have frequencies 4 and 6. Estimate the mean using midpoints. CHALLENGE

10.M9 A survey of 80 has 20 choosing A. Find the percentage and pie angle. APPLICATION

10.M10 A headline says “crime doubled” from 2 to 4 incidents. Give a balanced response. CHALLENGE

> FINAL REFLECTION
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 10 · PRACTICE · CRITERION B

Criterion B practice · investigating patterns

CRITERION B Investigating patterns

CRITERION B Investigating patterns

INVESTIGATION PRACTICE · EXPLORATION · HOW SHOULD DATA BE GROUPED?

10.B1 Find the six 60-minute frequencies. INVESTIGATION

10.B2 Find the three 120-minute frequencies. INVESTIGATION

10.B3 Which grouping shows the central cluster more clearly? Which loses more detail? INVESTIGATION

10.B4 Find mean, median, mode and range. How many students spend 240 minutes or more? APPLICATION

10.B5 Test “the typical student spends four hours a day” and recommend a newsletter display. CHALLENGE

CRITERION B Investigating patterns

INVESTIGATE FURTHER

10.B6 Investigate how the histogram of the 30 screen-time values changes when the interval width changes from 30 to 60 minutes. Which display would you show families, and why? INVESTIGATION

10.B7 A headline says “sales doubled” from 50 to 105 units (simulated data). Investigate the claim using percentage increase and the starting value, and write a corrected headline. INVESTIGATION

CHAPTER 10 · PRACTICE · CRITERION C

Criterion C practice · communicating

CRITERION C Communicating

CRITERION C Communicating

EXPLAIN, JUSTIFY AND CORRECT

10.C1 Explain the difference between a bar chart and a histogram, with one example of the data each is designed for. REASONING

10.C2 A report gives a mean of 8 and a median of 6.5 for the same data set. Explain what this suggests about the distribution. REASONING

10.C3 Write two sentences that explain why a pie chart with 14 similar slices is hard to read, and name a better display. REASONING

10.C4 Explain, with one calculation, why “90% of a small group” can be a smaller number of people than “50% of a large group”. REASONING

CHAPTER 10 · PRACTICE · CRITERION D

Criterion D practice · applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

CRITERION D Applying mathematics in real-life contexts

APPLICATION PRACTICE · APPLICATION · THE CANTEEN SURVEY

10.D1 Find the school-lunch percentage and the pie-chart angles for all three choices. APPLICATION

10.D2 Compare off-site eating in Year 9 and Year 7 using percentages. APPLICATION

10.D3 Find daily spending by lunch choice and total spending. How much extra canteen revenue if 24 students switch from packed to school lunch? APPLICATION

10.D4 Evaluate “off-site eating is out of control” and write a more accurate headline of at most twelve words. REASONING

CRITERION D Applying mathematics in real-life contexts

APPLY IN A NEW CONTEXT

10.D5 A school survey of 200 students (simulated data) finds lunch choices: school 96, packed 74, off-site 30. Prices: school 4.20, packed 3.10, off-site 5.50 (in pounds). Estimate total spending on lunches and the school's share of it. APPLICATION

10.D6 A shop claims “our busiest month” using a bar chart with an axis starting at 500 (simulated claim). The counts are 1,240 and 1,265. Evaluate the claim and write a fairer statement. APPLICATION

10.D7 A class records the number of steps taken by 12 students over a week (simulated data, in thousands): 42, 55, 55, 61, 63, 68, 70, 74, 78, 81, 90, 96. Choose a display, compute one summary and write a claim the data support. APPLICATION

CHAPTER 10 · REVIEW

Check your progress

Work through the checklist without looking back. Anything you cannot do yet is a target for the mixed practice.

I can now

- [ ] I can classify variables and organise data in frequency or two-way tables.
- [ ] I can construct and interpret bar charts, pie charts, dot plots and histograms.
- [ ] I can summarise a distribution with an appropriate average and describe its spread.
- [ ] I can describe a trend and detect a misleading scale or an unstated denominator.

Reflect

- Which display do you trust most when reading a claim, and why?
- Choose one item you solved confidently and one you had to revise. What changed in your method?

ATL FOCUS. In order for a student to process and communicate information, students must organise data, choose an appropriate representation, calculate summaries and justify a claim with its limitations.

CHAPTER 10 · ACTION AND PROJECTS

Project connections

Mathematics becomes service and action when it answers a need that other people have. Each connection below is student-led: choose one, plan it with your teacher and follow your school's requirements and supervision.

Service as Action

COMMUNITY NEED	School notices publish percentages without denominators, so families cannot judge them.
MATHEMATICAL CONNECTION	Data displays, averages and misleading graphs (sections 10.1–10.4).
PROPOSED ACTION	Audit 10 school notices, classify each figure as supported or unsupported, and present a short guide to the school office.
EXPECTED IMPACT	Notices state their denominators, so readers can judge the figures.
STUDENT REFLECTION	Record the figures you found, how you checked each one and what the office changed.

Community Project connection

PROJECT IDEA	A community data display on a topic chosen with local residents.
COMMUNITY NEED	Local decisions are argued with anecdote rather than evidence.
RESEARCH QUESTION	What can a small, well-designed survey show about a local question, and what can it not show?
MATHEMATICS	Variables, tables, displays, averages and claims.
ACTION AND OUTCOME	Design and run a survey of 30 to 50 respondents, publish a display with stated denominators and limitations.
BENEFICIARIES	Community members, local leaders and students.
EVIDENCE	Survey questions, data table, display and a report that states the limitations.

Personal Project connection · MYP 5

PERSONAL INTEREST	Journalism, public information or sport analysis.
LEARNING GOAL	To evaluate statistical claims in public reporting.
PRODUCT OR OUTCOME	A short evaluation of three published claims using data you collect or verify.
MATHEMATICS	Averages, displays, percentage change and misleading scales.
PROCESS	Select claims, collect or verify data, recompute the summaries and write the evaluation.
EVIDENCE	Product, process journal and an evaluation against the stated learning goal. This is an MYP 5-oriented extension: follow your school's Personal Project requirements and supervision.

Service as Action is not the same as completing an exercise: it requires action that benefits others and a reflection on that action. The Community Project connection suits the MYP community project completed by students finishing MYP in Year 3 or Year 4, and the Personal Project connection is an adaptable MYP 5-oriented extension. They are deliberately different from one another.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value

INVESTIGATE FURTHER

1.B6 **4-digit numbers: 9,000 (1,000 to 9,999). 5-digit numbers: 90,000. Rule: 9 × 10^(n−1).**
The first digit has 9 choices (1 to 9) and each remaining digit has 10 choices, so the count is 9 × 10^(n−1). For n = 1 this gives 9 and for n = 2 it gives 90, which matches a direct count.

1.B7 **Smallest 6.45; largest is any value just below 6.55, so 6.55 itself is not included.**
Values that round to 6.5 to the nearest tenth lie in the interval [6.45, 6.55). The lower endpoint rounds up because its hundredths digit is 5; the upper endpoint is excluded because it would round to 6.6.

EXPLAIN, JUSTIFY AND CORRECT

1.C1 **0 is a number with a precise position on the number line, and it holds places: 305 is not 35. A zero does not always change the value, because 0.4 = 0.40.**
A convincing reply uses evidence rather than assertion: a position on the number line, and one place-value example where the zero changes the value.

1.C2 **0.5 = 0.50, and 50 hundredths is greater than 45 hundredths, so 0.5 sits to the right of 0.45 on a number line.**
Comparing tenths first settles it: 5 tenths is more than 4 tenths. Writing both numbers with two decimal places makes the comparison visible.

1.C3 **Read the whole part from the left: 3 thousands, 0 hundreds, 4 tens, 8 ones. Then read the fractional part: 3 tenths, 0 hundredths, 7 thousandths. The two 3s are worth 3,000 and 0.3.**
A check names the place of every digit rather than only the answer; the two 3s differ by a factor of 10,000.

1.C4 **Any total from 45,000 to 54,999; for example 48,672, 50,000 and 54,120.**
Rounding to the nearest ten thousand keeps values within 5,000 of 50,000, so the range is [45,000, 55,000) and 54,999 is the largest whole number in it.

APPLY IN A NEW CONTEXT

1.D5 **For example: “3,142 students took part — about 3,100 to the nearest hundred.”**
Public notices need readable figures, but the exact count should still be available. Rounding to the nearest hundred keeps the error below 50.

1.D6 **About 50 jerrycans; exactly 1,250 ÷ 25 = 50.**
1,250 ÷ 25 = 50 exactly, so the estimate and the exact answer agree. The display value 1.3 kL = 1,300 L would give 52, so the unit on the display matters.

READY CHECK

1.R1 **4,060**
The 4 is in the thousands place and the 6 is in the tens place; zeros hold the hundreds and ones places.

1.R2 **0.5**
0.5 = 0.50, and 50 hundredths is greater than 45 hundredths.

1.R3 **56**
Eight groups of seven make 56.

1.R4 **3,050**
Adding one hundred to 29 hundreds gives 30 hundreds, so the thousands digit increases.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

READY CHECK

1.R5 **0.7**
Seven tenths is written 0.7.

1.R6 **−5, −2, 0, 3**
On a number line, values increase from left to right.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

1.1.1 **(a) natural, whole, integer, rational. (b) whole, integer, rational. (c) integer, rational. (d) rational. (e) rational.**
Natural numbers start at 1 in this book; whole numbers add 0; integers add negatives; rational numbers can be written as a fraction of integers with nonzero denominator.

1.1.2 **−3, −2, −1, 0, 1, 2**
Inclusive means both endpoints are included.

1.1.3 **(a) < (b) > (c) <**
The smaller number lies further left on the number line.

1.1.4 **−7 °C, −2 °C, 0 °C, 3 °C, 5 °C**
Coldest means least, so order the numbers from left to right on a number line.

1.1.5 **Not under this book's convention. Zero is whole, an integer and rational, but natural numbers start at 1.**
Some traditions include 0 in the natural numbers, so a definition must be stated. This book uses natural numbers = 1, 2, 3, ...

1.1.6 **For example, 5/4 = 1.25, which is greater than 1. Also 4/4 = 1.**
A fraction can be greater than or equal to 1 when the numerator is at least the denominator.

1.1.7 **Down 3 floors, then up 5 floors; final position is 2 floors above the start.**
1 to −2 is a change of −3; −2 to 3 is a change of +5; net change +2.

1.1.8 **2.5 lies halfway between 2 and 3. It is rational, not an integer.**
Integers are whole numbers and their negatives; halves between integers are rational numbers.

PRACTICE 1.2 · PLACE VALUE AND ZERO

1.2.1 **(a) 40 (b) 0.08 (c) 8,000 (d) 0**
The places are tens, hundredths, thousands and hundredths respectively. A zero digit contributes zero but still holds its place.

1.2.2 **4,632**
Combine the thousands, hundreds, tens and ones.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

PRACTICE 1.2 · PLACE VALUE AND ZERO

1.2.3 **(a) 362 (b) 548 (c) 92 (d) 0.073**
Multiplying by 10 shifts digits one place left; multiplying by 100 shifts two places. Division shifts digits right.

1.2.4 **It holds the tenths place. Removing it gives 2.6, which is 0.54 larger.**
2.06 has 0 tenths and 6 hundredths; 2.6 has 6 tenths. 2.60 − 2.06 = 0.54.

1.2.5 **3.5**
3.5 = 3.50. Both have 3 ones, but 5 tenths is greater than 0 tenths.

1.2.6 **2,400 pencils; the 2 is worth 2,000.**
24 × 100 = 2,400. The 2 is in the thousands place.

1.2.7 **4 → 40 multiplies by 10; 0.4 → 0.40 and 4.50 → 4.500 do not change the value. Correct rule: multiplying by 10 shifts every digit one place left. Appending a zero only guarantees this for whole numbers written without a decimal part.**
Trailing zeros after a decimal point add no value; they may record precision.

1.2.8 **Greatest 8,520; least 2,058.**
Put the largest digits in the highest places for the greatest number. For the least four-digit number, the thousands digit must be the smallest nonzero digit, 2.

EXPLORATION · HOW MANY NUMBERS CAN THE DIGITS WRITE?

1.B1 **9; 90; 900; 9,000.**
The first digit has 9 choices; each later digit has 10 choices.

1.B2 **It is multiplied by 10.**
Each existing number can be extended by any of 10 final digits.

1.B3 **9 × 10^(n−1); for n = 5 the count is 90,000.**
There are 9 choices for the first digit and 10 choices for each of the remaining n−1 digits.

1.B4 **The first digit cannot be 0, so it can be 1–9. Later digits can be 0–9.**
A leading zero would not create an n-digit positive whole number.

1.B5 **For n = 1, 2, 3 the counts are 4, 20, 100. Rule: 4 × 5^(n−1).**
The first digit has 4 nonzero choices; each later digit has 5 choices.

PRACTICE 1.3 · EXPANDED FORM

1.3.1 **(a) 2,000 + 500 + 6 (b) 30 + 0.7 + 0.08 (c) 0.4 + 0.009**
Each nonzero digit is multiplied by its place value; zero places may be omitted.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

PRACTICE 1.3 · EXPANDED FORM

1.3.2 **(a) 496 (b) 6.58 (c) 2,070.003**
Keep zeros in any missing places so the other digits stay in the correct positions.

1.3.3 **(a) 700 (b) 0.009**
The missing parts are 7 hundreds and 9 thousandths.

1.3.4 **The first expression is 0.25 and the second is 0.025. Each digit in the first is one place to the left, so each contributes ten times as much.**
0.25 ÷ 0.025 = 10.

1.3.5 **Student B shows the zero tens explicitly. Both forms have the same value, 407.3.**
Adding a zero term does not change a sum, but it can reveal structure.

1.3.6 **53.4 cm. The terms are 50 cm, 3 cm and 0.4 cm.**
5 tens of centimetres + 3 centimetres + 4 tenths of a centimetre.

1.3.7 **Sample: 1,602 becomes 1,062, a decrease of 540. Other valid starting numbers are possible.**
Moving 6 from hundreds to tens loses 600 − 60 = 540.

1.3.8 **1,000 = 1 × 1,000. The hundreds, tens and ones are all zero, but those positions are still needed to show the 1 is in the thousands place.**
Zero terms contribute no value but hold places.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.1 **(a) = (b) > (c) >**
Compare place by place; for negatives, the number further right is greater.

1.4.2 **0.07, 0.7, 0.707, 0.77**
Write them as 0.070, 0.700, 0.707, 0.770 to compare thousandths.

1.4.3 **(a) 8,600 (b) 8,650**
For hundreds, the tens digit 4 rounds down; for tens, the ones digit 7 rounds up.

1.4.4 **3.75 and 0.9**
The thousandths digit 6 rounds the hundredths up; the hundredths digit 9 rounds the tenths up with a carry.

1.4.5 **60.0. The hundredths digit 6 rounds 9 tenths up to 10 tenths, which is one whole.**
Keep the .0 to show the answer is to the nearest tenth.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.6 **4.56 km rounds to 4.6 km to the nearest tenth.**
The signs report the same length at different accuracies.

1.4.7 **Examples: 6.45, 6.50, 6.549. All values from 6.45 inclusive up to but not including 6.55 round to 6.5.**
6.45 is the lower midpoint; 6.55 would round to 6.6.

1.4.8 **Yes; 1,948 rounds to 2,000 to the nearest thousand. An accountability report should give the exact total, 1,948 CFA francs, and state any rounding used.**
The hundreds digit is 9, so 1,948 rounds up to 2,000 to the nearest thousand. “About 2,000” is reasonable for a quick headline, but an accountability report should keep the exact total and state the rounding.

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

1.5.1 **Estimate 700 + 3,100 = 3,800. Exact 3,833, so the estimate is 33 low.**
692 rounds up by 8; 3,141 rounds down by 41; the net change is −33.

1.5.2 **Estimate 60 × 30 = 1,800. Exact 1,798, so the estimate is 2 high.**
Rounding one factor up and one down does not guarantee the direction; the exact check shows it.

1.5.3 **Estimate 40; exact 41.16, so the estimate is below. Mixed rounding does not by itself guarantee the direction.**
Round 9.8 up to 10 and 4.2 down to 4: 10 × 4 = 40. Exactly, 9.8 × 4.2 = 41.16. One factor was rounded up and the other down, so the direction of the error is not guaranteed; the exact check shows this estimate is low.

1.5.4 **Estimate 20. Exact quotient is about 20.8, so the estimate is low.**
604 ÷ 29 = 20.793…; rounding the divisor down and dividend down has competing effects.

1.5.5 **Estimate 400 + 400 + 300 = 1,100. Exact 1,098; estimate is 2 high.**
The rounded values differ from the exact values by +3, −12 and +11.

1.5.6 **Yes. The exact total is 20.03 km, which rounds to 20 km to the nearest kilometre.**
The poster gives a useful whole-kilometre summary, not an exact distance.

1.5.7 **Sample: about 1,250,000 books, to the nearest ten thousand. Use the exact figure in an inventory or audit.**
The rounded figure is easier to read; the exact figure is needed when individual items must be accounted for.

1.5.8 **About 2.67 m is a sensible centimetre-level report. The extra digits come from calculation, not from measurement precision; a full accuracy claim would need the calculation and measurement uncertainties.**
One centimetre is 0.01 m. Do not imply the tape measured billionths of a metre.

CHECKPOINT · BEFORE THE APPLICATION

1.CK1 **0.07**
The 7 is in the hundredths place.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

CHECKPOINT · BEFORE THE APPLICATION

1.CK2 **406**
The digits shift two places left.

1.CK3 **40.0**
The hundredths digit 5 rounds the tenths up, carrying into the ones.

1.CK4 **About 50 × 20 = 1,000.**
The exact product is 1,029, so the estimate is close and slightly low.

APPLICATION · A SCHOOL NOTICEBOARD

1.D1 **Sample: enrolment about 1,300 for a headline; library books 3,680 for inventory; year opened 1998 exactly; visits about 10,000 for a headline; scholarship places 12 exactly because the number is small and consequential.**
Accuracy should match purpose. Small counts and identifiers should not be rounded merely for style.

1.D2 **Both appear as about 1,300, hiding a difference of 43 students. To the nearest ten, 1,280 and 1,330 preserve the comparison better.**
1,327 − 1,284 = 43. Rounding to hundreds maps both into the same hundred.

1.D3 **Whether it counts visits or unique visitors, the period covered, and whether repeat visits by the same person are counted more than once.**
A number is interpretable only when the counted unit and collection rule are clear.

1.D4 **The year opened, 1998, should not be rounded because it identifies a date. The 12 scholarship places are sensitive: rounding to the nearest ten gives 10, losing 2 places, about 17% of the total.**
The same rounding rule can be harmless for a large headline and misleading for a small decision-making count.

ASSESSMENT PRACTICE · A MUSEUM LABEL

1.S1 **8,000 and 6,000.**
The 8 is in the thousands place; the 6 is in the thousands place.

1.S2 **6,000 + 500 + 9, or 6 × 1,000 + 5 × 100 + 0 × 10 + 9 × 1.**
The zero tens may be omitted or shown to reveal the place structure.

1.S3 **49,000 objects and 210,000 visitors.**
The hundreds digit 6 rounds the thousands up; the thousands digit 4 rounds the ten-thousands down.

1.S4 **Sample: “The collection began in 1894. Today it holds about 49,000 objects, with 6,509 on display—fewer than one in seven. Last year the museum welcomed about 210,000 visitors. Come and see how a small display can tell a large story.”**
The label uses the opening year exactly, rounds the collection and visitor figures for readability, and keeps the display count exact. The comparison is valid because 6,509 is less than one seventh of 48,672. It stays within 70 words.

1.S5 **Sample: “I rounded the collection to the nearest thousand so the headline is quick to read. Rounding the display count too heavily could hide how few objects visitors can actually see.”**
The note connects accuracy to audience and meaning.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

MIXED PRACTICE 1

1.M1	**(a) 800 (b) 0.8 (c) 0.08** The places are hundreds, tenths and hundredths.
1.M2	**5,000 + 40 + 0.2 + 0.007** The hundreds, ones and hundredths digits are zero.
1.M3	**7,030.06** Keep zeros in the hundreds, ones and tenths places.
1.M4	**65,000; 64,800; 64,780** Inspect the digit immediately to the right of each target place.
1.M5	**0.50 and 13.0** Both involve a carry. Keep the final zero to show the requested place.
1.M6	**(a) 64 (b) 0.82 (c) 3,500** Shift digits left for multiplication by powers of ten and right for division.
1.M7	**−2.5, −0.5, 0.25, 2.05** Negative values lie left of zero; among negatives, the larger magnitude is smaller.
1.M8	**(a) < (b) = (c) <** Compare by place value and number-line position.
1.M9	**Examples: 4,650, 4,700, 4,749. Whole-number range: 4,650 to 4,749 inclusive.** The midpoint below is 4,650; the midpoint above is 4,750, which would round to 4,800.
1.M10	**6.09 < 6.1 because 6.1 = 6.10, and 9 hundredths is less than 10 hundredths.** Align place values instead of counting digits.
1.M11	**−3 °C is colder; the difference is 7 °C.** From −3 to 0 is 3 degrees, and from 0 to 4 is 4 degrees.
1.M12	**Underestimate by 480. “About 12,500” to the nearest hundred is more informative.** 12,480 − 12,000 = 480; 12,480 rounds to 12,500 to the nearest hundred.

ANSWERS AND REASONING

Chapter 1 · Number systems and place value (continued)

MIXED PRACTICE 1

1.M13	**In 4,507 the 5 is worth 500; in 457 it is worth 50. The zero in 4,507 keeps the 5 in the hundreds place.** The same digit has different value in different positions.
1.M14	**Estimate 4,000 + 4,000 + 3,000 = 11,000. Exact total 10,997, so the estimate is 3 high.** The rounded values differ by +88, −87 and +2, net +3.
1.M15	**8, giving 648.** The only even whole number strictly between 646 and 650 is 648.
1.M16	**The trailing zero adds no value, but it records accuracy to the nearest hundredth rather than the nearest tenth.** Numerical equality and reported precision are different ideas.
1.M17	**Sample: 3.512 and 3.501. Both lie between 3.500 and 3.600, and 3.501 is closer to 3.5.** Compare thousandths: 3.500 < 3.501 < 3.512 < 3.600.
1.M18	**3,900; 3,621; 3,342; 3,063.** Let the ones digit be u. Then tens is 2u and hundreds is 9 − 3u. The digit constraints allow u = 0, 1, 2, 3.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations

INVESTIGATE FURTHER

2.B7 **Thicknesses: 0.2, 0.4, 0.8, 1.6, 3.2, 6.4 mm. After 10 folds the model predicts 0.1 × 2^10 = 102.4 mm.**
Each fold doubles the thickness, so the model is geometric: t = 0.1 × 2^n mm. A real sheet cannot be folded many times because the layers must bend around each other and the paper tears.

2.B8 **4, 16, 36, 64 and 100 are all even, and the rule holds for every even number.**
An even number is 2k for some whole number k, so $(2k)^2 = 4k^2 = 2(2k^2)$, which is a multiple of 2. Examples suggest the rule; the structure proves it.

EXPLAIN, JUSTIFY AND CORRECT

2.C1 **3^4 means 3 × 3 × 3 × 3 = 81, because the exponent counts factors, not addends. 5^2 = 5 × 5 = 25, because squaring multiplies a number by itself.**
Both errors treat the index as an instruction to multiply by the exponent. The index counts copies of the base.

2.C2 **$\sqrt{(9 + 16)} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. The root asks for the number that was squared, and $7^2 = 49 \neq 25$.**
A clear explanation states what the symbol asks for, then shows the two values are different and checks the claim by squaring.

2.C3 **It means 2 × 1,000 = 2,000 mm, which is 2 m.**
10^3 = 1,000 and 1 m = 1,000 mm, so 2,000 mm = 2 m. Naming the power and the unit conversion keeps the explanation checkable.

2.C4 **18 is correct: multiplication comes before addition, so 8 + 10 = 18. The answer 50 comes from adding first, as if the expression were (8 + 2) × 5.**
Diagnosing the error is part of communicating mathematics: the wrong answer is a reasonable reading of a different expression, which is why brackets matter.

APPLY IN A NEW CONTEXT

2.D5 **About 4 GB; exactly 4 × 2^30 = 4,294,967,296 bytes.**
2^30 = 1,073,741,824 bytes is about 1.07 GB, so four files are a little over 4 GB. Stating both the estimate and the exact value keeps the claim honest.

2.D6 **Charge = 100 × 0.8^n %. After 5 hours: 100 × 0.8^5 = 32.768%, about 33%.**
Each hour multiplies the previous charge by 0.8, so the model is a power. 0.8^5 = 0.32768.

2.D7 **Estimate about 14 m^2; exactly 4.2 × 3.5 = 14.70 m^2.**
Measurements given to one decimal place cannot justify more than two decimal places in the product; 14.70 m^2 is the exact product and about 14 m^2 is the sensible estimate for ordering.

READY CHECK

2.R1 **42 and 81.**
Six groups of seven make 42. Nine groups of nine make 81.

2.R2 **4,536.**
The terms occupy the thousands, hundreds, tens and ones places.

2.R3 **24.**
Multiply first: 3 × 4 = 12. Then add: 12 + 12 = 24.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

READY CHECK

2.R4 **$3 \times 3 \times 3 = 27$.**
There are three factors of 3. First $3 \times 3 = 9$, then $9 \times 3 = 27$.

2.R5 **64 cm^2.**
Area of a square = side × side = $8 \times 8 = 64$ square centimetres.

2.R6 **$(2 + 4) \times 3 = 18$.**
The brackets make the addition happen first: $2 + 4 = 6$, then $6 \times 3 = 18$.

PRACTICE 2.1 · POWERS AND INDEX NOTATION

2.1.1 **(a) 4^3 (b) 7^2 (c) 10^5 (d) 2^7.**
The base is the repeated factor. The exponent counts how many copies of that factor are multiplied.

2.1.2 **(a) 8 (b) 9 (c) 125 (d) 1,000 (e) 64 (f) 64.**
Expand first if needed: $2\times2\times2=8$; $3\times3=9$; $5\times5\times5=125$; $10\times10\times10=1{,}000$; six 2s make 64; $4\times4\times4=64$.

2.1.3 **(a) 4 (b) 6 (c) 3 (d) 0.**
$3^4=81$, $2^6=64$ and $10^3=1{,}000$. For the nonzero base 7, the exponent 0 gives 1.

2.1.4 **$4^2 = 16$, but $4 \times 2 = 8$. The exponent counts equal factors; it is not a multiplier.**
4^2 means 4×4. The expression 4×2 means two groups of 4, or $4 + 4$.

2.1.5 **$9^2 = 81$ cm^2.**
A square's area is side × side. The unit is square centimetres because two lengths in centimetres are multiplied.

2.1.6 **$2^4 = 4^2 = 16$, but $2^3 = 8$ and $3^2 = 9$. The claim is false; one counterexample is enough.**
The first pair happens to match. The second pair does not, so a universal claim about swapping base and exponent cannot be true.

2.1.7 **Round 4: $3^4 = 81$ new people. Rounds 1–4: $3 + 9 + 27 + 81 = 120$ new people.**
Each round multiplies the number of new people by 3. The total asked for excludes the original person.

2.1.8 **2^{10} is larger. Exactly, $2^{10} = 1{,}024$ and $10^3 = 1{,}000$.**
One check is $2^{10} = 2^7 \times 2^3 = 128 \times 8$. Since $128 > 125$, multiplying both by 8 gives $128 \times 8 > 125 \times 8 = 1{,}000$. The exact value is 1,024.

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

2.2.1 **(a) 100 (b) 10,000 (c) 1,000,000 (d) 1.**
For positive whole-number exponents, 10^n is 1 followed by n zeros. The pattern of dividing by 10 gives $10^0 = 1$.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

2.2.2 **$8 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.**
The digits occupy thousands, hundreds, tens and ones. The zero tens term may be shown to make the place structure explicit.

2.2.3 **4,706.**
The terms are 4,000, 700, 0 and 6. The zero keeps the tens place empty.

2.2.4 **$9 \times 10^4 + 0 \times 10^3 + 0 \times 10^2 + 5 \times 10^1 + 0 \times 10^0$; the 5 is worth 50.**
The 5 is in the tens place. Zeros hold the thousands, hundreds and ones places.

2.2.5 **20,050. Zeros hold the thousands, hundreds and ones places.**
$2 \times 10^4 = 20{,}000$ and $5 \times 10^1 = 50$. Adding them gives 20,050; the missing places must not be collapsed.

2.2.6 **$1 \times 10^4 + 2 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 0 \times 10^0$. The 1 contributes 10,000, the largest share.**
Compare the values of the nonzero terms: 10,000, 2,000, 400 and 80. The ten-thousands term is largest.

2.2.7 **3,000 is 10 times 300.**
The digit 3 has moved one place to the left. Each place is ten times the place to its right.

2.2.8 **Sample: $12{,}690 = 1 \times 10^4 + 2 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$. The digits sum to 18 and there is exactly one zero term. Many answers are possible.**
Choose four nonzero digits that sum to 18, then place one zero in a non-leading position. The leading digit cannot be zero in a five-digit number.

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

2.3.1 **1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.**
These are 1^2 through 12^2. Each is a whole number multiplied by itself.

2.3.2 **(a) 169 (b) 225 (c) 216 (d) 1,000.**
$13 \times 13 = 169$; $15 \times 15 = 225$; $6 \times 6 \times 6 = 216$; $10 \times 10 \times 10 = 1{,}000$.

2.3.3 **(a) 7 (b) 10 (c) 13 (d) 2 (e) 3 (f) 5.**
Reverse the powers: $7^2=49$, $10^2=100$, $13^2=169$, $2^3=8$, $3^3=27$ and $5^3=125$.

2.3.4 **12 tiles.**
If each side has n tiles, the square contains n^2 tiles. Since $12^2 = 144$, $n = 12$.

2.3.5 **(a) 8 (b) 6 (c) 15.**
$8^2=64$, $6^3=216$ and $15^2=225$. The question asks for the positive value.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

2.3.6 **$\sqrt{81} + \sqrt{16} = 9 + 4 = 13$. $\sqrt{(81 + 16)} = \sqrt{97}$, which is not 13 because $13^2 = 169$. They are not equal.**
In the first expression the roots are taken separately and then added. In the second, the sum is formed first and the root applies to the whole sum.

2.3.7 **Between 7 and 8, closer to 7.**
7^2=49 and 8^2=64, so $7 < \sqrt{50} < 8$. Also 7.5^2=56.25, and $50 < 56.25$, so $\sqrt{50} < 7.5$ and is closer to 7.

2.3.8 **Edge length 7 cm. Doubled edge 14 cm gives $14^3 = 2{,}744$ cm³, which is 8×343. The factor is $2^3 = 8$.**
7^3=343. Doubling each of the three dimensions multiplies volume by $2 \times 2 \times 2$.

PRACTICE 2.4 · ORDER OF OPERATIONS

2.4.1 **18.**
Multiply first: $2 \times 5 = 10$. Then add: $8 + 10 = 18$.

2.4.2 **8.**
Multiply first: $3 \times 4 = 12$. Then subtract: $20 - 12 = 8$.

2.4.3 **(a) 22 (b) 100 (c) 17.**
(a) 4^2=16, then 6+16=22. (b) Brackets first: 10^2=100. (c) 8+9=17.

2.4.4 **(a) 12 (b) 3 (c) 9.**
(a) Division and multiplication have equal priority, so work left to right: 24÷4=6, then 6×2=12. (b) Brackets first: 4×2=8, then 24÷8=3. (c) 36÷6=6, then 6+3=9.

2.4.5 **$5 \times 8 - 9 = 40 - 9 = 31$.**
Evaluate the powers first: 2^3=8 and 3^2=9. Then multiply, then subtract.

2.4.6 **(a) $(3 + 5) \times 2 = 16$ (b) $(12 - 4) \times 2 = 16$ (c) $18 \div (3 + 6) = 2$.**
The brackets change which operation is completed first. Check each result after inserting them.

2.4.7 **The student added before dividing. Correct: $6 \div 3 = 2$, then $12 + 2 = 14$.**
Division has higher priority than addition unless brackets say otherwise.

2.4.8 **Samples: $2 + 3 \times 4 = 14$; $(2 + 3) \times 4 = 20$; $2^3 + 4 = 12$; $4^2 - 3 = 13$. Other valid expressions are possible.**
Each sample uses 2, 3 and 4 exactly once. The order of operations and brackets determine the value.

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.1 **(a) 18 (b) 50 (c) 200.**
Square n first, then multiply by 2: 2×9=18; 2×25=50; 2×100=200.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.2 **(a) 36 (b) 100 (c) 400. Each is double the matching result in 2.5.1.**
The brackets include the factor 2 in the base: $(2n)^2 = 4n^2$, while $2n^2$ is only $2n^2$. Therefore $(2n)^2$ is twice $2n^2$.

2.5.3 **$a^2 + b^2 = 9 + 16 = 25$. $(a + b)^2 = 7^2 = 49$. They are not equal.**
The first expression squares a and b separately. The second squares the whole sum.

2.5.4 **$A = 12^2 = 144$ cm².**
Substitute 12 for s. Multiplying centimetres by centimetres gives square centimetres.

2.5.5 **(a) 27 cm^3 (b) 125 cm^3.**
$3^3=27$ and $5^3=125$. Multiplying three lengths in centimetres gives cubic centimetres.

2.5.6 **(a) 20 m (b) 80 m. Doubling t multiplies t^2 by 4, so d quadruples in this model.**
$5\times2^2=20$ and $5\times4^2=80$. The model is simplified; it is not a precise measurement of a real fall.

2.5.7 **(a) $(2x)^2 = 36$ (b) $(a + b)^2 = 49$. The brackets put the whole product or sum in the base of the power.**
When x=3, $(2\times3)^2=6^2=36$. When a=2 and b=5, $(2+5)^2=7^2=49$.

2.5.8 **Side length 9 cm. Tripled side 27 cm gives new area $27^2 = 729$ $cm^2 = 9 \times 81$. The area is multiplied by $3^2 = 9$.**
Area depends on the square of the side length. Tripling the side multiplies area by 3×3.

EXPLORATION · THE DOUBLING SHEET

2.B1 **Physical results vary. Ideal model: layers 1, 2, 4, 8, 16, 32; thicknesses 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 mm for n = 0, 1, 2, 3, 4, 5.**
Start with one layer. Each fold doubles the layer count. Thickness = 0.1 × layers. The physical trial and the ideal model are different kinds of evidence.

2.B2 **The thickness doubles with each fold. $T = 0.1 \times 2^n$ mm.**
After n folds there are 2^n layers. At n=0, $2^0=1$, so the rule gives the original thickness 0.1 mm.

2.B3 **$T = 0.1 \times 2^{10} = 102.4$ mm. It is 1,024 times thicker, because $2^{10} = 1{,}024$.**
The starting thickness is 0.1 mm. Dividing 102.4 by 0.1 gives 1,024, the layer-count factor.

2.B4 **14 folds. At 13 folds, T = 819.2 mm; at 14 folds, T = 1,638.4 mm. Since 1 m = 1,000 mm, 14 is the first fold count above 1 m.**
Use $T = 0.1 \times 2^n$. Check n=13 and n=14 rather than guessing from the size of the exponent.

2.B5 **No. The table is an ideal model, not proof of physical feasibility. Assumptions include perfect folds, constant thickness and enough sheet area to keep folding. Improve the estimate by measuring a stack of many sheets and dividing by the number of sheets.**
Real paper resists folding because thickness grows, creases form and the available area shrinks. A mathematical prediction can be valid inside a model while the model stops matching reality.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

EXPLORATION · THE DOUBLING SHEET

2.B6 **Tripling model: $0.1 \times 3^5 = 24.3$ mm. Doubling model: $0.1 \times 2^5 = 3.2$ mm. The tripling model grows faster because each step multiplies by 3 rather than 2.**
Compare the growth factors, not only the starting value. Both start at 0.1 mm when n=0.

GUIDED PRACTICE · COPY AND COMPLETE

2.G1 **Base 8, exponent 3. $8 \times 8 \times 8 = 512$. If the exponent were 1, the value would be 8.**
The exponent counts factors of 8. It does not mean 8×3.

2.G2 **$\sqrt{169} = 13$; $\sqrt[3]{216} = 6$; side length = 14 cm.**
Roots reverse powers. Since $14^2=196$, a square with area 196 cm^2 has side 14 cm.

2.G3 **For x=5: $2x^2=50$ and $(2x)^2=100$. For x=2: $x^2+3^2=13$ and $(x+3)^2=25$. Brackets change the base of the power to include the whole product or sum.**
Without brackets, the exponent applies only to the quantity immediately before it. With brackets, it applies to everything inside the brackets.

2.G4 **$-3^2 = -9$, while $(-3)^2 = 9$. In the first expression the power is evaluated before the unary minus; in the second, the base is −3.**
-3^2 means $-(3^2)$. The brackets in $(-3)^2$ include the negative sign in the repeated factor: $(-3) \times (-3) = 9$.

CHECKPOINT · BEFORE THE APPLICATION

2.CK1 **Base 6, exponent 4; $6^4 = 1{,}296$.**
$6^4 = 6{\times}6{\times}6{\times}6 = 36{\times}36 = 1{,}296$.

2.CK2 **$5 \times 10^3 + 0 \times 10^2 + 2 \times 10^1 + 0 \times 10^0$.**
The zeros hold the hundreds and ones places.

2.CK3 **12 and 4.**
$12^2=144$ and $4^3=64$.

2.CK4 **0.**
$3^2=9$, then $2{\times}9=18$, then $18-18=0$.

APPLICATION · THE SCHOOL DATA STORE

2.D1 **10 TB = 10,000 GB = 10^4 GB.**
4+4+2=10 TB. With 1 TB=1,000 GB, multiply by 1,000: 10×1,000=10,000 GB.

2.D2 **1,416 complete projects, with 4 GB left over.**
15% means 15 out of 100, so the reserve is $10{,}000 \times 15 \div 100 = 1{,}500$ GB. Available space is 8,500 GB. Each project needs 6 GB. $8{,}500 \div 6 = 1{,}416$ remainder 4.

2.D3 **A 1,417th project would need $1{,}417 \times 6 = 8{,}502$ GB, but only 8,500 GB is available. Complete projects require rounding down.**
Nearest-whole rounding can create a capacity that does not exist. The context requires a feasible whole number of complete projects.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

APPLICATION · THE SCHOOL DATA STORE

2.D4 **10 TB = 10^{13} bytes. $10^{13} \div 2^{40} \approx 9.09$ TiB. No storage is missing; the units differ. Sample label: '10 TB decimal capacity (about 9.09 TiB in binary units).'**
A tebibyte is a larger unit than a terabyte, so the same number of bytes gives a smaller count in TiB. Honesty means naming the unit, not choosing the larger number.

ASSESSMENT PRACTICE · A POWER PRIMER FOR THE SCIENCE LAB

2.S1 **Sample: 'A power records repeated multiplication. In 5^3, 5 is the base and 3 is the exponent. It means $5 \times 5 \times 5 = 125$.'**
The definition names the repeated factor and the number of factors, and the example matches the notation.

2.S2 **Sample: '$5^3=125$, $5^2=25$, $5^1=5$. Each step divides by 5, so the next step gives $5^0=1$.'**
The pattern continues the division-by-the-base rule. The zero-exponent rule applies because the base 5 is nonzero.

2.S3 **Sample: draw a 4 by 4 grid and label it $4^2=16$. For 4^3, draw or describe 4 layers of 4 by 4 unit cubes: $4 \times 4 \times 4 = 64$ unit cubes.**
The square model shows two equal dimensions; the cube model shows three equal dimensions.

2.S4 **Brackets: $8-2=6$. Index: $6^2=36$. Multiply: $3 \times 36=108$. Divide: $108 \div 6=18$. Add: $4+18=22$.**
Brackets first, then indices, then multiplication/division from left to right, then addition/subtraction. The final value is 22.

2.S5 **When $x=2$, $3x^2=12$ and $(3x)^2=36$. The test expression is $2 + 5 \times 9 = 47$. Sample evaluation: strength—one step per line; limitation—symbols may be too small; revision—enlarge the 'indices before multiplication' label.**
The card should make the base of each power visible and support a reader who is learning the convention for the first time.

MIXED PRACTICE 2

2.M1 **(a) 81 (b) 128 (c) 1,000,000 (d) 216 (e) 121 (f) 256.**
Expand or use known squares/cubes: $3^4=81$; $2^7=128$; $10^6=1{,}000{,}000$; $6^3=216$; $11^2=121$; $4^4=256$.

2.M2 **(a) 9^3 (b) 12^2 (c) 10^5 (d) 5^7.**
Count the repeated factors. The base stays the same.

2.M3 **$6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.**
The zero terms hold the thousands and tens places.

2.M4 **80,205.**
The terms are 80,000, 200 and 5. Zeros hold the thousands and tens places.

2.M5 **(a) 9 (b) 14 (c) 3 (d) 10.**
$9^2=81$, $14^2=196$, $3^3=27$ and $10^3=1{,}000$.

2.M6 **(a) 16 (b) 40 (c) 26 (d) 6 (e) 9 (f) 25.**
Apply brackets, indices, multiplication/division left to right, then addition/subtraction left to right.

ANSWERS AND REASONING

Chapter 2 · Powers, expanded form and order of operations (continued)

MIXED PRACTICE 2

2.M7 **(a) 13 (b) 25 (c) 54 (d) 36.**
(a) 9+4=13. (b) 5^2=25. (c) 2×27=54. (d) 6^2=36.

2.M8 **(a) $(2 + 3) \times 4 = 20$ (b) $(24 - 8) \div 4 = 4$ (c) $(3 \times 2)^2 = 36$ (d) $(10 - 4)^2 = 36$.**
The brackets change the grouping and, in (c) and (d), the base of the power.

2.M9 **2^5 uses five factors of 2, giving 32. 5^2 uses two factors of 5, giving 25. Therefore 2^5 is larger. The base and exponent play different roles.**
Swapping base and exponent changes both the repeated factor and the number of factors. The correct comparison is 32 > 25.

2.M10 **$7^2 = 49$, not 14. The student multiplied 7 by 2 instead of using two factors of 7.**
The exponent counts factors: $7^2 = 7 \times 7$.

2.M11 **(a) 6 (b) 5 (c) 4 (d) 5.**
2^6=64; 10^5=100,000; 3^4=81; 5^3=125.

2.M12 **Each step down in exponent divides by 10: 1,000 → 100 → 10 → 1. Therefore $10^0 = 1$.**
The zero exponent continues the place-value pattern. It is not the same as multiplying by zero.

2.M13 **1,111. Each power of ten contributes a 1 in a different place: thousands, hundreds, tens and ones.**
1,000 + 100 + 10 + 1 = 1,111.

2.M14 **Side 13 cm, perimeter 52 cm. Doubled side 26 cm gives area 676 cm^2 and perimeter 104 cm. Area is multiplied by $4 = 2^2$; perimeter is multiplied by 2.**
√169=13. Perimeter is 4s, so it scales linearly. Area is s^2, so it scales by the square of the scale factor.

2.M15 **8 cm^3 and 64 cm^3. The second is 8 times larger because doubling each of three dimensions multiplies volume by $2^3 = 8$.**
2^3=8 and 4^3=64; 64÷8=8.

2.M16 **Samples: (a) $4 \times 3 + 2 = 14$ (b) $4 \times (3 + 2) = 20$.**
In (a), multiply before adding. In (b), brackets make the sum 5 before multiplying by 4.

2.M17 **(a) 81 new people. (b) $1 + 3 + 9 + 27 + 81 = 121$ people. (c) $3^5 = 243$ would describe the number of new people in round 5, not the total after four rounds.**
New people by round are 3^1, 3^2, 3^3, 3^4. The total includes the original person and stops after round 4.

2.M18 **$\sqrt{(36 + 64)} = \sqrt{100} = 10$, while $\sqrt{36} + \sqrt{64} = 6 + 8 = 14$. The example disproves the claim that a square root can be distributed over a sum.**
The root applies to the whole value under the radical. Adding first and then taking the root is different from taking roots first and then adding.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes

INVESTIGATE FURTHER

3.B6 **Squares (1, 4, 9, 16) have an odd number of factors; non-squares have an even number. $144 = 12^2$ has 15 factors.**

A factor pair contributes two factors unless the partners are equal, which happens only for squares. 144 = 2^4 × 3^2 has (4 + 1)(2 + 1) = 15 factors.

3.B7 **False. 12 and 36 are divisible by 4 and by 6 but not by 24.**

A counterexample settles a universal claim. The reason is that 4 and 6 share a factor 2, so their product over-counts: the correct condition is divisibility by 4 and by 3, or by 12.

EXPLAIN, JUSTIFY AND CORRECT

3.C1 **If 97 had two factors larger than 9, their product would exceed 9 × 9 = 81 and could still be 97; testing primes up to $\sqrt{97} \approx 9.8$ (2, 3, 5, 7) is enough because any composite factor would have a smaller prime factor.**

A clear explanation states the boundary (√n), the primes actually tested and why no factor can hide above the boundary.

3.C2 **$84 = 2^2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$, so the HCF is $2^2 \times 3 = 12$.**

A common factor can only use the powers that appear in both numbers, so the smaller power of each shared prime is taken; multiplying gives the largest number that divides both.

3.C3 **42 is not prime, so the splitting is unfinished: $84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$.**

A prime factorisation is complete only when every endpoint is prime; a quick check is to look for any remaining composite factor such as 42.

3.C4 **1 has exactly one positive factor, itself, so it has neither exactly two factors (prime) nor more than two (composite).**

The justification uses the definitions rather than a list of examples, and it explains why 1 must be excluded from both classes.

APPLY IN A NEW CONTEXT

3.D5 **Team sizes 12 (16 teams), 16 (12 teams) and 24 (8 teams) are excluded by the limits; 192 = 2^6×3, so team sizes 8 (24 teams) and 6 (32 teams) also fail. A size of 12 gives 16 teams, which fits.**

Team sizes are factors of 192. The factors between 7 and 23 that give 9 to 29 teams are limited, so 12 with 16 teams is a workable recommendation; the reason should mention supervision, not only arithmetic.

3.D6 **Day 75: 5 deliveries from the first supplier and 3 from the second.**

LCM(15, 25) = 75 because $15 = 3 \times 5$ and $25 = 5^2$, so the LCM is $3 \times 5^2 = 75$. Then 75 ÷ 15 = 5 and 75 ÷ 25 = 3.

3.D7 **For example 12 boxes per shelter with 84 bottles each, or 9 boxes per shelter with 112 bottles each.**

The number of boxes per shelter must divide 1,008 = 2^4×3^2×7. Two plans are $1{,}008 = 3 \times 336 = 3 \times 4 \times 84$ and $1{,}008 = 3 \times 9 \times 112$; the plan chosen should be justified by handling and storage.

READY CHECK

3.R1 **56 and 12.**

Use the inverse multiplications 8 × 7 and 8 × 12.

3.R2 **4, 8, 12, 16, 20.**

Multiply by 1 through 5.

3.R3 **10 × 10.**

10 is the square root of 100.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes (continued)

READY CHECK

3.R4 **$3^2 = 9$ is larger than $2^3 = 8$.**
Evaluate each power.

3.R5 **Yes: $7 \times 8 = 56$.**
The remainder is zero.

PRACTICE 3.1 · FACTOR PAIRS

3.1.1 **1×36; 2×18; 3×12; 4×9; 6×6.**
Each product is 36; the final pair repeats a factor.

3.1.2 **7, 14, 21, 28, 35.**
Multiply 7 by 1, 2, 3, 4 and 5.

3.1.3 **Yes to both: $8 \times 7 = 56$.**
The same exact multiplication links factors and multiples.

3.1.4 **Missing: 27 and 54. The pairs include 2×27 and 1×54.**
Search pairs systematically; do not stop at the square root without recording both partners.

3.1.5 **1×24, 2×12, 3×8 and 4×6.**
Each pair of side lengths has product 24.

3.1.6 **48. It is the only common multiple of 6 and 8 in the interval.**
List common multiples or use the least common multiple 24, then 48.

PRACTICE 3.2 · DIVISIBILITY

3.2.1 **2, 3, 4, 6 and 8.**
Use the last digits and digit sum 21; 728 is divisible by 8.

3.2.2 **3, 5 and 9: yes. 10: no.**
Digit sum 9; last digit 5, not 0.

3.2.3 **It is odd, so it fails the test for 2. Divisibility by 6 needs both 2 and 3.**
The digit-sum test establishes only the factor 3 here.

3.2.4 **Every hundred is divisible by 4, so only the last two digits can affect the remainder.**
Write the number as $100 \times a + b$, where b is the last two-digit number.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes (continued)

PRACTICE 3.2 · DIVISIBILITY

3.2.5 **6 packs: yes, 21 per pack. 9 packs: yes, 14 per pack.**
126 is even and has digit sum 9, so both tests pass.

3.2.6 **654 is not divisible by 4; the largest valid arrangement is 564.**
The possible even endings are 54, 64 and 46; only 64 and 56 work, and 564 is the largest valid arrangement.

PRACTICE 3.3 · PRIMES

3.3.1 **2, 3, 5, 7, 11, 13, 17, 19, 23, 29.**
Each has exactly two positive factors.

3.3.2 **1 neither; 2 prime; 21 composite; 29 prime.**
$21 = 3 \times 7$; 1 has only one factor.

3.3.3 **Test 2, 3, 5 and 7. None divides 97, so it is prime.**
The next prime, 11, is greater than $\sqrt{97}$.

3.3.4 **9 is odd but composite: $9 = 3 \times 3$.**
One counterexample disproves a universal claim.

3.3.5 **59 and 61; sum 120.**
Both are prime and differ by 2.

3.3.6 **No. It verifies one case, not a general proof.**
A general claim needs a reason covering every case.

PRACTICE 3.4 · PRIME PRODUCTS

3.4.1 **$36 = 2^2 \times 3^2$; $54 = 2 \times 3^3$; $72 = 2^3 \times 3^2$.**
Split until every factor is prime, then group equal factors.

3.4.2 **120.**
$8 \times 3 \times 5 = 120$.

3.4.3 **No: 4 is composite. Replace it by 2×2, giving $2^2 \times 3 \times 7$.**
A prime factorisation cannot contain 4 as a terminal factor.

3.4.4 **Prime factorisation is unique apart from order; the trees only choose different intermediate pairs.**
Both trees decompose the same number completely.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes (continued)

PRACTICE 3.4 · PRIME PRODUCTS

3.4.5 **$126 = 2 \times 3^2 \times 7 = 6 \times 21$.**
Group 2×3 as 6 and 3×7 as 21.

3.4.6 **$n = 5$, because $180 = 2^2 \times 3^2 \times 5$ and $900 = 30^2$.**
A square has even exponents in its prime factorisation.

PRACTICE 3.5 · HCF AND LCM

3.5.1 **HCF 6; LCM 36.**
Shared prime factors give 6; the smallest common multiple is 36.

3.5.2 **HCF 2; LCM 72.**
2 divides all three; 72 is the first positive multiple shared by all.

3.5.3 **12 cm; 9 pieces in total.**
HCF is 12; $48 \div 12 + 60 \div 12 = 4 + 5$.

3.5.4 **09:00, after 60 minutes.**
The next shared repeat is LCM(15,20) = 60 minutes.

3.5.5 **Use LCM: a shared future time must be a common multiple, and the next one is the lowest.**
HCF describes a largest common divisor, not a future repeat.

3.5.6 **(6,36) and (12,18).**
Write the numbers as 6a and 6b with coprime a,b and ab = 6.

EXPLORATION · WHY DO SQUARES HAVE AN ODD FACTOR COUNT?

3.B1 **3, 3, 5, 3, 9.**
List pairs and count a repeated middle factor once.

3.B2 **1, 4, 9, 16 and 25: the perfect squares.**
Each square has one pair whose two entries coincide.

3.B3 **Its factors form distinct pairs $a \times b$, with a different from b.**
No unpaired middle factor occurs.

3.B4 **49 has 3 factors. 24 has 8; 25 has 3. A larger number need not have more factors.**
$49 = 7^2$; compare complete factor lists.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes (continued)

EXPLORATION · WHY DO SQUARES HAVE AN ODD FACTOR COUNT?

3.B5 **4, 9, 25 and 49: squares of primes. p^2 has factors 1, p and p^2.**
A three-factor number must have one repeated middle pair and no other prime factor.

CHECKPOINT

3.CK1 **1, 2, 3, 4, 6, 8, 12, 24.**
Use four factor pairs.

3.CK2 **No: 91 = 7 × 13.**
Test primes through √91, including 7.

3.CK3 **$2 \times 3^2 \times 5$.**
$90 = 9 \times 10$.

3.CK4 **HCF 6; LCM 72.**
$18 = 2 \times 3^2$; $24 = 2^3 \times 3$.

APPLICATION · THE FAIRNESS RELAY

3.D1 **12 teams of 16: 7 from A, 5 from B and 4 from C.**
The team count must divide all three totals; use HCF.

3.D2 **7 + 5 + 4 = 16 teams.**
Divide each year total by 12.

3.D3 **Equal mixed composition and equal team size within each year are different fairness rules. Both can be valid for different purposes.**
State the criterion before judging the allocation.

3.D4 **No. 16 does not divide 84, 60 or 48 exactly.**
A feasible team count must be a common factor.

ASSESSMENT PRACTICE · THE NUMBER MUSEUM

3.S1 **1×84, 2×42, 3×28, 4×21, 6×14, 7×12; $84 = 2^2 \times 3 \times 7$.**
There are twelve factors; the prime product checks to 84.

3.S2 **HCF 12; LCM 420. HCF answers largest equal groups.**
Use shared prime factors for HCF and highest powers for LCM.

3.S3 **Sample: "84 can be arranged in six different rectangular arrays. Its prime building blocks are 2, 2, 3 and 7." Question: "How many factors does 84 have?" Answer: 12.**
A clear label connects a representation to a verified fact.

ANSWERS AND REASONING

Chapter 3 · Factors, multiples and primes (continued)

MIXED PRACTICE 3

3.M1 **1, 2, 3, 6, 7, 14, 21, 42.**
Pair 1×42, 2×21, 3×14 and 6×7.

3.M2 **9, 18, 27, 36, 45, 54.**
Multiply 9 by 1 through 6.

3.M3 **All eight listed numbers.**
360 is a common multiple of each.

3.M4 **It has exactly one positive factor, not two or more than two.**
Prime and composite definitions both apply to numbers greater than 1.

3.M5 **$96 = 2^5 \times 3$; $150 = 2 \times 3 \times 5^2$.**
Group repeated primes.

3.M6 **72 seconds.**
Use LCM, not HCF.

3.M7 **14 cm.**
Use HCF(42,70).

3.M8 **No. 28 has 6 factors; 29 is prime and has 2.**
A prime can be larger than a highly divisible composite.

3.M9 **60; it has 12 factors.**
Multiply 4×3×5; exponents 2,1,1 give (2+1)(1+1)(1+1).

3.M10 **20. HCF(12,20)=4 and LCM(12,20)=60.**
For two positive integers, HCF × LCM equals their product.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals

INVESTIGATE FURTHER

4.B6 **In simplest form, a fraction terminates exactly when the denominator has only the prime factors 2 and 5. 7/8 = 0.875 terminates; 9/12 = 3/4 = 0.75 terminates.**
A terminating decimal can be written over a power of ten, whose only prime factors are 2 and 5. Testing 7/8 and 9/12 confirms the rule, and the prime-factor argument explains why it must hold.

4.B7 **Counterexamples: 1/2 × 1/3 = 1/6 < 1/2 and 5 × 2/3 = 10/3 ≈ 3.33 < 5. Correct version: multiplying by a number greater than 1 makes a positive number bigger; multiplying by a proper fraction makes it smaller.**
Two counterexamples are enough to reject a universal claim, and the corrected statement separates the cases that the original claim confused.

EXPLAIN, JUSTIFY AND CORRECT

4.C1 **2/5 = 0.4 is less than 1/2 = 0.5, but adding a positive amount to 1/2 must give more than 1/2.**
An inequality check is often faster than a calculation, and it shows understanding of what addition does rather than how to perform it.

4.C2 **9 ÷ 6 = 3/2, so the oil needed is 2/3 × 3/2 = 1 cup. The amount is not doubled because 9 people is 1.5 times 6 people, not 2 times.**
A clear explanation names the scale factor first, then applies it to each ingredient, and states why an intuitive guess is wrong.

4.C3 **3/8 = 0.375 inch and 0.4 inch = 4/10 = 2/5 inch, so 0.4 inch is larger by 0.025 inch. A work card should give the decimal to match the measuring tool.**
Converting to a common form makes the comparison exact; the recommendation then depends on the tool, which is a communication decision.

4.C4 **0.5 names half of a stated whole. Half of a small pizza is a different amount from half of a large one, so a fraction is only meaningful once the whole is identified.**
The explanation separates the number from the quantity it describes, which is the central idea behind equivalent fractions.

APPLY IN A NEW CONTEXT

4.D5 **Total cost 360 + 300 = 660 XOF; cost per kg = 660 ÷ 1.25 = 528 XOF per kg.**
3/4 × 480 = 360 and 1/2 × 600 = 300; the mixture weighs 1.25 kg, so dividing the total cost by the total weight gives the unit cost.

4.D6 **1,600 × (3/4 − 2/5) = 1,600 × 7/20 = 560 L, which is 7/20 of the tank.**
The added amount is the difference of the two fractions of the same whole: 3/4 − 2/5 = 15/20 − 8/20 = 7/20.

4.D7 **0.75 km = 3/4 km is further by 1/8 km = 125 m. A training log should record both distances in the same unit.**
5/8 = 0.625 and 0.75 = 0.750, so the difference is 0.125 km; converting to metres keeps the log readable.

READY CHECK

4.R1 **3/4.**
Divide both parts by 2.

4.R2 **4 and 56.**
Use known multiplication facts.

4.R3 **0.3.**
3 tenths is 0.30, greater than 0.25.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

READY CHECK

4.R4 **9.**
18÷2 = 9.

4.R5 **7/7.**
Seven sevenths make one whole.

PRACTICE 4.1 · EQUIVALENCE

4.1.1 **8/12.**
Multiply numerator and denominator by 4.

4.1.2 **3/4, 5/7 and 3/4.**
Divide by HCFs 6, 7 and 12.

4.1.3 **No. 3/5 = 0.6, while 5/7 ≈ 0.714. Multiply or divide both parts by the same nonzero number.**
Adding equal amounts does not preserve a ratio.

4.1.4 **Examples: 2/3, 8/12 and 12/18.**
Simplify by 2, or scale both parts.

4.1.5 **3/4.**
Simplify 12/16 by 4.

4.1.6 **24.**
40÷5 = 8, so multiply 3 by 8.

PRACTICE 4.2 · COMPARE

4.2.1 **3/7 < 3/5.**
Equal numerators: fifths are larger pieces than sevenths.

4.2.2 **3/4 < 5/6.**
9/12 < 10/12.

4.2.3 **3/5, 2/3, 7/10.**
Use thirtieths: 18/30, 20/30, 21/30.

4.2.4 **8/9 is closer: its gap is 1/9, smaller than 1/8.**
Compare the missing parts, not just the numerators.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

PRACTICE 4.2 · COMPARE

4.2.5 **Ben: 2/3 = 16/24, while 5/8 = 15/24.**
The books are the same size, so the fractions can be compared.

4.2.6 **5/12.**
1/3 = 4/12 and 1/2 = 6/12.

PRACTICE 4.3 · ADD AND SUBTRACT

4.3.1 **4/8 = 1/2; 5/9.**
Like denominators: combine numerators.

4.3.2 **19/12 = 1 7/12.**
Use twelfths: 9/12 + 10/12.

4.3.3 **31/12 = 2 7/12.**
16/3 − 11/4 = 64/12 − 33/12.

4.3.4 **2/5 is less than 1/2, but adding a positive amount to 1/2 must increase it. The correct sum is 5/6.**
Use a benchmark to detect an unreasonable result.

4.3.5 **13/12 litres = 1 1/12 litres.**
Add using twelfths.

4.3.6 **7/10.**
11/10 − 4/10 = 7/10.

PRACTICE 4.4 · MULTIPLY AND DIVIDE

4.4.1 **2/5 and 1/2.**
Multiply across, then simplify.

4.4.2 **6 and 1/2.**
Multiply by the reciprocal of the divisor.

4.4.3 **3/8 cup.**
Half of 3/4 is 1/2 × 3/4.

4.4.4 **6 cups.**
3/2 ÷ 1/4 = 3/2 × 4.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

PRACTICE 4.4 · MULTIPLY AND DIVIDE

4.4.5 **There are two halves in each whole, so six wholes contain 12 halves.**
Division counts groups of size 1/2.

4.4.6 **11/10 = 1 1/10.**
First add to get 11/12, then multiply by 6/5.

PRACTICE 4.5 · DECIMALS

4.5.1 **7 hundredths = 0.07.**
The first digit after the point is tenths; the second is hundredths.

4.5.2 **0.305, 0.35, 0.4, 0.45.**
Compare place values; 0.35 = 0.350 and 0.4 = 0.400.

4.5.3 **4.18 and 2.64.**
Align decimal points; write 5 as 5.00.

4.5.4 **0.24 and 8.**
6×4 = 24 with two decimal places; 24 tenths ÷ 3 tenths.

4.5.5 **1.00. The zeros show accuracy to two decimal places.**
The 5 causes a carry through both 9s.

4.5.6 **4.65 ≤ length < 4.75 cm.**
Values at 4.65 round up; 4.75 rounds to 4.8.

PRACTICE 4.6 · CONVERT AND CONNECT

4.6.1 **0.375, 0.35 and 0.4.**
Divide, or scale to thousandths/hundredths/tenths.

4.6.2 **6/25 and 9/8 (= 1 1/8).**
24/100 and 1125/1000, then simplify.

4.6.3 **5/6 repeats. Its reduced denominator contains a factor 3.**
Terminating denominators contain only 2s and 5s.

4.6.4 **0.4 inch, by 0.025 inch.**
0.375 < 0.400.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

PRACTICE 4.6 · CONVERT AND CONNECT

4.6.5 **180 mL and 360 mL.**
Multiply 240 by 3/4 and by 3/2.

4.6.6 **0.85 = 17/20.**
Use 3/4 + 2/5 − 3/10 with denominator 20.

EXPLORATION · WHICH FRACTIONS STOP?

4.B1 **0.375 and 0.35 terminate; 0.8333… and 0.4666… repeat.**
Divide numerator by denominator and mark repetition.

4.B2 **$8=2^3$; $20=2^2\times5$; $6=2\times3$; $15=3\times5$.**
Prime factors explain the stopping behaviour.

4.B3 **It terminates exactly when its denominator has no prime factors other than 2 and 5.**
Powers of 10 contain only 2 and 5.

4.B4 **13/40 terminates (0.325); 13/30 repeats (0.4333…).**
$40=2^3\times5$; $30=2\times3\times5$.

4.B5 **A non-reduced denominator may hide a cancelling factor. 3/6 = 1/2 terminates even though 6 contains 3.**
The rule concerns the reduced denominator.

CHECKPOINT

4.CK1 **2/3.**
Divide by HCF 12.

4.CK2 **5/6.**
Use sixths.

4.CK3 **3/2 = 1 1/2.**
Multiply by 9/4.

4.CK4 **0.625.**
5÷8 = 0.625.

APPLICATION · THE SCALING KITCHEN

4.D1 **Milk 9/8 = 1 1/8 cups; rice 9/4 = 2 1/4 cups; oil 1 cup.**
Multiply every amount by 9/6 = 3/2.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

APPLICATION · THE SCALING KITCHEN

4.D2 **Milk 270 mL; oil 240 mL.**
Multiply cups by 240.

4.D3 **Each ingredient must be multiplied by the same scale factor 3/2. Adding 3/2 changes the amount by a fixed quantity, not proportionally.**
Equivalent ratios preserve the recipe.

4.D4 **Not enough. The recipe needs 1/2 cup = 120 mL, so 100 mL is 20 mL short.**
Scale first, then convert.

ASSESSMENT PRACTICE · THE MEASUREMENT EXPLAINER

4.S1 **0.375 inch < 0.4 inch; difference 0.025 inch.**
Use a common decimal representation.

4.S2 **11/12. A 3-by-4 rectangle has 8 shaded columns for 2/3 and 3 more rows for 1/4, covering 11 of 12 cells.**
A common denominator makes parts equal.

4.S3 **Sample warning: “Do not add denominators.” Check: 1/2 + 1/4 = 3/4, not 2/6.**
The answer should expose the error and show a correct alternative.

MIXED PRACTICE 4

4.M1 **3/4.**
Divide by 15.

4.M2 **5/9 < 4/7.**
35/63 < 36/63.

4.M3 **13/4 = 3 1/4.**
10/4 + 3/4.

4.M4 **9/20.**
14/20 − 5/20.

4.M5 **2/3 and 6/5 = 1 1/5.**
Multiply; for division use the reciprocal.

4.M6 **1 and 6.**
Use place value or equivalent fractions.

ANSWERS AND REASONING

Chapter 4 · Fractions and decimals (continued)

MIXED PRACTICE 4

4.M7 **2.68.**
The third decimal digit is 5.

4.M8 **1/3 = 0.3333…; stopping after three digits loses the remaining amount.**
0.333 = 333/1000, which is slightly smaller.

4.M9 **24.**
18 ÷ 3/4 = 24.

4.M10 **1.**
First 5/6, then multiply by 6/5.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications

INVESTIGATE FURTHER

5.B6 **No. A price of 200 becomes 240 and then 192, so it ends 4% lower; the multipliers are 1.2 × 0.8 = 0.96.**
The decrease is applied to the larger amount, so the two changes do not cancel. The product 0.96 explains the 4% loss for every starting price.

5.B7 **Answers depend on the chosen data; for example reductions of 50, 20, 15, 10 and 5 give an average of 20%, far below the headline 50%.**
An average reduction summarises the whole offer, while "up to" describes only the best case. Comparing the two measures shows why the claim can mislead.

EXPLAIN, JUSTIFY AND CORRECT

5.C1 **The multipliers are 1.25 and 0.75, and 1.25 × 0.75 = 0.9375, so the final price is 93.75% of the original — a 6.25% loss.**
Naming the base of each change is the key: the decrease applies to the increased price, not to the original one.

5.C2 **Both are 12, so the claim happens to be true here, but the reason is that a% of b = b% of a because both equal ab/100.**
A single example cannot justify a general rule; writing both as fractions of 100 shows why the symmetry always holds.

5.C3 **Gross profit is revenue minus the cost of goods: 480 − 264 = 216. Net profit also subtracts fixed costs: 216 − 90 = 126.**
The explanation names which costs are included in each figure, so a reader can tell which one a headline percentage refers to.

5.C4 **For example: original 500 XOF, 20% off = 100 XOF, final price 400 XOF. A reader checks by computing 20% of 500 and subtracting, or by confirming 400 is 80% of 500.**
A checkable card states the base, the change and the result, and gives the reader a second route to the same answer.

APPLY IN A NEW CONTEXT

5.D5 **Revenue 42,000; goods cost 25,200; gross profit 16,800; net profit 7,800; net profit is 18.57% of revenue.**
Goods cost is 0.6 × 42,000 = 25,200, so gross profit is 16,800; subtracting fixed costs gives 7,800, and 7,800 ÷ 42,000 ≈ 0.1857.

5.D6 **Adding tax gives 80,000 × 1.15 = 92,000 XOF; the tax-inclusive display shows the same final price directly, which is clearer.**
Both displays end at the same amount when the base is stated, but the tax-inclusive display avoids asking the customer to compute a percentage.

5.D7 **Books 75,000; repairs 112,500; transport 62,500. Total 250,000.**
30% + 45% = 75%, so transport is 25% of 250,000 = 62,500; the three amounts sum to the total, which is the check.

READY CHECK

5.R1 **9.**
Divide by 10.

5.R2 **0.5 and 50%.**
Half of 100 is 50.

5.R3 **10.**
One fifth of 50.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications (continued)

READY CHECK

5.R4 **48.**
Subtract the reduction.

5.R5 **3/4.**
Divide by 25.

PRACTICE 5.1 · PER CENT

5.1.1 **0.37, 0.05 and 1.25.**
Divide by 100.

5.1.2 **60%, 35% and 12.5%.**
Multiply each fraction by 100%.

5.1.3 **6/25 and 3/2.**
24/100 and 150/100, then simplify.

5.1.4 **0.5% = 0.005. 0.5 = 50%.**
Per cent means divide by 100.

5.1.5 **45%.**
18/40 × 100 = 45.

5.1.6 **Yes. If the reference is 50, then 60 is 120% of it.**
Percentages above 100 represent more than the reference whole.

PRACTICE 5.2 · PERCENTAGE OF

5.2.1 **24, 60 and 36.**
Use a tenth, a quarter, and a tenth plus half a tenth.

5.2.2 **40 and 7.5.**
12.5% = 1/8; 1% means divide by 100.

5.2.3 **28. Use 0.35×80, or 25% (20) + 10% (8).**
Both methods use the same reference whole.

5.2.4 **Reduction £12; sale price £48.**
20% of 60 is 12; subtract from 60.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications (continued)

PRACTICE 5.2 · PERCENTAGE OF

5.2.5 **Yes, both are 15. Multiplication can be done in either order.**
30/100×50 = 50/100×30.

5.2.6 **180.**
27 ÷ 0.15 = 180.

PRACTICE 5.3 · CHANGE

5.3.1 **92 and 68.**
Use multipliers 1.15 and 0.85.

5.3.2 **20%.**
Change 10 divided by original 50, times 100.

5.3.3 **£180 versus £175; the single 30% offer is £5 cheaper.**
250×0.8×0.9 = 180; 250×0.7 = 175.

5.3.4 **£96. The 20% decrease is £24, larger than the original £20 increase.**
The base changed from 100 to 120.

5.3.5 **No. 0.9×0.8 = 0.8×0.9 = 0.72.**
Multiplication is commutative.

5.3.6 **25%.**
From 80 back to 100, the increase is 20/80 = 25%.

PRACTICE 5.4 · MONEY DECISIONS

5.4.1 **£8.10 and £3.00.**
Use multipliers 0.9 and 0.75.

5.4.2 **£44.**
Tax £4; total 40×1.1.

5.4.3 **Gross £105; net £42.**
Gross = 395−290; net = 105−63.

5.4.4 **£3 profit: 50% of cost, 33 1/3% of selling price.**
The denominator changes the percentage.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications (continued)

PRACTICE 5.4 · MONEY DECISIONS

5.4.5 **23 jars cost £138; 30 at £5.40 cost £162, £24 more.**
The lower unit price requires buying 7 extra jars.

5.4.6 **£36.96.**
New fixed costs 63×1.08 = £68.04; net = 395−290−68.04.

EXPLORATION · DO TWO DISCOUNTS ADD UP?

5.B1 **A: £180, £28.80, £57.60. B: £175, £28, £56.**
Use 0.8×0.9 = 0.72 and 0.7.

5.B2 **28%; order does not matter because 0.8×0.9 = 0.9×0.8.**
The product of multipliers is unchanged by order.

5.B3 **40%.**
0.75×0.8 = 0.60 remains.

5.B4 **Yes: 4% then 25% (or 25% then 4%), since 0.96×0.75 = 0.72. The pair 10% and 20% also works but is excluded by the question.**
Solve (100−a)(100−b)=7200; factor pairs in range include 96×75 and 80×90.

5.B5 **Remaining multiplier (1−a/100)(1−b/100); combined discount is 1 minus this product. The second discount uses the reduced price.**
Multiplying multipliers tracks the changing base.

CHECKPOINT

5.CK1 **30.**
10% is 20; 5% is 10.

5.CK2 **56.**
50×1.12.

5.CK3 **25%.**
20/80×100.

5.CK4 **0.72, equivalent to 28% off.**
0.9×0.8.

APPLICATION · THE COMMUNITY MARKET STALL

5.D1 **Honey £8.10, profit £2.10; soap £3.00, profit £0.50; jam £5.00, profit £1.80.**
Apply each discount, then subtract cost.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications (continued)

APPLICATION · THE COMMUNITY MARKET STALL

5.D2 **Revenue £395; goods £290; gross £105; net £42.**
Revenue 162+108+125; goods 120+90+80; fixed 63.

5.D3 **23 honey, 42 soap, 29 jam; cost £335.80.**
Ceil(20×1.15)=23, ceil(36×1.15)=42, ceil(25×1.15)=29.

5.D4 **Not on cost alone: £162 versus £138, so £24 more is tied up in stock. Accept only if extra demand or storage justifies it.**
Compare total outlay and risk, not just unit price.

ASSESSMENT PRACTICE · THE FAIR-PRICE CARD

5.S1 **£60 versus £60.80; 25% off is 80p cheaper.**
80×0.75 = 60; 80×0.8×0.95 = 60.8.

5.S2 **24%.**
1−0.8×0.95 = 0.24.

5.S3 **Sample: “The second discount applies to the reduced price, so 20% and 5% do not combine to 25%.”**
The warning identifies the changing base.

MIXED PRACTICE 5

5.M1 **8%.**
Multiply by 100.

5.M2 **21.**
0.35×60.

5.M3 **126.**
120×1.05.

5.M4 **£35.**
50×0.7.

5.M5 **25%.**
10/40×100.

5.M6 **The final price is 25% of the original: 0.5×0.5 = 0.25.**
The second half is taken from the reduced price.

ANSWERS AND REASONING

Chapter 5 · Percentages and applications (continued)

MIXED PRACTICE 5

5.M7 **£55; 27.5%.**
55/200×100.

5.M8 **£80.**
88÷1.1.

5.M9 **0.99, a 1% decrease.**
1.1×0.9.

5.M10 **£13.50 for 3 packs.**
18÷6 = 3; 3×4.50.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations

INVESTIGATE FURTHER

6.B6 **Stage 10 is 41 using the rule 4n + 1. A different rule can fit the first four terms, for example one built from a table of differences, so the first few terms do not prove a unique rule.**
First differences are constant at 4, so a linear rule fits; but infinitely many rules agree on any finite list of terms, which is why the structure must justify the rule.

6.B7 **1 + 2 = 3, 2 + 3 = 5, 3 + 4 = 7 and 10 + 11 = 21 are all odd, and the claim is true for every pair.**
Consecutive whole numbers can be written as n and n + 1, so their sum is 2n + 1, which is odd by definition. The examples suggest the rule and the algebra justifies it.

EXPLAIN, JUSTIFY AND CORRECT

6.C1 **3n is shorthand for multiplication: with n = 5 it means 15, not 8. Writing the multiplication sign when substituting (3 × 5) prevents the error.**
The explanation names the convention and gives a practical habit, which is what makes it useful to another student.

6.C2 **Dividing first gives n + 5/3 = 20/3, which is harder to finish and easy to get wrong. The correct order is to subtract 5 (3n = 15) and then divide by 3, giving n = 5.**
Both routes can work in principle, but the second keeps the arithmetic in whole numbers; explaining the choice is part of communicating a method.

6.C3 **An expression names a quantity, such as 4x + 3. An equation states that two expressions are equal, such as 4x + 3 = 19, and can be solved.**
The distinction is about what the symbols claim: a value, or an equality between values.

6.C4 **Substitute: 4 × 10 + 4 = 44, which is true. The check is not optional because it catches slips in the manipulation.**
A check is an independent test of the answer against the original statement, not a repeat of the same steps.

APPLY IN A NEW CONTEXT

6.D5 **Slabs = 4n ÷ 0.5 + 4 = 8n + 4; for n = 6 the gardener needs 52 slabs.**
A square of side n has a perimeter of 4n metres, and one slab every 0.5 m gives 8n slabs; adding the 4 corners gives 8n + 4.

6.D6 **Fare = 500 + 180k XOF; for 12 km the fare is 500 + 2,160 = 2,660 XOF. The fixed part is 500 XOF and the variable part is 180 XOF per km.**
A formula separates the constant term from the rate, which makes it easy to check and to adapt to a new fare structure.

6.D7 **3,000 + 25m = 4,500 gives 25m = 1,500, so m = 60 minutes.**
Setting the two expressions equal models the break-even point; checking at m = 60 gives 4,500 for both plans.

READY CHECK

6.R1 **17.**
Multiply before adding.

6.R2 **14.**
Add 3.

6.R3 **28 and 7.**
Inverse operations.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations (continued)

READY CHECK

6.R4 **5x.**
Add coefficients of like terms.

6.R5 **8.**
15−7.

PRACTICE 6.1 · PATTERNS

6.1.1 **16, 19; add 3.**
The difference is constant.

6.1.2 **3n + 1; 61.**
3×20 + 1.

6.1.3 **The first square needs 4; each extra square adds 3, giving 4+3(n−1)=3n+1.**
Shared sides prevent counting four new matches for every square.

6.1.4 **4n + 4; 32 slabs.**
The outer square is $(n+2)^2$; subtract n^2.

6.1.5 **No. 3n+1=50 gives n=49/3, not a whole number.**
The number of squares must be a positive integer.

6.1.6 **33 squares, using 100 matches exactly.**
$3n+1 \le 100$; $n \le 33$.

PRACTICE 6.2 · SUBSTITUTE

6.2.1 **17 and 35.**
Substitute, then multiply before adding.

6.2.2 **16.**
10+6.

6.2.3 **$x^2=4$; $-x^2=-4$.**
The first squares −2; the second applies the minus after squaring.

6.2.4 **3+2d; £19.**
3+2×8.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations (continued)

PRACTICE 6.2 · SUBSTITUTE

6.2.5 **$4n=12$, while $4+n=7$. Juxtaposition means multiplication.**
Substitution exposes the difference.

6.2.6 **$n=9$; $2\times9+7=25$.**
Subtract 7, then divide by 2.

PRACTICE 6.3 · SIMPLIFY AND EXPAND

6.3.1 **$6x-2$ and $5a+4$.**
Combine like terms only.

6.3.2 **$6x+12$ and $5y-10$.**
Multiply every inside term by the outside factor.

6.3.3 **$x+x=2x$. For $x=3$, this is 6, while $x^2=9$.**
Adding two copies differs from multiplying a number by itself.

6.3.4 **$2((2x+3)+4)=4x+14$.**
Perimeter is twice length plus width.

6.3.5 **No. Correct expansion is $3x+6$; at $x=1$, 9 is not equal to 5.**
The outside factor multiplies both terms.

6.3.6 **$6(2x+3)$.**
The HCF of 12 and 18 is 6.

PRACTICE 6.4 · SOLVE AND CHECK

6.4.1 **$n=8$ and $n=7$.**
Undo addition or multiplication.

6.4.2 **$n=5$; $3\times5+5=20$.**
Subtract 5, divide by 3, substitute.

6.4.3 **10 m.**
$4n+4=44$ gives $n=10$.

6.4.4 **$d=8$ km.**
$19-3=16$; $16\div2=8$.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations (continued)

PRACTICE 6.4 · SOLVE AND CHECK

6.4.5 **Yes. x=6 gives 2×9=18. An alternative method divides by 2 first: x+3=9, x=6.**
Both balanced methods agree.

6.4.6 **n=6; both sides equal 23.**
Subtract 2n, add 7, divide by 3.

EXPLORATION · THE MATCHSTICK CHAIN

6.B1 **4, 7, 10, 13, 16.**
Each extra square adds 3 matches.

6.B2 **Start at 4 and add 3; matches = 3n+1.**
The first square has 4; each later square adds 3.

6.B3 **61. The rule gives the answer directly from the position.**
3×20+1.

6.B4 **33 squares uses 100 exactly. 50 does not fit: (50−1)/3 is not whole.**
Solve $3n+1 \le 100$ and test divisibility by 3.

6.B5 **First square: 4 matches. Each additional square shares one side, so adds 3: 4+3(n−1)=3n+1.**
The shared-side argument applies to every stage.

CHECKPOINT

6.CK1 **31.**
3×10+1.

6.CK2 **15.**
12+3.

6.CK3 **4x+8.**
Multiply both terms by 4.

6.CK4 **n=5.**
Subtract 3, divide by 2.

APPLICATION · THE COURTYARD POND

6.D1 **4n+4; 32 and 52 slabs.**
$(n+2)^2-n^2 = 4n+4$.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations (continued)

APPLICATION · THE COURTYARD POND

6.D2 **n=10 m. Cost = 3.5(4n+4)+25 = 14n+39 pounds.**
Solve 4n+4=44; multiply slabs by 3.50 and add delivery.

6.D3 **n=11 m costs £193; n=12 costs £207, so 11 is largest.**
14n+39 ≤ 200 gives n ≤ 161/14 ≈ 11.5.

6.D4 **48 slabs take 9 h 36 min; plus 2 h gives 11 h 36 min.**
4×11+4=48; 48×12=576 min=9 h 36 min.

ASSESSMENT PRACTICE · THE PATTERN EXPLAINER

6.S1 **$(n+2)^2-n^2$ and 4n+4. The first subtracts pond area from outer area; the second counts four sides plus four corners.**
Both count each border slab once.

6.S2 **40 slabs; $11^2-9^2 = 121-81 = 40$.**
A second representation checks the rule.

6.S3 **Sample: "Do not count corners twice: four sides of n slabs plus four corner slabs gives 4n+4." Check n=3: 16 slabs.**
The warning names the error and the check verifies a case.

MIXED PRACTICE 6

6.M1 **21, 25.**
Add 4.

6.M2 **21.**
24−3.

6.M3 **2x+9.**
Combine like terms.

6.M4 **6x−10.**
Distribute the 2.

6.M5 **n=9 and n=20.**
Undo multiplication or subtraction.

6.M6 **n=7.**
Subtract 2, divide by 3.

ANSWERS AND REASONING

Chapter 6 · Patterns, expressions and equations (continued)

MIXED PRACTICE 6

6.M7 **2x and 3 are not like terms; at x=1 they give 5, while 5x gives 5, but at x=2 they give 7 and 10.**
Test a second value to expose the error.

6.M8 **13.**
(31−5)÷2.

6.M9 **n=9.**
Divide by 4, then add 2.

6.M10 **33.**
3n ≤ 99.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions

ANGLE LANGUAGE AND TURN FACTS

7.1.01 **35° acute; 90° right; 125° obtuse; 180° straight; 210° reflex.**
Acute is between 0° and 90°; obtuse between 90° and 180°; reflex between 180° and 360°. The exact values 90° and 180° have their own names.

7.1.02 **The vertex is B; the rays are BA and BC. Sketches may vary but the angle at B must be obtuse.**
The middle letter names the vertex. Many correct sketches are possible.

7.1.03 **Straight line: 180°; full turn: 360°; vertically opposite angles: equal.**
A straight line is a half-turn; a point is a full turn; crossing lines make equal opposite pairs.

7.1.04 **x = 115°.**
180° − 65° = 115°; check 65° + 115° = 180°.

7.1.05 **85°.**
360° − (90° + 110° + 75°) = 85°; the four angles sum to 360°.

7.1.06 **x = 65°, y = 115°, z = 65°.**
y is vertically opposite 115°. x and z are adjacent to 115° on straight lines, so each is 180° − 115°.

7.1.07 **x = 35°; the angles are 65°, 70° and 45°.**
(x+30)+2x+(x+10)=180, so 4x+40=180 and x=35. Check 65+70+45=180.

7.1.08 **x = 15°; each marked angle is 75°; an adjacent angle is 105°.**
4x+15=2x+45 gives 2x=30, x=15. Adjacent angles on a line sum to 180°.

7.1.09 **78°.**
Vertical is perpendicular to level ground, so the ground–post angle is 90°. The ramp splits this right angle into 12° and x, so x = 90° − 12° = 78°.

7.1.10 **60°; no, the order does not change the total.**
360° − (118° + 97° + 85°) = 60°. Addition is commutative, so the missing part of a full turn is unchanged.

7.1.11 **36°, 72°, 108°, 144°; no pair is complementary.**
The ratio has 10 parts, so one part is 360°÷10=36°. Complementary angles sum to 90°; no two of these angles do.

7.1.12 **Claim 2 is wrong: angles around a point sum to 360°, not 180°. For example 90°, 90°, 100° and 80° meet at a point.**
A half-turn is 180°; a full turn is 360°. Many counterexamples are possible.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions (continued)

ANGLE LANGUAGE AND TURN FACTS

7.1.13 **True.**
If each angle is x, then 2x=180°, so x=90°. The sketch should show two right angles on a line.

7.1.14 **2x + (x+40) + 80 = 360; 3x = 240; x = 80°. Angles are 160°, 120° and 80°.**
Collect like terms: 3x+120=360. Check 160+120+80=360.

7.1.15 **x = 50°; angles 50°, 70°, 100°, 140°; largest is 140°, which is obtuse because it is between 90° and 180°.**
x+(x+20)+2x+(3x−10)=360 gives 7x+10=360.

7.1.16 **Every correct diagram has angles 70°, 110°, 70°, 110°. Orientation and the lengths of the drawn segments may change.**
Vertically opposite angles are equal and adjacent angles are supplementary. The angle family is fixed even when the drawing is rotated.

PARALLEL LINES AND TRANSVERSALS

7.2.01 **Examples: corresponding angle 1 and angle 5; alternate angle 4 and angle 5; co-interior angle 4 and angle 6. Corresponding and alternate pairs are equal; co-interior pairs are supplementary.**
Other correct pairs include angle 2/angle 6, angle 3/angle 6 and angle 3/angle 5. The parallel marks are essential.

7.2.02 **False. For example angle 4 and angle 6 are co-interior and supplementary, not necessarily equal.**
Only corresponding and alternate pairs are equal. A counterexample is any non-right co-interior pair.

7.2.03 **a = 68°, b = 112°, c = 68°.**
a is alternate to 68°; b is co-interior with 68°; c corresponds to 68° (angle 8).

7.2.04 **angle 5 = 52°, angle 6 = 128°, angle 7 = 128°.**
angle 1 is adjacent to angle 2 on a line, so angle 1=52°; angle 5 corresponds to angle 1. angle 6 corresponds to angle 2; angle 7 is vertically opposite angle 6.

7.2.05 **Yes: 112° + 68° = 180°, so the lines are parallel by the converse of the co-interior angle fact.**
Equal alternate angles or supplementary co-interior angles can establish parallel lines. A sketch without those facts does not prove parallelism.

TRIANGLE AND EXTERIOR-ANGLE FACTS

7.2.06 **60°; acute scalene.**
180°−35°−85°=60°. All angles are acute and different, so the sides are different.

7.2.07 **x = 30°; angles 30°, 60°, 90°; right triangle.**
x+2x+3x=180°, so 6x=180°.

7.2.08 **70° each.**
Equal side ticks show an isosceles triangle. (180°−40°)÷2=70°.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions (continued)

TRIANGLE AND EXTERIOR-ANGLE FACTS

7.2.09 **67° and 65°.**
Other opposite angle =115°−48°=67°. Adjacent interior angle =180°−115°=65°.

7.2.10 **Apex 50°; exterior 115°; isosceles, so the rafters are equal.**
180°−65°−65°=50°; 180°−65°=115°. Equal base angles imply equal opposite sides.

PARALLEL LINES AND TRANSVERSALS

7.2.11 **x = 114°.**
Co-interior angles between parallel lines sum to 180°: 180°−66°.

TRIANGLE AND EXTERIOR-ANGLE FACTS

7.2.12 **Marked base angle 70°; whole left-tip angle 140°.**
Each base angle is (180°−40°)÷2=70°. The lower triangle is a mirror copy of the upper triangle, so it has the same base angle; the whole tip is 70°+70°.

7.2.13 **Top interior angle 76°; exterior angle 104°.**
180°−90°−14°=76°. The exterior angle is supplementary to 76°, so 104°.

7.2.14 **No.**
Two obtuse angles already sum to more than 180°, leaving no positive third angle. A sketch will fail to close as a triangle.

7.2.15 **The exterior angle equals the sum of the opposite interior angles: 110°=60°+50°. The adjacent interior angle is 70°.**
The exterior angle and the adjacent interior angle are supplementary. The two opposite interior angles are not added to the exterior angle to make 180°.

7.2.16 **The angle sets are the same (40°, 60°, 80°); the sizes are different. Equal angles do not force equal side lengths.**
Angle sum controls the angles, not the overall scale. The diagrams are a counterexample to “same angles means same side lengths”.

7.2.17 **Base angles 50° each; apex 80°.**
2x+(x+30)=180 gives x=50. Checks: 50+50+80=180; apex exceeds a base angle by 30°.

7.2.18 **52°, 78° and 50°; acute scalene.**
Opposite interior angles sum to 130°. Five ratio parts give 26° each: 52° and 78°. Adjacent interior angle is 180° −130°=50°.

PARALLEL LINES AND TRANSVERSALS

7.2.19 **Angles 1, 4, 5 and 8 are 73°; angles 2, 3, 6 and 7 are 107°. Four angles are acute and four are obtuse.**
At each intersection the acute angles are vertically opposite and the obtuse angles are vertically opposite; adjacent angles sum to 180°. Parallel-line facts transfer the values to the second intersection.

7.2.20 **x = 25°; marked angles 60°; co-interior partner 120°.**
Alternate angles are equal: 2x+10=3x−15, so x=25. Co-interior angles sum to 180°.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions (continued)

POLYGON ANGLE SUMS

7.3.01 **Pentagon 5; hexagon 6; octagon 8. A regular polygon has all sides equal and all interior angles equal.**
Polygon names count sides. Both equal sides and equal angles are required for regularity.

7.3.02 **Both have six sides and interior angle sum 720°. A regular hexagon has equal sides and equal 120° angles; an irregular hexagon need not.**
The angle-sum rule depends on the number of sides, not regularity. Sketches may vary.

7.3.03 **Quadrilateral: 2 triangles, 360°; pentagon: 3, 540°; hexagon: 4, 720°.**
An n-gon splits into n−2 triangles from one vertex; multiply by 180°.

7.3.04 **Heptagon 900°; nonagon 1260°; decagon 1440°.**
(7−2)×180=900; (9−2)×180=1260; (10−2)×180=1440.

7.3.05 **x = 80°.**
360°−(110°+75°+95°)=80°.

7.3.06 **8 sides; interior 135°.**
360°÷45°=8; interior and exterior are supplementary, so 180°−45°=135°.

7.3.07 **Regular octagon. No: 360°÷135° is not a whole number, so whole tiles cannot fill a full turn exactly.**
Exterior angle is 45°, giving 8 sides. Two tiles leave 90°; three overlap. This is an angle argument, not a claim about every possible tiling with other shapes.

7.3.08 **Total turn is 360° in each row because the walker returns to the starting direction after one full circuit.**
The number of turns changes, but one complete circuit is always one full turn.

TRIANGLE INEQUALITY

7.3.09 **3,4,8: no (7<8); 5,5,9: yes (10>9); 6,7,10: yes (13>10).**
A triangle needs the two shorter sides together to exceed the longest. Equality would give a straight line.

7.3.10 **Rows: yes, no, no (borderline), yes. Student row may vary; a borderline row has shorter sum equal to longest.**
Compare a+b with c when c is longest. Strictly greater gives a triangle; equal gives zero area; less gives a gap.

7.3.11 **Yes, it already is a triangle because 7+9>15. No change is needed.**
The question checks whether you test before changing. The shortest-sum test is 16>15.

7.3.12 **Impossible: the two shorter sides give 4+4<9. The arcs do not meet; there is a 1 cm gap.**
The relevant test compares the two shorter sides with the longest. A correct sketch shows the failure to close.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions (continued)

TRIANGLE INEQUALITY

7.3.13 **5+8=13 gives a degenerate straight line. With 12 cm, 5+8>12, so a triangle is possible.**
Equality places the apex on the base and gives zero area. Reducing the base lets the arcs meet above it.

7.3.14 **5, 6, 7, ..., 17 cm.**
The third side x must satisfy 11−7 < x < 11+7, so 4<x<18. Endpoints are excluded.

7.3.15 **5 m.**
2x>8, so x>4. At x=4 the rafters lie flat along the base.

7.3.16 **If the base is c, arcs of radii a and b meet above the base only when a+b>c. Equality meets on the base; less leaves a gap.**
The arc intersection is the possible apex. No intersection above the base means no non-degenerate triangle.

7.3.17 **12 sides.**
Interior + exterior =180°. If interior=5e, then 6e=180°, e=30°, n=360°÷30°=12.

BISECTORS AND LOCI

7.4.01 **Perpendicular bisector for two points; angle bisector for the angle between two roads.**
A perpendicular bisector compares distances to endpoints. An angle bisector compares perpendicular distances to the two arms.

7.4.02 **They record equal distances from the named centres. Changing the radius destroys the equal-distance evidence.**
The construction is justified by the defining property, not by appearance.

7.4.03 **ii, iv, i, iii; radius > 4 cm.**
The arcs must meet at two points. The line through those points is perpendicular to AB and passes through its midpoint.

7.4.04 **34° each.**
The same radius from the two arm-intersection points locates a point on the bisector; the two half-angles are equal.

7.4.05 **x = 12°.**
Each half is 42°, so 3x+6=42 and x=12. Check 3(12)+6=42.

7.4.06 **The perpendicular bisector of AB. Check that the point is the same distance from A and B.**
Every point on the perpendicular bisector is equidistant from the endpoints.

7.4.07 **Radius must be greater than 3 cm. A 2 cm radius is too small; 4 cm works.**
Equal-radius arcs meet at two points only when the radius exceeds half the segment length.

ANSWERS AND REASONING

Chapter 7 · Lines, angles, triangles and constructions (continued)

BISECTORS AND LOCI

7.4.08 **No. Ask for the equal-radius arcs or re-construct them; check midpoint, right angle and equal distances to A and B.**
A drawing is not proof. The defining properties and construction evidence justify the result.

CONSTRUCTING TRIANGLES

7.4.09 **Each half is 3.6 cm; the bisector meets AB at 90°.**
Use one radius greater than 3.6 cm from both endpoints. Keep arcs visible.

7.4.10 **32° each.**
Draw an arc from the vertex, then equal-radius arcs from the two arm intersections. Join the vertex to their intersection.

7.4.11 **A correct SSS construction; for example base 6 cm, arcs 4 cm and 5 cm from the endpoints.**
Check the triangle inequality first: 4+5>6. The arc intersection fixes the third vertex.

7.4.12 **All sides 5 cm; all angles 60°.**
Equal sides give equal opposite angles; 180°÷3=60°.

7.4.13 **The apex is equidistant from the base endpoints, so it lies on the perpendicular bisector.**
Use radius 6 cm from both base endpoints. The perpendicular bisector is the locus of points equidistant from the endpoints.

7.4.14 **The arcs do not meet because 3+4<8.**
No apex exists; the two shorter sides cannot reach across the base.

7.4.15 **Model base 12 cm; real rafters 1.8 m.**
2.4 m=240 cm; 240÷20=12 cm. Model sides 9 cm represent 9×20=180 cm=1.8 m.

7.4.16 **The angle bisector; 28° to each road. For two points, use the perpendicular bisector of the segment joining them.**
Distance to lines is measured perpendicular to the lines; the angle bisector is the locus. Distance to points uses the perpendicular bisector.

ANSWERS AND REASONING

Chapter 8 · Sets and classification

INVESTIGATE FURTHER

8.B6 **Smallest overlap 3 (because 46 + 37 = 83 exceeds 80 by 3); largest overlap 37.**
The overlap cannot be smaller than the excess of the two totals over the universal set, and it cannot exceed the smaller group. Both endpoints are attainable by arranging the groups accordingly.

8.B7 **The claim is true for every choice of sets.**
Every element of A is in B, and every element of B is in C, so every element of A is in C. The examples support it and the chain of membership proves it.

EXPLAIN, JUSTIFY AND CORRECT

8.C1 **$a \in \{a, b\}$ says the element a belongs to the set. $\{a\} \subset \{a, b\}$ says every element of {a} — just a — also belongs to {a, b}.**
The first symbol relates an element to a set; the second relates one set to another. Naming the two kinds of object is the key.

8.C2 **Place the overlap, subtract it from each total to fill the "only" regions, then subtract the union from the universal set to find the outside region. The overlap is placed first because it belongs to both totals.**
The order prevents double counting, which is the most common error in survey problems.

8.C3 **The total number of members is needed, because a percentage is a part of a stated whole; with 80 members the figure is 46/80 = 57.5%.**
A count without its denominator cannot be compared with another count, which is why survey reports must state the population.

8.C4 **Both equal one half, and a rational number is any number that can be written as a fraction of two integers; the form it is written in does not change its classification.**
Classification depends on properties, not appearance, so equivalent representations stay in the same sets.

APPLY IN A NEW CONTEXT

8.D5 **18 + 15 − 6 = 27 have a preference, so 5 have none; 5/32 = 15.625%.**
The union counts each student once, so the overlap is subtracted; the remainder is measured against the class total of 32.

8.D6 **Fiction only 40, non-fiction only 25, both 30, neither 25. Exactly one type: 65 of 120 = 54.17%.**
The overlap is subtracted from each total, and the union of 95 leaves 25 borrowers in neither region.

8.D7 **The overlap must be between 3 and 41; to leave at least 20 using neither, the overlap must be at least 52 + 41 − 70 = 23.**
Neither = 90 − (52 + 41 − overlap) ≥ 20, so overlap ≥ 23. Any value from 23 to 41 satisfies the target, so a recommendation should also consider usage.

READY CHECK

8.R1 **2, 4, 6, 8.**
Apply the rule.

8.R2 **11 and 5.**
Subtract the overlap and then the union.

8.R3 **8.**
2×2×2.

ANSWERS AND REASONING

Chapter 8 · Sets and classification (continued)

READY CHECK

8.R4 **−3, 0, 2.**
Negatives lie left of zero.

8.R5 **7.5%.**
6÷80×100.

PRACTICE 8.1 · SETS AND SUBSETS

8.1.1 **{2, 4, 6, 8, 10}.**
Apply the membership rule once to each candidate.

8.1.2 **8 and 16.**
Use 2^n.

8.1.3 **Yes. Repetition and order do not matter; both contain exactly 1, 2 and 3.**
A set records membership, not listing order.

8.1.4 **True, true, true, false (unless {a} itself were listed as an element).**
Distinguish elements from subsets.

8.1.5 **32.**
2^5.

8.1.6 **16.**
Fix 1 as included; choose freely among the other four.

PRACTICE 8.2 · VENN OPERATIONS

8.2.1 **{1,2,3,4,5} and {3,4}.**
Union collects all; intersection keeps shared elements.

8.2.2 **F only 11; both 6; C only 8; neither 5.**
Subtract the overlap from each total; subtract the union from 30.

8.2.3 **The intersection is counted twice, so subtract it once.**
Use inclusion–exclusion.

8.2.4 **7.**
Union 22+18−7=33; 40−33.

ANSWERS AND REASONING

Chapter 8 · Sets and classification (continued)

PRACTICE 8.2 · VENN OPERATIONS

8.2.5 **A and B.**
Every element of A is already in B.

8.2.6 **4.**
12+9−17.

PRACTICE 8.3 · NUMBER FAMILIES

8.3.1 **−4 integer; 0 whole; 5 natural; 1/2 rational. All are real.**
Use the nesting of number sets.

8.3.2 **√2 and π.**
0.25=1/4 and 1/3 are ratios of integers.

8.3.3 **Yes; no. For example −3=−3/1 is rational, but 1/2 is not an integer.**
Integers are a subset of rationals.

8.3.4 **−2 < 0 < 1/2 < 2.**
Negative values lie left of zero.

8.3.5 **It equals 1/3, a ratio of integers.**
Repeating decimals are rational.

8.3.6 **Sample: 3/2 and √2.**
1.5 lies between; √2≈1.414 also lies between.

PRACTICE 8.4 · SURVEY COUNTS

8.4.1 **26, 18, 19, 17.**
Subtract overlap, then subtract union from 80.

8.4.2 **78.75%.**
63/80×100.

8.4.3 **Yes for all gym users: 37/80=46.25%. No for gym only: 19/80=23.75%.**
The category definition changes the claim.

8.4.4 **$26×4 + 19×3 + 18×6 = $269.**
Use the exactly-one and both regions.

ANSWERS AND REASONING

Chapter 8 · Sets and classification (continued)

PRACTICE 8.4 · SURVEY COUNTS

8.4.5 **The same count can mean different proportions in different populations.**
Percentages and “neither” depend on the total and the denominator.

8.4.6 **Both 20; vanilla only 14.**
Union 43; overlap 34+29−43.

EXPLORATION · HOW MUCH OVERLAP IS POSSIBLE?

8.B1 **11, 6, 8, 5; at least one 25.**
Subtract the overlap and use the total 30.

8.B2 **31−x and x−1.**
17+14−x; 30−(31−x).

8.B3 **x=1. If x=0, the union would be 31, more than the 30 students.**
The union cannot exceed the universal set.

8.B4 **x=14. All concert helpers also help at the fair; C is a subset of F.**
The overlap cannot exceed either set total.

8.B5 **Yes, 25 helpers. Across x=1 to 14, the union ranges from 30 down to 17, so for x≥12 there would be fewer than 20.**
Union = 31−x; solve 31−x≥20.

CHECKPOINT

8.CK1 **16.**
2^4.

8.CK2 **{2,3}.**
Keep shared elements.

8.CK3 **15.**
10+8−3.

8.CK4 **Rational: 3/5.**
It is a ratio of integers.

APPLICATION · THE CENTRE AUDIT

8.D1 **26, 18, 19, 17; sum 80.**
Pool only 44−18; gym only 37−18; neither 80−63.

ANSWERS AND REASONING

Chapter 8 · Sets and classification (continued)

APPLICATION · THE CENTRE AUDIT

8.D2 **$269; 78.75%.**
26×4+19×3+18×6; 63/80×100.

8.D3 **All gym users: true (46.25%). Gym only: false (23.75%).**
State whether the overlap is included.

8.D4 **Sample: members save $1; the centre may receive less per dual user but gain simplicity and uptake. Recommend tracking visits and uptake before changing price.**
A fairness judgment needs a stated criterion and data.

ASSESSMENT PRACTICE · THE CLASSIFICATION POSTER

8.S1 **Sample: $2 \in \{1,2,3\}$; $\{1,2\} \subseteq \{1,2,3\}$.**
The first relates an element; the second relates sets.

8.S2 **Integer, whole, rational, irrational; all real. $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.**
Use the most specific listed family and wider membership.

8.S3 **Sample: n(A)=10, n(B)=8, overlap 3. Union? 15, not 18.**
Subtract the overlap once.

MIXED PRACTICE 8

8.M1 **32.**
2^5.

8.M2 **{2,4,6,8}; {4,6}.**
Collect all; keep shared.

8.M3 **9.**
50−(28+22−9).

8.M4 **Yes; repeating decimals are ratios of integers (12/99 = 4/33).**
A repeating block can be converted to a fraction.

8.M5 **√9=3 is natural/rational; √10 is irrational.**
Simplify perfect squares first.

8.M6 **7 and 12.**
A is entirely inside B.

ANSWERS AND REASONING

Chapter 8 · Sets and classification (continued)

MIXED PRACTICE 8

8.M7 **13.**
Union 50; overlap 35+28−50.

8.M8 **5 to 15.**
max(0,20+15−30) ≤ x ≤ min(20,15).

8.M9 **Sets ignore repetition; both have exactly elements 1 and 2.**
Membership defines a set.

8.M10 **32.**
Fix it in; choose among the other five: 2^5.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume

INVESTIGATE FURTHER

9.B6 **The 6 cm by 6 cm square has the greatest area, 36 cm^2; the 1 cm by 11 cm rectangle has area 11 cm^2.**
For a fixed perimeter the area is greatest when the sides are as close as possible, because the product $l \times w$ is largest near $l = w$. Testing the whole-number pairs confirms it.

9.B7 **1×1×24 has surface area 98 cm^2; 1×2×12 has 76 cm^2; 2×2×6 has 56 cm^2; 2×3×4 has 52 cm^2, which is the smallest.**
Surface area = $2(lw + lh + wh)$. Holding the volume fixed, the most compact box has the smallest surface area because the terms lw, lh and wh are most nearly equal.

EXPLAIN, JUSTIFY AND CORRECT

9.C1 **Perimeter = $2(4 + 3) = 14$ cm; area = $4 \times 3 = 12$ cm^2. The perimeter is a length around the boundary and the area is a surface measure.**
Naming the units makes the distinction visible: cm for the boundary and cm^2 for the surface.

9.C2 **Area is measured in square units, so the answer should be 12 cm^2; "12 cm" describes a length.**
Units are part of the answer, and using the wrong unit changes what the number claims to measure.

9.C3 **The net shows all six faces as rectangles, so the surface area is the sum of their areas. With no top, one face is omitted, so the surface area is smaller by the area of that face.**
The explanation links the flat pattern to the solid, and identifies exactly which face is missing in the open-topped case.

9.C4 **1 m = 100 cm, so 1 m^3 = 100 × 100 × 100 = 1,000,000 cm^3, and 1 cm^3 = 1 mL, so 1 m^3 = 1,000 L.**
The calculation shows the 100× scale factor for each dimension, which is why volume conversions use 1,000,000 rather than 100.

APPLY IN A NEW CONTEXT

9.D6 **Area 96 m^2, so about 384 tiles; with 10% extra about 423 tiles, costing about 761,400 XOF.**
96 ÷ 0.25 = 384 tiles; adding 10% gives 384 × 1.1 = 422.4, so 423 tiles are bought, and 423 × 1,800 = 761,400 XOF.

9.D7 **Volume 0.8 m^3; surface area of the four sides = $2(2 \times 0.4) + 2(1 \times 0.4) = 2.4$ m^2.**
Volume is length × width × height. The four side areas are 0.8, 0.8, 0.4 and 0.4 m^2, which sum to 2.4 m^2; the base and the open top are excluded.

9.D8 **Volume ≈ $3.14 \times 0.25 \times 1.2 = 0.942$ m^3, which is about 942 L.**
Cross-section area is $\pi r^2 \approx 0.785$ m^2 and volume = area × height. Using $\pi \approx 3.14$ makes the result accurate to about 0.05%.

READY CHECK

9.R1 **28 and 22.**
Area and perimeter of a 7 by 4 rectangle.

9.R2 **20.**
½×40.

9.R3 **36.**
6×6.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume (continued)

READY CHECK

9.R4 **2,000 mL.**
1 L=1,000 mL.

9.R5 **0.72.**
Multiply as whole numbers, then place two decimal digits.

PRACTICE 9.1 · AREA AND PERIMETER

9.1.1 **28 cm^2; 22 cm.**
Area 7×4; perimeter 2(7+4).

9.1.2 **20 cm^2.**
½×8×5.

9.1.3 **32 cm^2.**
½(6+10)×4.

9.1.4 **The 5 cm side is slant, not perpendicular height. Height is 4 cm; area 12 cm^2.**
Half the base is 3; 3–4–5 right triangle.

9.1.5 **216 m^2; $1,620.**
18×12; multiply by 7.50.

9.1.6 **6 cm by 8 cm; perimeter 28 cm.**
Compare factor pairs of 48; the closest pair minimises perimeter.

PRACTICE 9.2 · CIRCLES AND COMPOSITES

9.2.1 **31.4 cm; 78.5 cm^2.**
$2\pi r$ and πr^2.

9.2.2 **6 m; 37.68 m; 113.04 m^2.**
$r=d/2$; $C=\pi d$; $A=\pi r^2$.

9.2.3 **56.52 m^2; 18.84 m.**
Half the circle area; half the circumference.

9.2.4 **272.52 m^2; 66.84 m.**
Area 216+56.52; boundary 18+18+12+18.84.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume (continued)

PRACTICE 9.2 · CIRCLES AND COMPOSITES

9.2.5 **It is an internal join between lawn and stage, not an outer edge.**
Fencing follows the outside boundary only.

9.2.6 **Circumference doubles; area becomes four times as large.**
C is proportional to r; A to r^2.

PRACTICE 9.3 · SURFACE AREA

9.3.1 **52 cm^2.**
2(12+8+6).

9.3.2 **216 cm^2.**
6×6^2.

9.3.3 **2.52 m^2.**
Base 0.72 + long sides 1.20 + short sides 0.60.

9.3.4 **The top area lw.**
The top is not an outside face to paint.

9.3.5 **0.84 L needed; buy 1 tin.**
$3 \times 2.52 = 7.56$ m^2; $7.56 \div 9 = 0.84$; round tins up.

9.3.6 **98 cm^2 and 52 cm^2. The compact box uses less material.**
Same volume does not imply same surface area.

PRACTICE 9.4 · VOLUME AND CAPACITY

9.4.1 **1,000 cm^3 = 1 L.**
20×10×5.

9.4.2 **280 cm^3.**
½×8×5×14.

9.4.3 **60 L.**
60,000 $cm^3 \div 1{,}000$.

9.4.4 **2,500 cm^3; 450 cm^3; 800 L.**
Use 1 L=1,000 cm^3 and 1 m^3=1,000 L.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume (continued)

PRACTICE 9.4 · VOLUME AND CAPACITY

9.4.5 **1 m^3=1,000,000 cm^3=1,000 L. Volume scales in three dimensions.**
Cube the length conversion.

9.4.6 **12 cm; volume becomes 8 times as large.**
Length=V÷A; scale factor 2^3=8.

EXPLORATION · THE PACKAGING PROBLEM

9.B1 **98, 76, 70, 68, 56, 52 cm^2.**
Use 2(lw+lh+wh).

9.B2 **Least 2×3×4 (52 cm^2); most 1×1×24 (98 cm^2).**
Compare the completed surface areas.

9.B3 **Surface areas 54, 78, 110 cm^2; yes, the cube uses least.**
Compact dimensions reduce exposed face area.

9.B4 **Sample: 3×3×4 has area 66 cm^2; 1×6×6 has area 96 cm^2. Recommend 3×3×4 for less cardboard.**
Both volumes are 36; compare surface areas.

9.B5 **Sample: product shape, stacking, opening strength or shelf display may matter. The pattern compares closed boxes with whole-number edges and ignores waste, tabs and thickness.**
A mathematical optimum is not the only design criterion.

CHECKPOINT

9.CK1 **30 cm^2.**
½×10×6.

9.CK2 **28.26 cm^2.**
3.14×9.

9.CK3 **96 cm^2.**
6×16.

9.CK4 **1 L.**
1,000 cm^3.

APPLICATION · THE COURTYARD PROJECT

9.D1 **216 m^2; 56.52 m^2; 272.52 m^2.**
Rectangle plus semicircle of radius 6 m.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume (continued)

APPLICATION · THE COURTYARD PROJECT

9.D2 **66.84 m; turf $1,620; paving $2,373.84; fencing $1,871.52.**
Boundary 18+18+12+18.84; multiply each quantity by its price.

9.D3 **864 L; 22 bags; $209.**
One bed 0.36 m^3=360 L; 80% is 288 L; three beds 864 L; ceil(864/40)=22.

9.D4 **7.56 m^2 outside area; 0.84 L paint; buy 1 tin for $22.**
One bed 2.52 m^2; three beds 7.56; divide by 9 and round tins up.

9.D5 **$6,096.36, over by $96.36. With 10% contingency: $6,706.00 (to the nearest cent), over by $706.00.**
Add 1620+2373.84+1871.52+209+22; multiply by 1.10.

ASSESSMENT PRACTICE · THE MEASURE EXPLAINER

9.S1 **24 cm^3; 52 cm^2.**
4×3×2; 2(12+8+6).

9.S2 **Cardboard and paint use surface area; filling uses volume or capacity.**
Boundary length would use perimeter.

9.S3 **Sample: "Area uses square units, volume cubic units; 1,000 cm^3 = 1 L."**
Units reveal whether a formula matches the quantity.

MIXED PRACTICE 9

9.M1 **36 cm^2.**
base×height.

9.M2 **31.4 cm.**
πd.

9.M3 **62 cm^2.**
2(10+15+6).

9.M4 **20 L.**
20,000 cm^3÷1,000.

9.M5 **Use the perpendicular distance from the base to the opposite vertex; a slant side is not the height unless it is perpendicular.**
Area requires base×perpendicular height÷2.

ANSWERS AND REASONING

Chapter 9 · Perimeter, area, surface area and volume (continued)

MIXED PRACTICE 9

9.M6 **25.12 m²; 12.56 m.**
½πr² and πr.

9.M7 **5 cm.**
60÷12.

9.M8 **Surface area ×4; volume ×8.**
Area scales with the square, volume with the cube.

9.M9 **3 tins.**
20÷9≈2.22; round up.

9.M10 **3×3×4: 66 cm² versus 146 cm², saving 80 cm².**
Use 2(lw+lh+wh).

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs

INVESTIGATE FURTHER

10.B6 **Wider intervals give a smoother, simpler picture with fewer bars; narrower intervals show more detail but noisier bars. For families, a 30-minute interval with stated boundaries is usually clearer.**

Grouping changes the picture without changing the data, so the display must be chosen for the audience and the boundaries must be stated. The justification should reference the shape of the distribution.

10.B7 **The increase is 55 units, which is 110% of 50, so sales more than doubled. A corrected headline could be "sales rose 110%".**

Percentage change is measured against the original value, so the claim must be checked against 50, not against the new total.

EXPLAIN, JUSTIFY AND CORRECT

10.C1 **A bar chart compares separate categories with gaps between bars; a histogram groups numerical data into intervals with touching bars. Lunch choice suits a bar chart; screen-time values suit a histogram.**

The explanation names what is on each axis and why the bars touch or are separated, which is the practical difference.

10.C2 **The mean is larger than the median, which suggests a few high values or an outlier pulling the average upwards.**

Comparing the two summaries diagnoses the shape of the distribution, and naming the cause makes the explanation useful.

10.C3 **Fourteen similar slices have similar angles, so the eye cannot rank them, and labels crowd together. A bar chart sorted by frequency is easier to read.**

The explanation refers to what the reader can actually perceive, which is the purpose of choosing a display.

10.C4 **90% of 20 is 18 people while 50% of 200 is 100 people, so the percentage alone says nothing about the amount.**

The calculation shows that the denominator decides the amount, which is why every percentage needs its whole.

APPLY IN A NEW CONTEXT

10.D5 **Spending ≈ 96 × 4.20 + 74 × 3.10 + 30 × 5.50 = 403.20 + 229.40 + 165.00 = 797.60; the school share is 403.20 ÷ 797.60 ≈ 50.5%.**

Each category is multiplied by its price and the parts are summed; the share is then measured against the total spending, not against the number of students.

10.D6 **The difference is 25 units, about 2% of 1,240, so the claim exaggerates a small change. A fairer statement is "January was slightly busier than December, by about 2%."**

Percentage change against the original value shows the true size of the difference, and the axis choice explains why the graph looked dramatic.

10.D7 **A dot plot or histogram with intervals of 10; the median is (68 + 70) ÷ 2 = 69 thousand steps; a supported claim is "half the class walked at least 69 thousand steps".**

The median is stated with the sample size, and the claim is limited to what the display and summary actually show.

READY CHECK

10.R1 **200.**

Add the category counts.

10.R2 **48%.**

Divide and multiply by 100.

10.R3 **4.**

Sum 12÷3.

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs (continued)

READY CHECK

10.R4 **90°.**
A quarter turn.

10.R5 **2, 3, 5, 8.**
Sort before finding a median.

PRACTICE 10.1 · DATA AND VARIABLES

10.1.1 **Nominal categorical; ordinal categorical; discrete numerical; continuous numerical.**
Name, order, count and measurement are different variable types.

10.1.2 **96.**
Add the three counts.

10.1.3 **It is an ordered category or label; an average year code may not describe a meaningful category.**
Variable type determines appropriate summaries.

10.1.4 **70; 38/70 ≈ 54.3%.**
Row total 70; use the row as denominator.

10.1.5 **No. They are category labels, not counts or measurements.**
Nominal codes have no numerical order or distance.

10.1.6 **30; 15%.**
200−96−74; 30/200×100.

PRACTICE 10.2 · BAR AND PIE CHARTS

10.2.1 **172.8°, 133.2°, 54°.**
Multiply each share by 360°.

10.2.2 **48%.**
96/200×100.

10.2.3 **Bar height encodes frequency; a truncated axis exaggerates differences.**
Start the frequency axis at zero so the visual ratio matches the data ratio.

10.2.4 **A clustered bar chart (or a well-labelled two-way table).**
It keeps year groups visible side by side.

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs (continued)

PRACTICE 10.2 · BAR AND PIE CHARTS

10.2.5 **Yes, they sum to 100%.**
A pie chart partitions the whole.

10.2.6 **30.**
54/360×200.

PRACTICE 10.3 · DISTRIBUTION AND AVERAGES

10.3.1 **Mean 4; median 3; mode 3; range 6.**
Sum 16÷4; middle two 3 and 3; 8−2.

10.3.2 **Mean 158; median 137.5; mode 120; range 315 minutes.**
Use the definitions directly.

10.3.3 **A tail of high values pulls the mean upward; the median depends only on position.**
The distribution is right-skewed.

10.3.4 **1, 9, 9, 5, 3, 3.**
Count each value once using the stated boundary rule.

10.3.5 **A value on a boundary must belong to exactly one interval; otherwise counts can be duplicated or omitted.**
Equal widths also make heights comparable.

10.3.6 **168 minutes.**
Weighted midpoint sum ÷30 = 168. A grouped estimate need not equal the raw mean (158 minutes).

PRACTICE 10.4 · TRENDS AND CLAIMS

10.4.1 **Overall +8; largest fall 16 from year 1 to year 2.**
96−88=8; 88−72=16.

10.4.2 **About 18.2%.**
16/88×100.

10.4.3 **False: 80% increase, not 100%.**
Doubling would require 100.

10.4.4 **The area becomes four times as large, exaggerating a twofold value.**
Area scales with the square of linear size.

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs (continued)

PRACTICE 10.4 · TRENDS AND CLAIMS

10.4.5 **20% versus about 11.4%; Year 9 is higher.**
Use each year group as its denominator.

10.4.6 **Yes, 100% increase, but ask for the time period, population size and longer trend; small counts can make a dramatic headline.**
A relative change needs context.

EXPLORATION · HOW SHOULD DATA BE GROUPED?

10.B1 **1, 9, 9, 5, 3, 3.**
Count each value once.

10.B2 **10, 14, 6.**
Combine adjacent 60-minute intervals.

10.B3 **60-minute intervals show the cluster around 60–180 more clearly; 120-minute intervals lose more detail.**
Narrower equal intervals preserve more shape.

10.B4 **Mean 158; median 137.5; mode 120; range 315 minutes. Six students are at 240 or more.**
Use the raw ordered data; count 240,240,270,300,330,360.

10.B5 **Not supported: mean 158 min, median 137.5 min; only 6/30=20% reach 240 min. Recommend a labelled 60-minute histogram with median and a note about the high-value tail.**
Choose an average and grouping that match the claim and audience.

CHECKPOINT

10.CK1 **Discrete numerical.**
It counts separate whole values.

10.CK2 **54°.**
30/200×360.

10.CK3 **3.**
The middle ordered value.

10.CK4 **80%.**
40/50×100.

APPLICATION · THE CANTEEN SURVEY

10.D1 **48%; angles 172.8°, 133.2°, 54°.**
Use denominator 200.

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs (continued)

APPLICATION · THE CANTEEN SURVEY

10.D2 **Year 9: 20%; Year 7: about 11.4%.**
13/65 and 8/70.

10.D3 **School $403.20; packed $229.40; off-site $165.00; total $797.60. Extra canteen revenue $100.80.**
Multiply counts by prices; the switch adds 24×4.20 to canteen revenue.

10.D4 **Sample headline: "Off-site lunches are 15% overall, higher in Year 9." The claim is not supported by one share alone; compare year groups and time data.**
15% overall and 20% in Year 9 are descriptive, not proof of a crisis.

ASSESSMENT PRACTICE · THE DATA EXPLAINER

10.S1 **A clustered bar chart or two-way table; it preserves year-group comparisons.**
A single pie chart hides the year structure.

10.S2 **48%; angles sum to 360° (172.8+133.2+54).**
The whole is 200 students.

10.S3 **Sample claim: "School lunch is the most common choice (48%)." Warning: a bar axis not starting at zero can exaggerate differences.**
The claim quotes a denominator and the warning names a visual distortion.

MIXED PRACTICE 10

10.M1 **Continuous numerical.**
It is measured on a scale.

10.M2 **133.2°.**
74/200×360.

10.M3 **Mean 6; median 5; mode 4; range 6.**
Sum 24÷4; middle (4+6)/2; 10−4.

10.M4 **36%.**
18/50×100.

10.M5 **It exaggerates visual differences by hiding the zero baseline.**
Bar height should be proportional to frequency.

10.M6 **20%.**
18/90×100.

ANSWERS AND REASONING

Chapter 10 · Data, tables and graphs (continued)

MIXED PRACTICE 10

10.M7	**A high-value tail or outliers pulling the mean upward.** The median is resistant to extreme values.
10.M8	**11.** (4×5+6×15)/10.
10.M9	**25%; 90°.** 20/80×100 and ×360.
10.M10	**The 100% increase is correct, but the base is tiny; ask for population, time period and longer trend before judging risk.** Relative change without context can mislead.

FORMULA BANK

Formula bank (1 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

CHAPTER 1 · NUMBER SYSTEMS AND PLACE VALUE

Number families

natural ⊂ whole ⊂ integer ⊂ rational

Place value (base ten)

each place is worth 10× the place to its right

Rounding to a stated place

look at the digit to the right; 5 or more rounds up

Expanded form

3,048.307 = 3×1000 + 0×100 + 4×10 + 8×1 + 3×0.1 + 0×0.01 + 7×0.001

Powers of ten

×10^n shifts digits n places left; ÷10^n shifts them right

CHAPTER 2 · POWERS, EXPANDED FORM AND ORDER OF OPERATIONS

Index notation

a^n means n copies of the base a multiplied together

Special powers

a^1 = a and a^0 = 1 for any nonzero base a

Powers of ten

10^0 = 1, 10^1 = 10, 10^2 = 100, 10^3 = 1,000

Square and cube roots

√49 = 7 because $7^2 = 49$; $\sqrt[3]{64} = 4$ because $4^3 = 64$

Order of operations

brackets, then indices, then × and ÷ left to right, then + and − left to right

FORMULA BANK

Formula bank (2 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

CHAPTER 3 · FACTORS, MULTIPLES AND PRIMES

Factor and multiple

a is a factor of b when $b = a \times k$ for a whole number k

Prime and composite

a prime has exactly two factors; 1 is neither

Square-root boundary

a composite n has a prime factor $\leq \sqrt{n}$

HCF and LCM

HCF takes the lower shared prime powers; LCM takes the highest

HCF × LCM

$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

CHAPTER 4 · FRACTIONS AND DECIMALS

Equivalent fractions

$a/b = (a \times k)/(b \times k)$ for any nonzero k

Simplest form

divide numerator and denominator by their HCF

Addition and subtraction

rewrite with a common denominator, then combine numerators

Multiplication

$(a/b) \times (c/d) = ac/bd$

Division

$(a/b) \div (c/d) = (a/b) \times (d/c)$

Terminating decimals

a fraction in simplest form terminates exactly when the denominator has only the prime factors 2 and 5

FORMULA BANK

Formula bank (3 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

CHAPTER 5 · PERCENTAGES AND APPLICATIONS

Per cent

% means “out of 100”; 45% = 45/100 = 0.45

Percentage of a quantity

p% of A = (p/100) × A

Increase and decrease

multiply by 1 + p/100 or 1 − p/100

Successive changes

multipliers compose: (1 + r)(1 + s)

Profit

gross profit = revenue − goods cost; net profit = gross profit − fixed costs

CHAPTER 6 · PATTERNS, EXPRESSIONS AND EQUATIONS

Like terms

only terms with the same letters and powers combine

Distributive law

$a(b + c) = ab + ac$

Product of brackets

$(x + a)(x + b) = x^2 + (a + b)x + ab$

Solving an equation

apply the same inverse operation to both sides

Checking

substitute the solution into the original statement

FORMULA BANK

Formula bank (4 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

CHAPTER 7 · LINES, ANGLES, TRIANGLES AND CONSTRUCTIONS

Angles on a straight line

sum = 180°

Angles around a point

sum = 360°

Parallel lines

corresponding and alternate angles are equal; co-interior angles are supplementary

Triangle

angle sum = 180°; exterior angle = sum of the two opposite interior angles

Polygons

interior sum = (n − 2) × 180°; exterior sum = 360°

Triangle inequality

the sum of any two sides exceeds the third

Constructions

equal radii from two points locate points that are equidistant from them

CHAPTER 8 · SETS AND CLASSIFICATION

Element and subset

$a \in A$ means a belongs to A; $B \subset A$ means every element of B is in A

Number of subsets

a set with n elements has 2^n subsets

Union and intersection

$A \cup B$ counts each element once; $A \cap B$ is the overlap

Counting formula

$|A \cup B| = |A| + |B| - |A \cap B|$

Number sets

whole $\subset$ integer $\subset$ rational; irrational numbers are outside them

FORMULA BANK

Formula bank (5 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

CHAPTER 9 · PERIMETER, AREA, SURFACE AREA AND VOLUME

Rectangle

$P = 2(l + w)$; $A = lw$

Triangle

$A = \frac{1}{2} \times$ base $\times$ perpendicular height

Parallelogram and trapezoid

$A =$ base $\times$ height; $A = \frac{1}{2}(a + b)h$

Circle

$C = \pi d = 2\pi r$; $A = \pi r^2$

Cuboid

surface area $= 2(lw + lh + wh)$; $V = lwh$

Prism

$V =$ cross-section area $\times$ length; $1\ m^3 = 1{,}000$ L

CHAPTER 10 · DATA, TABLES AND GRAPHS

Frequency table

organise categories; counts sum to the population

Pie chart

sector angle = (category ÷ total) × 360°

Mean, median, mode

mean = total ÷ count; median is the middle value; mode is the most frequent

Range

range = largest value − smallest value

Percentage change

(change ÷ original value) × 100%

GLOSSARY

Glossary (1 of 4)

Acute angle (ch. 7)
An angle greater than 0° and less than 90°.

Alternate angles (ch. 7)
Angles in matching positions on opposite sides of a transversal; equal when the lines are parallel.

Associative law (ch. 6)
Grouping does not change a sum or product: (a + b) + c = a + (b + c).

Bar chart (ch. 10)
A display using equal-width bars to compare categories; the frequency axis starts at zero.

Base (of a power) (ch. 2)
The number that is used repeatedly as a factor.

Benchmark (ch. 4)
A familiar value such as 1/2 used to judge the size of another quantity.

Categorical variable (ch. 10)
A variable that names groups; nominal if unordered, ordinal if ordered.

Co-interior angles (ch. 7)
Angles between parallel lines on the same side of a transversal; supplementary.

Common denominator (ch. 4)
A shared multiple of two or more denominators, used to add or compare fractions.

Common factor (ch. 3)
A number that divides each of two or more numbers exactly.

Commutative law (ch. 6)
Order does not change a sum or product: a + b = b + a.

Composite number (ch. 3)
A whole number greater than 1 with more than two positive factors.

Continuous data (ch. 10)
Numerical data that can take any value in an interval, such as height.

Corresponding angles (ch. 7)
Angles in the same relative position at each intersection; equal when the lines are parallel.

Cross-section (ch. 9)
The shape exposed by cutting a solid parallel to a face; its area times the length gives a prism's volume.

Cube number (ch. 2)
A whole number multiplied by itself twice, such as 8 = 2 × 2 × 2.

Cuboid (ch. 9)
A box shape with six rectangular faces.

Decimal place (ch. 1)
A position to the right of the decimal point; each place is one tenth of the place to its left.

Denominator (ch. 4)
The number of equal parts a whole is divided into.

Diameter (ch. 9)
A chord passing through the centre of a circle; twice the radius.

Discrete data (ch. 10)
Numerical data that counts separate values, such as the number of siblings.

Distributive law (ch. 6)
a(b + c) = ab + ac.

Divisibility test (ch. 3)
A shortcut that uses digits to decide whether one whole number divides another.

Element (ch. 8)
An object that belongs to a set, written a ∈ A.

Equation (ch. 6)
A statement that two expressions are equal.

Equivalent fractions (ch. 4)
Fractions with the same value, such as 1/2 and 3/6.

GLOSSARY

Glossary (2 of 4)

Estimate (ch. 1)
A value obtained using friendly numbers, used to predict or check a result.

Exponent (ch. 2)
The number of times a base is used as a factor.

Expression (ch. 6)
A combination of numbers, letters and operations that names a quantity.

Exterior angle (ch. 7)
The angle between one side of a polygon and the extension of an adjacent side.

Factor (ch. 3)
A number that divides another number exactly.

Factor tree (ch. 3)
A diagram that splits a number into factors until every endpoint is prime.

Frequency (ch. 10)
The number of times a value or category occurs.

Gross profit (ch. 5)
Revenue minus the cost of goods.

Highest common factor (HCF) (ch. 3)
The largest number that divides each of two or more numbers exactly.

Histogram (ch. 10)
A display for grouped numerical data with touching bars and stated interval boundaries.

Index notation (ch. 2)
A compact way of writing repeated multiplication, such as 2^5.

Integer (ch. 1)
A whole number that may be positive, negative or zero.

Intersection (ch. 8)
The elements common to two sets, written $A \cap B$.

Irrational number (ch. 8)
A number that cannot be written as a fraction of two integers, such as $\sqrt{2}$.

Isosceles triangle (ch. 7)
A triangle with two equal sides and two equal base angles.

Like terms (ch. 6)
Terms with the same letters and the same powers, which may be combined.

Lowest common multiple (LCM) (ch. 3)
The smallest positive number that is a multiple of each of two or more numbers.

Mean (ch. 10)
The total divided by the number of values.

Median (ch. 10)
The middle value when the data are ordered.

Mixed number (ch. 4)
A whole number written with a fraction, such as $2\frac{1}{2}$.

Mode (ch. 10)
The most frequent value in a data set.

Multiple (ch. 3)
The product of a number and a whole number.

Natural number (ch. 1)
A counting number: 1, 2, 3, …

Net (ch. 9)
A flat pattern that folds into a solid.

Numerator (ch. 4)
The number of equal parts taken.

Obtuse angle (ch. 7)
An angle greater than 90° and less than 180°.

GLOSSARY

Glossary (3 of 4)

Order of operations (ch. 2)
Brackets, indices, multiplication and division, then addition and subtraction.

Parallel lines (ch. 7)
Lines in the same plane that never meet.

Percentage (ch. 5)
A number of parts out of one hundred.

Perimeter (ch. 9)
The total boundary length of a shape.

Perpendicular bisector (ch. 7)
The line that cuts a segment in half at right angles.

Pie chart (ch. 10)
A display using sectors to show parts of one whole.

Place value (ch. 1)
The value a digit has because of its position in a number.

Polygon (ch. 7)
A closed shape with straight sides.

Prime factorisation (ch. 3)
Writing a number as a product of primes.

Prime number (ch. 3)
A whole number greater than 1 with exactly two positive factors.

Prism (ch. 9)
A solid with a constant cross-section.

Profit (ch. 5)
What remains when costs are subtracted from revenue.

Proper fraction (ch. 4)
A fraction whose numerator is smaller than its denominator.

Radius (ch. 9)
The distance from the centre of a circle to any point on it.

Range (ch. 10)
The largest value minus the smallest value.

Rational number (ch. 1)
A number that can be written as a fraction of two integers.

Reciprocal (ch. 4)
The number that multiplies with another to give 1, such as 3/4 and 4/3.

Reflex angle (ch. 7)
An angle greater than 180° and less than 360°.

Regular polygon (ch. 7)
A polygon with all sides and all angles equal.

Revenue (ch. 5)
The total money taken from sales.

Root (ch. 2)
The number that, raised to a given power, gives a stated value.

Rounding (ch. 1)
Replacing a number by a nearby value at a stated place.

Scalene triangle (ch. 7)
A triangle with three different side lengths.

Sector angle (ch. 10)
The angle of a pie-chart sector: (category ÷ total) × 360°.

Sequence (ch. 6)
An ordered list of numbers or shapes following a rule.

Simplest form (ch. 4)
A fraction whose numerator and denominator have no common factor greater than 1.

GLOSSARY

Glossary (4 of 4)

Square number (ch. 2)
A whole number multiplied by itself, such as $9 = 3 \times 3$.

Square root (ch. 2)
The non-negative number that squares to give a stated value.

Subset (ch. 8)
A set whose every element belongs to another set, written $B \subset A$.

Substitution (ch. 6)
Replacing a letter by its value in an expression.

Surface area (ch. 9)
The total area of the outside faces of a solid.

Terminating decimal (ch. 4)
A decimal that stops after a finite number of places.

Transversal (ch. 7)
A line that crosses two or more lines.

Triangle inequality (ch. 7)
The sum of any two sides of a triangle exceeds the third side.

Two-way table (ch. 10)
A table that classifies data by two variables at once.

Union (ch. 8)
The elements in either of two sets, written $A \cup B$.

Universal set (ch. 8)
The set of all elements under consideration in a problem.

Variable (ch. 10)
A quantity that can take different values.

Venn diagram (ch. 8)
A diagram using overlapping regions to show set relationships.

Vertically opposite angles (ch. 7)
Angles formed opposite each other when two lines cross; equal.

Volume (ch. 9)
The amount of space inside a solid, in cubic units.

Whole number (ch. 1)
A natural number together with zero.

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- First-edition MYP Mathematics 1 Student Coursebook (repository archive copy, SHA-256 131a3a99…b3f0). Used as the principal content reference for chapter scope, teaching order, examples and exercise coverage. Preserved unchanged.

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